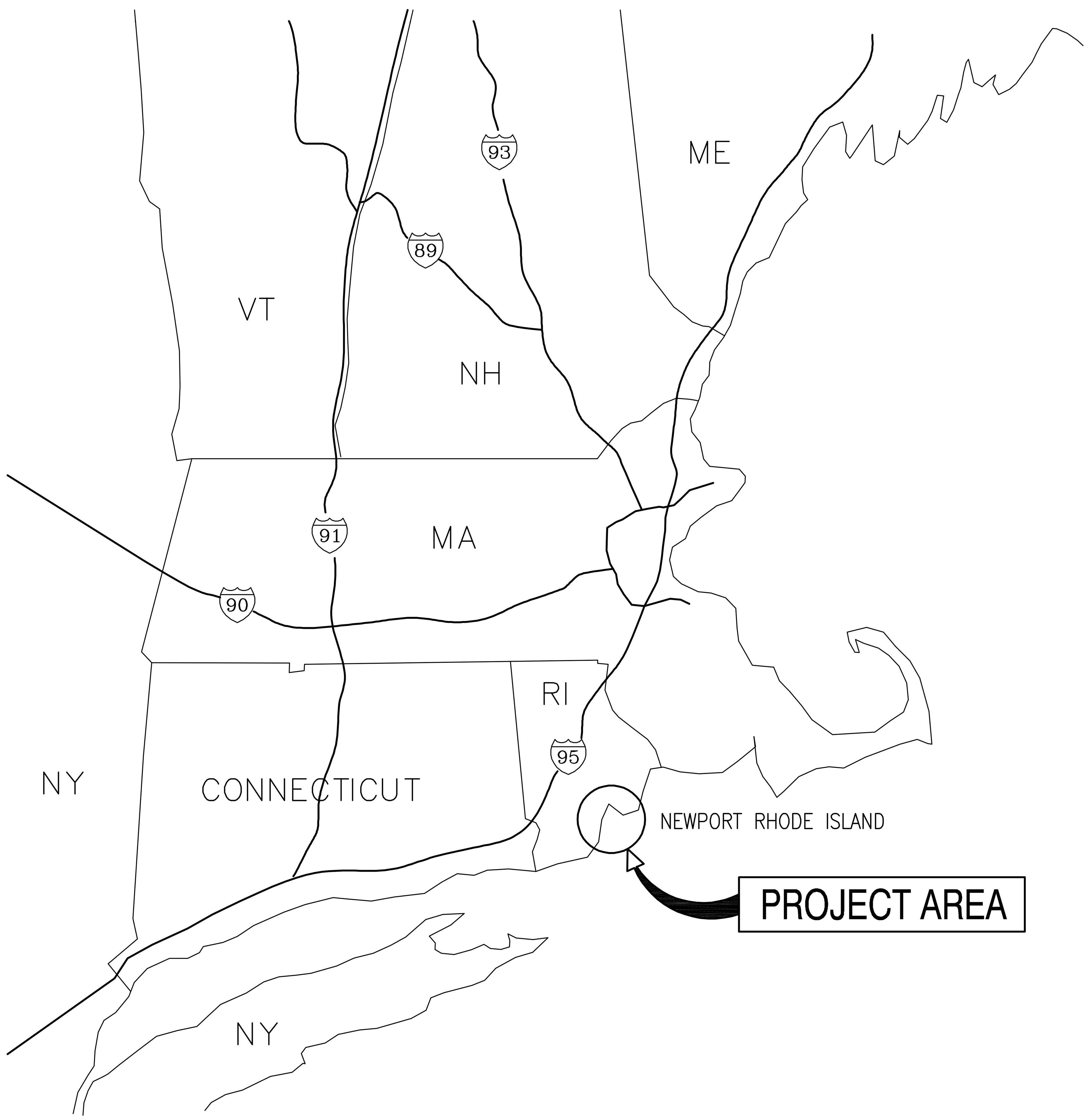
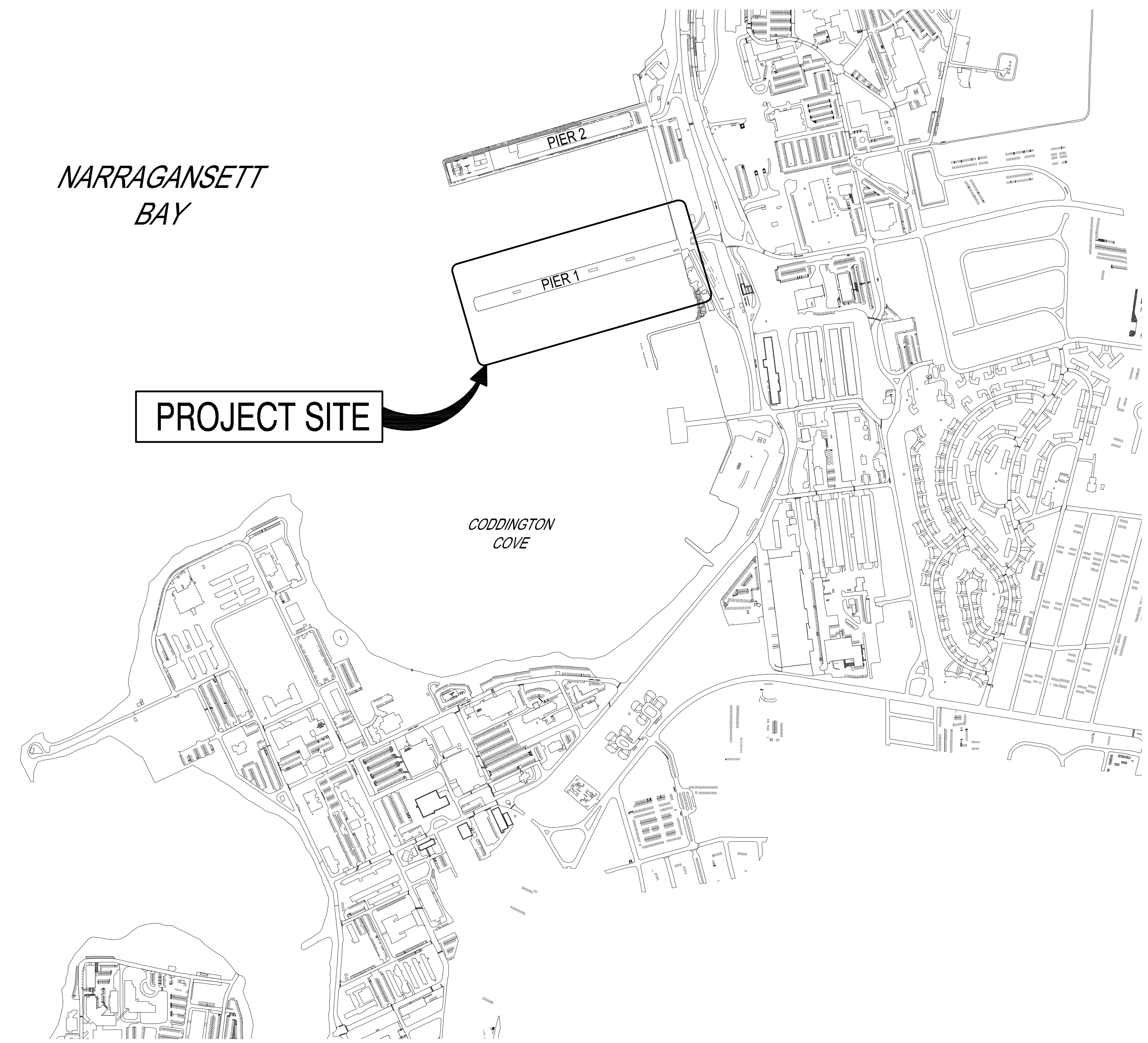
USCG OCR HOMEPORT INITIATIVE
OPC PIER
NAVAL STATION NEWPORT
NEWPORT, RHODE ISLAND



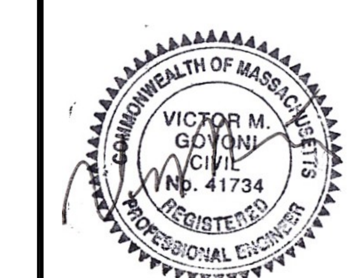
VICINITY MAP
SCALE: NTS

PLAN NORTH



LOCATION MAP
SCALE: NTS

PLAN NORTH



SYM | DESCRIPTION | DATE | APPR

A/E INFO

APPROVED
James E. Donahue, R.A.
FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES CHG	DRW HBF	CHK MN
PM/DM	MRI/CWJ
BRANCH MANAGER	NEK
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
TITLE SHEET

SCALE:	AS NOTED
EPROJECT NO.:	1826479
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12936570
SHEET 1 OF 247	

G-001

DRAWFORM REVISION: 25 AUGUST 2020

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SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
VICTOR M. GOYONI
CIVIL
No. 41734
REGISTERED PROFESSIONAL ENGINEER
SEAL

Stantec

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY

Approved by Frank Cole USCG MASI
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CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
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SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936571
SHEET 2 OF 247
G-002

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Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

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LEGEND

#	KEY NOTE
#	CABLE DESIGNATION-SEE SCHEDULE
#	LUMINAIRE SYMBOL.
	FEATURES TO BE PROVIDED UNDER THIS CONTRACT SHOWN THUS
	EXISTING FEATURES SHOWN THUS
	DISCONNECT, REMOVE AND DISPOSE OF EXISTING ELECTRICAL EQUIPMENT/UTILITIES AND ASSOCIATED APPURTENANCES SHOWN ON PLAN.
K	KEY INTERLOCK
	TRANSFORMER
	DELTA CONNECTION, 3∅, 3 W.
	GROUNDED WYE CONNECTION, 3∅., 4W.
	600V POWER AIR CIRCUIT BREAKER, DRAWOUT TYPE
	SHIP-TO-SHORE RECEPTACLE, 450V, 500A, 3∅, 3P. MIL. SPEC MIL. C24368
	EQUIPMENT RECEPTACLE
	FUSE - DRAWOUT TYPE
T	DRY TYPE TRANSFORMER
	FUSE
A AS	AMMETER, AMMETER SWITCH
WHD	DEMAND WATTHOUR METER
V VS	VOLTMETER, VOLTMETER SWITCH
CS	CONTROL SWITCH
64	GROUND DETECTION RELAY
3P 30A 30A	FUSED SAFETY SWITCH, NO. OF POLES, SIZE OF SW. AND FUSE SIZE AS SHOWN, 480V. U.O.N. STAINLESS STEEL ENCLOSURE,NEMA 4X. NF INDICATES NON-FUSED
R G	LED INDICATOR LIGHT (PUSH-TO-TEST WHERE SHOWN) R = RED, G = GREEN
WP	RECEPTACLE-125V., 15A SINGLE CONVENIENCE NEMA 5-15, WEATHERPROOF.
SST	SOLID STATE TRIP
	CURRENT TRANSFORMER
SPD	SURGE PROTECTIVE DEVICE
	HIGH MAST LIGHTING POLE
	NAVIGATION LIGHT POLE
C	COMMUNICATIONS MANHOLE
P	POWER MANHOLE
LIM	LINE INSULATION MONITORING SYSTEM
M	METER SOCKET AND AMI METER
TPC	TRI-PHASE CONTROLLER
PLR	PHASE LOSS RELAY
	LIGHTING POLE, TYPE AS SHOWN
	LED LIGHTING FIXTURE
S_{WP}	SINGLE POLE 20A, 125V, SWITCH "WP" INDICATES WEATHERPROOF
	HOMERUN TO PANELBOARD, SIZE AND NUMBER OF CONDUCTORS AND CONDUIT AS SHOWN ON THE PLANS. CONDUITS IN DECK MUST BE PVC (SCH. 40). CONDUITS RUN EXPOSED (UNDER DECK) MUST BE PVC COATED RIGID GALVANIZED STEEL.
G	GROUND CONDUCTOR (BCSD) UON
	GROUNDING CONNECTION. EXOTHERMIC WELDING PROCESS
	GROUND ROD CONNECTION
	HEAT TRACE
	600V CIRCUIT BREAKER, MOLDED CASE, 3 PH. U.O.N.
2 480V/120V	POTENTIAL TRANSFORMER WITH NUMBER AND RATIO AS INDICATED.
3 500:5	CURRENT TRANSFORMER WITH NUMBER AND RATIO AS INDICATED.
NO NC	AUXILIARY CONTACTS N0-NORMALLY OPEN NC-NORMALLY CLOSED
	LIGHTNING ARRESTER
	GROUND ROD
PC	PHOTOCELL
	PUSH BUTTON-MOMENTARY
86	LOCKOUT RELAY
	TELEPHONE TERMINAL CABINET
A X / 1 2	EXISTING CABLE DESIGNATION (NOT USED); DUCTBANK SECTION LOOKING IN DIRECTION OF ARROWS; SPARE DUCT SAME SIZE AS ADJACENT DUCT; PROPOSED CABLE DESIGNATION (TYPICAL). SEE CABLE SCHEDULE; HEAVY LINE INDICATES BOTTOM OF DUCTBANK
	DUCTBANK
T	EXISTING TELEPHONE DUCTBANK
	DETAIL NUMBER; DRAWING NUMBER WHERE SHOWN
F	FIBER ENCLOSURE WITH FIBER PATCH PANEL
J J	JUNCTION BOX
H	HEAT TRACE MULTIPLE ENTRY CONNECTION BOX. HT-X DENOTES HEAD TRACE PANEL, CIRCUIT NUMBER "X"
T	THERMOSTAT
	LIGHTNING PROTECTION AIR TERMINAL

GENERAL NOTES:

1. PROVIDE NECESSARY COMPONENTS REQUIRED FOR MAKING FINAL CONNECTION OF ALL EQUIPMENT INSTALLED OR MODIFIED AS PART OF THIS CONTRACT.
2. ALARM INDICATION AND CONTROL WIRING IN JUNCTION BOXES MUST BE WIRED TO NUMBERED TERMINAL STRIPS AND IDENTIFIED AS TO START AND END OF RUN.
3. ELECTRICAL EQUIPMENT INSTALLED AGAINST CONCRETE OR MASONRY WALLS MUST BE INSTALLED WITH A 1/4" SPACE BETWEEN THE EQUIPMENT AND THE MOUNTING SURFACE. SPACERS MUST BE STAINLESS STEEL, PVC OR NYLON.
4. PROVIDE TEST PITS (AND COORDINATE WITH CIVIL AS REQUIRED) TO DETERMINE LOCATION OF EXISTING UTILITIES BEFORE CONSTRUCTION OF DUCTBANK. DUCTBANK MUST COORDINATE WITH EXISTING AND OTHER PROPOSED UTILITIES. MODIFY PROFILES TO SUIT EXISTING CONDITIONS.
5. UTILITIES AND ELECTRICAL EQUIPMENT SHOWN ON PIER ACCESS ROAD WERE EXISTING AT THE TIME OF FIELD SURVEY AND MAY HAVE BEEN ALTERED BEFORE CONSTRUCTION BEGINS. CONTRACTOR MUST VERIFY INFORMATION SHOWN ON THE DRAWINGS. COORDINATE WITH THE GOVERNMENT BEFORE REMOVAL OF SAME.
6. EXTERIOR ENCLOSURES MUST BE NEMA 3RX OR 4X RATED AND UTILIZE TYPE 316 STAINLESS STEEL HARDWARE. NEMA 3RX ENCLOSURES MUST BE CONSTRUCTED OF TYPE 316 STAINLESS STEEL.
7. SEE USCG ELECTRICAL SHORE TIES CSTO REV B (SILC-CSTO-36 11 21 27 11-04) SECTION 2.4.6 FOR ADDITIONAL ELECTRICAL INSTALLATION AND ACCEPTANCE TESTING REQUIREMENTS THAT ARE NOT COVERED IN THE PROJECT SPECIFICATIONS.
8. CONDUIT TYPES MUST BE:
   A. LANDSIDE IN CONCRETE ENCASED DUCTBANK: PVC SCHEDULE 40
   B. ON PIER IN CONCRETE ENCASED DUCTBANK: PVC SCHEDULE 80
   C. EXPOSED BELOW PIER, ON PIER EXTERIOR, AND TURNING UP INTO TOPSIDE EQUIPMENT: RTRC TYPE XW.
   D. EXPOSED IN SUBSTATION P1E AND P1W: RTRC TYPE XW
   E. EXPOSED IN UTILITY TRENCH: RTRC TYPE XW.

ABBREVIATIONS

A, AMP	AMPERES
AC	ALTERNATING CURRENT
AF	AMPERE FRAME
AFF	ABOVE FINISHED FLOOR
AHU	AIR HANDLING UNIT
AIC	AMPERES INTERRUPTING CAPACITY
AM	AMMETER
AMI	ADVANCED METERING INFRASTRUCTURE
AT	AMPERE TRIP
ATS	AUTOMATIC TRANSFER SWITCH
AUX	AUXILIARY
AWG	AMERICAN WIRE GAUGE
BCHD	BARE COPPER HARD DRAWN
BCSD	BARE COPPER SOFT DRAWN
BKR	BREAKER
BLDGS	BUILDINGS
C	CONDUIT
CAB	CABINET
CASCON	CASUALTY CONTROL
CKT	CIRCUIT
COMB.	COMBINATION
COMM	COMMUNICATIONS
CONT.	CONTINUITY
CT	CURRENT TRANSFORMER
CU.	COPPER
DIA.	DIAMETER
DIST.	DISTRIBUTION
DLO	DIESEL LOCOMOTIVE CABLE
DN.	DOWN
DPDT	DOUBLE POLE, DOUBLE THROW
DT	DRY TYPE
DWG.	DRAWING
EA	EACH
ELEV	ELEVATION
EMER	EMERGENCY
EMH	ELECTRIC MANHOLE
ENCL.	ENCLOSURE
E.O.	ELECTRICALLY OPERATED
EQ.	EQUAL
EQUIP.	EQUIPMENT
ETS	EMERGENCY TRIP SWITCH
EX., EXIST.	EXISTING
EXP.	EXPANSION
F	FAHRENHEIT
FDRS	FEEDERS
FR	FRAME
FT	FEET, FOOT
G, GND	GROUND
GA	GAUGE
GND FAT	GROUND FAULT
GOV'T	GOVERNMENT
GRS	GALVANIZED RIGID STEEL
HD'RM	HEADROOM
HMI	HUMAN MACHINE INTERFACE
HOA	HAND OFF AUTO
HPS	HIGH PRESSURE SODIUM
HR	HOUR
HT	HEAT TRACE
HV	HIGH VOLTAGE
HVAC	HEATING VENTILATING AND AIR CONDITIONING
HZ	HERTZ
IC	INTERRUPTING CAPACITY
ID	IDENTIFICATION
INST	INSTANTANEOUS
KCMIL	THOUSANDS OF CIRCULAR MILLS
KV	KILOVOLTS
KVA	KILOVOLT AMP
LG	LONG TIME
LLF	LUMEN LOSS FACTOR
LT	LIGHT
LTG	LIGHTING
LV	LOW VOLTAGE
MAX	MAXIMUM
MCB	MAIN CIRCUIT BREAKER
MFR	MANUFACTURER
MH	MANHOLE
MHW	MEAN HIGH WATER
MIL	MILITARY
MIN.	MINIMUM
MISC.	MISCELLANEOUS
MLW	MEAN LOW WATER
MTG.	MOUNTING
MTL	MATERIAL
MV	MEDIUM VOLTAGE
N.C.	NORMALLY CLOSED
NEC	NATIONAL ELECTRICAL CODE
NEMA	NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
NEUT.	NEUTRAL
N.O.	NORMALLY OPEN
NO., #	NUMBER
NP	NAMEPLATE
OC	ON CENTER
OCT	BREAKER OVERCURRENT TRIP ALARM
OW	OILY WASTE
P	POLE
PC	PHOTOCELL
PH, ∅	PHASE
PNL	PANEL
PRI.	PRIMARY
PT	POTENTIAL TRANSFORMER
PVC	POLYVINYL CHLORIDE
PW	POTABLE WATER
RCPT	RECEPTACLE
RD	ROAD
REF	REFERENCE
REQ'D	REQUIRED
RMS	ROOT MEANS SQUARED
SCH	SCHEDULE
SEC.	SECONDARY
SEP	SEPARATE
SHT	SHORT TIME
SPEC	SPECIFICATION
STA	STATION
S.S.	STAINLESS STEEL
SW	SWITCH
SWGR	SWITCHGEAR
SYM	SYMMETRICAL
T	THERMOSTAT
TCS	TRIP CONTROL SWITCH (ONE PER BREAKER)
TEL	TELEPHONE
TR	TRIP
TS	TEST SWITCH
TVSS	TRANSIENT VOLTAGE SURGE SUPPRESSOR
TYP	TYPICAL
UL	UNDERWRITERS LABORATORY
UON	UNLESS OTHERWISE NOTED
V	VOLT
VM	VOLTMETER
W	WIRE, WATTS
WHD	WATT-HOUR DEMAND METER
WP	WEATHER PROOF
XFMR	TRANSFORMER
Z	IMPEDANCE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL LEGEND, ABBREVIATIONS & GENERAL NOTES

SCALE: AS NOTED

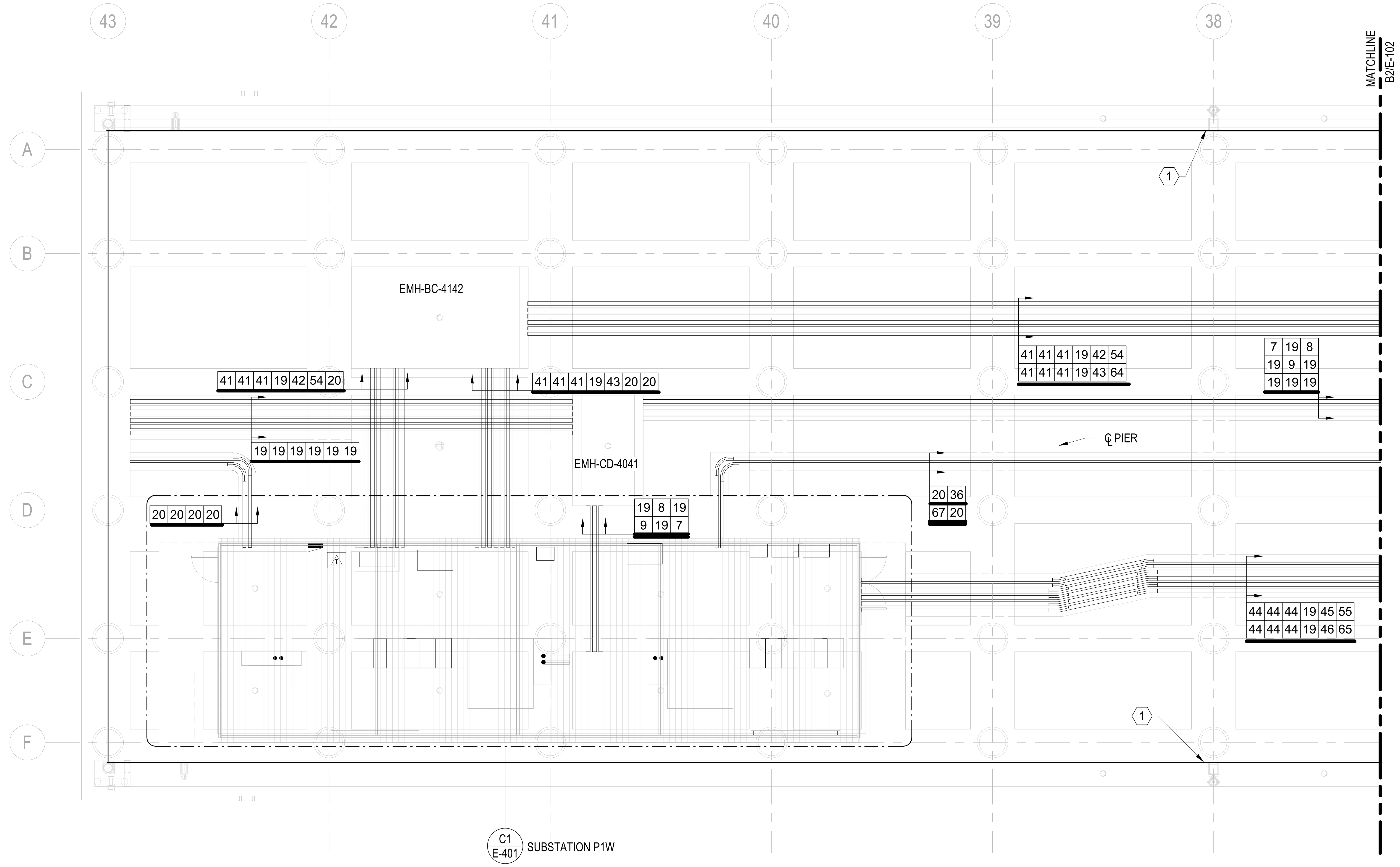
EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936759

SHEET 190 OF 247

E-001



B2 ELECTRICAL DECK PLAN - SHEET 1 OF 7

SCALE: 1/8" = 1'-0"

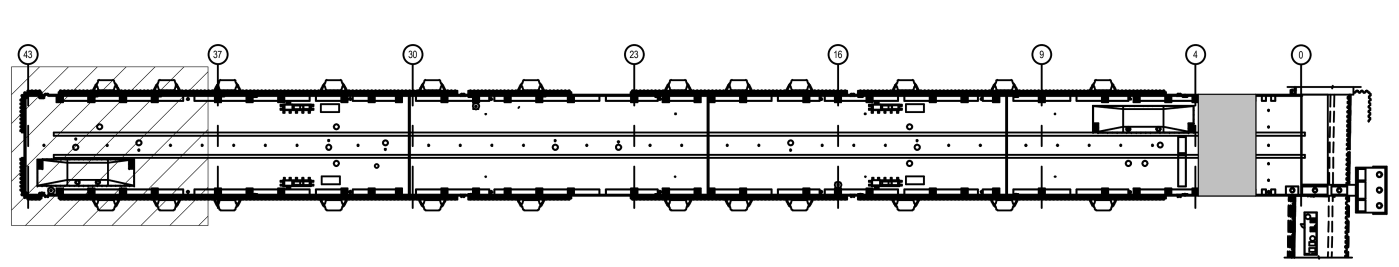
PLAN NORTH

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, E-603 AND E-604. FOR FEEDER SCHEDULE SEE DRAWINGS E-614 AND E-615.
2. PROVIDE EXPANSION/DEFLECTION FITTINGS ON ALL CONDUITS CROSSING EXPANSION JOINTS. SEE STRUCTURAL DRAWINGS FOR LOCATIONS.

SHEET KEYNOTES:

1. PROVIDE HEAT TRACING CABLE ON FRESH WATER PIPING IN UTILITY TRENCH. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.



KEY PLAN

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
REGISTERED PROFESSIONAL ENGINEER
SEAL

Stantec

A/E INFO

APPROVED James E. Donahue, R.A.
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF / DRW GLV / CHK RJW	
PM/DM	MRI/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL DECK PLAN - SHEET 1 OF 7

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936760
SHEET 191 OF 247

E-101

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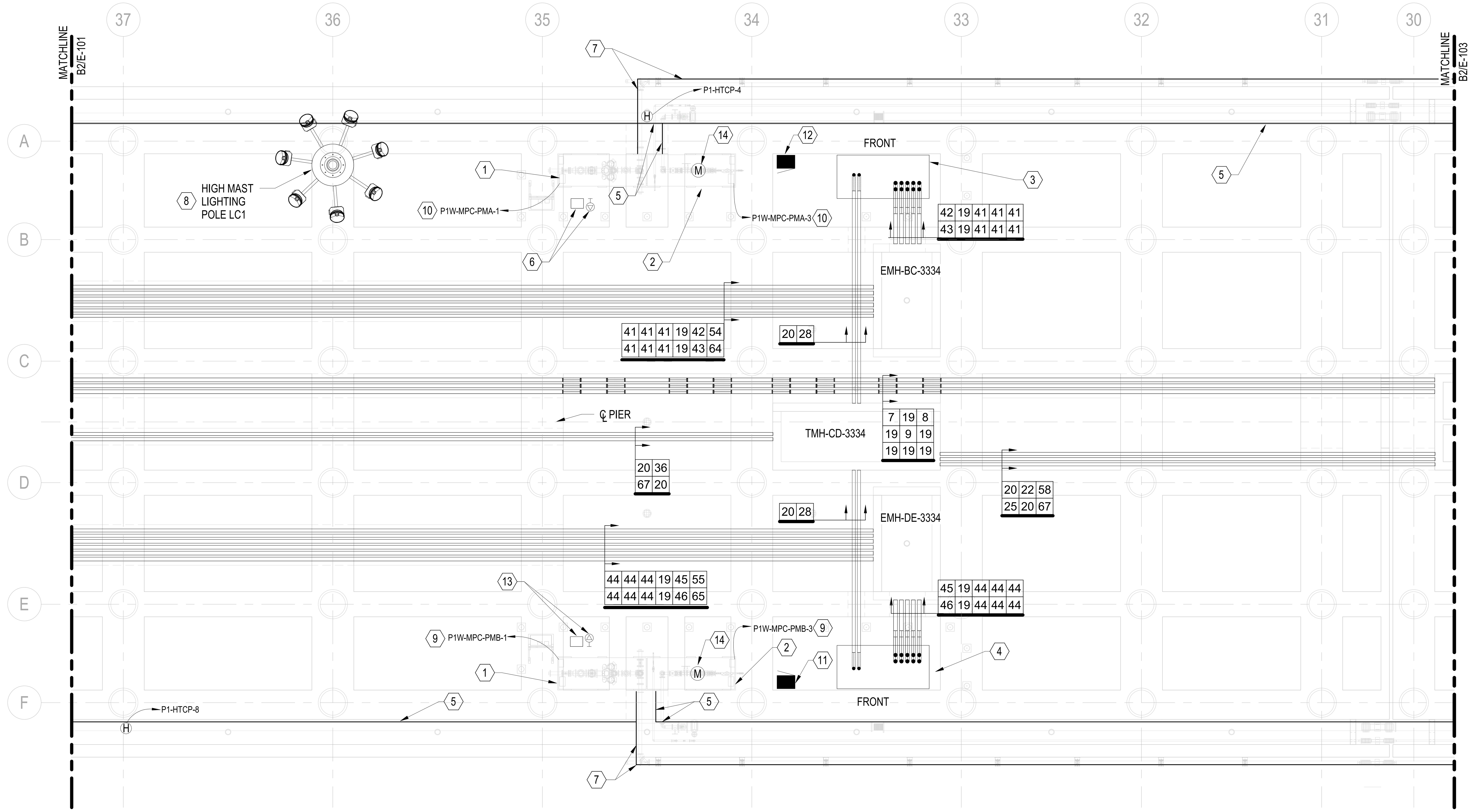
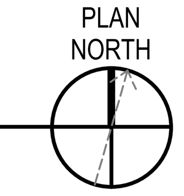
E-101



B2 ELECTRICAL DECK PLAN - SHEET 2 OF 7

SCALE: 1/8" = 1'-0"

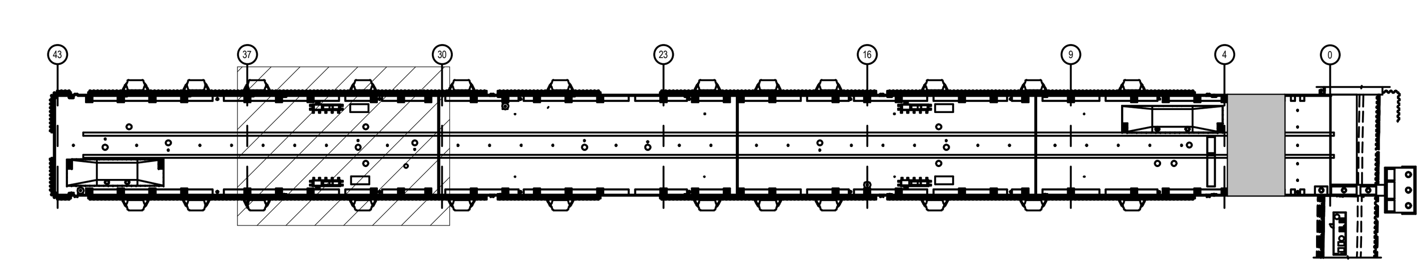


GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, E-603 AND E-604. FOR FEEDER SCHEDULE SEE DRAWINGS E-614 AND E-615.
2. PROVIDE EXPANSION/DEFLECTION FITTINGS ON ALL CONDUITS CROSSING EXPANSION JOINTS. SEE STRUCTURAL DRAWINGS FOR LOCATIONS.

SHEET KEYNOTES:

1. SANITARY SEWER HEATED ENCLOSURE. SEE MECHANICAL DRAWINGS FOR DETAILS.
2. COMPRESSED AIR AND FRESH WATER HEATED ENCLOSURE. SEE MECHANICAL DRAWINGS FOR DETAILS.
3. POWER MOUND P1W-PMA. SEE DETAILS ON DRAWINGS E-503 AND E-504.
4. POWER MOUND P1W-PMB. SEE DETAILS ON DRAWINGS E-503 AND E-504.
5. PROVIDE HEAT TRACING CABLE ON FRESH WATER PIPING IN UTILITY TRENCH. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.
6. MANUAL TRANSFER SWITCH AND PORTABLE EQUIPMENT RECEPTACLE SERVING BACKUP POWER TO SEWER PUMP CONTROLLER. MOUNT EQUIPMENT TO TYPE 316 STAINLESS STEEL STRUT SUPPORTS. SEE FEEDER 31 ON FEEDER SCHEDULE FOR CONDUIT AND WIRE TO EXTEND FROM EMH-BC-3334 TO TRANSFER SWITCH.
7. PROVIDE HEAT TRACING CABLE ON EXPOSED SEWER PIPING. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.
8. HIGH MAST LIGHTING POLE LC1. FOR HIGH MAST LIGHT POLE SCHEDULE AND DETAILS SEE DRAWINGS E-508 AND E-509. SEE FEEDER 64 FOR CONDUIT AND WIRING FROM EMH-BC-4142 TO POLE. EMH-BC-4142 IS SHOWN ON DRAWING E-101.
9. SEE P1W-MPC-PMB PANEL SCHEDULE ON DRAWING E-606 FOR CONDUIT AND WIRING FROM PANEL ENCLOSURE UNIT HEATER AND ALARM BOX.
10. SEE P1W-MPC-PMA PANEL SCHEDULE ON DRAWING E-606 FOR CONDUIT AND WIRING FROM PANEL ENCLOSURE UNIT HEATER AND ALARM BOX.
11. PANEL P1W-MPC-PMB. SEE PANEL SCHEDULE ON DRAWING E-606. SEE FEEDER 61 FOR CONDUIT AND WIRING FROM EMH-DE-3334 TO PANEL. MOUNT EQUIPMENT TO TYPE 316 STAINLESS STEEL STRUT SUPPORTS.
12. PANEL P1W-MPC-PMA. SEE PANEL SCHEDULE ON DRAWING E-606. SEE FEEDER 60 FOR CONDUIT AND WIRING FROM EMH-BC-3334 TO PANEL. MOUNT EQUIPMENT TO TYPE 316 STAINLESS STEEL STRUT SUPPORTS.
13. MANUAL TRANSFER SWITCH AND PORTABLE EQUIPMENT RECEPTACLE SERVING BACKUP POWER TO SEWER PUMP CONTROLLER. MOUNT EQUIPMENT TO TYPE 316 STAINLESS STEEL STRUT SUPPORTS. SEE FEEDER 31 ON FEEDER SCHEDULE FOR CONDUIT AND WIRE TO EXTEND FROM EMH-DE-3334 TO TRANSFER SWITCH.
14. SEE COMMUNICATIONS ONE LINE DIAGRAM AND FEEDER SCHEDULE FOR CONDUIT AND WIRING FROM WATER ENCLOSURE AMI METER TO COMMUNICATIONS MANHOLE SYSTEM.



KEY PLAN

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
REGISTERED PROFESSIONAL ENGINEER
SEAL

Stantec

A/E INFO

APPROVED
James E. Donahue, R.A.
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV	CHK RJW
PM/DM		MRI/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL DECK PLAN - SHEET 2 OF 7

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936761
SHEET 192 OF 247

E-102

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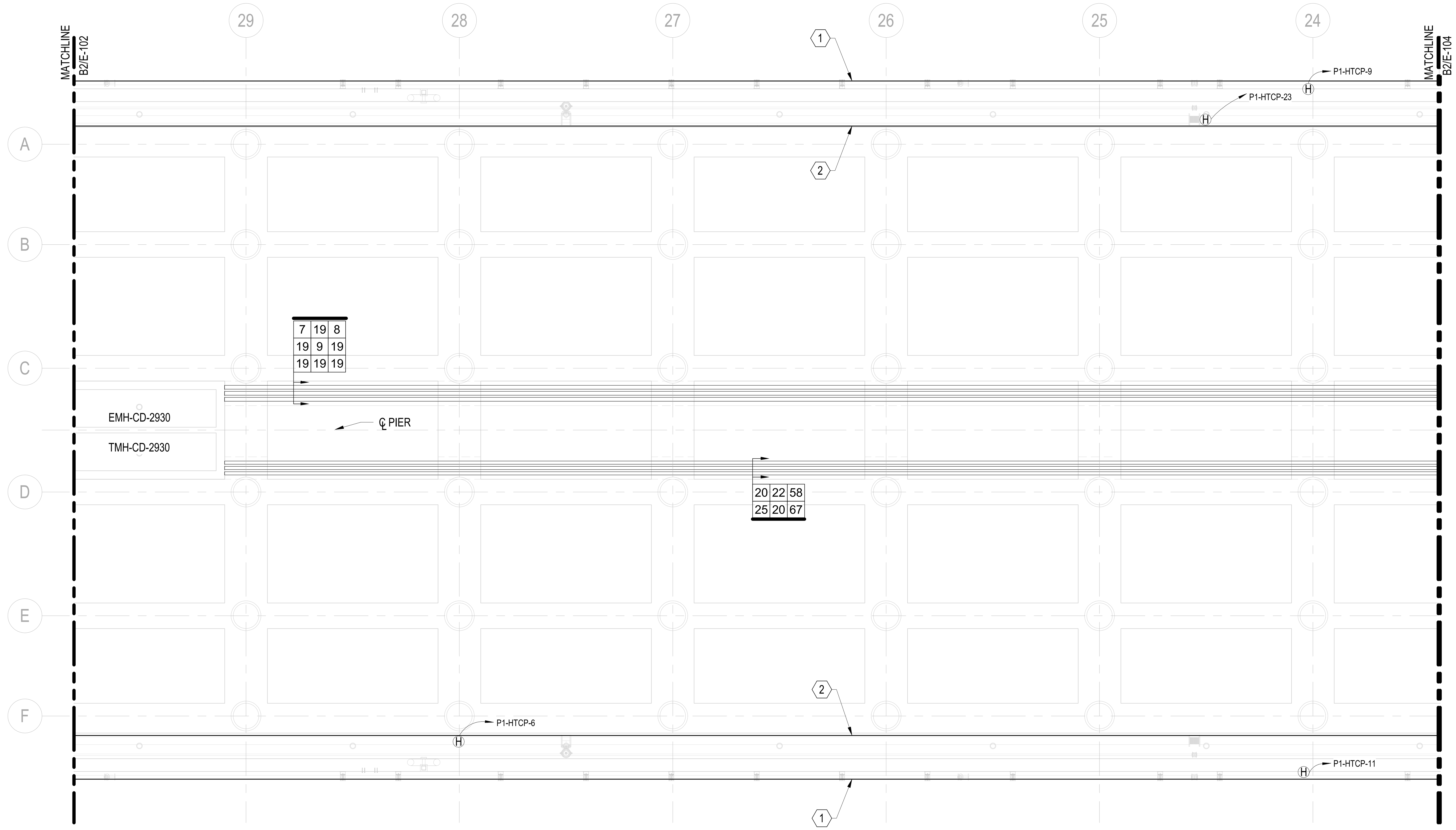
E-102

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, E-603 AND E-604. FOR FEEDER SCHEDULE SEE DRAWINGS E-614 AND E-615.
2. PROVIDE EXPANSION/DEFLECTION FITTINGS ON ALL CONDUITS CROSSING EXPANSION JOINTS. SEE STRUCTURAL DRAWINGS FOR LOCATIONS.

SHEET KEYNOTES:

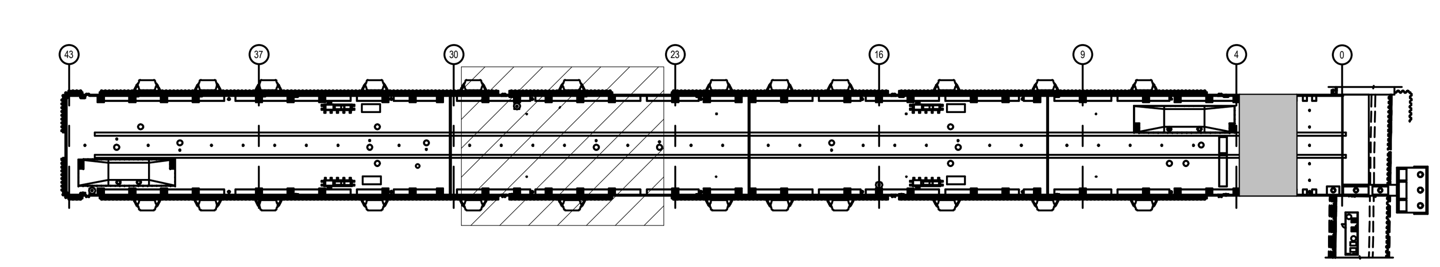
1. PROVIDE HEAT TRACING CABLE ON EXPOSED SEWER PIPING. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.
2. PROVIDE HEAT TRACING CABLE ON FRESH WATER PIPING IN UTILITY TRENCH. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.



B2 ELECTRICAL DECK PLAN - SHEET 3 OF 7

SCALE: 1/8" = 1'-0"

PLAN NORTH



KEY PLAN

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

DAVID P FERGUSON ELECTRICAL No. 52642 COMMONWEALTH OF MASSACHUSETTS PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES	DPF	DRW	GLV	CHK	RJW
PM/DM					MRI/CWJ
BRANCH MANAGER					WRB
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL DECK PLAN - SHEET 3 OF 7

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936762
SHEET 193 OF 247

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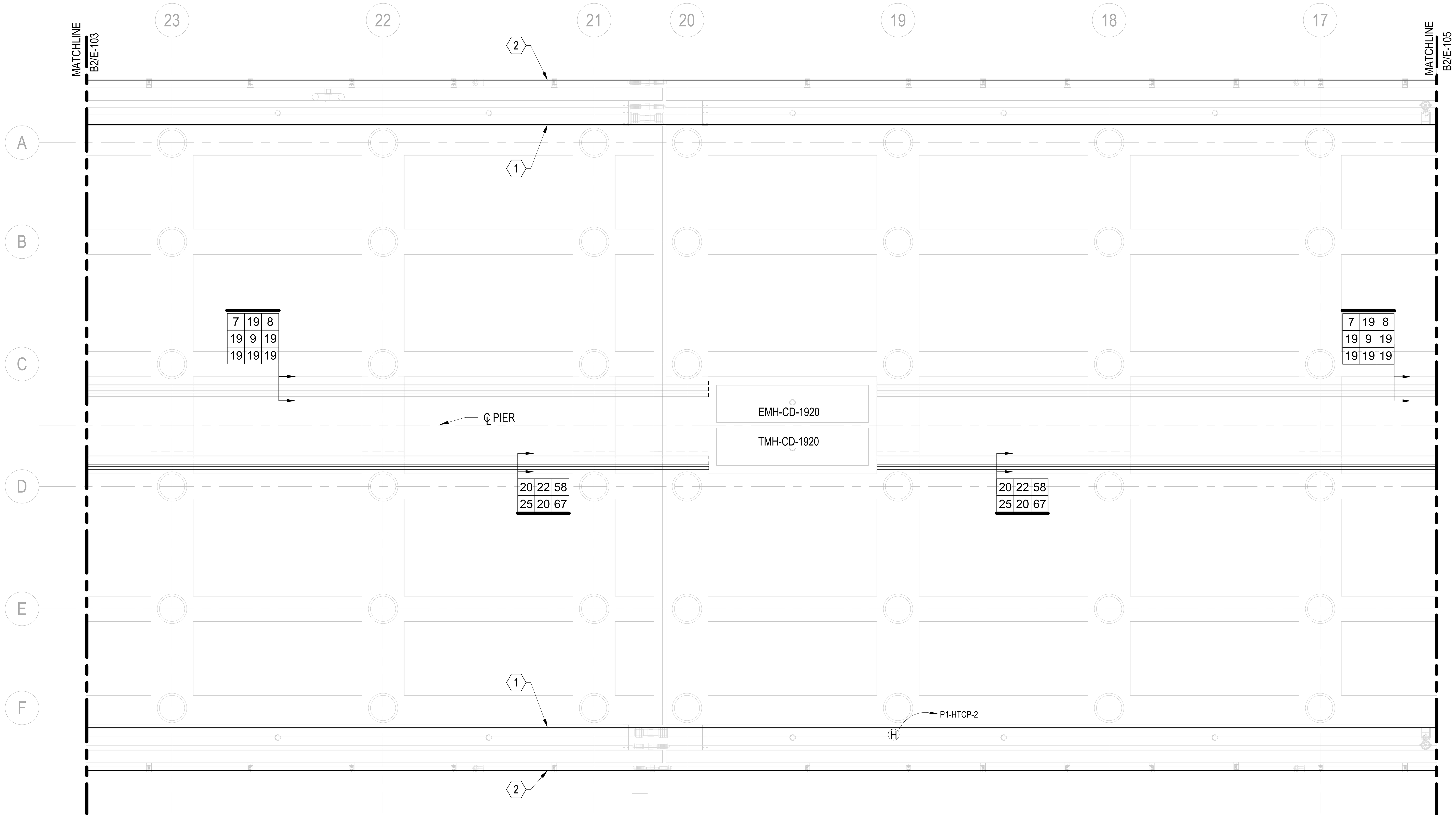
E-103

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, E-603 AND E-604. FOR FEEDER SCHEDULE SEE DRAWINGS E-614 AND E-615.
2. PROVIDE EXPANSION/DEFLECTION FITTINGS ON ALL CONDUITS CROSSING EXPANSION JOINTS. SEE STRUCTURAL DRAWINGS FOR LOCATIONS.

SHEET KEYNOTES:

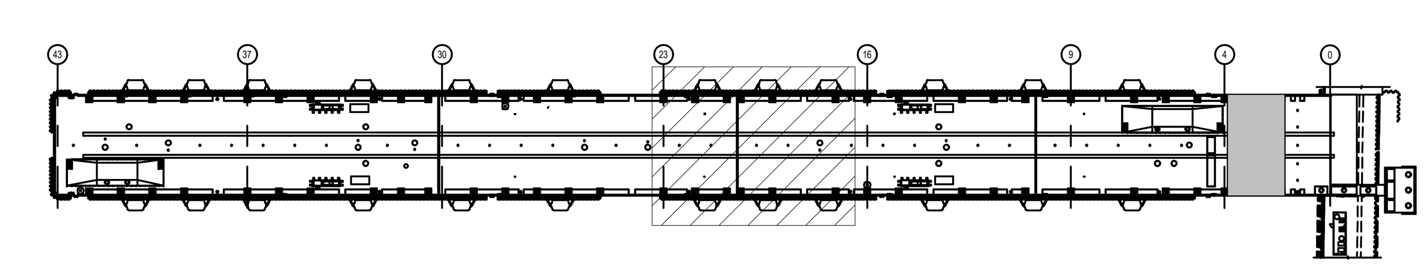
1. PROVIDE HEAT TRACING CABLE ON FRESH WATER PIPING IN UTILITY TRENCH. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.
2. PROVIDE HEAT TRACING CABLE ON EXPOSED SEWER PIPING. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.



B2 ELECTRICAL DECK PLAN - SHEET 4 OF 7

SCALE: 1/8" = 1'-0"

PLAN NORTH



KEY PLAN

1/8" = 1' - 0"   0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS DAVID P FERGUSON ELECTRICAL No. 52642 PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER

ELECTRICAL DECK PLAN - SHEET 4 OF 7

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936763

SHEET 194 OF 247

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DRAWFORM REVISION: 25 AUGUST 2020

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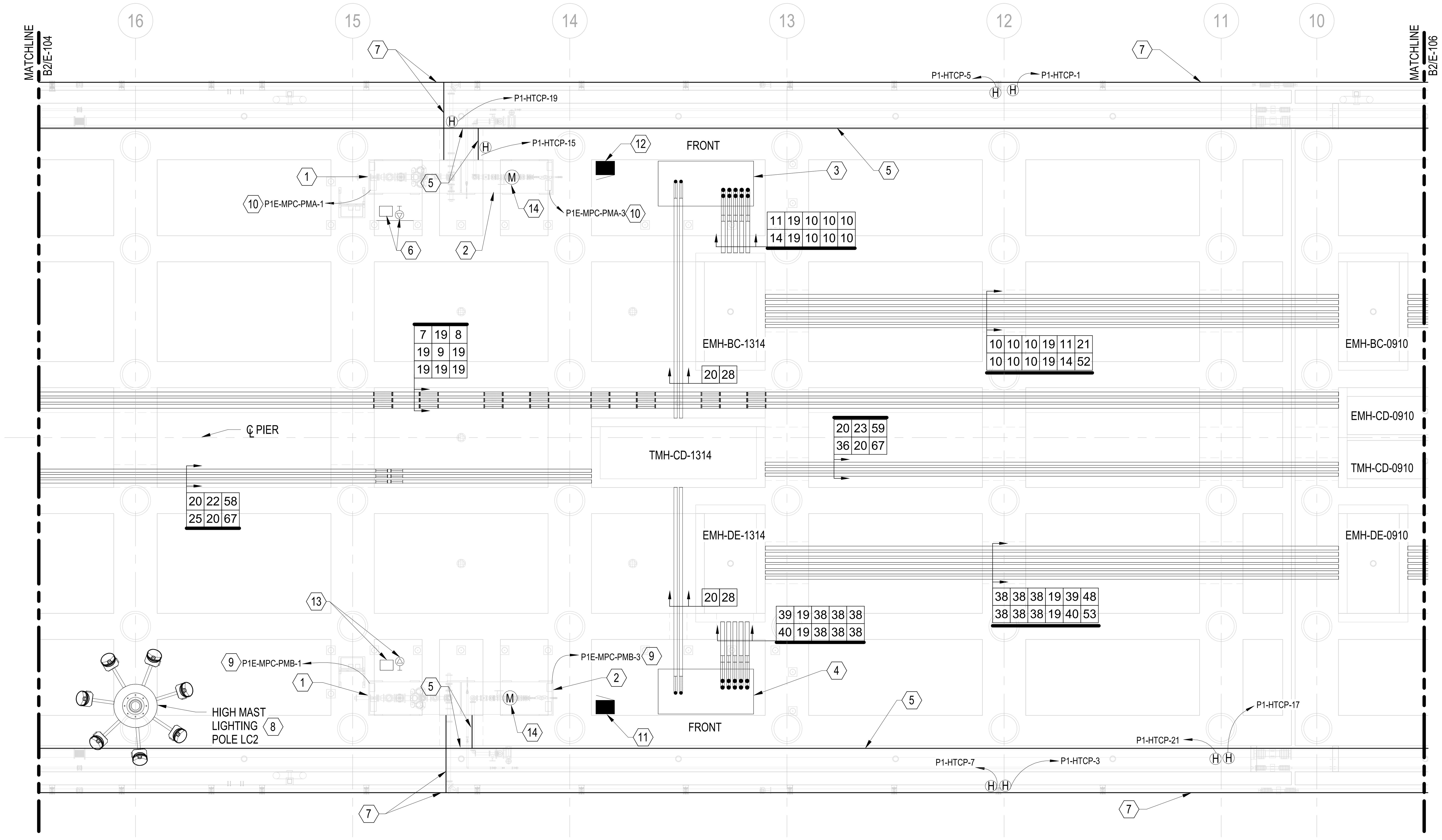
E-104

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, E-603 AND E-604. FOR FEEDER SCHEDULE SEE DRAWINGS E-614 AND E-615.
2. PROVIDE EXPANSION/DEFLECTION FITTINGS ON ALL CONDUITS CROSSING EXPANSION JOINTS. SEE STRUCTURAL DRAWINGS FOR LOCATIONS.

SHEET KEYNOTES:

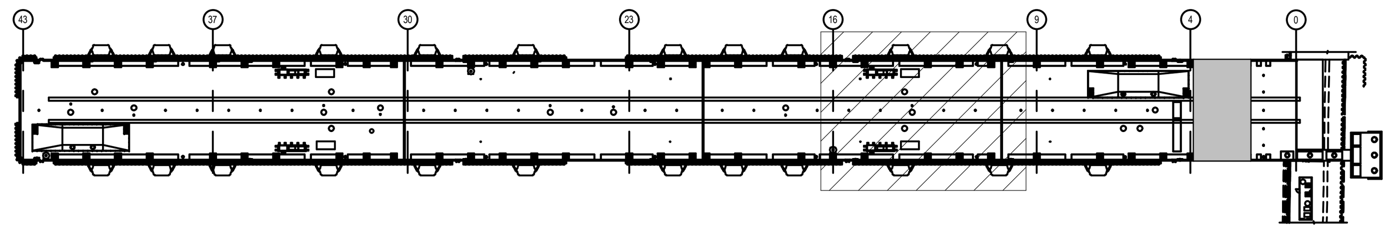
1. SANITARY SEWER HEATED ENCLOSURE. SEE MECHANICAL DRAWINGS FOR DETAILS.
2. COMPRESSED AIR AND FRESH WATER HEATED ENCLOSURE. SEE MECHANICAL DRAWINGS FOR DETAILS.
3. POWER MOUND P1E-PMA. SEE DETAILS ON DRAWINGS E-503 AND E-504.
4. POWER MOUND P1E-PMB. SEE DETAILS ON DRAWINGS E-503 AND E-504.
5. PROVIDE HEAT TRACING CABLE ON FRESH WATER PIPING IN UTILITY TRENCH. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.
6. MOUNT TRANSFER SWITCH AND PORTABLE EQUIPMENT RECEPTACLE SERVING BACKUP POWER TO SEWER PUMP CONTROLLER. MOUNT EQUIPMENT TO TYPE 316 STAINLESS STEEL STRUT SUPPORTS. SEE FEEDER 32 ON FEEDER SCHEDULE FOR CONDUIT AND WIRE TO EXTEND FROM EMH-BC-1314 TO TRANSFER SWITCH.
7. PROVIDE HEAT TRACING CABLE ON EXPOSED SEWER PIPING. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.
8. FOR HIGH MAST LIGHT POLE SCHEDULE AND DETAILS SEE DRAWINGS E-508 AND E-509. SEE FEEDER 65 FOR CONDUIT AND WIRING FROM EMH-DE-3334 TO POLE. EMH-DE-3334 IS SHOWN ON DRAWING E-102.
9. SEE P1E-MPC-PMB PANEL SCHEDULE ON DRAWING E-606 FOR CONDUIT AND WIRING FROM PANEL ENCLOSURE UNIT HEATER AND ALARM BOX.
10. SEE P1E-MPC-PMA PANEL SCHEDULE ON DRAWING E-606 FOR CONDUIT AND WIRING FROM PANEL ENCLOSURE UNIT HEATER AND ALARM BOX.
11. PANEL P1E-MPC-PMB. SEE PANEL SCHEDULE ON DRAWING E-606. SEE FEEDER 63 FOR CONDUIT AND WIRING FROM EMH-BC-1314 TO PANEL. MOUNT EQUIPMENT TO TYPE 316 STAINLESS STEEL STRUT SUPPORTS.
12. PANEL P1E-MPC-PMA. SEE PANEL SCHEDULE ON DRAWING E-606. SEE FEEDER 62 FOR CONDUIT AND WIRING FROM EMH-DE-1314 TO PANEL. MOUNT EQUIPMENT TO TYPE 316 STAINLESS STEEL STRUT SUPPORTS.
13. MANUAL TRANSFER SWITCH AND PORTABLE EQUIPMENT RECEPTACLE SERVING BACKUP POWER TO SEWER PUMP CONTROLLER. MOUNT EQUIPMENT TO TYPE 316 STAINLESS STEEL STRUT SUPPORTS. SEE FEEDER 32 ON FEEDER SCHEDULE FOR CONDUIT AND WIRE TO EXTEND FROM EMH-DE-1314 TO TRANSFER SWITCH.
14. SEE COMMUNICATIONS ONE LINE DIAGRAM AND FEEDER SCHEDULE FOR CONDUIT AND WIRING FROM WATER ENCLOSURE AMI METER TO COMMUNICATIONS MANHOLE SYSTEM.



B2 ELECTRICAL DECK PLAN - SHEET 5 OF 7

SCALE: 1/8" = 1'-0"

PLAN NORTH



KEY PLAN

1/8" = 1' - 0"   0 5' 10' 15' 30'

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL DECK PLAN - SHEET 5 OF 7

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER

Stantec

SYM	DESCRIPTION	DATE	APPR

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES	DPF	DRW	GLV	CHK	RJW
PM/DM	MRI/CWJ				
BRANCH MANAGER	WRB				
CHIEF ENG/ARCH	JWJ				
FIRE PROTECTION	HPN				

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936764
SHEET 195 OF 247

E-105

DRAWFORM REVISION: 25 AUGUST 2020

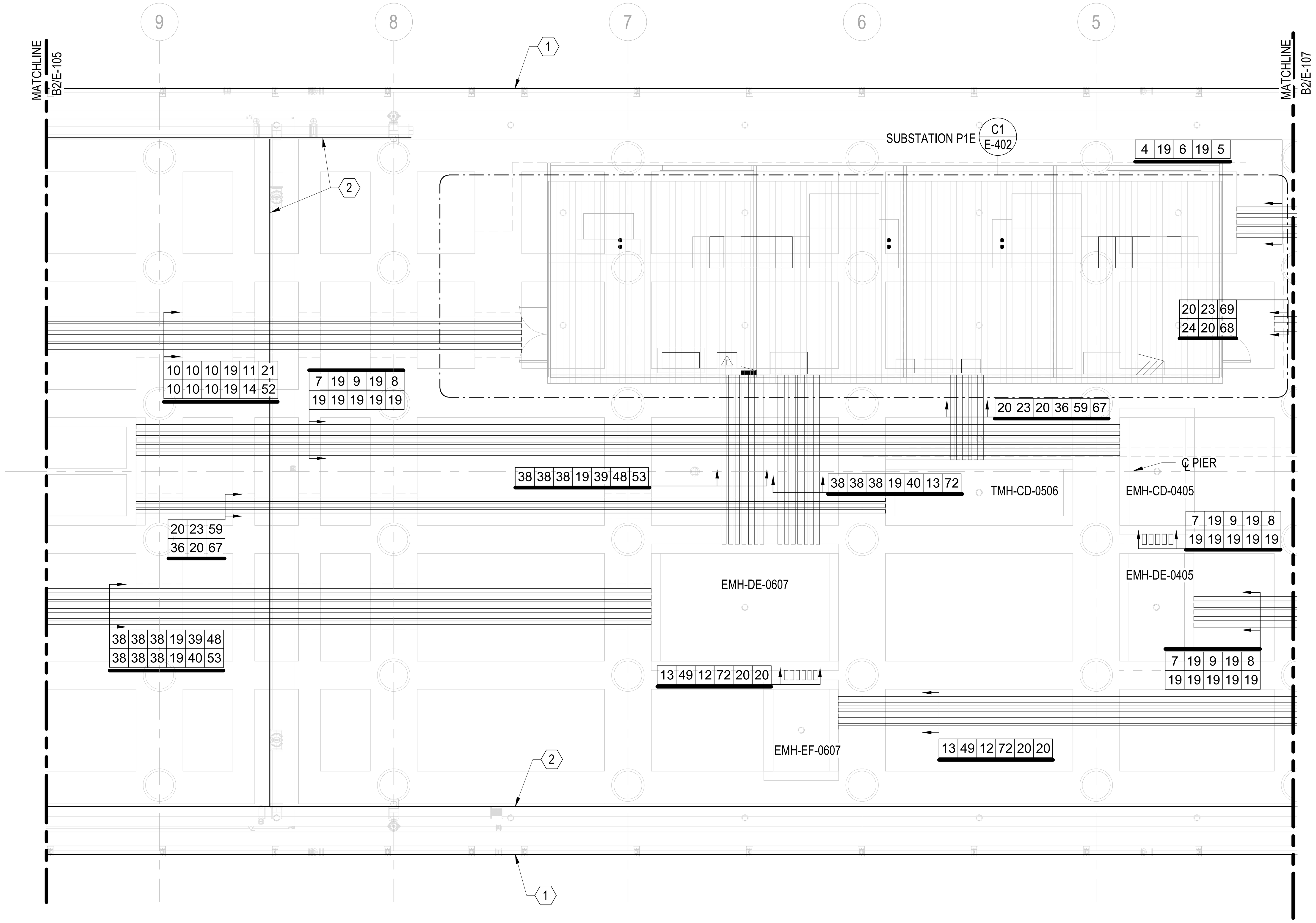
3/2/2026 5:15:50 PM E-105
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, E-603 AND E-604. FOR FEEDER SCHEDULE SEE DRAWINGS E-614 AND E-615.
2. PROVIDE EXPANSION/DEFLECTION FITTINGS ON ALL CONDUITS CROSSING EXPANSION JOINTS. SEE STRUCTURAL DRAWINGS FOR LOCATIONS.

SHEET KEYNOTES:

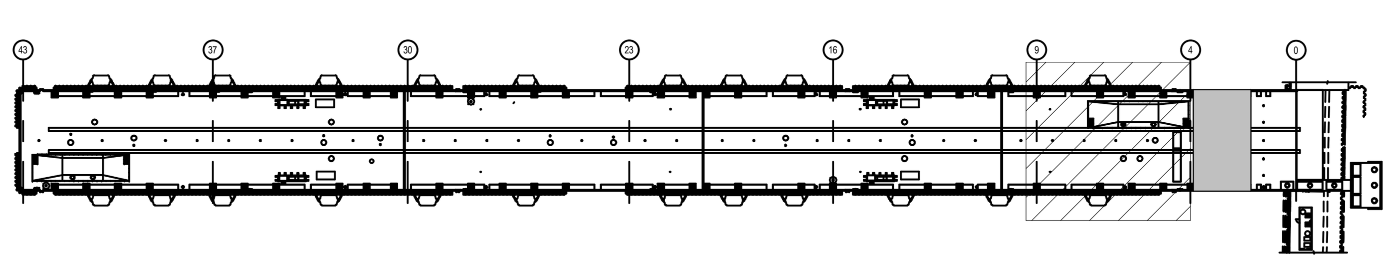
1. PROVIDE HEAT TRACING CABLE ON EXPOSED SEWER PIPING. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.
2. PROVIDE HEAT TRACING CABLE ON FRESH WATER PIPING IN UTILITY TRENCH. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.



B2 ELECTRICAL DECK PLAN - SHEET 6 OF 7

SCALE: 1/8" = 1'-0"

PLAN NORTH



KEY PLAN

1/8" = 1' - 0"   0   5'   10'   15'   30'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS — DAVID P FERGUSON ELECTRICAL No. 52642 PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES	DPF	DRW	GLV	CHK	RJW
PM/DM					MRI/CWJ
BRANCH MANAGER					WRB
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL DECK PLAN - SHEET 6 OF 7

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936765

SHEET 196 OF 247

E-106

DRAWFORM REVISION: 20 AUGUST 2020

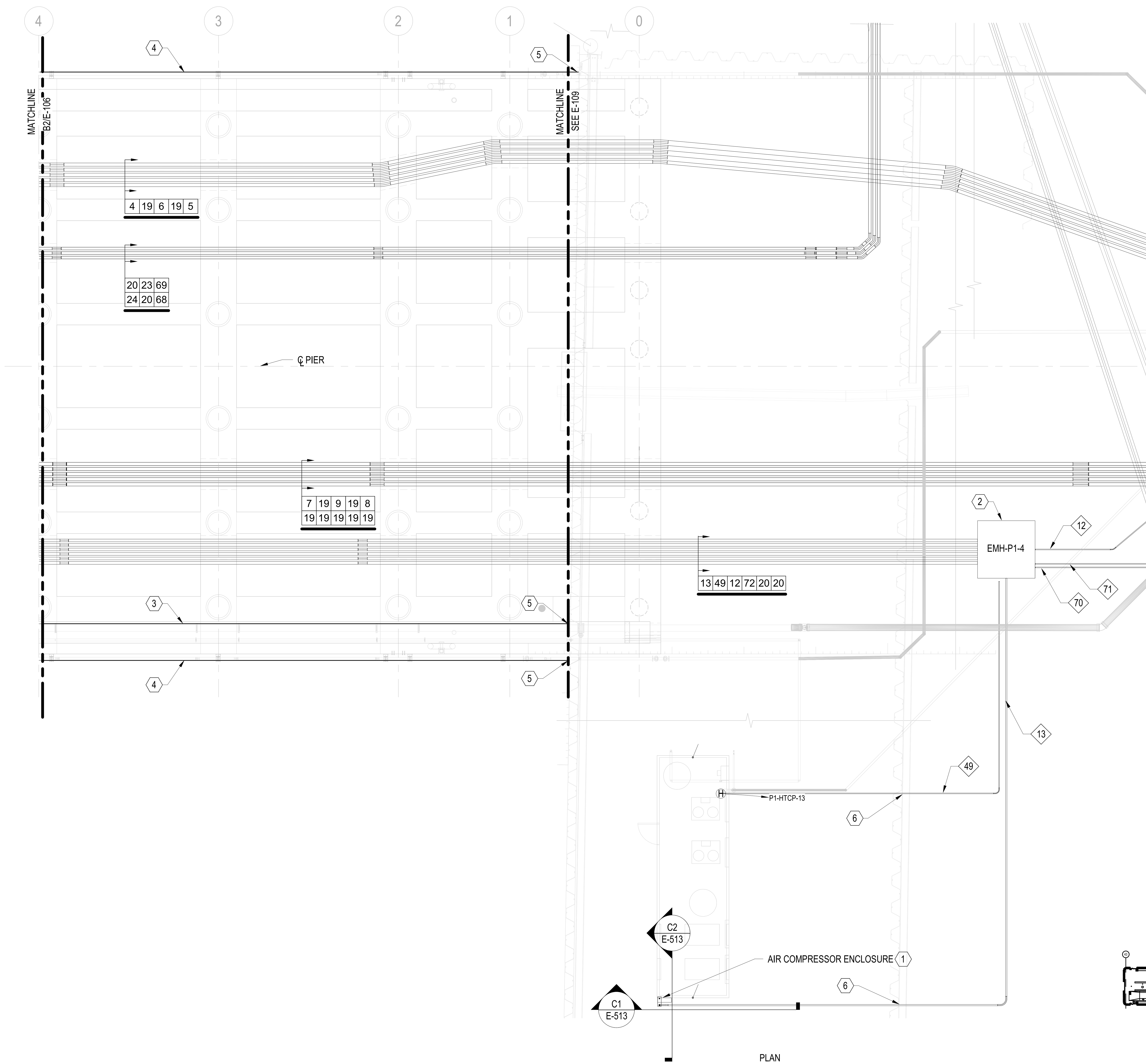
3/2/2026 5:15:56 PM
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt
E-106

GENERAL SHEET NOTES:

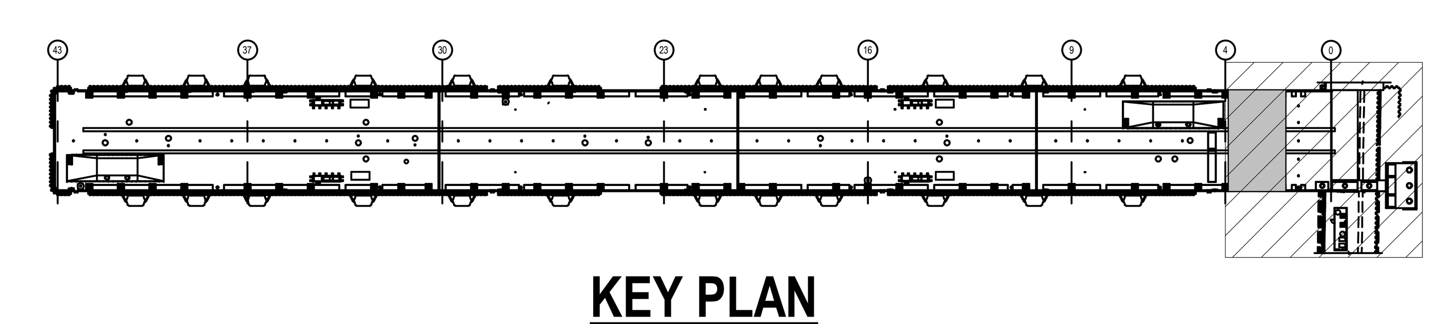
1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, E-603 AND E-604. FOR FEEDER SCHEDULE SEE DRAWINGS E-614 AND E-615.
2. PROVIDE EXPANSION/DEFLECTION FITTINGS ON ALL CONDUITS CROSSING EXPANSION JOINTS. SEE STRUCTURAL DRAWINGS FOR LOCATIONS.

SHEET KEYNOTES:

1. MAKE CONNECTION TO EXTERIOR DISCONNECT SWITCH PROVIDED WITH ELEVATED AIR COMPRESSOR. SEE MECHANICAL DRAWINGS FOR DETAILS OF AIR COMPRESSOR ENCLOSURE. SEE DRAWING E-513 FOR ADDITIONAL DETAILS ON CONDUIT RUN FROM UNDERGROUND UP TO DISCONNECT SWITCH.
2. PROVIDE ELECTRICAL MANHOLE. SEE DETAIL ON E-501.
3. PROVIDE HEAT TRACING CABLE ON FRESH WATER PIPING IN UTILITY TRENCH. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.
4. PROVIDE HEAT TRACING CABLE ON EXPOSED SEWER PIPING. FOR ADDITIONAL INFORMATION SEE DRAWINGS E-514, E-608, AND E-612.
5. TERMINATE HEAT TRACING CIRCUIT AT PIPE PENETRATION THROUGH BULKHEAD.
6. PROVIDE WATERTIGHT CONDUIT PENETRATION THROUGH EXISTING BULKHEAD.



A2 ELECTRICAL DECK PLAN - SHEET 7 OF 7
SCALE: 1/8" = 1'-0"



KEY PLAN

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS DAVID P FERGUSON ELECTRICAL No. 52642 PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL DECK PLAN - SHEET 7 OF 7

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936766

SHEET 197 OF 247

E-107

DRAWFORM REVISION: 25 AUGUST 2020

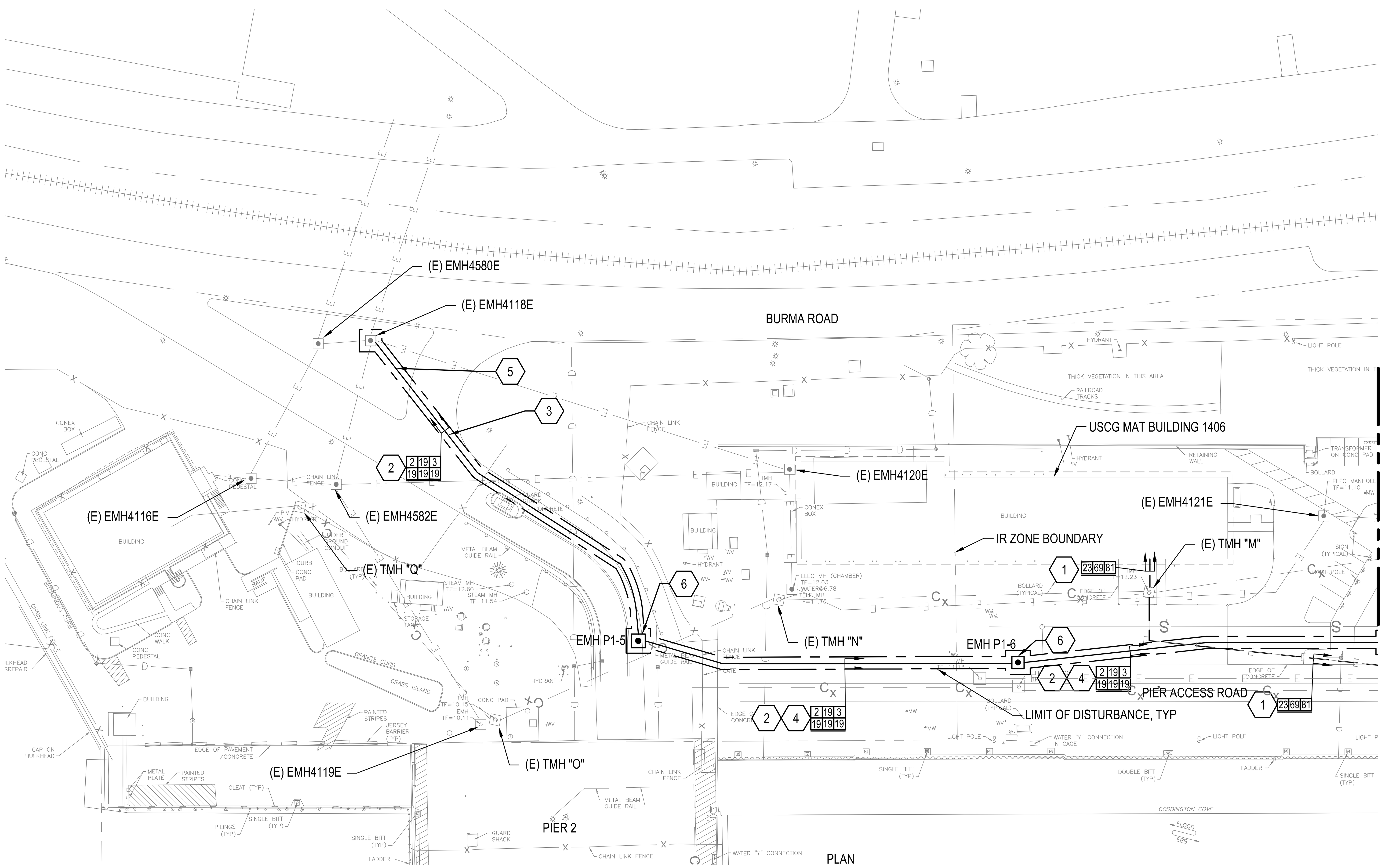
3/2/2026 5:16:05 PM E-107
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt

GENERAL SHEET NOTES:

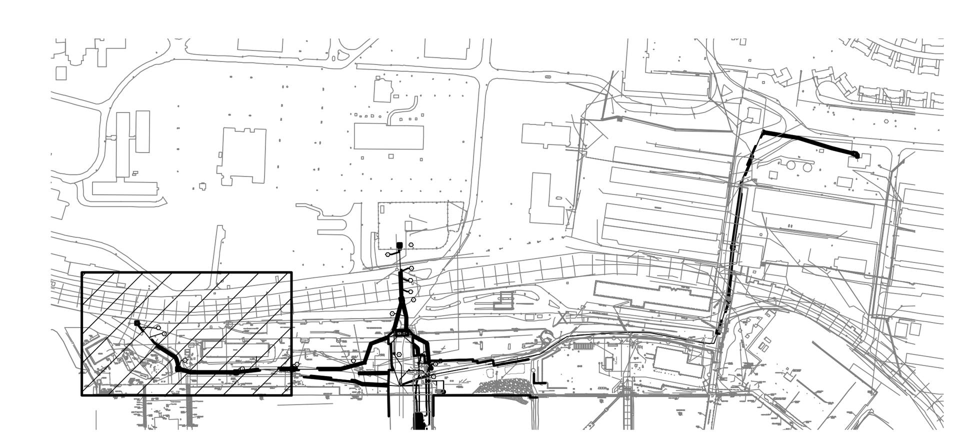
1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES SEE DRAWINGS E-001.
2. FOR SITE POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601 AND E-604. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.

SHEET KEYNOTES:

1. ROUTE CABLING IN EXISTING SPARE DUCTS.
2. PROVIDE CONCRETE ENCASED DUCTBANK.
3. SEE DETAIL B1/C-502 FOR TRENCH REPAIR WORK.
4. SEE DETAIL B2/C-502 FOR TRENCH REPAIR WORK.
5. SEE DETAIL C2/C-502 FOR TRENCH REPAIR WORK.
6. PROVIDE ELECTRICAL MANHOLE.



A2 ELECTRICAL SITE PLAN - PIER 2
SCALE: 1" = 50'



KEY PLAN

1" = 50' — 50' 0 50' 100'

APPROVED	
FOR COMMANDER NAVFAC	3/5/2026
ACTIVITY	Approved by Frank Cole USCG MASI
SATISFACTORY TO	DATE 06 FEB 2026
DES DPF	DRW GLV · CHK RJW
PM/DM	MRI/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DAVID P FERGUSON ELECTRICAL No. 52642 — COMMONWEALTH OF MASSACHUSETTS REGISTERED PROFESSIONAL ENGINEER

NAVFAC

Stantec

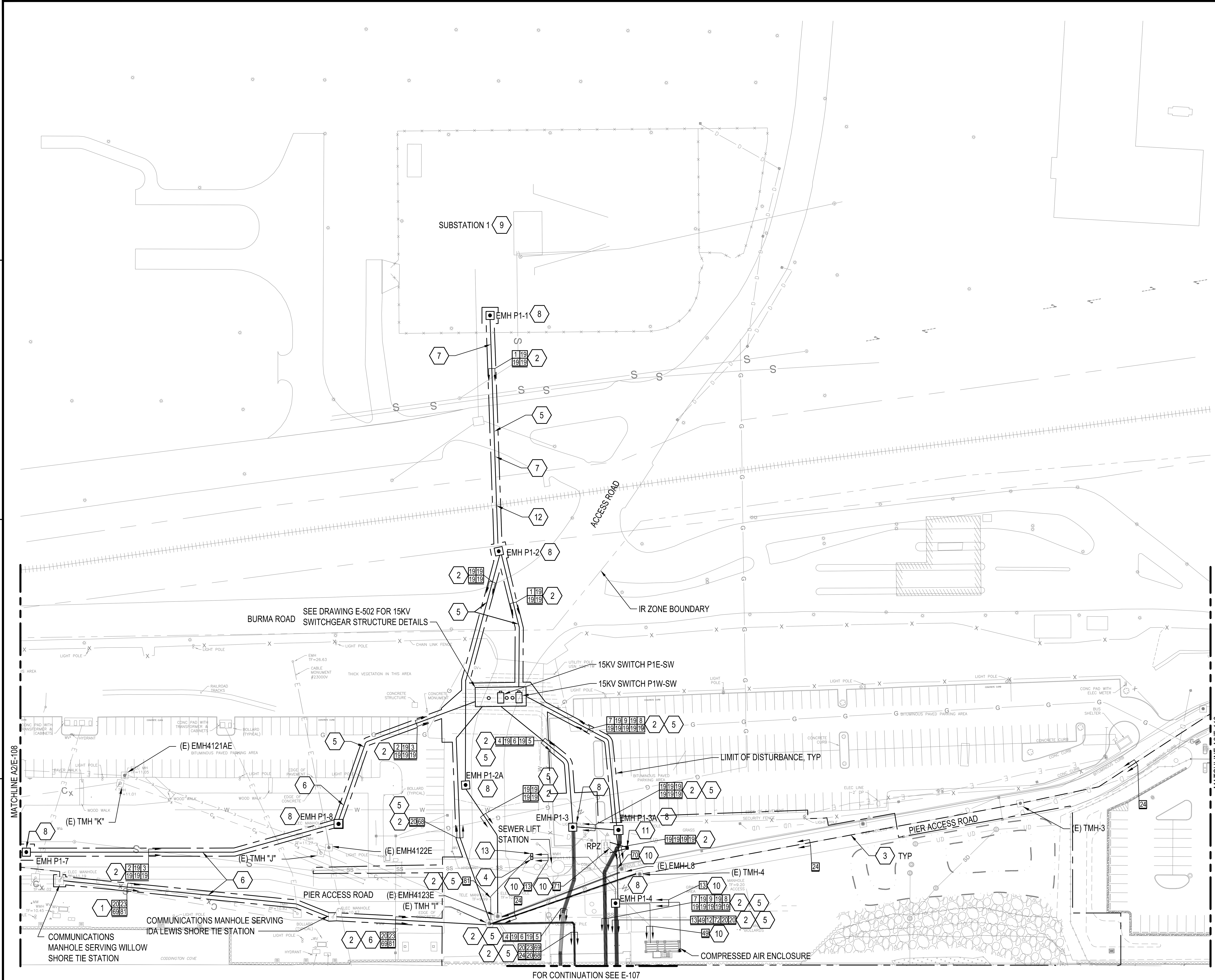
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL SITE PLAN - PIER 2

SCALE:	AS NOTED
EPROJECT NO.	1826479
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12936767
SHEET	198 OF 247

E-108

DRAWFORM REVISION: 25 AUGUST 2020

FILE NAME: C:\Users\dpferguson\DC\ACCDocs\Stantec Consulting\224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1\Project Files\01_WIP\13_electrical\sheet_files\E-108 ELECTRICAL SITE PLANS.dwg LAYOUT NAME: E-108 PLOTTED: Monday, March 02, 2026 - 5:11pm USER: dpferguson



A2 ELECTRICAL SITE PLAN - PIER 1

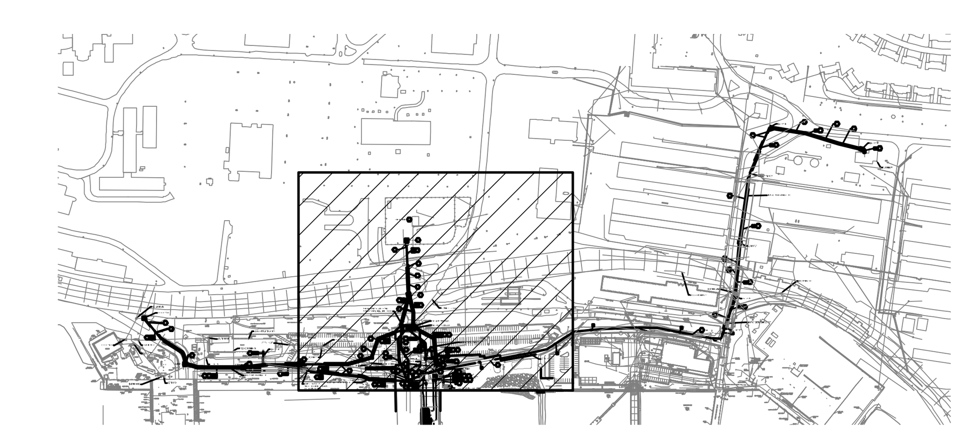
SCALE: 1" = 50'

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES SEE DRAWINGS E-001.
2. FOR SITE POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601 AND E-604. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.

SHEET KEYNOTES:

1. PROVIDE CONCRETE ENCASED DUCTBANK PARALLEL TO EXISTING DUCTBANK BETWEEN MANHOLES.
2. PROVIDE CONCRETE ENCASED DUCTBANK.
3. ROUTE FEEDER 24 THROUGH EXISTING SPARE DUCTS FROM TMH "I" TO TMH "F", WHICH IS SHOWN ON DRAWING E-110.
4. STAND MOUNTED DUPLEX PUMP CONTROL PANEL FOR LIFT STATION. SEE DETAIL C4/E-513.
5. SEE DETAIL B1/C-502 FOR TRENCH REPAIR WORK.
6. SEE DETAIL B2/C-502 FOR TRENCH REPAIR WORK.
7. SEE DETAIL C2/C-502 FOR TRENCH REPAIR WORK.
8. PROVIDE ELECTRICAL MANHOLE.
9. FOR ADDITIONAL WORK IN SUBSTATION 1 SEE ES DRAWINGS.
10. PROVIDE DIRECT BURIED CONDUIT.
11. PROVIDE PANEL P1E-MPC-RPZ. SEE PANEL SCHEDULE ON E-606. PROVIDE DIRECT BURIED CONDUITS FROM PANEL TO RPZ ENCLOSURE AND LIFT STATION COMMUNICATIONS ENCLOSURE, SEE PANEL SCHEDULE FOR CONDUIT & WIRE. SEE DETAIL B4/E-513 FOR PANEL MOUNTING.
12. COORDINATE DUCTBANK RAIL CROSSING WITH RIDOT. THE CONTRACTOR WILL NOTIFY THE CONTRACTING OFFICER IN WRITING A MINIMUM OF FIVE (5) WORKING DAYS PRIOR TO PERFORMING ANY WORK WITHIN OR ADJACENT TO THE RAILROAD EASEMENT AREA. NO WORK WILL BE PERFORMED BY THE CONTRACTOR WITHIN THE RAILROAD EASEMENT AREA UNTIL THE REQUIRED EASEMENT OR AUTHORIZATION HAS BEEN OBTAINED BY THE GOVERNMENT AND THE CONTRACTOR HAS RECEIVED WRITTEN NOTICE FROM THE CONTRACTING OFFICER AUTHORIZING WORK WITHIN THE EASEMENT AREA.
13. STAND MOUNTED COMMUNICATIONS ENCLOSURE FOR LIFT STATION. ENCLOSURE TO BE 36"W x 36"T x 10"D. ENCLOSURE TO HAVE DUPLEX RECEPTACLE, AUTODIALER (SEE SPEC SECTION 33 32 16), 50 PAIR 110 BLOCK, AND 25 PAIR 710/110 BUILDING ENTRANCE PROTECTOR. SEE DETAIL C4/E-513.



KEY PLAN

50' 0 50' 100'

1" = 50'

SYM | DESCRIPTION | DATE | APPR

COMMONWEALTH OF MASSACHUSETTS DAVID P FERGUSON ELECTRICAL No. 52642 PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV	CHK RJW
PM/DM		MRI/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL SITE PLAN - PIER 1

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936768
SHEET 199 OF 247

E-109

DRAWFORM REVISION: 25 AUGUST 2020

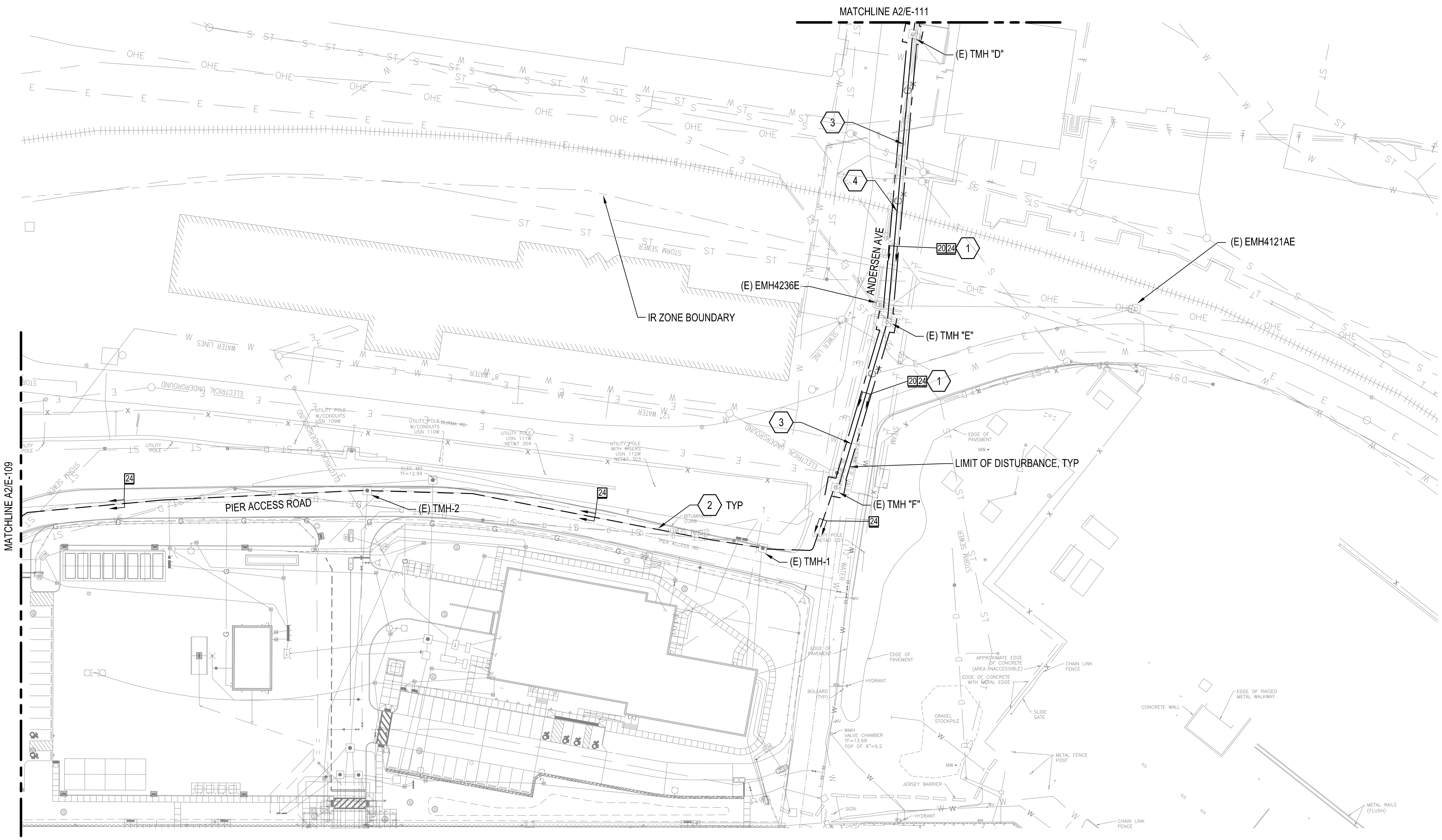
FILE NAME: C:\Users\dpferguson\DC\ACCDocs\Stantec Consulting\224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1\Project Files\01_WIP\13_electrical\sheet_files\E-108 ELECTRICAL SITE PLANS.dwg LAYOUT NAME: E-109 PLOTTED: Monday, March 02, 2026 - 5:11pm USER: dpferguson

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES SEE DRAWINGS E-001.
2. FOR SITE POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601 AND E-604. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.
3. BACKGROUND UTILITIES SHOWN ARE IMPORTED FROM THE NOAA PIER PROJECT DRAWINGS AND WERE NOT DEVELOPED BY THE SURVEY UNDER THIS CONTRACT.

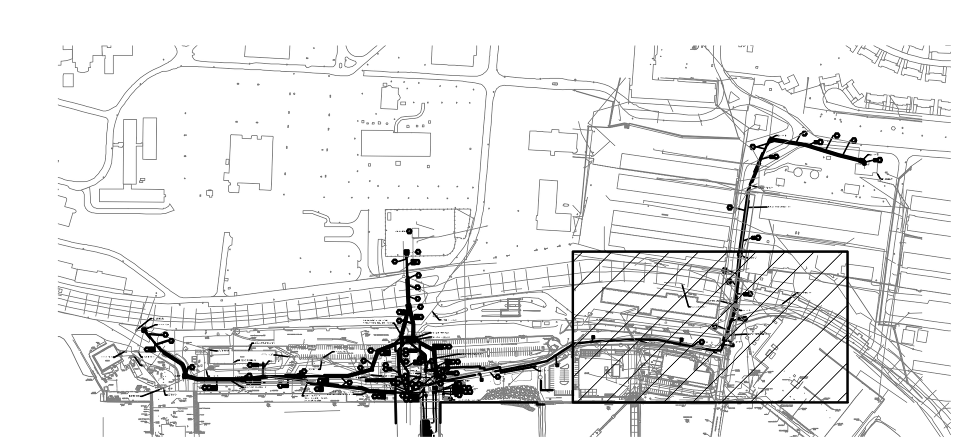
SHEET KEYNOTES:

1. PROVIDE CONCRETE ENCASED DUCTBANK PARALLEL TO EXISTING DUCTBANK BETWEEN MANHOLES.
2. ROUTE FEEDER 24 THROUGH EXISTING SPARE DUCTS FROM TMH "F" TO TMH "I, WHICH IS SHOWN ON DRAWING E-109.
3. SEE DETAIL B1/C-502 FOR TRENCH REPAIR WORK.
4. COORDINATE DUCTBANK RAIL CROSSING WITH RIDOT. THE CONTRACTOR WILL NOTIFY THE CONTRACTING OFFICER IN WRITING A MINIMUM OF FIVE (5) WORKING DAYS PRIOR TO PERFORMING ANY WORK WITHIN OR ADJACENT TO THE RAILROAD EASEMENT AREA. NO WORK WILL BE PERFORMED BY THE CONTRACTOR WITHIN THE RAILROAD EASEMENT AREA UNTIL THE REQUIRED EASEMENT OR AUTHORIZATION HAS BEEN OBTAINED BY THE GOVERNMENT AND THE CONTRACTOR HAS RECEIVED WRITTEN NOTICE FROM THE CONTRACTING OFFICER AUTHORIZING WORK WITHIN THE EASEMENT AREA.



A2 ELECTRICAL SITE PLAN - NOAA BUILDING
SCALE: 1" = 50'

PLAN NORTH



KEY PLAN

50' 0 50' 100'
1" = 50'

APPROVED	FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY	Approved by Frank Cole USCG MASI
SATISFACTORY TO	DATE 06 FEB 2026
DES DPF	DRW GLV CHK RJW
PM/DM	MRI/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
CI CORE - HAMPTON ROADS — NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT — NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL SITE PLAN - NOAA BUILDING

SCALE:	AS NOTED
EPROJECT NO.	1826479
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12936769
SHEET	200 OF 247

E-110

DAVID P FERGUSON ELECTRICAL No. 52642 — COMMONWEALTH OF MASSACHUSETTS — PROFESSIONAL ENGINEER

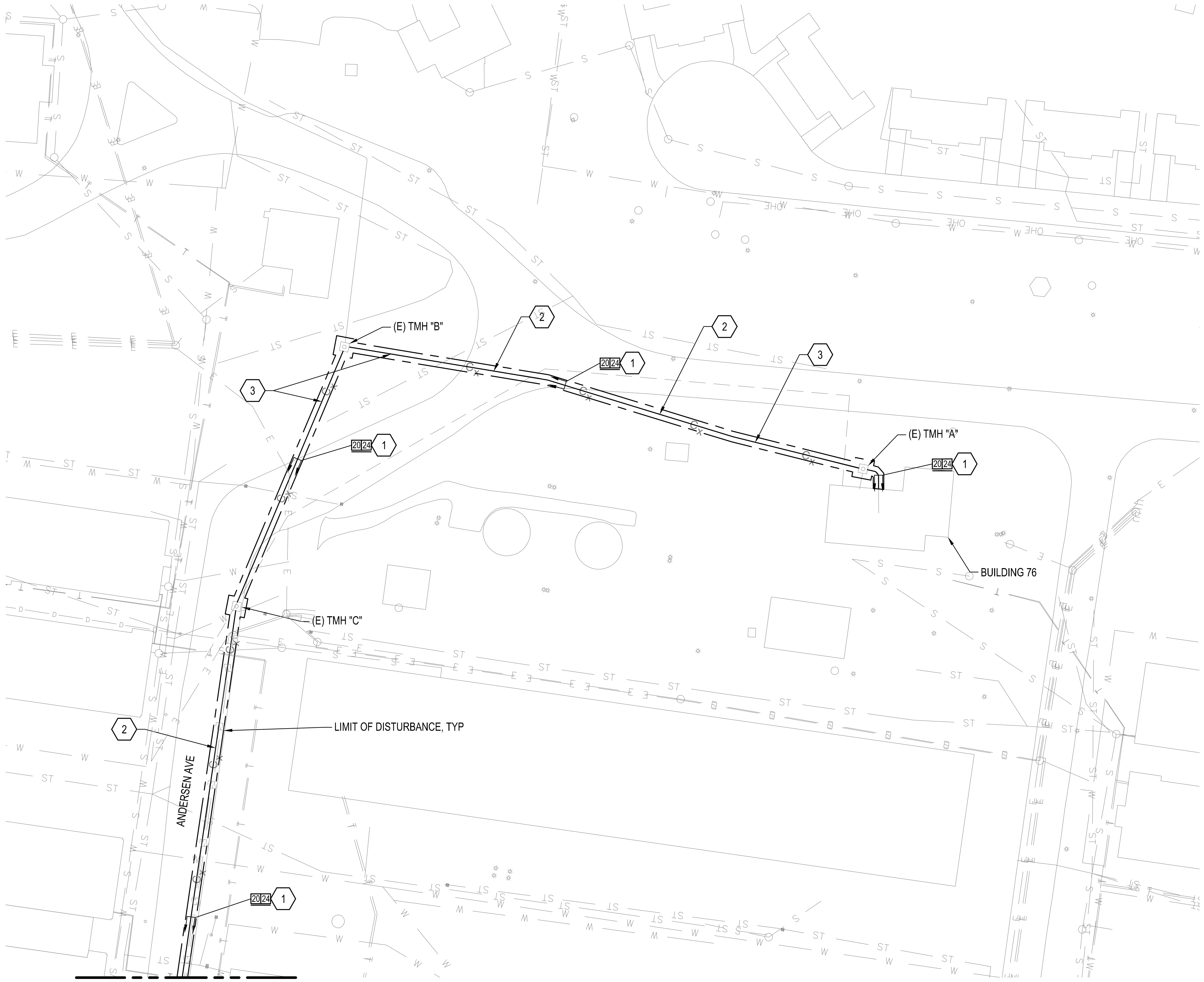
Stantec

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES SEE DRAWINGS E-001.
2. FOR SITE POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601 AND E-604. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.
3. BACKGROUND UTILITIES SHOWN ARE IMPORTED FROM THE NOAA PIER PROJECT DRAWINGS AND WERE NOT DEVELOPED BY THE SURVEY UNDER THIS CONTRACT.

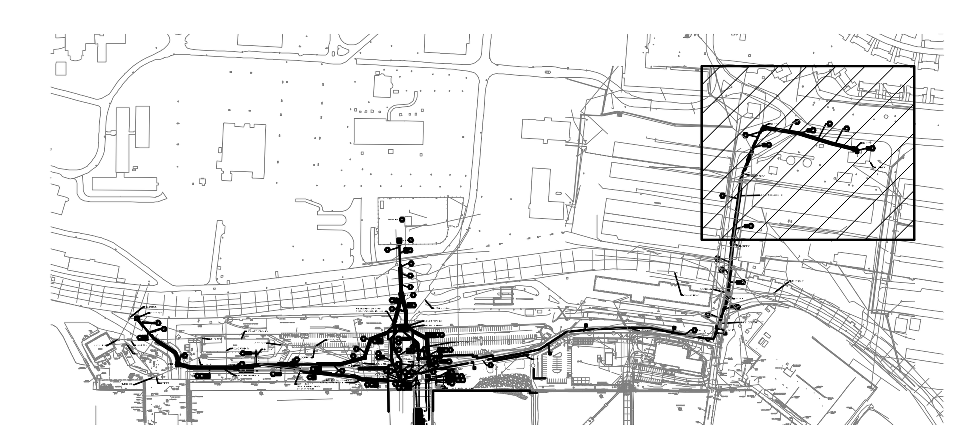
SHEET KEYNOTES:

1. PROVIDE CONCRETE ENCASED DUCTBANK PARALLEL TO EXISTING DUCTBANK BETWEEN MANHOLES.
2. SEE DETAIL B1/C-502 FOR TRENCH REPAIR WORK.
3. SEE DETAIL C2/C-502 FOR TRENCH REPAIR WORK.



A2 ELECTRICAL SITE PLAN - BUILDING 76
SCALE: 1" = 50'

PLAN NORTH



KEY PLAN

50' 0 50' 100'
1" = 50'

APPR | DATE | DESCRIPTION | SYM

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
SEAL

Stantec
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV / CHK RJW
PM/DM	MRI/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

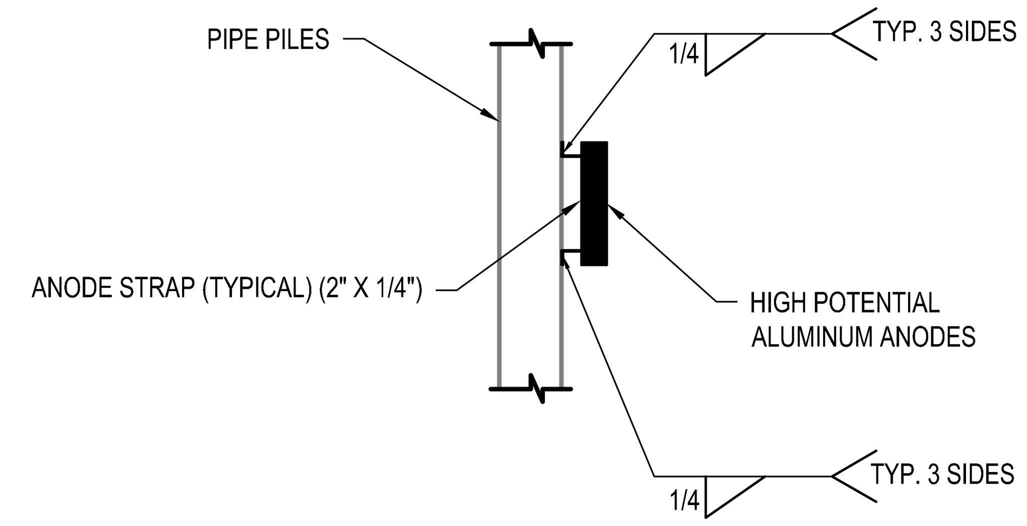
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
CI CORE - HAMPTON ROADS
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL SITE PLAN - BUILDING 76

SCALE:	AS NOTED
EPROJECT NO.:	1826479
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12936770
SHEET	201 OF 247

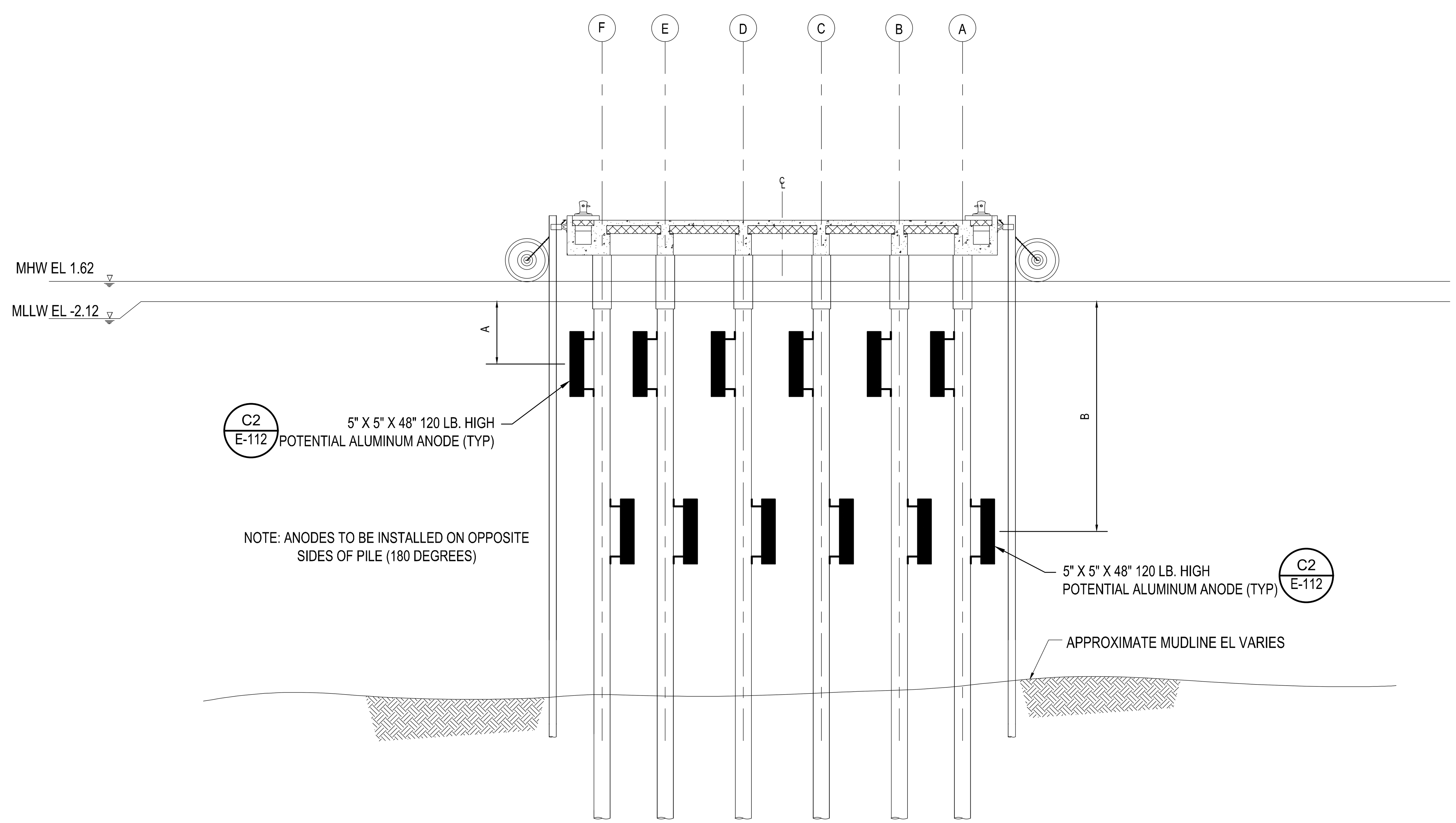
E-111

DRAWFORM REVISION: 25 AUGUST 2020

FILE NAME: C:\Users\dpferguson\DC\ACCDocs\Stantec Consulting\224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1\Project Files\01_WIP\13_electrical\sheet_files\E-108 ELECTRICAL SITE PLANS.dwg LAYOUT NAME: E-111 PLOTTED: Monday, March 02, 2026 - 5:12pm USER: dpferguson



C2 OFFSET MOUNT ANODE INSTALLATION - TYPICAL
NOT TO SCALE E-112



A2 ANODE LOCATION DETAIL - TYPICAL BENT
NOT TO SCALE E-112

BENT	ANODE QUANTITY PER PILE	LOCATION A (FT)	LOCATION B (FT)
1	2	*	*
2	2	*	*
3	2	*	*
4	2	*	*
5	2	*	*
6	2	*	*
7	2	6	6
8	2	8	8
9	2	10	20
10	2	10	20
11	2	12	22
12	2	12	22
13	2	12	22
14	2	12	22
15	2	12	22
16	2	12	22
17	2	12	22
18	2	12	22
19	2	12	22
20	2	12	22
21	2	12	22
22	2	12	22
23	2	12	22
24	2	12	22
25	2	12	22
26	2	12	22
27	2	12	22
28	2	12	22
29	2	12	22
30	2	12	22
31	2	12	22
32	2	12	22
33	2	12	22
34	2	12	22
35	2	12	22
36	2	12	22
37	2	12	22
38	2	12	22
39	2	12	22
40	2	12	22
41	2	12	22
42	2	12	22
43	2	12	22

* DENOTES TO PLACE BOTTOM OF ANODE 1 FOOT FROM MUDLINE

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER

Stantec

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES JC DRW DT CHK JC
PM/DM MRI/CWJ
BRANCH MANAGER NEK
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
CATHODIC PROTECTION PILE PLAN

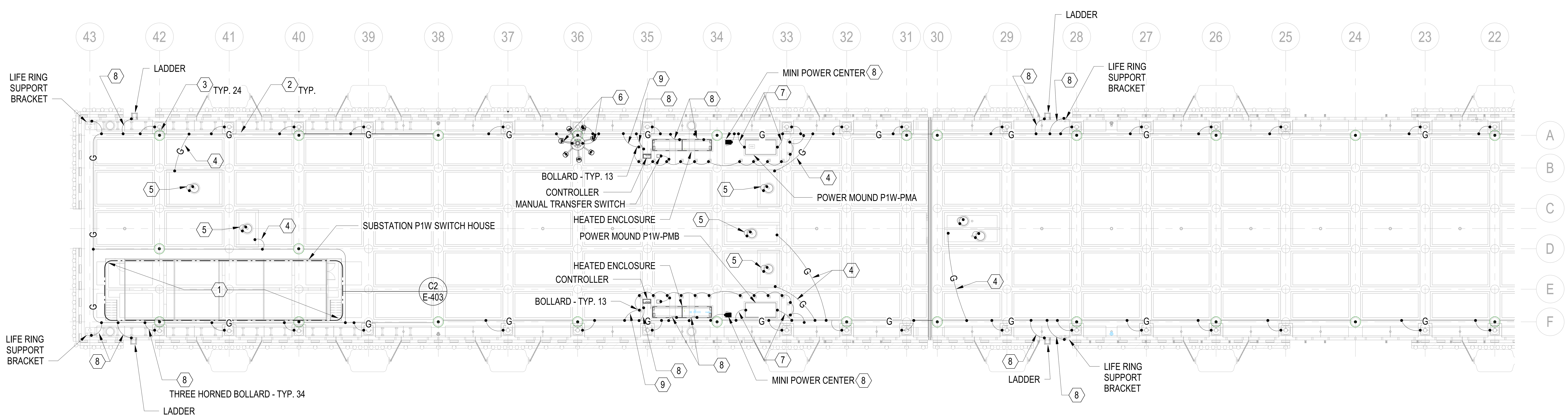
SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936771
SHEET 202 OF 247
E-112

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.
2. ALL CONNECTIONS TO PIER PERIMETER GROUND GRID MUST BE EXOTHERMICALLY WELDED.
3. ALL BCSD GROUND CONDUCTORS MUST BE ROUTED IN-SLAB WITH A MINIMUM COVER OF 3".
4. TERMINATE TO EQUIPMENT BEING BONDED IN-SLAB WHERE GROUNDING TERMINATION POINTS ARE PROVIDED FOR THIS PURPOSE OR WHERE BONDING TO EQUIPMENT STRUCTURAL MEMBERS, WHICH ARE IN-SLAB, CAN BE DONE. WHERE CONCEALED BONDING CAN'T BE ACCOMPLISHED BOND AT THE NEAREST POINT ABOVE SLAB, ARRANGE BONDING CONDUCTORS NEATLY ALONG EQUIPMENT SUPPORTS OR BASE.

SHEET KEYNOTES:

1. PROVIDE CONNECTIONS FROM PIER GROUND GRID TO SUBSTATION POINTS AS SHOWN ON DWG E-403.
2. PROVIDE #4/0 BCSD GROUND CONDUCTOR ENCASED IN CONCRETE AROUND PERIMETER OF PIER.
3. PROVIDE #4/0 BCSD GROUND CONDUCTOR FROM PERIMETER GROUND TO PIER PILE CAP, SEE DETAILS A1 AND A3 ON DWG E-513.
4. PROVIDE #4/0 BCSD GROUND CONDUCTOR FROM PERIMETER GROUND TO MANHOLE TO GROUND MANHOLE METALLIC COMPONENTS INCLUDING COVER, COVER FRAME, LADDERS, RACKS, ETC.
5. PROVIDE A FLEXIBLE CONNECTION WITH A #2 AWG BRAIDED COPPER GROUND CABLE FROM MANHOLE COVER FRAME TO THE COVER. PROVIDE A MINIMUM OF 6'-0" OF CABLE TO ALLOW EASY REMOVAL OF COVER.
6. PROVIDE #4/0 BCSD GROUND CONDUCTOR FROM PERIMETER GROUND GRID TO HIGH MAST LIGHT POLE LIGHTNING PROTECTION CONDUCTORS, SEE DWG E-115 FOR CONTINUATION.
7. PROVIDE A #2/0 BCSD GROUND CONDUCTOR FROM PIER GROUND GRID TO POWER MOUND.
8. PROVIDE A #2 BCSD GROUND CONDUCTOR FROM PERIMETER GROUND GRID TO EQUIPMENT INDICATED.
9. PROVIDE A #6 BCSD GROUND CONDUCTOR FROM PERIMETER GROUND GRID TO BOLLARDS, LOOP BETWEEN BOLLARDS AS SHOWN.



A2 GROUNDING PLAN - PIER WEST
SCALE: 1" = 20'-0"

PLAN NORTH

1" = 20' 0 10' 20' 40' 80'

APPR | DATE | DESCRIPTION | SYM

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
SEAL

Stantec
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV	CHK RJW
PM/DM		MRI/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
GROUNDING PLAN - PIER WEST

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936772
SHEET 203 OF 247
E-113

DRAWFORM REVISION: 25 AUGUST 2020

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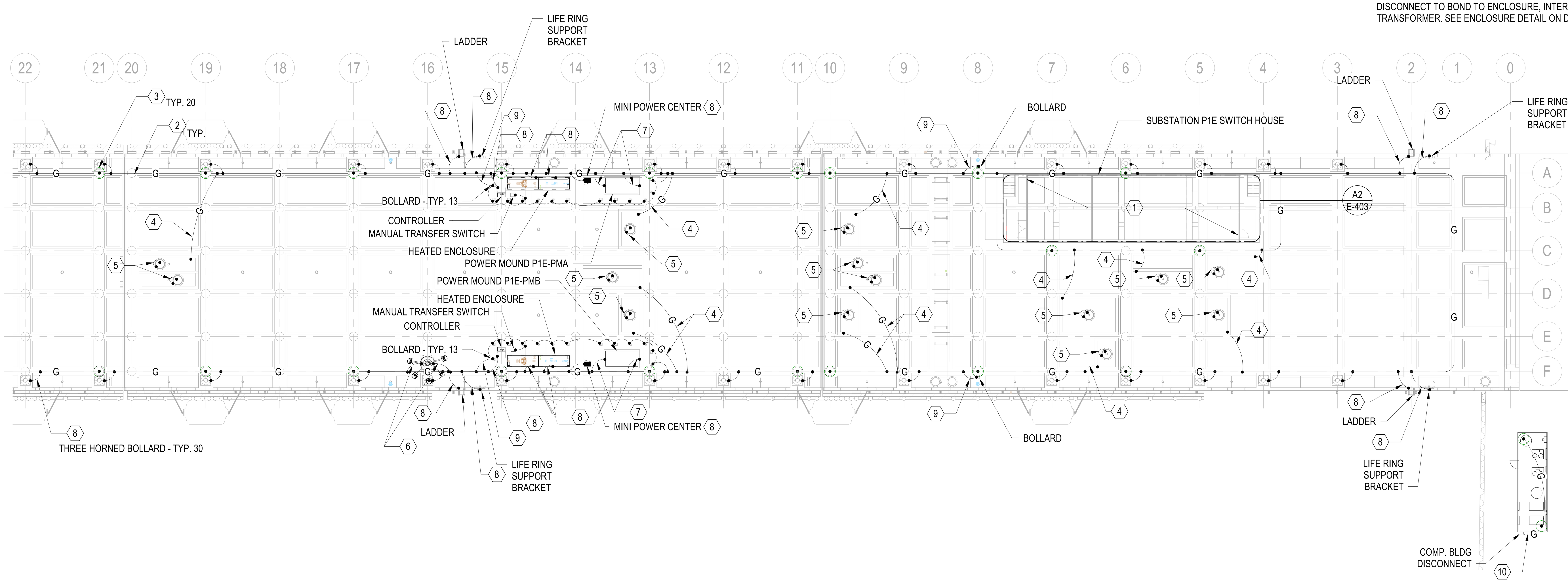
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt
E-113

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.
2. ALL CONNECTIONS TO PIER PERIMETER GROUND GRID MUST BE EXOTHERMICALLY WELDED.
3. ALL BCSD GROUND CONDUCTORS MUST BE ROUTED IN-SLAB WITH A MINIMUM COVER OF 3".
4. TERMINATE TO EQUIPMENT BEING BONDED IN-SLAB WHERE GROUNDING TERMINATION POINTS ARE PROVIDED FOR THIS PURPOSE OR WHERE BONDING TO EQUIPMENT STRUCTURAL MEMBERS, WHICH ARE IN-SLAB, CAN BE DONE. WHERE CONCEALED BONDING CAN'T BE ACCOMPLISHED BOND AT THE NEAREST POINT ABOVE SLAB, ARRANGE BONDING CONDUCTORS NEATLY ALONG EQUIPMENT SUPPORTS OR BASE.

SHEET KEYNOTES:

1. PROVIDE CONNECTIONS FROM PIER GROUND GRID TO SUBSTATION POINTS AS SHOWN ON DWG E-403.
2. PROVIDE #4/0 BCSD GROUND CONDUCTOR ENCASED IN CONCRETE AROUND PERIMETER OF PIER.
3. PROVIDE #4/0 BCSD GROUND CONDUCTOR FROM PERIMETER GROUND TO PIER PILE CAP, SEE DETAILS A1 AND A3 ON DWG E-513.
4. PROVIDE #4/0 BCSD GROUND CONDUCTOR FROM PERIMETER GROUND TO MANHOLE TO GROUND MANHOLE METALLIC COMPONENTS INCLUDING COVER, COVER FRAME, LADDERS, RACKS, ETC.
5. PROVIDE A FLEXIBLE CONNECTION WITH A #2 AWG BRAIDED COPPER GROUND CABLE FROM MANHOLE COVER FRAME TO THE COVER. PROVIDE A MINIMUM OF 6'-0" OF CABLE TO ALLOW EASY REMOVAL OF COVER.
6. PROVIDE #4/0 BCSD GROUND CONDUCTOR FROM PERIMETER GROUND GRID TO HIGH MAST LIGHT POLE LIGHTNING PROTECTION CONDUCTORS, SEE DWG E-116 FOR CONTINUATION.
7. PROVIDE A #2/0 BCSD GROUND CONDUCTOR FROM PERIMETER GROUND GRID TO POWER MOUND.
8. PROVIDE A #2 BCSD GROUND CONDUCTOR FROM PERIMETER GROUND GRID TO EQUIPMENT INDICATED.
9. PROVIDE A #6 BCSD GROUND CONDUCTOR FROM PERIMETER GROUND GRID TO BOLLARDS, LOOP BETWEEN BOLLARDS AS SHOWN.
10. PROVIDE A #2 BCSD GROUND CONDUCTOR FROM DISCONNECT TO PILE CAPS BENEATH STRUCTURE AS SHOWN. CONNECT TO PILE CAPS AS SHOWN ON DETAIL A1 & A3 ON DWG E-513. EXTEND #2 BCSD GROUND CONDUCTOR FROM DISCONNECT TO BOND TO ENCLOSURE, INTERIOR PANELBOARDS AND TRANSFORMER. SEE ENCLOSURE DETAIL ON DRAWING M-401.



PLAN NORTH

A2 GROUNDING PLAN - PIER EAST

SCALE: 1" = 20'-0"

1" = 20' 0 10' 20' 40' 80'

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
GROUNDING PLAN - PIER EAST

NAVFAC

Stantec

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER

SYM	DESCRIPTION	DATE	APPR

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF / DRW GLV / CHK RJW	
PM/DM	MRI/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN
SCALE:	AS NOTED
EPROJECT NO.	1826479
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12936773
SHEET 204 OF 247	

E-114

DRAWFORM REVISION: 25 AUGUST 2020

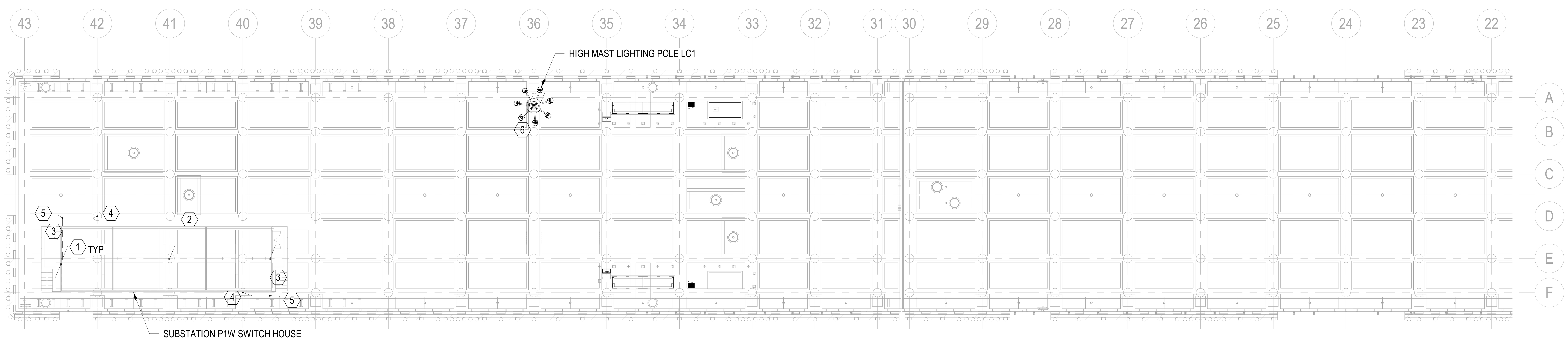
3/2/2026 5:13:23 PM E-114
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.
2. SEE ASSOCIATED SUBSTATION GROUNDING PLAN ON DWG. E403.

SHEET KEYNOTES:

1. PROVIDE A 12" HIGH COPPER LIGHTNING PROTECTION AIR TERMINAL.
2. PROVIDE A CLASS 1 LIGHTNING PROTECTION SYSTEM ON STRUCTURE AS SHOWN WITH ALL REQUIRED ACCESSORIES FOR A UL LISTED SYSTEM. MAIN AND DOWN CONDUCTORS MUST BE TINNED COPPER, SIZED AS REQUIRED, WITH ANTI-CORROSIVE COPPER AIR TERMINALS.
3. ROUTE DOWN CONDUCTORS ON EXTERIOR OF STRUCTURE INTO CONCRETE SLAB BELOW.
4. BOND DOWN CONDUCTOR TO PILE REBAR SYSTEM WITH EXOTHERMIC WELDS OR PRESSURE CONNECTORS.
5. BOND DOWN CONDUCTOR TO STRUCTURE GROUNDING SYSTEM IN CONCRETE SLAB WITH A #4/0 BARE COPPER GROUND WIRE.
6. PROVIDE #4/0 BARE TINNED COPPER LIGHTNING PROTECTION CONDUCTOR FROM LIGHT POLE DOWN CONDUCTORS TO BASE OF POLE. SEE DETAIL A1/E-509 FOR ARRANGEMENT OF AIR TERMINALS ON POLE AND DOWN CONDUCTORS. BOND EACH DOWN CONDUCTOR TO GROUNDING SYSTEM WITH A #4/0 BARE COPPER GROUND WIRE, SEE DWG E-113 FOR CONNECTION TO GROUND GRID.



A2 LIGHTNING PROTECTION PLAN - PIER WEST

SCALE: 1" = 20'-0"

PLAN NORTH

1" = 20' 0 10' 20' 40' 80'

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
LIGHTNING PROTECTION PLAN - PIER WEST

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS DAVID P FERGUSON ELECTRICAL No. 52642 PROFESSIONAL ENGINEER David Ferguson

SEAL

Stantec

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES DPF DRW GLV CHK RJW
PM/DM MRI/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936774
SHEET 205 OF 247

E-115

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 5:13:25 PM
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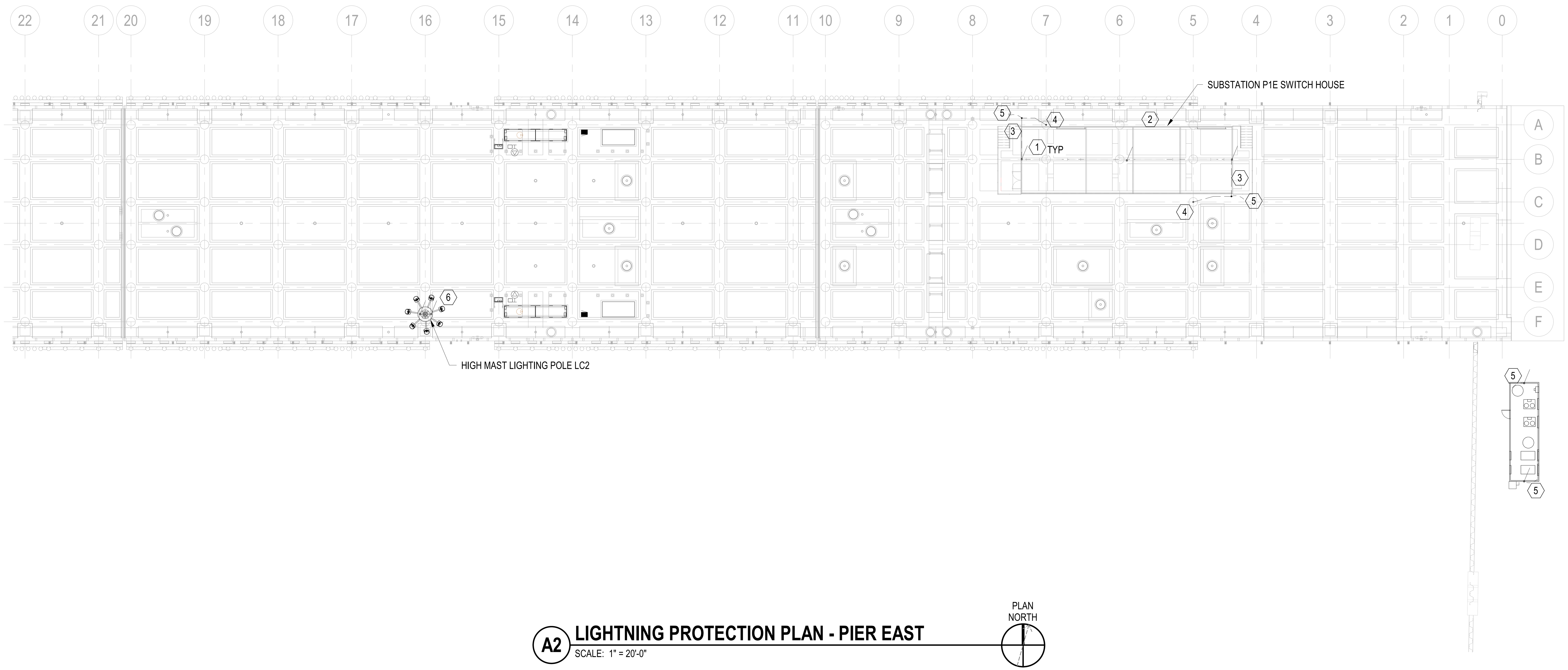
E-115

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.
2. SEE ASSOCIATED SUBSTATION GROUNDING PLAN ON DWG. E-403.

SHEET KEYNOTES:

1. PROVIDE A 12" HIGH COPPER LIGHTNING PROTECTION AIR TERMINAL.
2. PROVIDE A CLASS 1 LIGHTNING PROTECTION SYSTEM ON STRUCTURE AS SHOWN WITH ALL REQUIRED ACCESSORIES FOR A UL LISTED SYSTEM. MAIN AND DOWN CONDUCTORS MUST BE TINNED COPPER, SIZED AS REQUIRED, WITH ANTI-CORROSIVE COPPER AIR TERMINALS.
3. ROUTE DOWN CONDUCTORS ON EXTERIOR OF STRUCTURE INTO CONCRETE SLAB BELOW.
4. BOND DOWN CONDUCTOR TO PILE REBAR SYSTEM WITH EXOTHERMIC WELDS OR PRESSURE CONNECTORS.
5. BOND DOWN CONDUCTOR TO STRUCTURE GROUNDING SYSTEM IN CONCRETE SLAB WITH A #4/0 BARE COPPER GROUND WIRE.
6. PROVIDE #4/0 BARE TINNED COPPER LIGHTNING PROTECTION CONDUCTOR FROM LIGHT POLE DOWN CONDUCTORS TO BASE OF POLE. SEE DETAIL A1/E-509 FOR ARRANGEMENT OF AIR TERMINALS ON POLE AND DOWN CONDUCTORS. BOND EACH DOWN CONDUCTOR TO GROUNDING SYSTEM WITH A #4/0 BARE COPPER GROUND WIRE, SEE DWG E-114 FOR CONNECTION TO GROUND GRID.



A2 LIGHTNING PROTECTION PLAN - PIER EAST
SCALE: 1" = 20'-0"

1" = 20' 0 10' 20' 40' 80'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON ELECTRICAL No. 52642
PROFESSIONAL ENGINEER

Stantec

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES	DPF	DRW	GLV	CHK	RJW
PM/DM					MRI/CWJ
BRANCH MANAGER					WRB
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
LIGHTNING PROTECTION PLAN - PIER EAST

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936775
SHEET 206 OF 247

E-116

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 5:13:26 PM E-116
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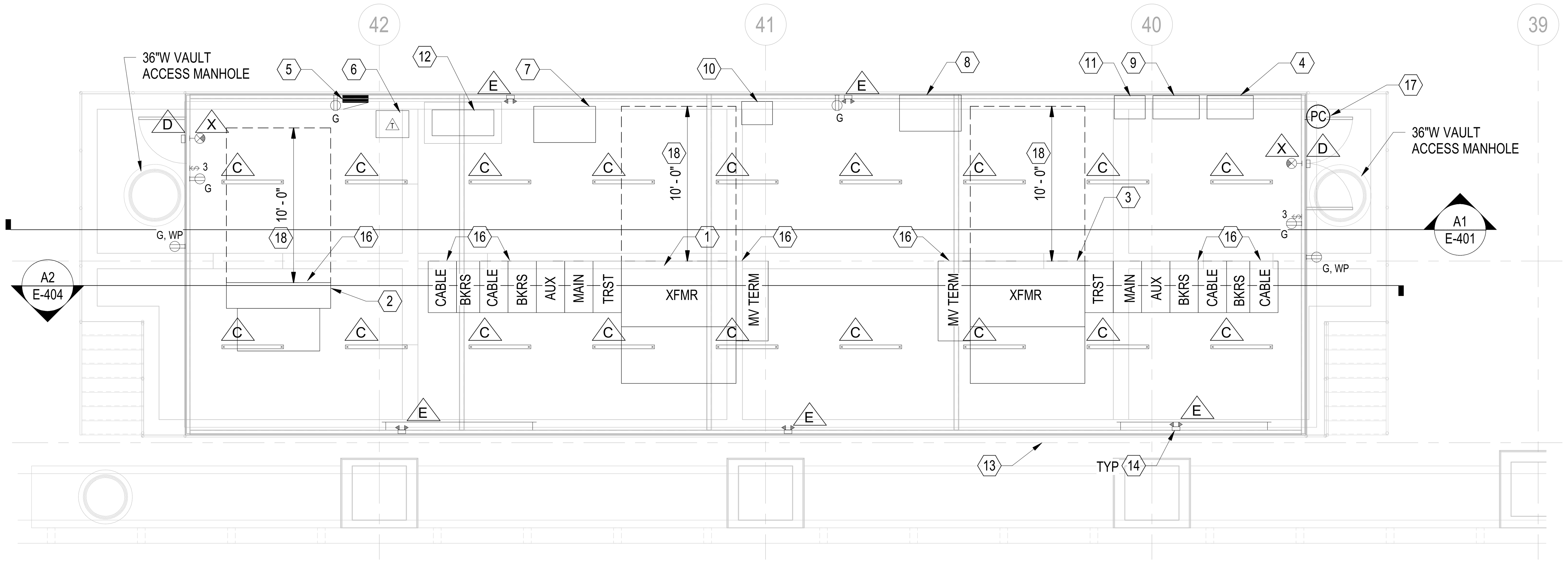
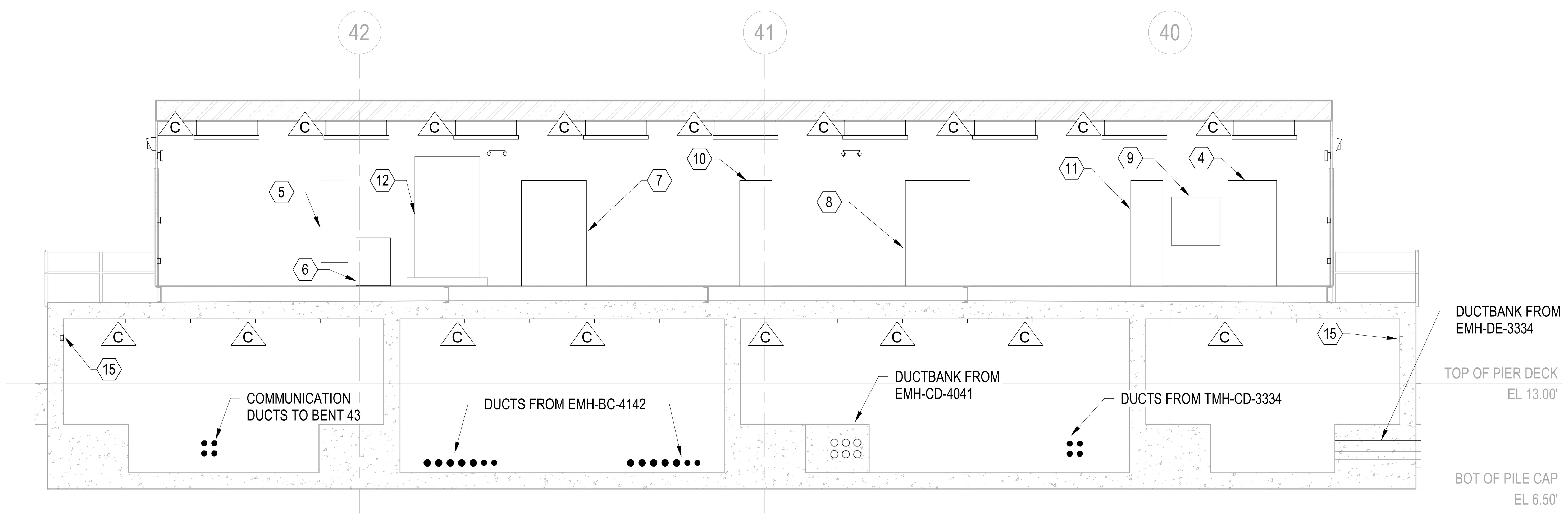


C1 SUBSTATION P1W - PART PLAN

3/16" = 1'-0"

PLAN NORTH



A1 SUBSTATION P1W FRONT ELEVATION

3/16" = 1'-0"

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR LUMINAIRE SCHEDULE SEE DRAWING E-508. FOR POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, E-603 AND E-604. FOR CIRCUITING, RACEWAYS AND LIGHTING FOR DEVICES AND LIGHTING SEE DRAWING E-613. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.
2. UNIT SUBSTATION TRANSFORMERS AND PAD MOUNTED TRANSFORMER TO BE FURNISHED WITH INTEGRAL CONTAINMENT RATED FOR 110% OF TRANSFORMER FLUID VOLUME. FLUID CONTAINMENT AREA MUST MAINTAIN A MINIMUM OF 3' OF CLEARANCE FROM THE PREFABRICATED SUBSTATION ENCLOSURE WALLS. FOR VISUAL CLARITY THE CONTAINMENT AREAS ARE NOT SHOWN.
3. COORDINATE CONDUIT RISER EXACT LOCATIONS WITH EQUIPMENT MANUFACTURERS. AVOID HOUSE FLOOR STRUCTURAL MEMBERS AND ELECTRICAL EQUIPMENT CONFLICTING WITH CABLE INSTALLATION.
4. FOR VAULT PLAN AND REAR ELEVATION SEE DRAWING E-404.
5. SEE STRUCTURAL DRAWINGS FOR ADDITIONAL INFORMATION ON SUBSTATION VAULT STRUCTURE.

SHEET KEYNOTES:

1. SUBSTATION P1W-SPA.
2. TRANSFORMER P1W-IP-XFMR.
3. SUBSTATION P1W-SPB.
4. LIGHTING CONTROL CABINET.
5. PANEL P1W-RP.
6. 30KVA, 480V TO 120/208V TRANSFORMER.
7. POWER FACTOR CORRECTION CAPACITOR BANK P1W-SPA-PFC.
8. POWER FACTOR CORRECTION CAPACITOR BANK P1W-SPB-PFC.
9. FIBER ENCLOSURE P1W-FIB.
10. METERING ENCLOSURE P1W-SPA-MET.
11. METERING ENCLOSURE P1W-SPB-MET.
12. DISTRIBUTION PANEL P1W-IP-MDP.
13. PREFABRICATED SUBSTATION EQUIPMENT ENCLOSURE MANUFACTURER MUST PROVIDE COMPLETE ASSEMBLY INCLUDING HOUSE SKID, INTERIOR EQUIPMENT AND ASSOCIATED CABLING. ENCLOSURE LENGTH MUST BE 72'-7" MINIMUM. ENCLOSURE WIDTH MUST BE 22'-0". ENCLOSURE MUST BE INSULATED, AIR SEALED AND ENVIRONMENTALLY CONTROLLED.
14. WALL MOUNTED EMERGENCY BATTERY LIGHTING UNIT. UNIT MUST HAVE TWO HEADS, A TEST SWITCH AND HAVE A FIBERGLASS ENCLOSURE.
15. THREE WAY WATERPROOF LIGHT SWITCH TO CONTROL VAULT LIGHTING. LOCATE ADJACENT TO VAULT MANHOLE ENTRANCE.
16. PROVIDE 5" RTRC-XW CONDUIT SLEEVES UON WITH COUPLINGS ON BOTH ENDS. SLEEVES MUST EXTEND 6" ABOVE EQUIPMENT BASE. QUANTITY AND LOCATION OF SLEEVES TO BE COORDINATED WITH EQUIPMENT. SEAL PENETRATIONS WITH DUX SEAL. CAP SPARE CONDUITS WITH MANUFACTURED END PLUG ON BOTH SIDES. SLEEVES MUST PENETRATE CONCRETE STRUCTURE, EQUIPMENT ENCLOSURE AND EQUIPMENT. AFTER CABLES ARE INSTALLED SEAL AROUND INSIDE OF CONDUITS WITH DUX SEAL. NO MOISTURE MUST ENTER EQUIPMENT ENCLOSURES.
17. PROVIDE PHOTOCELL ON HOUSE ROOF. PROVIDE 3/4"C WITH WIRING AS REQUIRED TO LIGHTING CONTROL CABINET.
18. WORKING SPACE CLEARANCE IN FRONT OF TRANSFORMER MUST BE 10'-0".

3/16" = 1' - 0" 0 5' 10' 20'

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER

Stantec

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

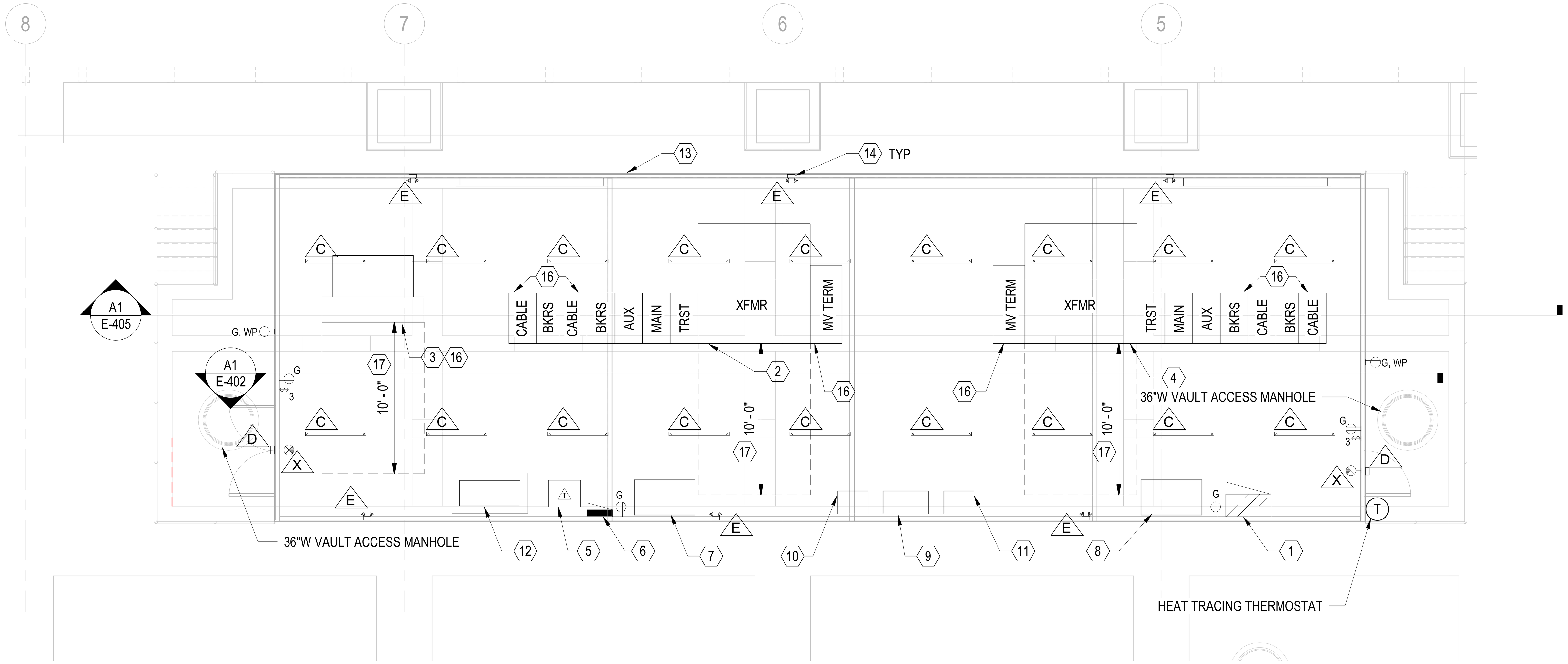
DES DPF	DRW GLV	CHK RJW	
PM/DM			MRI/CWJ
BRANCH MANAGER			WRB
CHIEF ENG/ARCH			JWJ
FIRE PROTECTION			HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
SUBSTATION P1W PART PLAN & ELEVATION

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936776
SHEET 207 OF 247
E-401

DRAWFORM REVISION: 25 AUGUST 2020

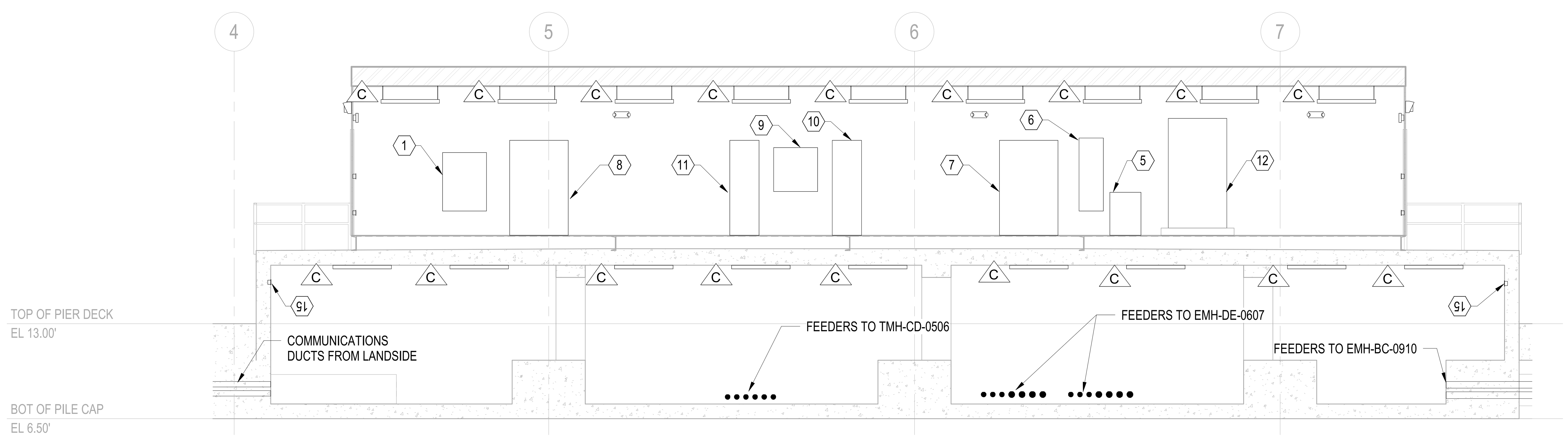
3/2/2026 5:13:31 PM E-401
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt



PLAN NORTH

C1 SUBSTATION P1E - PART PLAN

3/16" = 1'-0"



A1 SUBSTATION P1E - FRONT ELEVATION

3/16" = 1'-0"

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR LUMINAIRE SCHEDULE SEE DRAWING E-508. FOR POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, E-603 AND E-604. FOR CIRCUITING, RACEWAYS AND LIGHTING FOR DEVICES AND LIGHTING SEE DRAWING E-612. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.
2. UNIT SUBSTATION TRANSFORMERS AND PAD MOUNTED TRANSFORMER TO BE FURNISHED WITH INTEGRAL CONTAINMENT RATED FOR 110% OF TRANSFORMER FLUID VOLUME. FLUID CONTAINMENT AREA MUST MAINTAIN A MINIMUM OF 3' OF CLEARANCE FROM THE PREFABRICATED SUBSTATION ENCLOSURE WALLS. FOR VISUAL CLARITY THE CONTAINMENT AREAS ARE NOT SHOWN.
3. COORDINATE CONDUIT RISER EXACT LOCATIONS WITH EQUIPMENT MANUFACTURERS. AVOID HOUSE FLOOR STRUCTURAL MEMBERS AND ELECTRICAL EQUIPMENT CONFLICTING WITH CABLE INSTALLATION.
4. FOR VAULT PLAN AND REAR ELEVATION SEE DRAWING E-405.
5. SEE STRUCTURAL DRAWINGS FOR ADDITIONAL INFORMATION ON ELEVATED SUBSTATION STRUCTURE.

SHEET KEYNOTES:

1. HEAT TRACE CONTROL PANEL.
2. SUBSTATION P1E-SPA.
3. TRANSFORMER P1E-IP-XFMR. CLEARANCE SPACE IN FRONT OF TRANSFORMER MUST BE 10'-0" MINIMUM.
4. SUBSTATION P1E-SPB.
5. 30KVA, 480V TO 120/208V TRANSFORMER.
6. PANEL P1E-RP.
7. POWER FACTOR CORRECTION CAPACITOR BANK P1E-SPA-PFC.
8. POWER FACTOR CORRECTION CAPACITOR BANK P1E-SPB-PFC.
9. FIBER ENCLOSURE P1E-FIB.
10. METERING ENCLOSURE P1E-SPA-MET.
11. METERING ENCLOSURE P1E-SPB-MET.
12. DISTRIBUTION PANEL P1E-IP-MDP.
13. PREFABRICATED SUBSTATION EQUIPMENT ENCLOSURE MANUFACTURER MUST PROVIDE COMPLETE ASSEMBLY INCLUDING HOUSE SKID, INTERIOR EQUIPMENT AND ASSOCIATED CABLING. ENCLOSURE LENGTH MUST BE 72'-1" MINIMUM. ENCLOSURE WIDTH MUST BE 23'-0". ENCLOSURE MUST BE INSULATED, AIR SEALED AND ENVIRONMENTALLY CONTROLLED.
14. WALL MOUNTED EMERGENCY BATTERY LIGHTING UNIT. UNIT MUST HAVE TWO HEADS, A TEST SWITCH AND HAVE A FIBERGLASS ENCLOSURE.
15. THREE WAY WATERPROOF LIGHT SWITCH TO CONTROL VAULT LIGHTING. LOCATE ADJACENT TO VAULT MANHOLE ENTRANCE.
16. PROVIDE 5" RTRC-XW CONDUIT SLEEVES UON WITH COUPLINGS ON BOTH ENDS. SLEEVES MUST EXTEND 6" ABOVE EQUIPMENT BASE. QUANTITY AND LOCATION OF SLEEVES TO BE COORDINATED WITH EQUIPMENT. SEAL PENETRATIONS WITH DUX SEAL. CAP SPARE CONDUITS WITH MANUFACTURED END PLUG ON BOTH SIDES. SLEEVES MUST PENETRATE CONCRETE STRUCTURE, EQUIPMENT ENCLOSURE AND EQUIPMENT. AFTER CABLES ARE INSTALLED SEAL AROUND INSIDE OF CONDUITS WITH DUX SEAL. NO MOISTURE MUST ENTER EQUIPMENT ENCLOSURES.
17. WORKING SPACE CLEARANCE IN FRONT OF TRANSFORMER MUST BE 10'-0".

3/16" = 1' - 0" 0 5' 10' 20'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER

Stantec

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES	DPF	DRW	GLV	CHK	RJW
PM/DM	MRI/CWJ				
BRANCH MANAGER	WRB				
CHIEF ENG/ARCH	JWJ				
FIRE PROTECTION	HPN				

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER

SUBSTATION P1E PART PLAN & ELEVATION

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936777
SHEET 208 OF 247

E-402

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 5:13:36 PM
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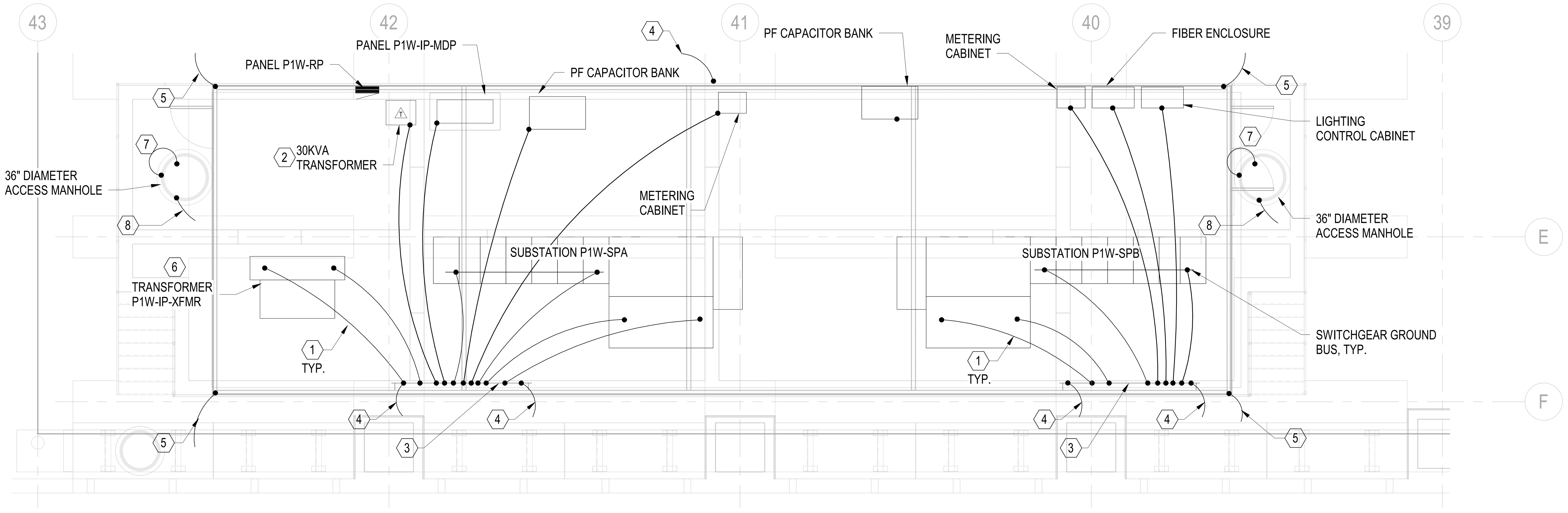
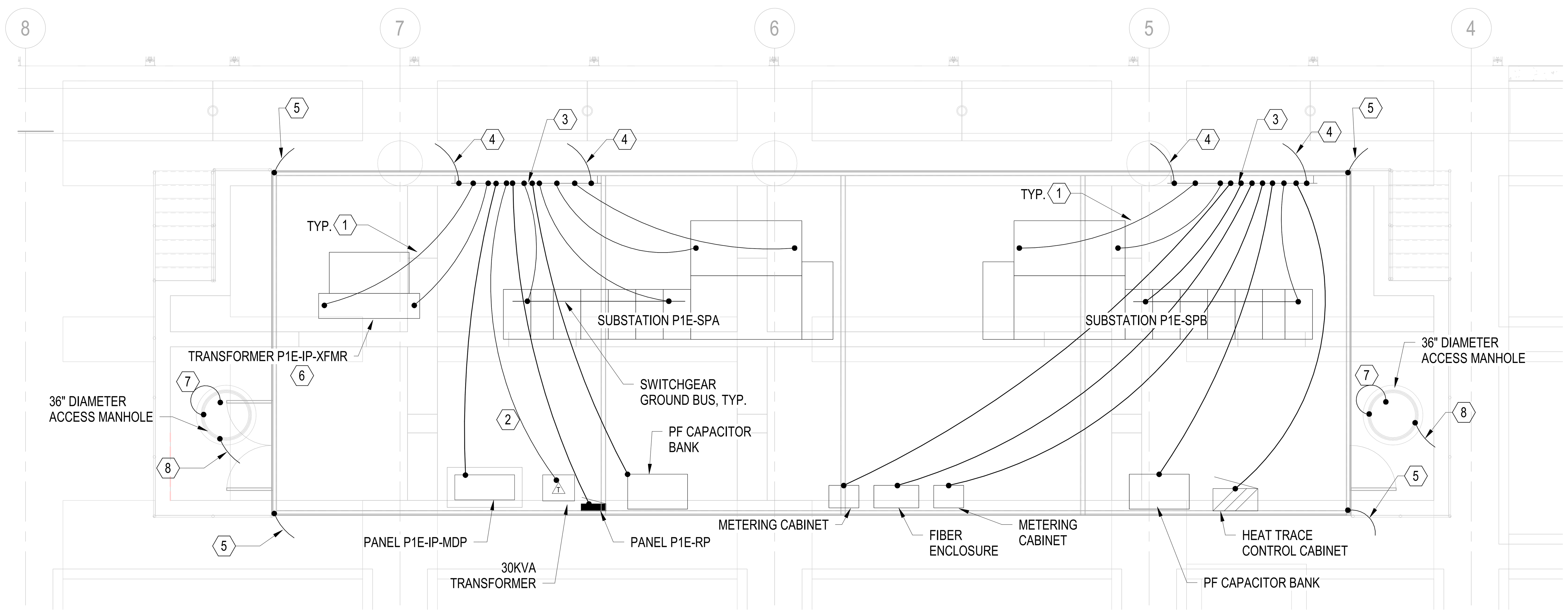
E-402



C2 SUBSTATION P1W - GROUNDING PART PLAN

3/16" = 1'-0" (E-113)

PLAN NORTH



A2 SUBSTATION P1E - GROUNDING PART PLAN

3/16" = 1'-0" (E-114)

PLAN NORTH

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.
2. ALL CONNECTIONS TO PIER GROUND GRID MUST BE EXOTHERMICALLY WELDED.
3. ALL METALLIC OBJECTS IN VAULTS BELOW SUBSTATION, WHICH ARE NOT ALREADY BONDED BY APPROVED NEC MEANS, ARE TO BE BONDED TO ONE OF THE WALL MOUNTED SUBSTATION GROUND BARS. THIS INCLUDES BUT IS NOT LIMITED TO CABLE TRAYS, WALL MOUNTED RACKS, AND PERMANENT ACCESS LADDERS.

SHEET KEYNOTES:

1. PROVIDE A #4/0 BCSD GROUND CONDUCTOR ENCASED IN CONCRETE FROM EQUIPMENT GROUND BAR TO WALL MOUNTED GROUND BAR.
2. PROVIDE #2 BCSD TO TRANSFORMER NEUTRAL/GROUND TO WALL MOUNTED COPPER GROUND BAR.
3. PROVIDE COPPER GROUND BAR ON PERIMETER WALL: 4" x 1/4" x 10'-0" LONG. SUPPORT BAR AT 3'-0" INTERVALS AT 3'-0" AFF.
4. CONNECT WALL MOUNTED COPPER GROUND BAR TO PIER GROUND GRID WITH A #4/0 BSCD GROUNDING CONDUCTOR ENCASED IN CONCRETE. SEE PIER GROUNDING PLAN ON DRAWING E-113.
5. GROUND SWITCH HOUSE STEEL AT CORNERS WITH A #4/0 BCSD GROUND CONDUCTOR TO PIER GROUND GRID.
6. PROVIDE #2/0 BCSD FROM TRANSFORMER NEUTRAL TO TRANSFORMER GROUND BUS.
7. PROVIDE A FLEXIBLE CONNECTION WITH A #2 AWG BRAIDED COPPER GROUND CABLE FROM MANHOLE COVER FRAME TO THE COVER. PROVIDE A MINIMUM OF 6'-0" OF CABLE TO ALLOW EASY REMOVAL OF COVER.
8. PROVIDE A #4/0 BCSD FROM MANHOLE COVER FRAME TO PIER GROUND GRID.

3/16" = 1' - 0" 0 5' 10' 20'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON ELECTRICAL No. 52642 REGISTERED PROFESSIONAL ENGINEER
SEAL

Stantec

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES DPF DRW GLV CHK RJW
PM/DM MRI/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
SUBSTATION GROUNDING PART PLANS

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936778
SHEET 209 OF 247
E-403
DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 5:13:39 PM E-403
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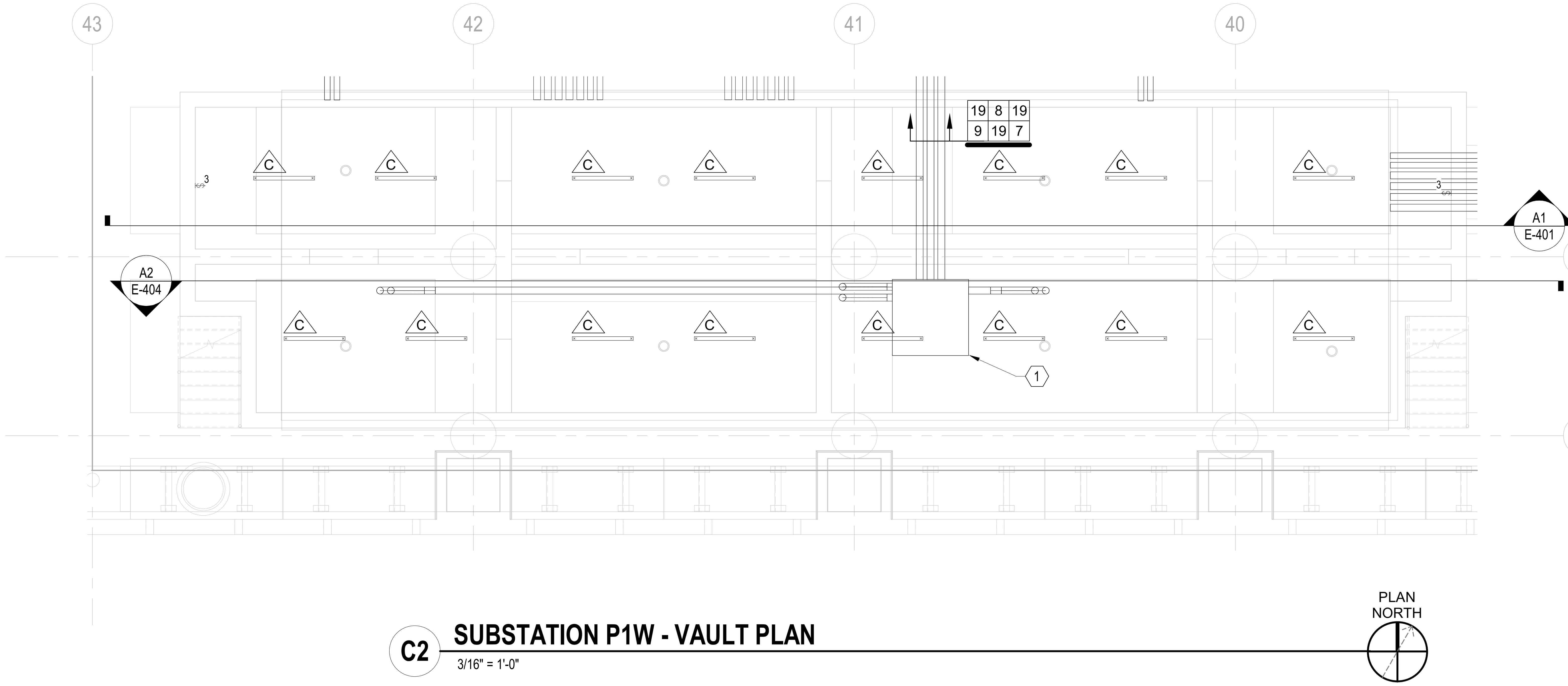
GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR LUMINAIRE SCHEDULE SEE DRAWING E-508. FOR POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, E-603 AND E-604. FOR CIRCUITING, RACEWAYS AND WIRING FOR VAULT LIGHTING SEE DRAWING E-613. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.
2. COORDINATE CONDUIT RISER EXACT LOCATIONS WITH EQUIPMENT MANUFACTURERS. AVOID HOUSE FLOOR STRUCTURAL MEMBERS AND ELECTRICAL EQUIPMENT CONFLICTING WITH CABLE INSTALLATION.

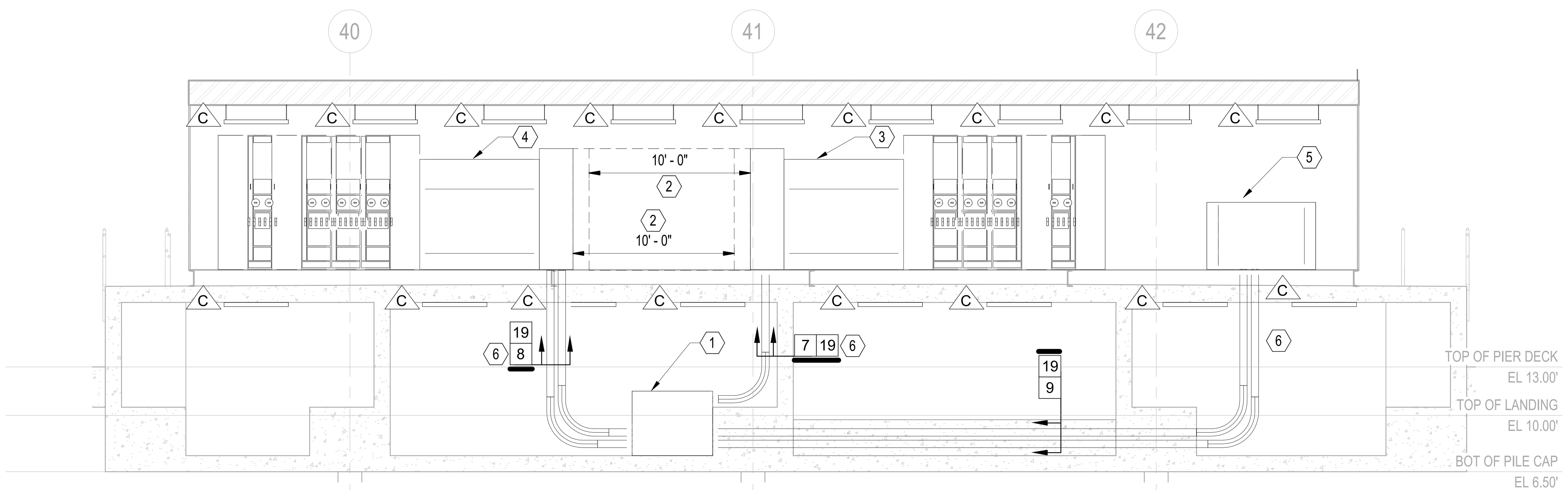
SHEET KEYNOTES:

1. PROVIDE 60"L x 60"W x 48"T PULLBOX.
2. CLEARANCE SPACE FOR TERMINATION CHAMBER ACCESS DOOR MUST BE 10'-0" MINIMUM.
3. SUBSTATION P1W-SPA.
4. SUBSTATION P1W-SPB.
5. TRANSFORMER P1W-IP-XFMR.
6. EXTEND FEEDERS UP INTO EQUIPMENT WITHOUT CONCRETE ENCASEMENT.



C2 SUBSTATION P1W - VAULT PLAN

3/16" = 1'-0"



A2 SUBSTATION P1W - REAR ELEVATION

3/16" = 1'-0"

3/16" = 1' - 0" 0 5' 10' 20'

NAVFAC

DAVID P FERGUSON ELECTRICAL No. 52642 COMMONWEALTH OF MASSACHUSETTS PROFESSIONAL ENGINEER

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES	DPF	DRW	GLV	CHK	RJW
PM/DM					MRI/CWJ
BRANCH MANAGER					WRB
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
SUBSTATION P1W VAULT PART PLAN & ELEVATION

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936779
SHEET 210 OF 247

E-404

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 5:13:43 PM E-404
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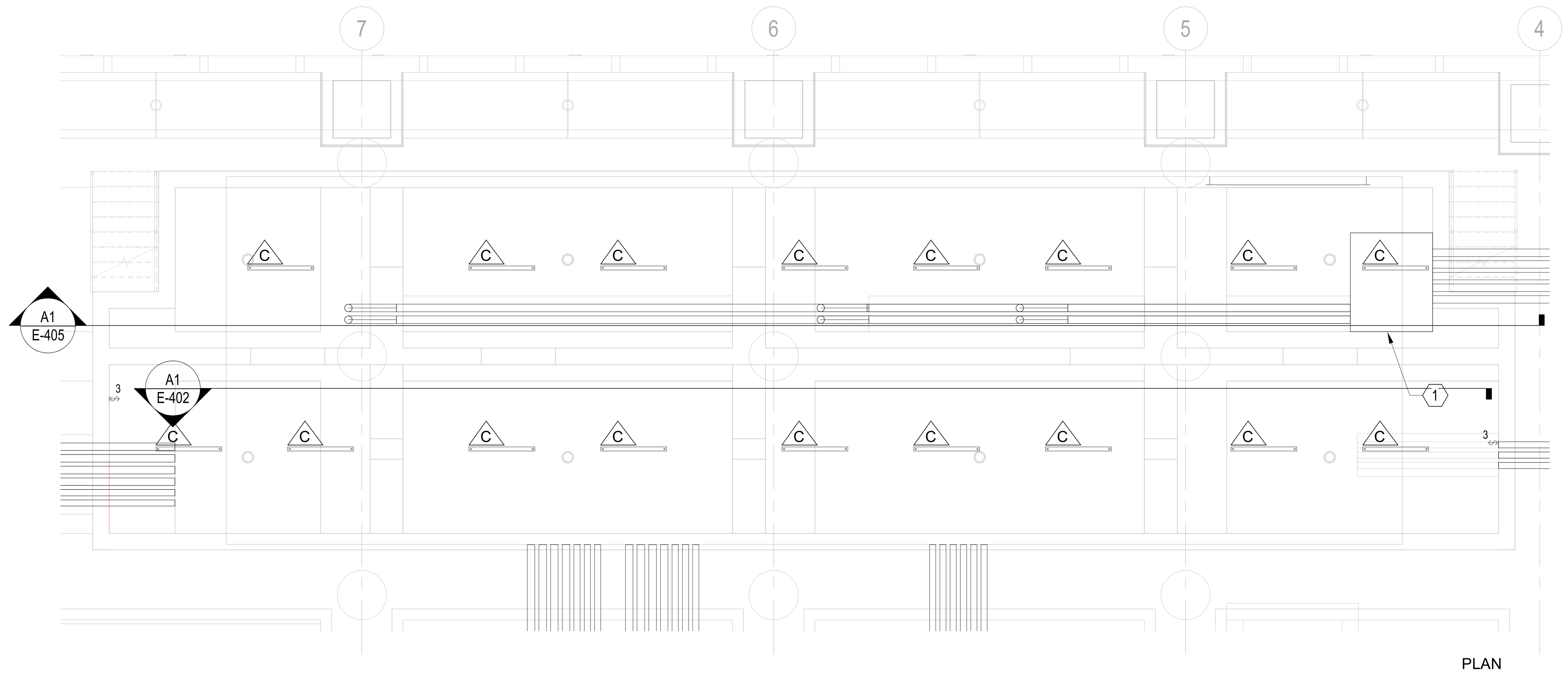
GENERAL SHEET NOTES:

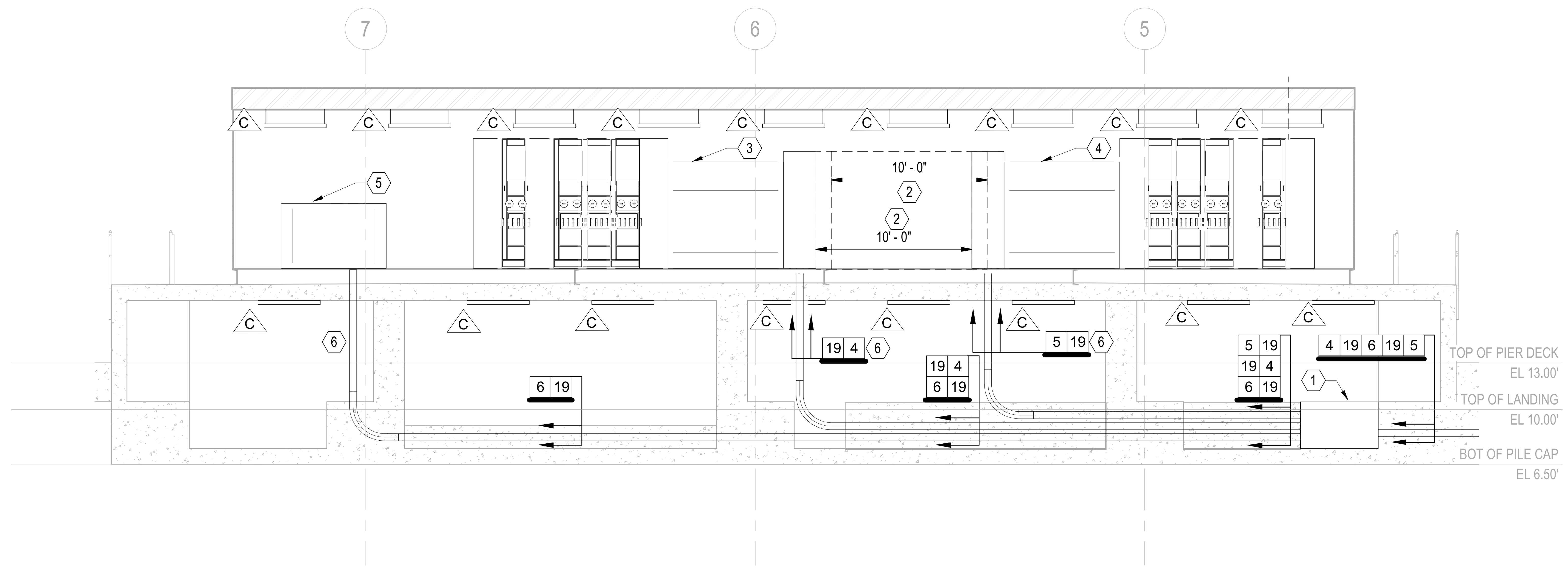
1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR LUMINAIRE SCHEDULE SEE DRAWING E-508. FOR POWER AND COMMUNICATIONS ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, E-603 AND E-604. FOR CIRCUITING, RACEWAYS AND WIRING TO VAULT LIGHTING SEE DRAWING E-612. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.
2. COORDINATE CONDUIT RISER EXACT LOCATIONS WITH EQUIPMENT MANUFACTURERS. AVOID HOUSE FLOOR STRUCTURAL MEMBERS AND ELECTRICAL EQUIPMENT CONFLICTING WITH CABLE INSTALLATION.

SHEET KEYNOTES:

1. PROVIDE 72"L x 60"W x 36"T PULLBOX.
2. CLEARANCE SPACE FOR TERMINATION CHAMBER ACCESS DOOR MUST BE 10'-0" MINIMUM.
3. SUBSTATION P1E-SPA.
4. SUBSTATION P1E-SPB.
5. TRANSFORMER P1E-IP-XFMR.
6. EXTEND FEEDER CONDUIT(S) UP INTO EQUIPMENT WITHOUT CONCRETE ENCASEMENT.



C1 **SUBSTATION P1E - VAULT PLAN**
3/16" = 1'-0"



A1 **SUBSTATION P1E - REAR ELEVATION**
NOT TO SCALE (E-402)

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
SEAL

Stantec
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV	CHK RJW
PM/DM	MRI/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
SUBSTATION P1E VAULT PART PLAN & ELEVATION

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936780
SHEET 211 OF 247

E-405

DRAWFORM REVISION: 25 AUGUST 2020

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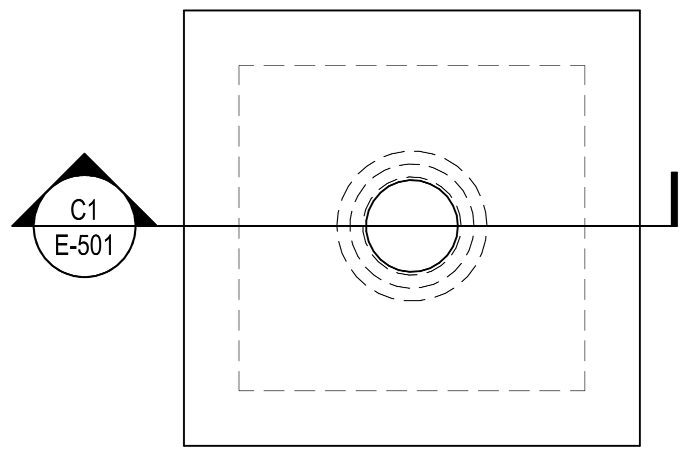
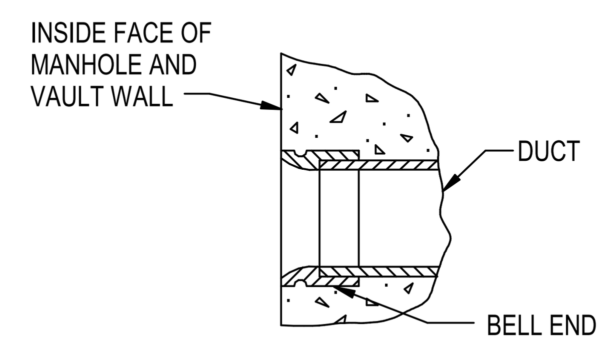
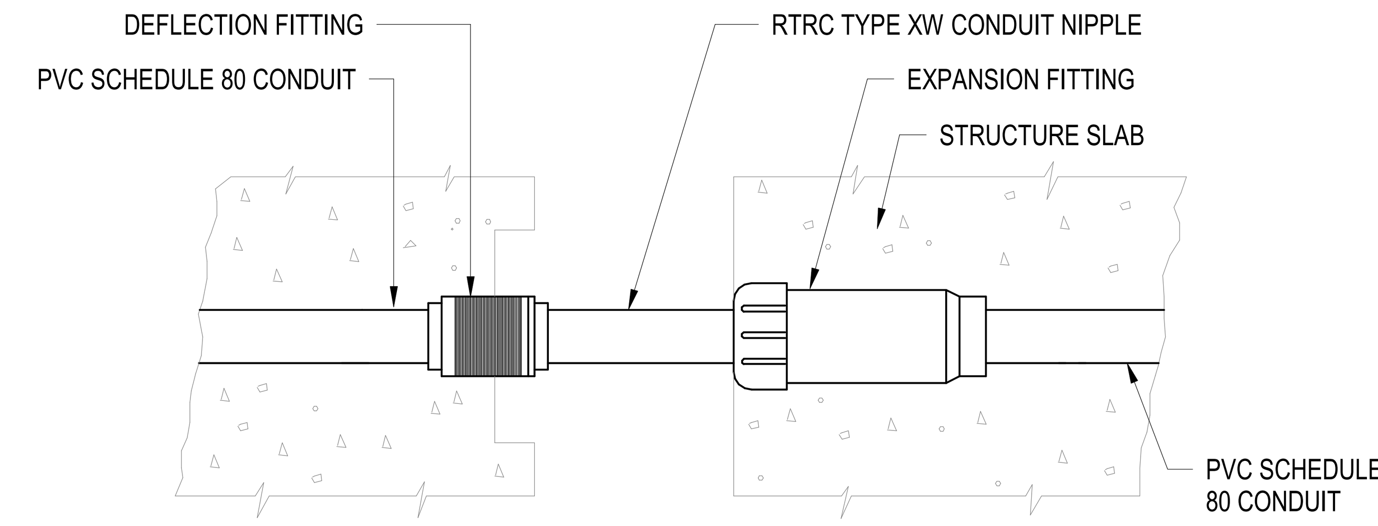
E-405



D1 PLAN - PIER MANHOLES 1 4 6
NOT TO SCALE



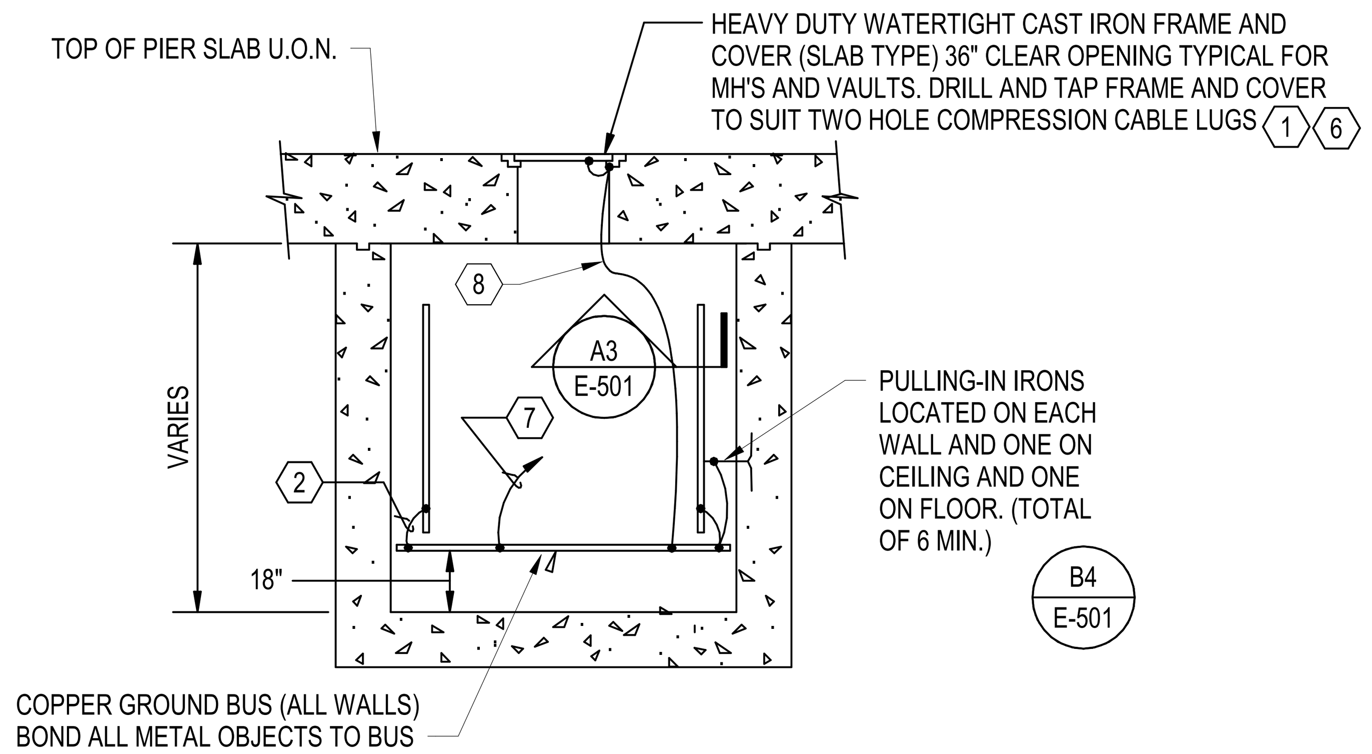
D2 BELL END DETAIL 5
NOT TO SCALE



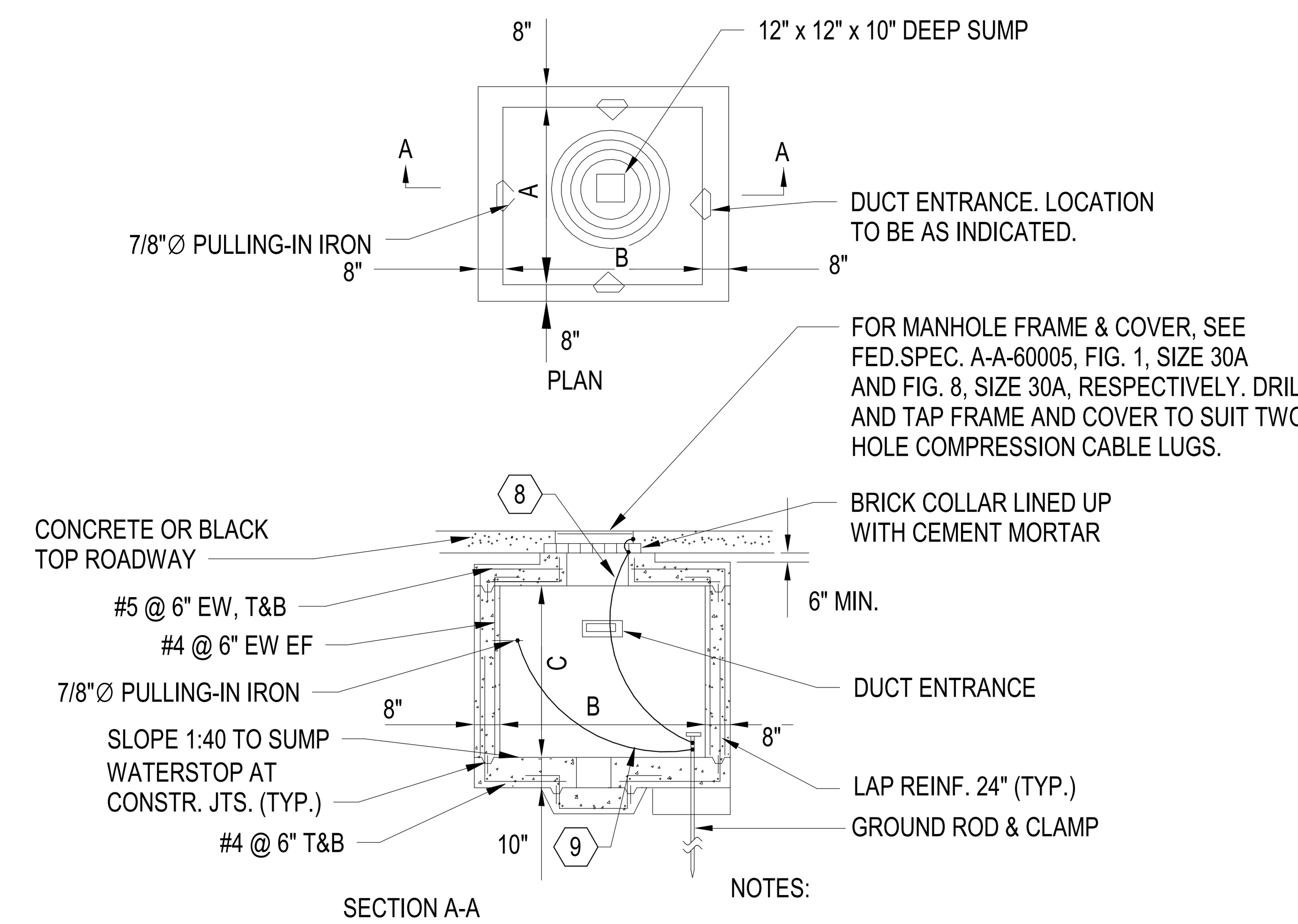
NOTE:

1. WRAP ENTIRE ASSEMBLY IN BURLAP PRIOR TO CONCRETE POUR

C3 COMBINATION EXPANSION-DEFLECTION FITTING
NOT TO SCALE



C1 SECTION OF PIER MANHOLE - TYP 1 4 6
NOT TO SCALE



NOTES:

1. MANHOLE AND COVERS ARE DESIGNED FOR MAXIMUM WHEEL LOAD IN ACCORDANCE WITH AASHTO HS20-44.
2. SEE DETAILS OF CABLE RACKS, DUCT ENTRANCE AND PULLING-IN IRONS ON THIS SHEET.
3. MINIMUM CONCRETE COMPRESSIVE STRENGTH MUST BE 3000 PSI.

MANHOLE DIMENSIONS			
TYPE	A	B	C (AT HIGH PT.)
3	6'-0"	6'-0"	6'-6"
4	6'-0"	8'-0"	6'-6"
4A	7'-0"	7'-0"	6'-6"

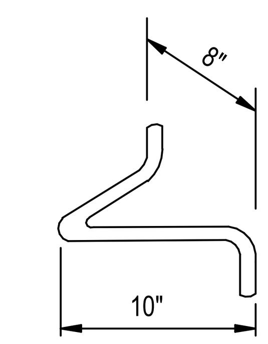
B3 LANDSIDE MANHOLE DETAIL 1 4 6
NOT TO SCALE

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.

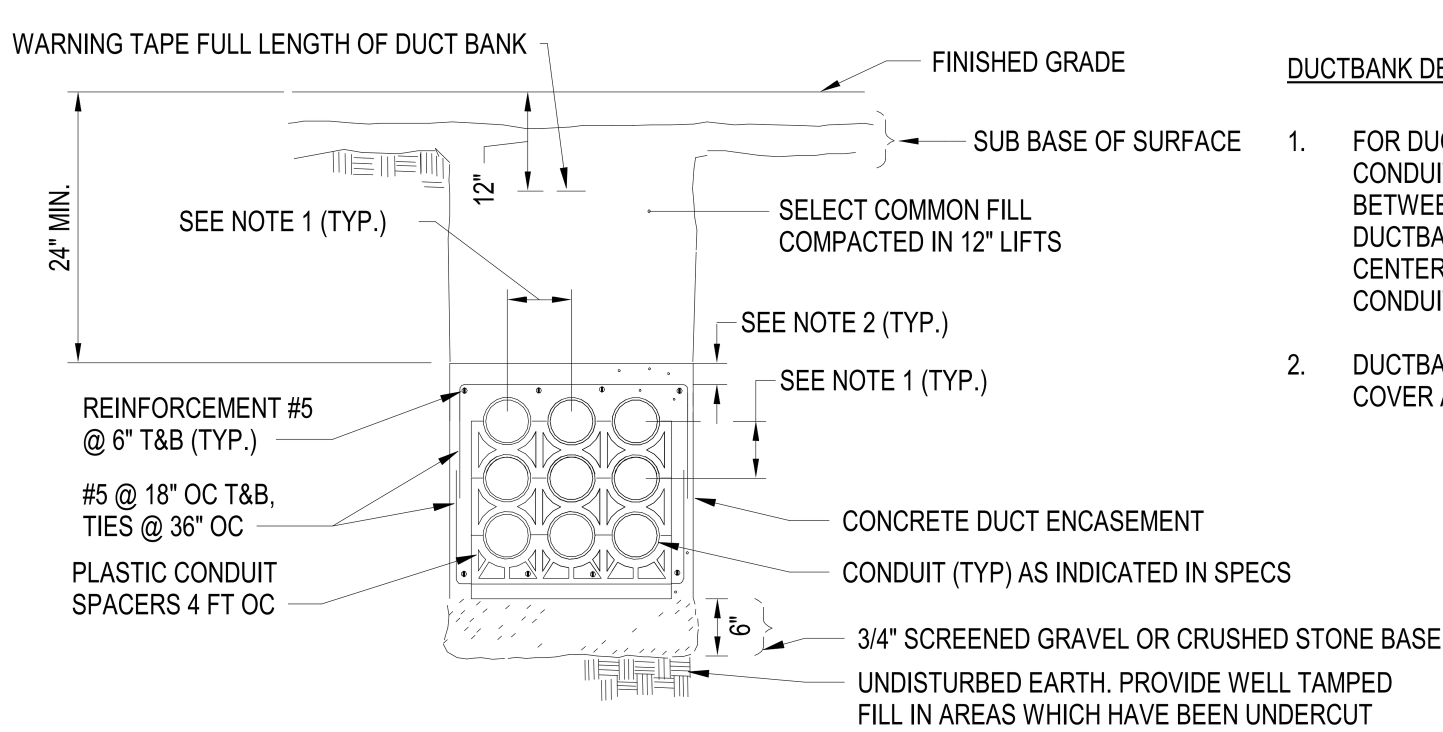
SHEET KEYNOTES:

1. MANHOLES DETAILS SHOWN FOR REFERENCE ONLY. SEE SPECIFICATION SECTIONS 337102 AND 338200.
2. BOND COPPER GROUND BUS AT CORNERS FOR A CONTINUOUS GROUND. GROUND BUS EQUIVALENT TO #4/0 CONDUCTOR MINIMUM.
3. MAXIMUM SPACING 3 FT. (HORIZONTAL) BETWEEN RACKS AND 1 FT FROM RACKS TO CLOSEST WALL. RACKS MUST BE 48" LONG.
4. FOR DETAILS AND LOCATIONS OF MANHOLES SEE ELECTRICAL DRAWINGS.
5. DETAILS A1, D2, B4, AND A3 ARE ALSO APPLICABLE FOR SUBSTATION P1E AND P1W VAULT EXTERIOR WALLS.
6. COVERS ON MANHOLES MUST HAVE A STENCIL PROVIDED BY USING WELD BEAD, FIELD INSTALLED, INDICATING THE MANHOLE IDENTITY AS PROVIDED BY THE GOVERNMENT. EACH CHARACTER MUST BE A MINIMUM OF 3" TALL BY 2" WIDE. THIS IS IN ADDITION TO THE MANHOLE STANDARD CAST IDENTIFICATIONS.
7. #4/0 BCSD TO GROUND NETWORK.
8. PROVIDE #6AWG BONDING JUMPER TO CONNECT MANHOLE COVER AND FRAME TO GROUND ROD.
9. PROVIDE #6AWG BONDING JUMPER TO CONNECT PULLING-IN IRON GROUND ROD.



ALL METAL PARTS MUST BE HOT DIP GALVANIZED

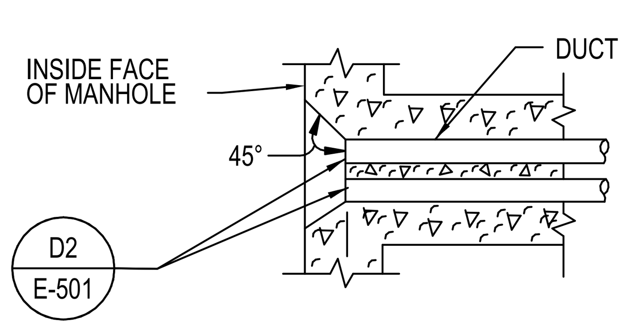
B4 PULLING-IN IRON DETAIL 5
NOT TO SCALE



DUCTBANK DETAIL NOTES:

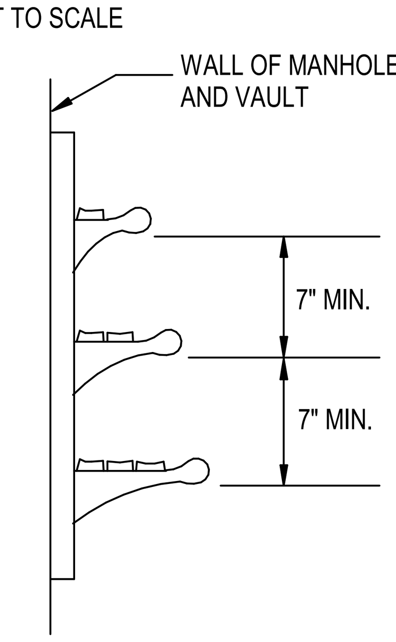
1. FOR DUCTBANKS WITH 4" OR SMALLER CONDUITS, THE CENTERLINE SPACING BETWEEN CONDUITS MUST BE 7.5". FOR DUCTBANKS WITH 5" CONDUITS, THE CENTERLINE SPACING BETWEEN CONDUITS MUST BE 8.5".
2. DUCTBANKS MUST HAVE 3" CONCRETE COVER AROUND THE CONDUITS.

B1 TYPICAL LANDSIDE DUCTBANK DETAIL
NOT TO SCALE



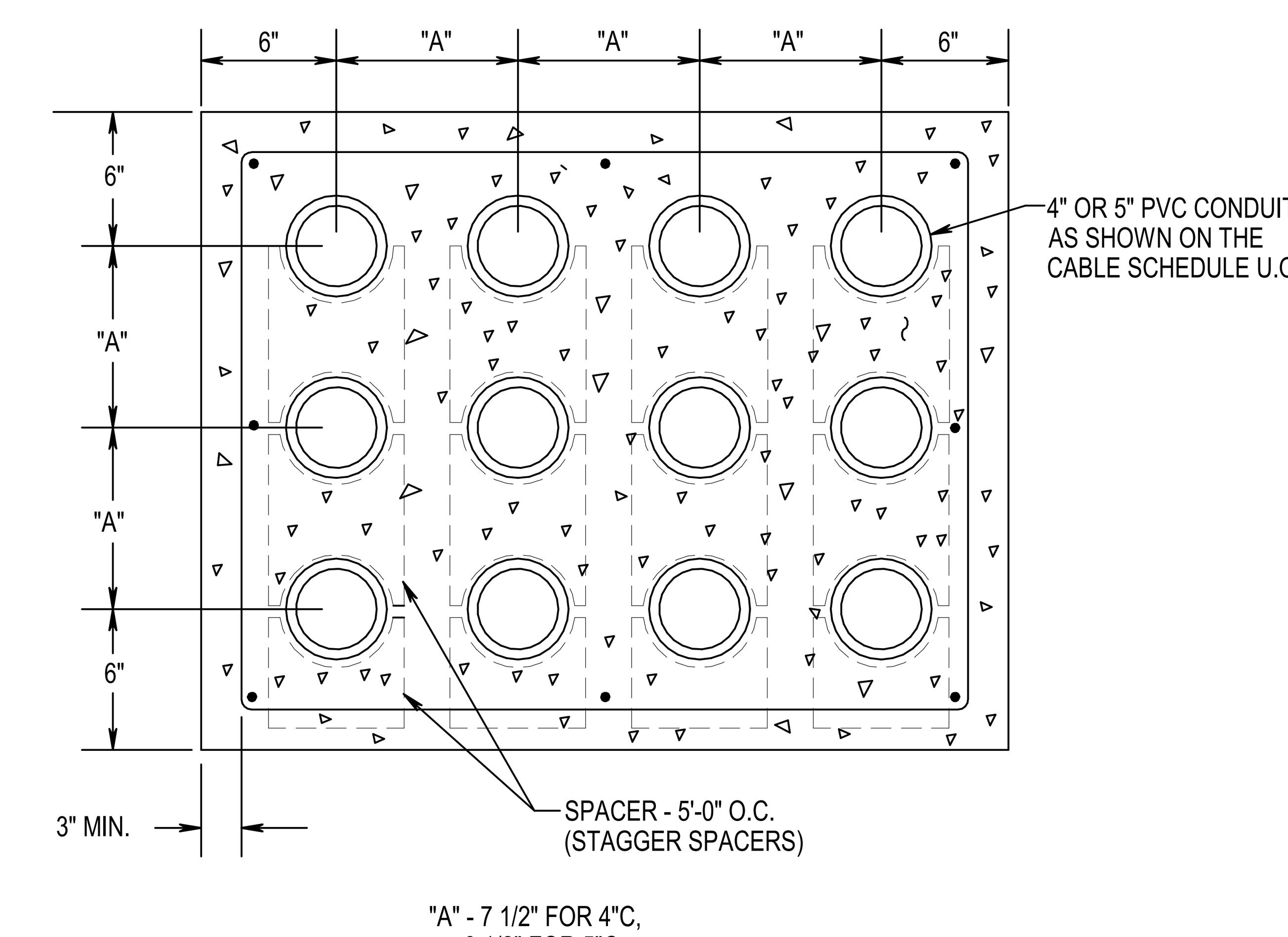
A1 DUCTBANK TO MANHOLE CONDUIT ENTRY DETAIL
NOT TO SCALE

5



NUMBER OF RACKS AS REQUIRED. TO INCLUDE ONE SPARE RACK ARM PER 337102 SPECIFICATIONS.

A3 CABLE RACK DETAIL 3 5
NOT TO SCALE



COORDINATE EFFORTS WITH STRUCTURAL FOR PIER DUCTBANK INSTALLATION. SEE STRUCTURAL DRAWINGS. (ACTUAL ARRANGEMENT VARIES, SEE DRAWINGS, E-101 THRU E-107.)

A4 TYPICAL PIER DUCT BANK SECTION
NOT TO SCALE

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
REGISTERED PROFESSIONAL ENGINEER
SEAL

Stantec
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES	DPF	DRW	GLV	CHK	RJW
PM/DM					MRI/CWJ
BRANCH MANAGER					WRB
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
CI CORE - HAMPTON ROADS
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
MANHOLE & DUCTBANK DETAILS

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936781
SHEET 212 OF 247
E-501

DRAWFORM REVISION: 25 AUGUST 2020

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E-501
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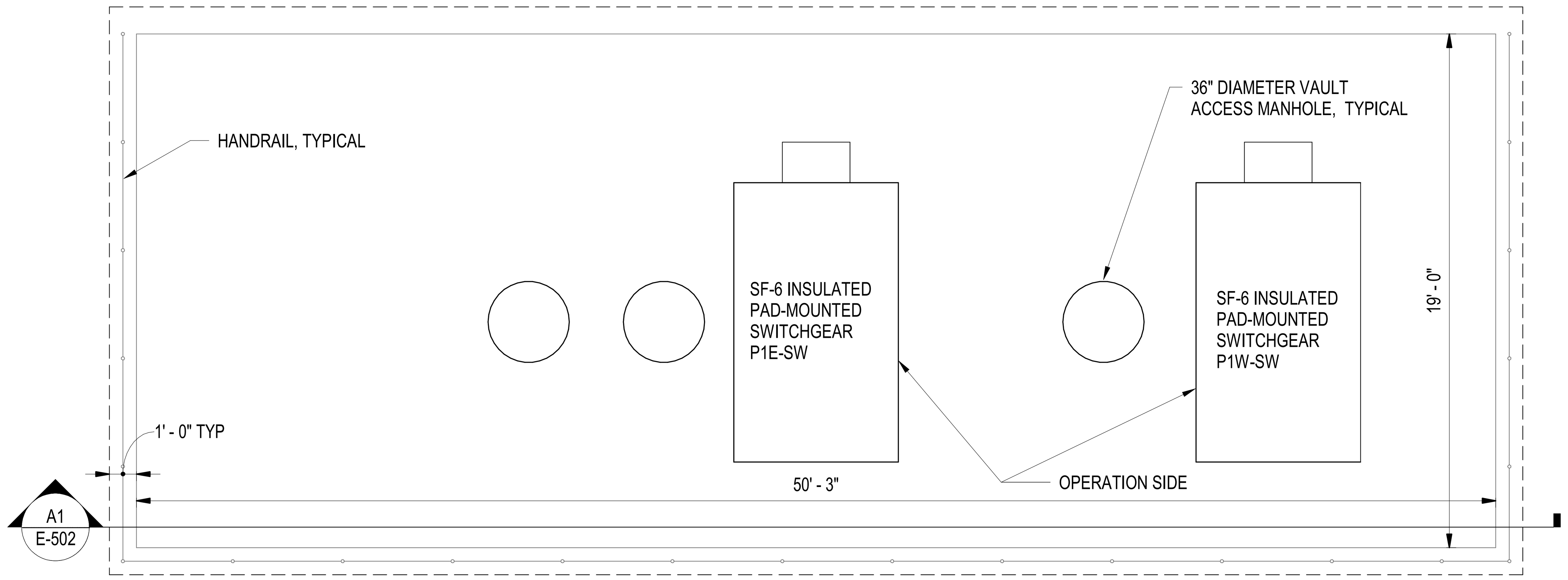
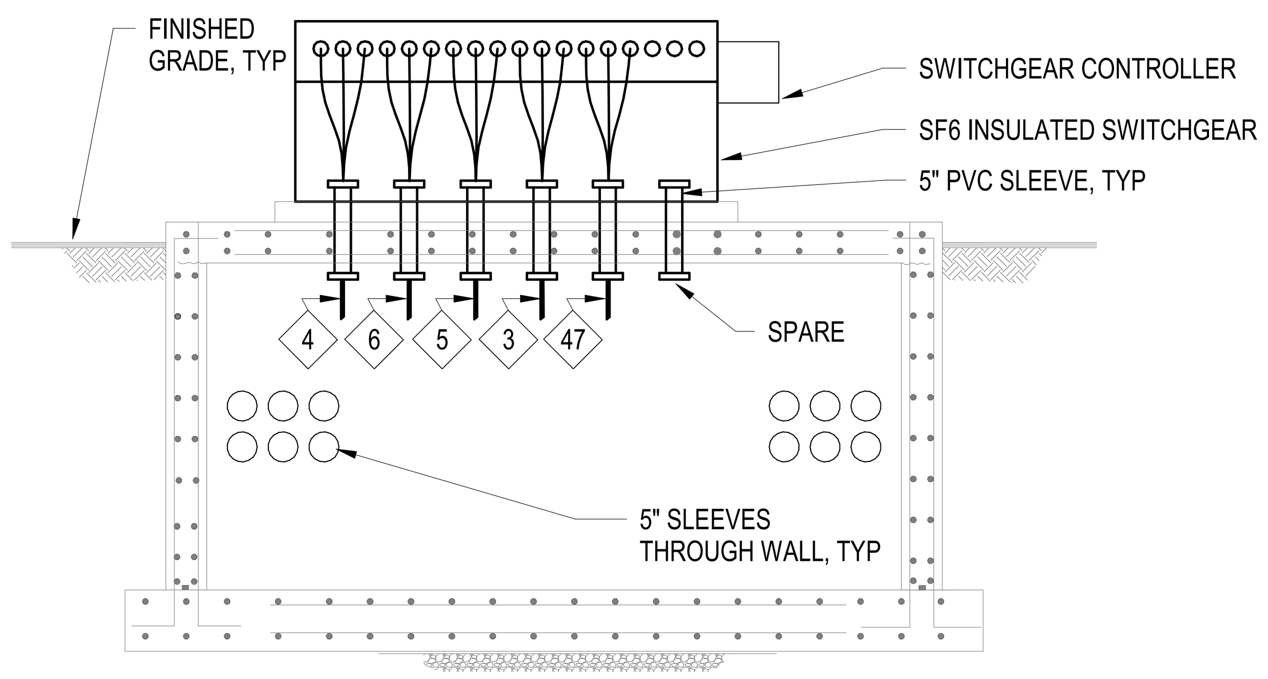
GENERAL SHEET NOTES:

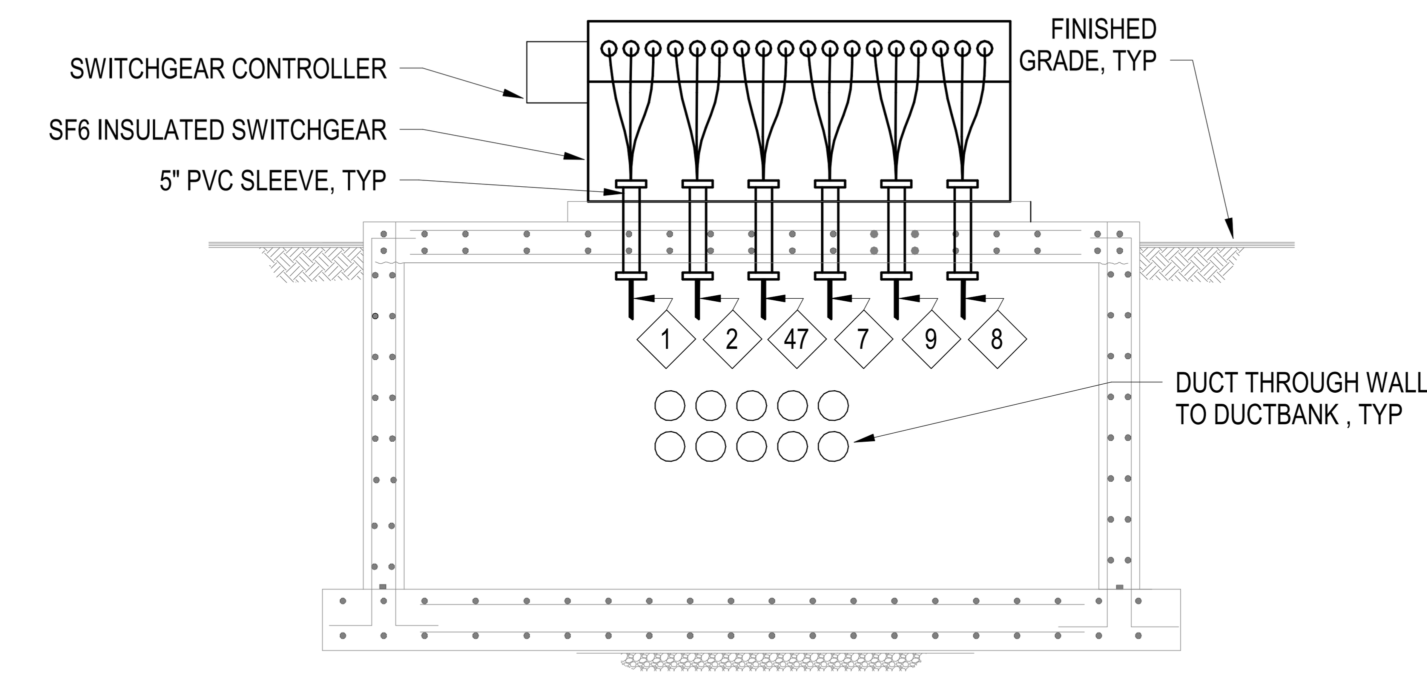
1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR SWITCH PLATFORM LOCATION SEE DRAWING E-109. FOR ONE LINE DIAGRAM SEE DRAWING E-601. SEE FEEDER SCHEDULE ON DRAWINGS E-614 AND E-615
2. COORDINATE SWITCH AND DUCTBANK INSTALLATION SLEEVES AND OPENINGS WITH STRUCTURAL.
3. SEE STRUCTURAL DRAWINGS FOR ADDITIONAL DETAILS FOR STRUCTURE.



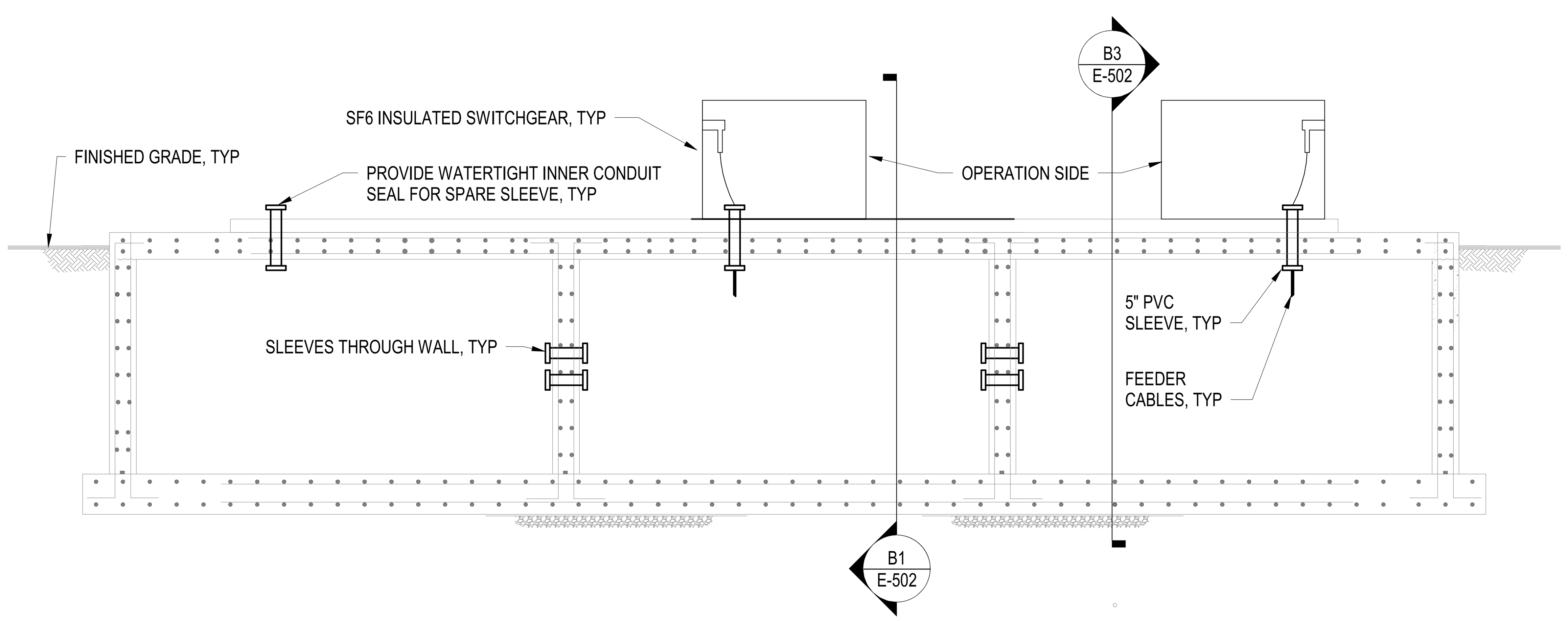
C1 PART PLAN - 15KV SWITCHGEAR STRUCTURE
1/4" = 1'-0"



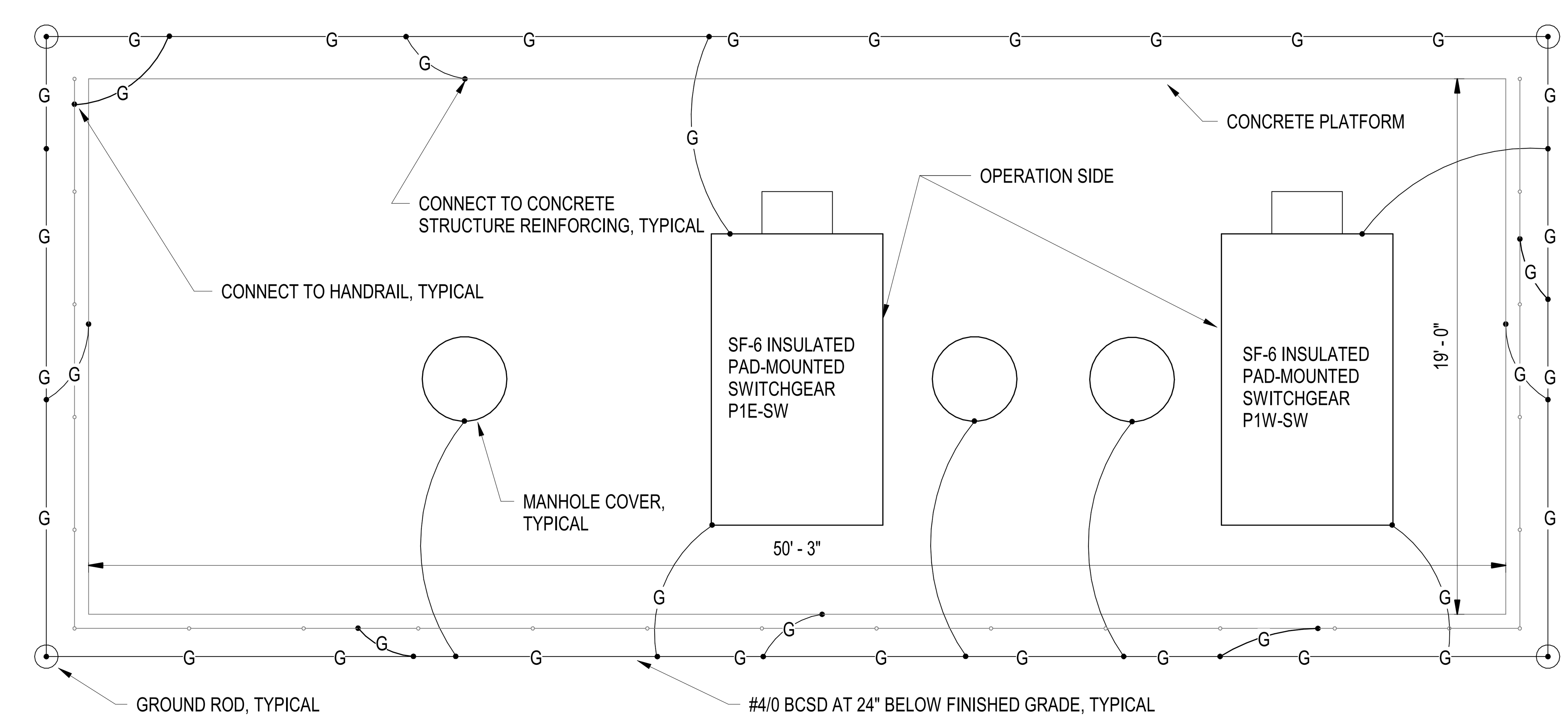
B1 TRANSVERSE SECTION - 15KV SWITCHGEAR STRUCTURE - P1E-SW
1/4" = 1'-0" (E-502)



B3 TRANSVERSE SECTION - 15KV SWITCHGEAR STRUCTURE - P1W-SW
1/4" = 1'-0" (E-502)



A1 LONGITUDINAL SECTION - 15KV SWITCHGEAR STRUCTURE
1/4" = 1'-0" (E-502)



A3 PART PLAN - 15KV SWITCHGEAR STRUCTURE GROUNDING PLAN
1/4" = 1'-0"

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
David Ferguson
SEAL

Stantec
A/E INFO

APPROVED
James E. Donahue, R.A.
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV	CHK RJW
PM/DM		MRI/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

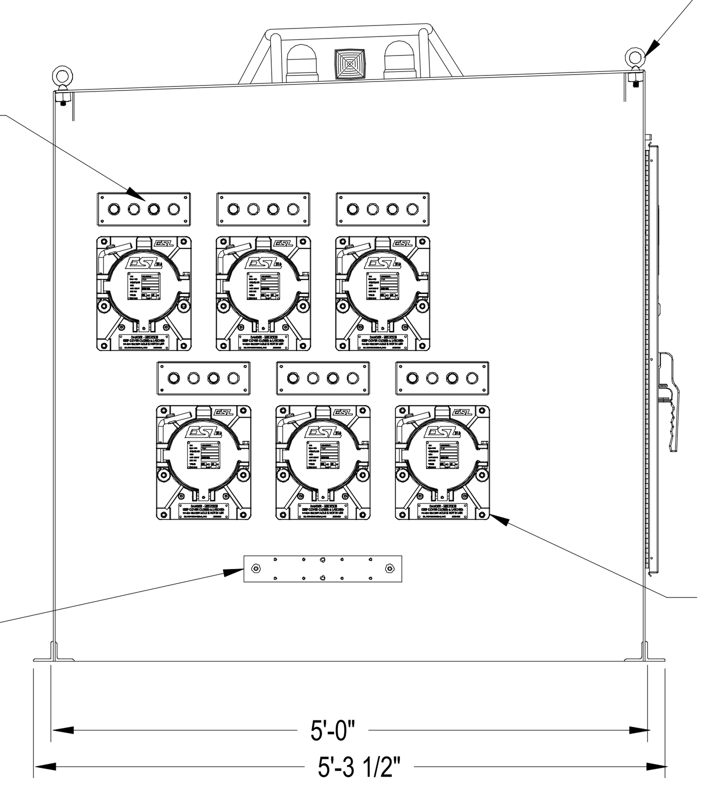
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
CI CORE - HAMPTON ROADS
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
15KV SWITCHGEAR STRUCTURE DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936782
SHEET 213 OF 247
E-502
DRAWFORM REVISION: 25 AUGUST 2020

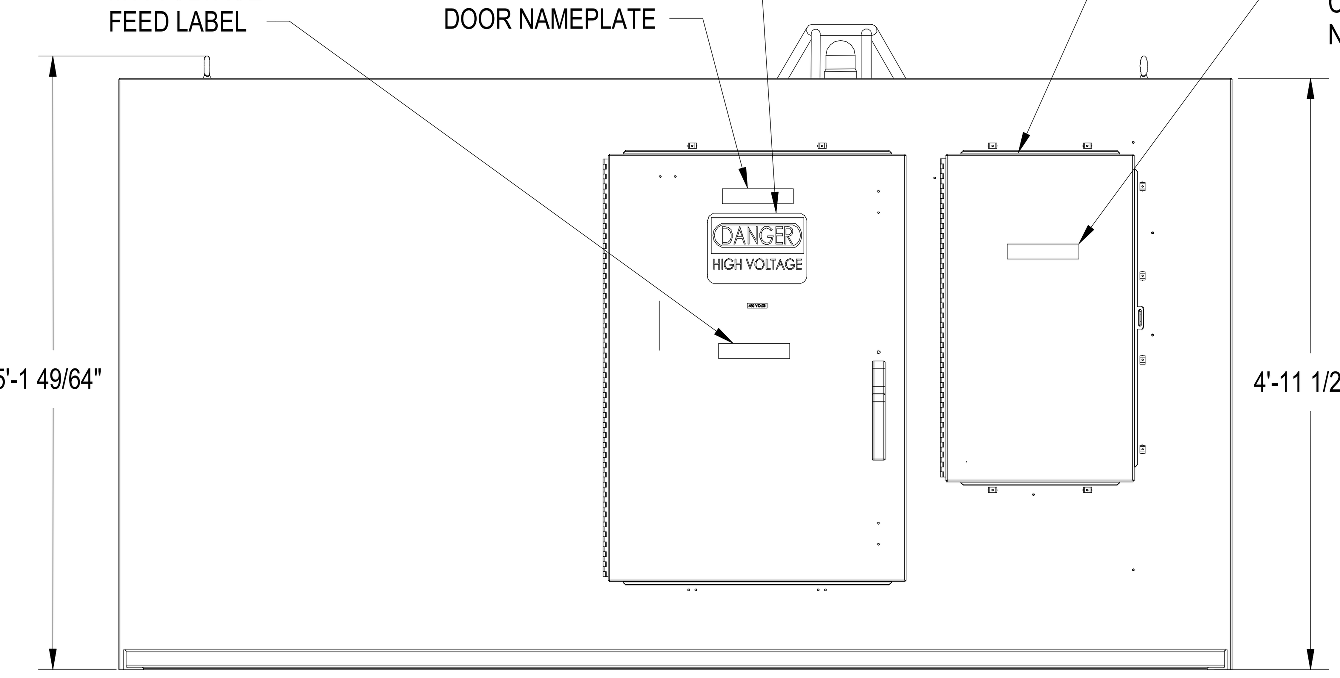
3/2/2026 5:13:51 PM
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt
E-502

POWER MOUND DETAIL NOTES:

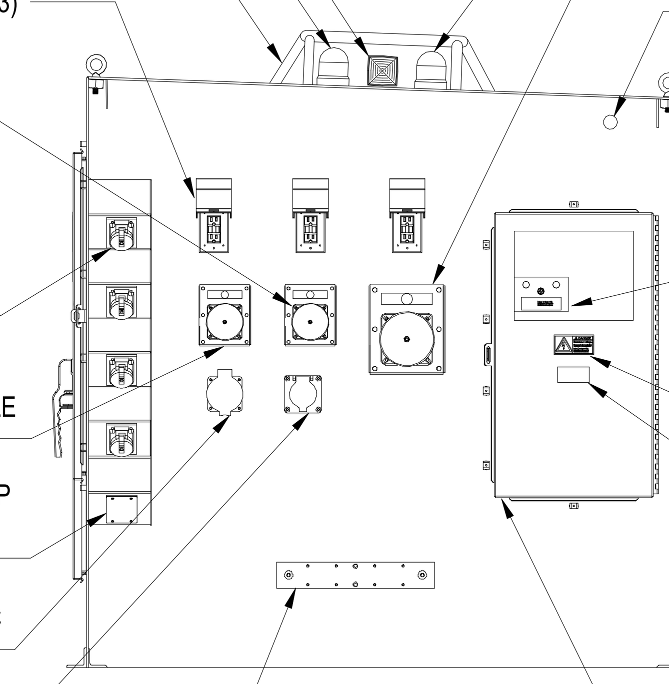
1. ORIENTATION OF THE RIGHT AND LEFT VIEWS ARE SHOWN FOR POWER MOUNDS P1E-PMA AND P1W-PMA. MIRROR THE RIGHT AND LEFT VIEWS FOR POWER MOUNDS P1E-PMB AND P1W-PMB.
2. PROVIDE PLACARD ON EXTERIOR OF POWER MOUND WHICH STATES "ONLY ONE SHIP MUST BE FED FROM THIS SHORE POWER MOUND AT A TIME. NON-COMPLIANCE MAY CAUSE DANGEROUS ELECTRICAL CURRENT BETWEEN HULLS OR IN WATER."
3. DIMENSIONS ARE SHOWN IN FEET & INCHES.



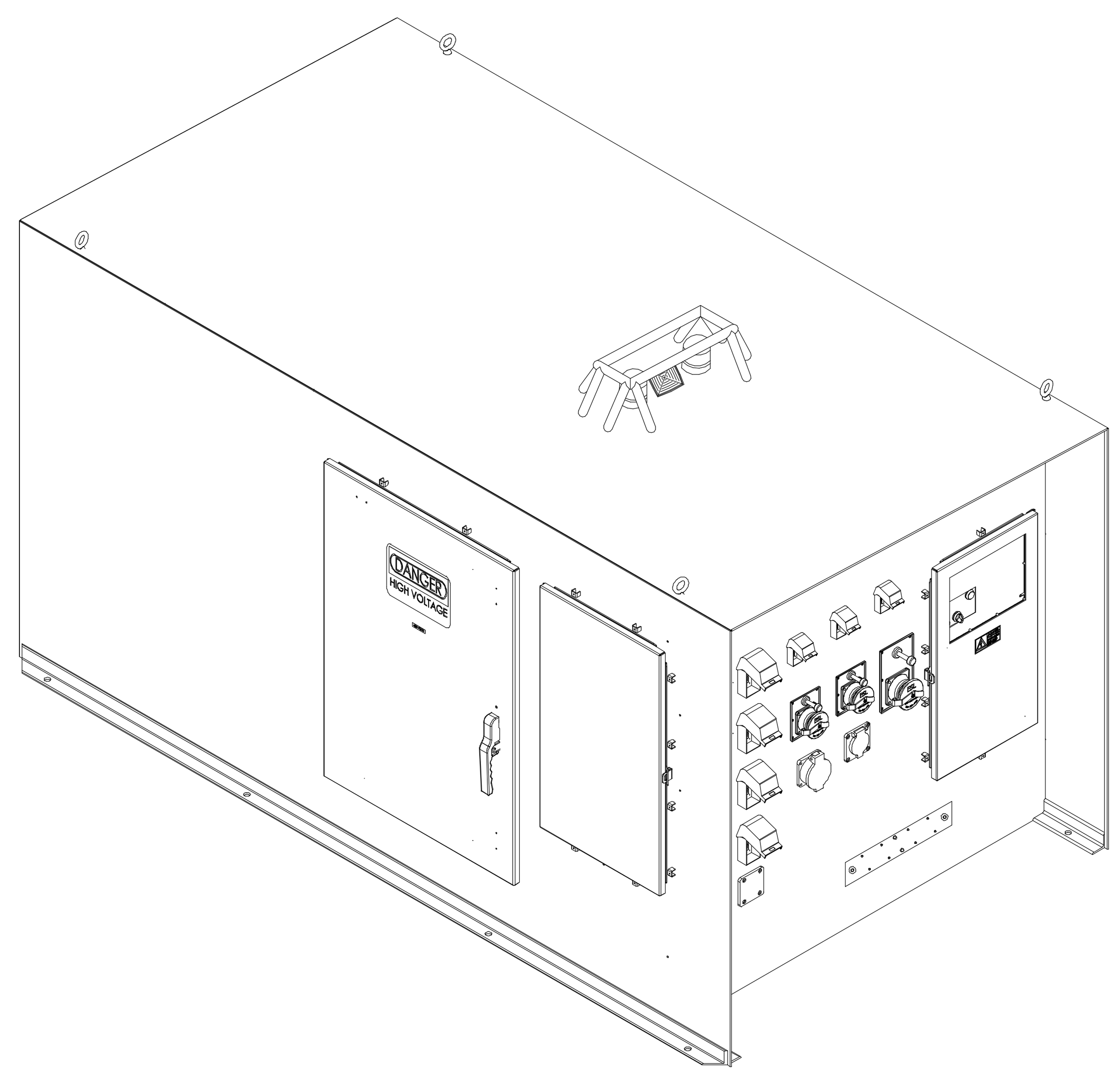
LEFT VIEW

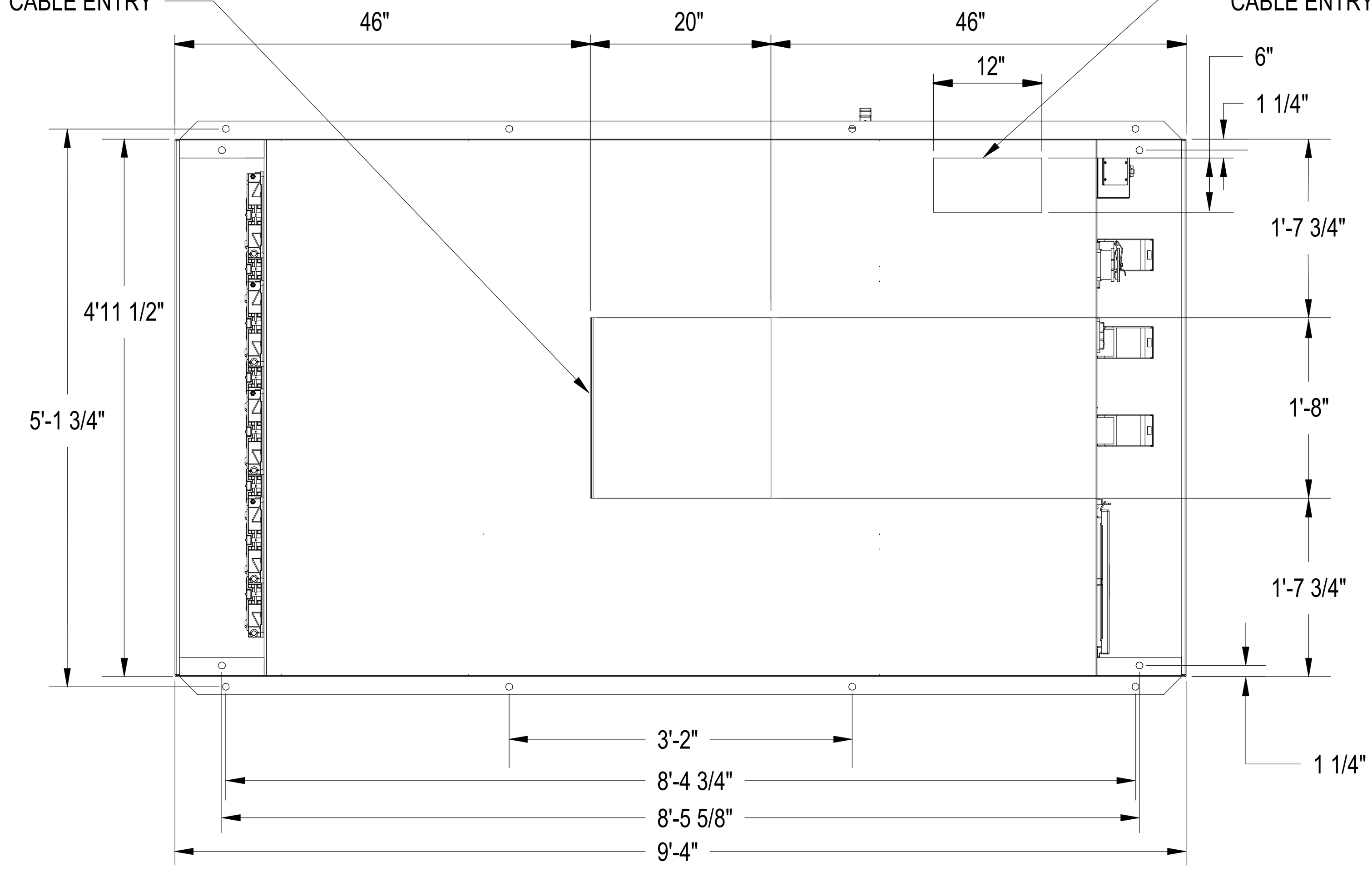


FRONT VIEW



RIGHT VIEW





BOTTOM VIEW

A1 **POWER MOUND DETAIL**

NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
REGISTERED PROFESSIONAL ENGINEER
David Ferguson

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
POWER MOUND DETAILS - SHEET 1 OF 2

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

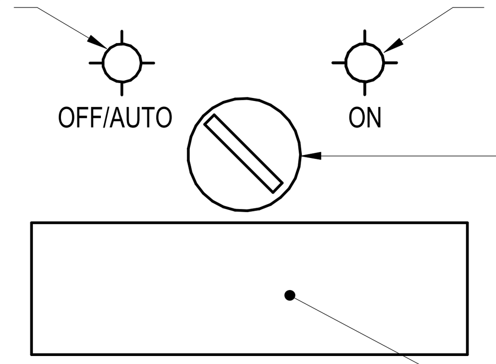
NAVFAC DRAWING NO. 12936783

SHEET 214 OF 247

E-503

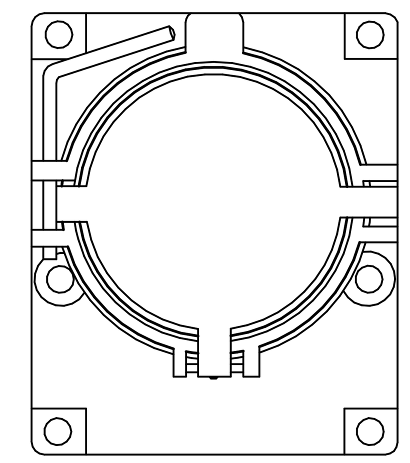
DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 5:13:51 PM E-503
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt

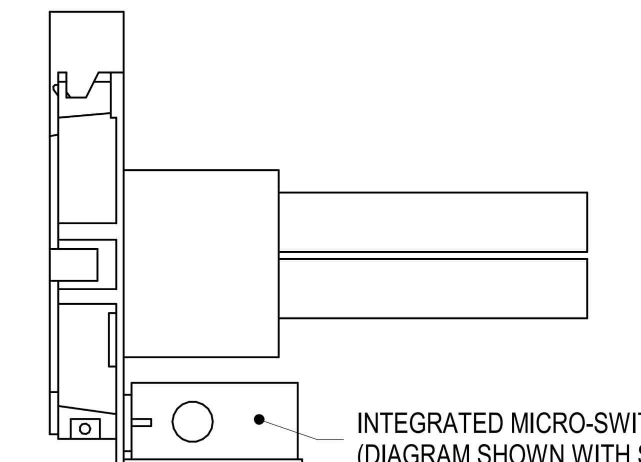


C1 POWER MOUND LIM OVERRIDE SWITCH DETAIL

NOT TO SCALE

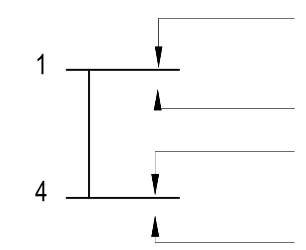


FRONT VIEW

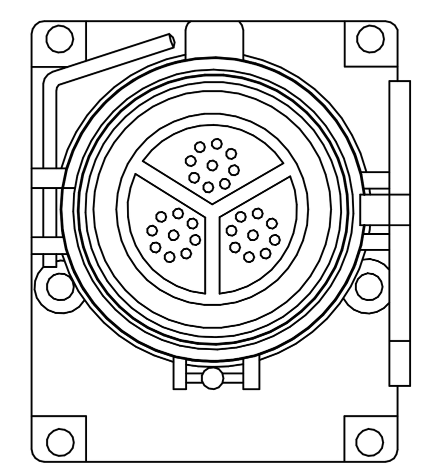


SIDE VIEW

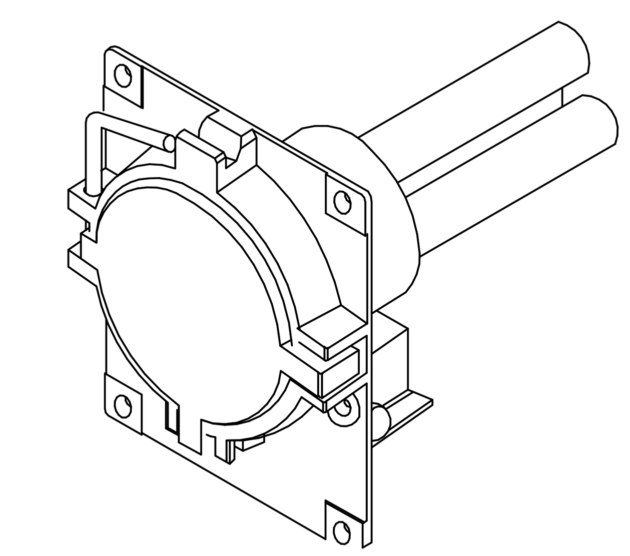
	2	1	3	5	4	6
DOOR CLOSED		—			—	
DOOR OPEN	—			—		
PLUG INSERTED		—			—	

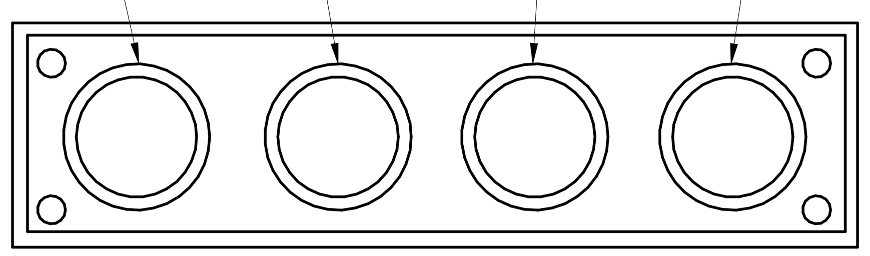


INTEGRATED MICRO-SWITCH



FRONT VIEW
COVER OPEN





PUSHBUTTON 1	PILOT LIGHT 1	PUSHBUTTON 2	PILOT LIGHT 2
RECEPTACLE 1	RECEPTACLE 1	RECEPTACLE 1	RECEPTACLE 1
CLOSE	CLOSED	OPEN	OPEN

ENGRAVED NAMEPLATE

B1 SHORE POWER RECEPTACLE CONTROL FACEPLATE

NOT TO SCALE

B3 MIL-C-24368 RECEPTACLE DETAIL

NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
David Ferguson
SEAL

Stantec
A/E INFO

APPROVED
James E. Donahue, R.A.
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV	CHK RJW
PM/DM		MRI/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
POWER MOUND DETAILS - SHEET 2 OF 2

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936784
SHEET 215 OF 247
E-504
DRAWFORM REVISION: 25 AUGUST 2020

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E-504
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METER SYMBOLS

E — ELECTRIC

ELECTRICAL SYMBOLS

C — COMMUNICATIONS ENCLOSURE

J — JUNCTION BOX

F — FIBER ENCLOSURE WITH FIBER PATCH PANEL

E — METER CABINET

ELECTRICAL LINES

CONDUIT, RUN BURIED OR SURFACE MOUNTED IN VAULT/SUBSTATION

CONDUIT ROUTED VERTICALLY

COMMUNICATIONS ABBREVIATIONS

SM - SINGLE MODE FIBER

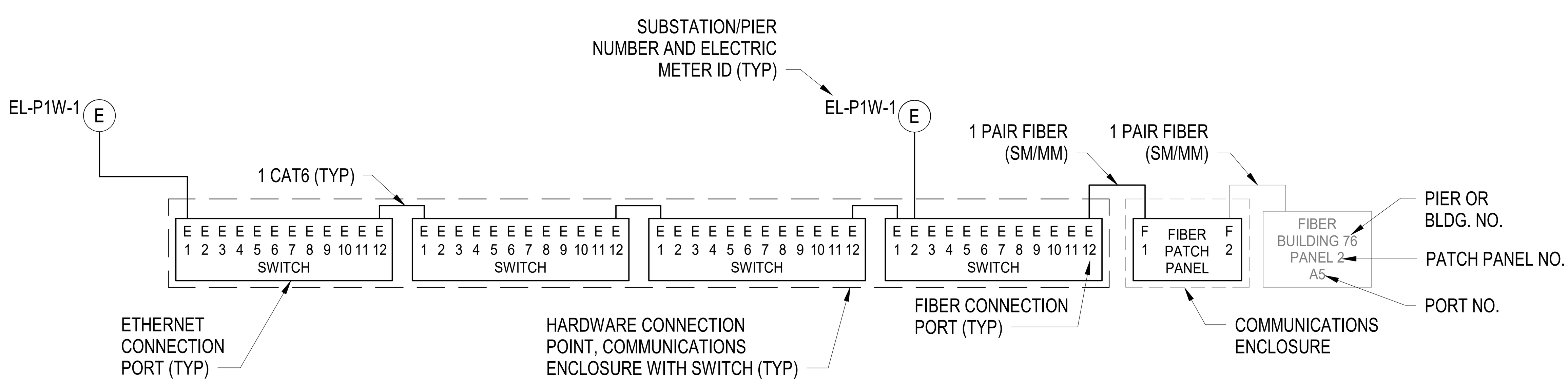
MM - MULTI-MODE FIBER

FPP - FIBER PATCH PANEL

UON - UNLESS OTHERWISE NOTED

NOTES:

1. ELECTRICAL WORK PERFORMED MUST BE IN ACCORDANCE WITH THE LATEST EDITION OF IEEE C2 (NATIONAL ELECTRIC SAFETY CODE), NFPA 70 (NATIONAL ELECTRICAL CODE), CONTRACT SPECIFICATIONS, INSTALLATION DIAGRAMS, AND APPLICABLE LOCAL CODES UNLESS DIRECTED OTHERWISE BY THE CONTRACTING OFFICER.
2. THE ELECTRICAL CONTRACTOR MUST COORDINATE THE WORK WITH THE GOVERNMENT AND EXISTING AMI INFRASTRUCTURE.
3. WORK MUST BE PERFORMED BY A LICENSED ELECTRICAL CONTRACTOR IN A FIRST CLASS WORKMANLIKE MANNER.
4. ELECTRICAL METER, CURRENT (CT) AND POTENTIAL (PT) TRANSFORMER (IF REQUIRED) AND POWER WIRING TO THE COMMUNICATIONS ENCLOSURES MUST BE IN ACCORDANCE WITH THE BASE AMI INFRASTRUCTURE. COORDINATE ALL REQUIREMENTS WITH THE GOVERNMENT AND EXISTING CONDITIONS.
5. NON-COMMUNICATION/NON-SIGNAL WIRING MUST BE TYPE RATED SIS AND XHHW OR XHHW-2 COPPER OR APPROVED EQUAL MINIMUM SIZE #12 AWG FOR POWER/VOLTAGE REFERENCE WIRING AND #10 AWG FOR CURRENT TRANSFORMER/METERING CIRCUIT WIRING.
6. AN EQUIPMENT GROUNDING CONDUCTOR MUST BE PROVIDED IN CONDUITS WHETHER OR NOT THE GROUNDING WIRING IS SHOWN ON THE DRAWINGS.
7. EQUIPMENT GROUNDING AND NEUTRAL CONDUCTORS MUST BE THE SAME SIZE AWG AS THE LARGEST CURRENT CARRYING CONDUCTOR UON.
8. COMMUNICATION CABLES INSTALLED MUST BE IN ACCORDANCE WITH TIA-568-C.1, ANSI/TIA/EIA-568-B.2-2, TIA-568-C.3, TIA-569-B, NFPA 70, AND UL STANDARDS AS APPLICABLE.
9. FIBER CABLE MUST BE INDOOR/OUTDOOR RATED.
10. WIRING AND CABLES MUST BE RATED FOR THE ENVIRONMENT THEY ARE INSTALLED IN.
11. REGARDLESS OF LOCATION (E.G. PLENUM AREAS) RACEWAY PENETRATING WALLS AND FLOORS MUST BE INSTALLED INSIDE CONDUIT AND PROPERLY SEALED. FIRE RATED WALL PENETRATIONS MUST BE FIRESTOPPED IN ACCORDANCE WITH THE PROJECT SPECIFICATIONS. ALL OUTSIDE WALL PENETRATIONS MUST BE WATERTIGHT.
12. WIRING MUST BE LABELED AT BOTH ENDS.
13. CIRCUITS ABOVE 50VAC/VDC MUST BE INSTALLED IN A SEPARATE RACEWAY NOT CONTAINING ANY COMMUNICATIONS OR SIGNALING CABLES UON.
14. CIRCUITING AND RACEWAYS SHOWN ARE DIAGRAMMATICAL. EXACT LOCATION AND ROUTING IS TO BE DETERMINED IN THE FIELD WITH JUNCTION AND PULL BOXES INSTALLED AS REQUIRED.
15. CONTRACTOR MUST PROVIDE RACEWAY SUPPORTS, FITTINGS, JUNCTION AND PULL BOXES AS REQUIRED.
16. SUBSTATION MOUNTING SURFACES WILL BE METAL UON.
17. MINIMUM CONDUIT SIZE MUST BE 3/4" UON.
18. COMMUNICATIONS AND SIGNAL CABLES MUST BE INSTALLED IN CONDUIT UON.
19. TREATED MATERIALS THAT ARE CUT/SCRATCHED/FIELD MODIFIED MUST HAVE THEIR FINISHES TOUCHED-UP IN THE FIELD WITH THE REQUIRED TREATMENTS.
20. METER CABINET PLACEMENT MUST NOT OBSTRUCT ACCESS TO SERVICEABLE ELECTRICAL COMPONENTS SUCH AS HINGED PRIMARY COMPARTMENTS, HINGED DOORS, OR TRANSFORMER INSTRUMENTS.



B1 TYPICAL COMMS INTERCONNECTION DIAGRAM

NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT
NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

METERING SYSTEM DETAILS - SHEET 1 OF 3

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936785

SHEET 216 OF 247

E-505

DRAWFORM REVISION: 20 AUGUST 2020

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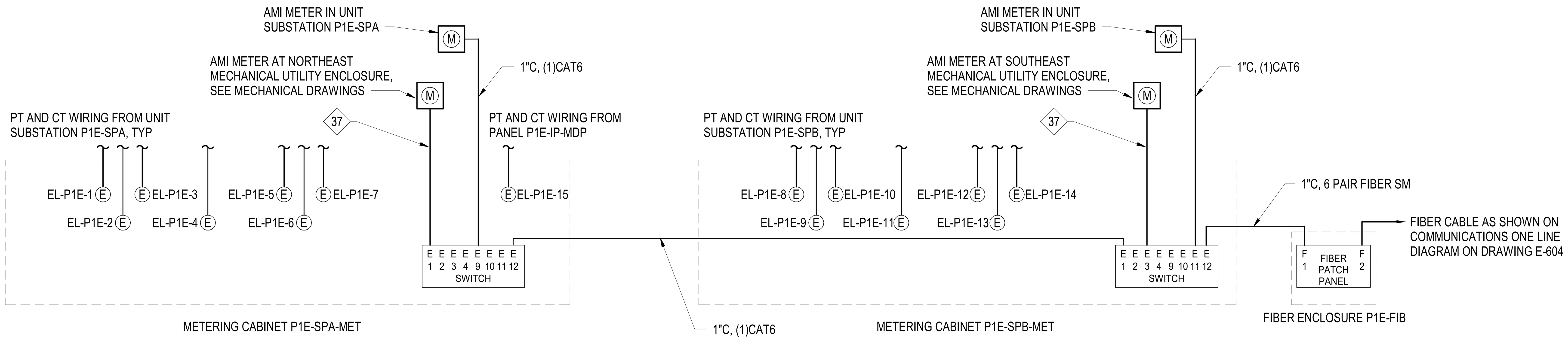
1 2 3 4 5

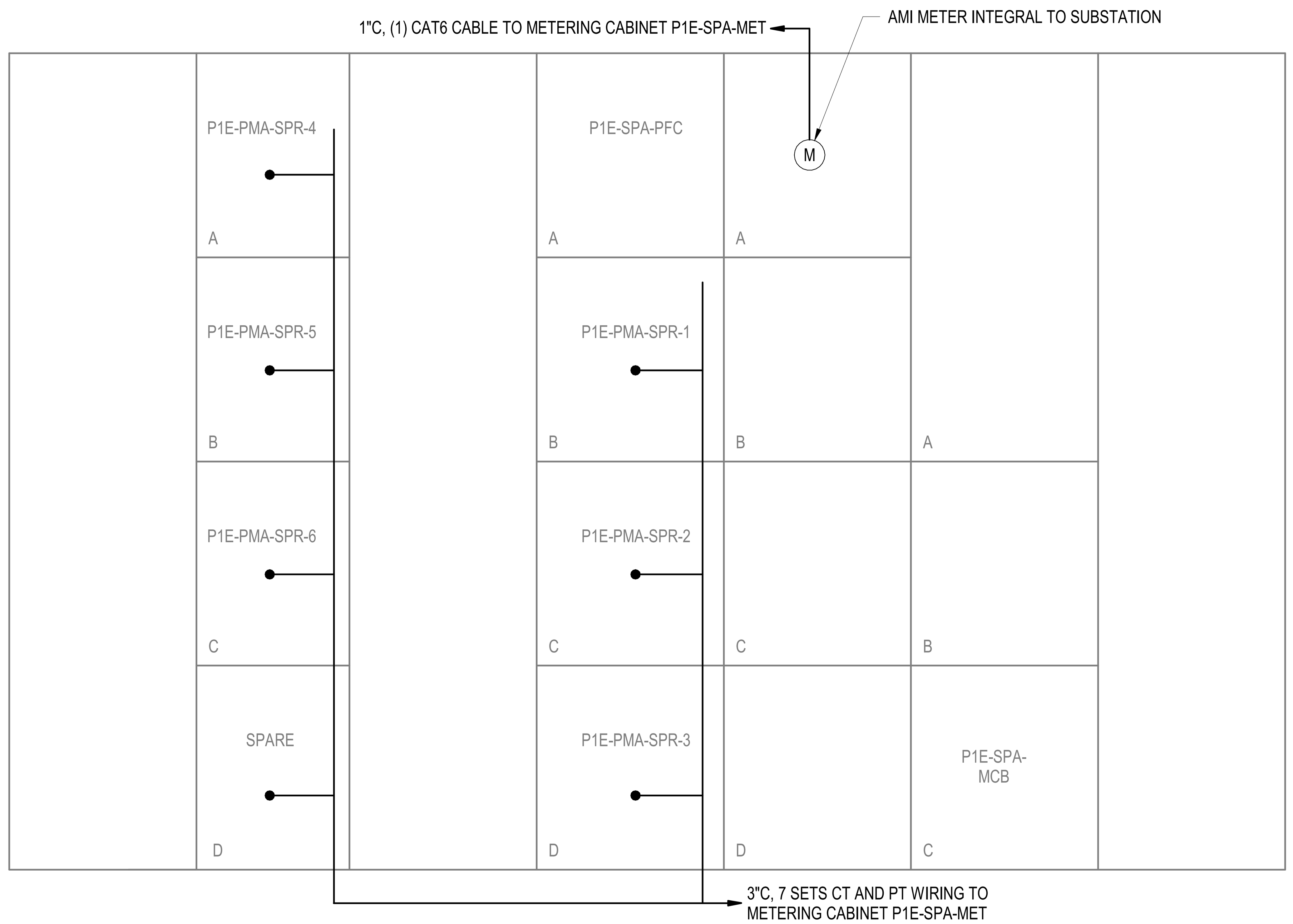
GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR SUBSTATION P1W PART PLAN SEE DRAWING E-401. FOR COMMUNICATIONS ONE LINE DIAGRAM SEE DRAWING E-604. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.
2. METERS OWNED BY USCG.
3. AMI METERS REPORT TO NAVSTA AMI SYSTEM. METERS SERVING INDIVIDUAL SHORE POWER FEEDER CIRCUITS DO NOT REPORT TO A REMOTE LOCATIONS.
4. METERS MUST BE SHARK 270/250 TYPE.



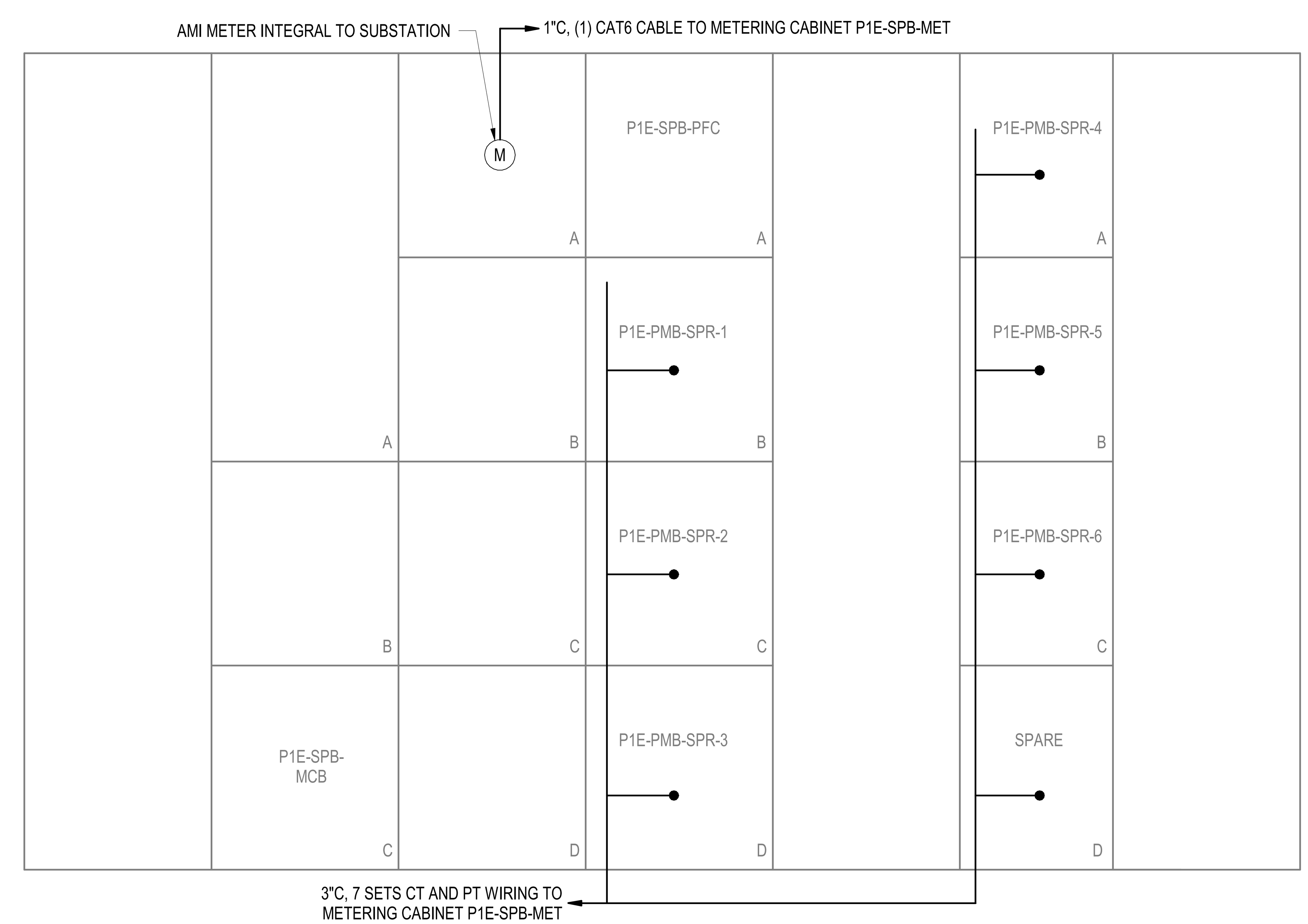
C1 P1E METERING INTERCONNECTION DIAGRAM

NOT TO SCALE



A1 ELEVATION - SWITCHGEAR P1E-SPA - METERING

NOT TO SCALE



A3 ELEVATION - SWITCHGEAR P1E-SPB - METERING

NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR

DAVID P FERGUSON ELECTRICAL No. 52642 COMMONWEALTH OF MASSACHUSETTS PROFESSIONAL ENGINEER

Stantec

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
METERING SYSTEM DETAILS - SHEET 2 OF 3

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

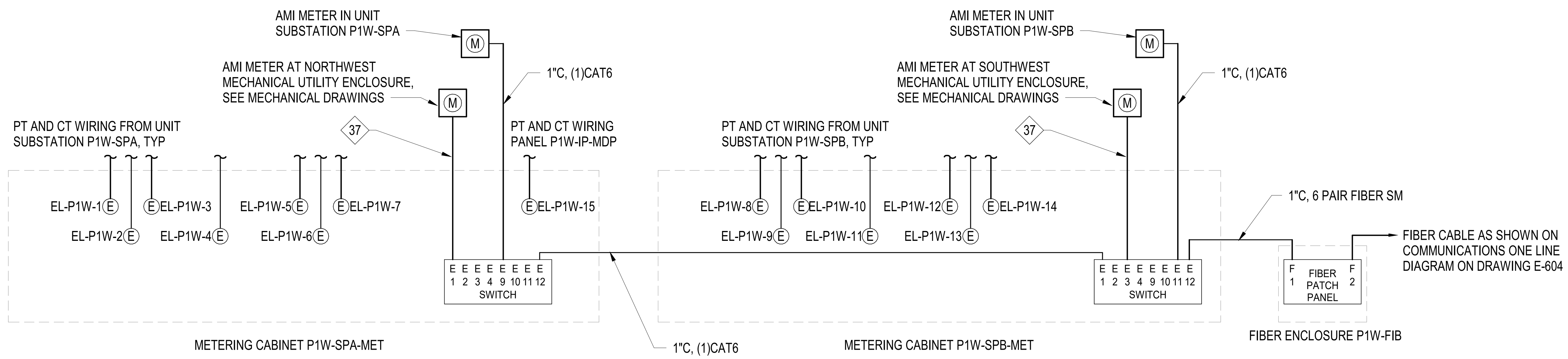
NAVFAC DRAWING NO. 12936786

SHEET 217 OF 247

E-506

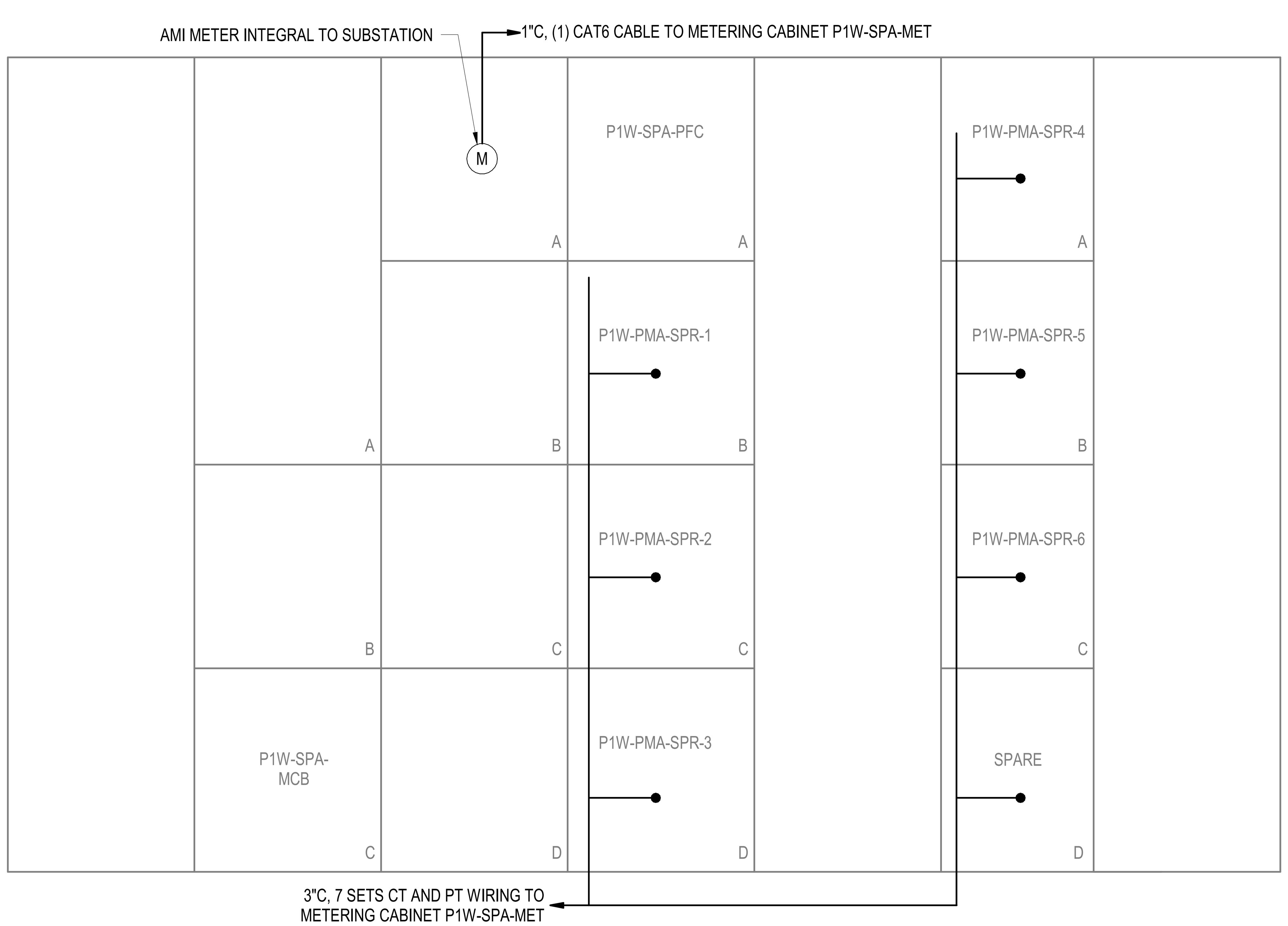
GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR SUBSTATION P1W PART PLAN SEE DRAWING E-401. FOR COMMUNICATIONS ONE LINE DIAGRAM SEE DRAWING E-604. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.
2. METERS OWNED BY USCG.
3. AMI METERS REPORT TO NAVSTA AMI SYSTEM. METERS SERVING INDIVIDUAL SHORE POWER FEEDER CIRCUITS DO NOT REPORT TO A REMOTE LOCATIONS.
4. METERS MUST BE SHARK 270/250 TYPE.



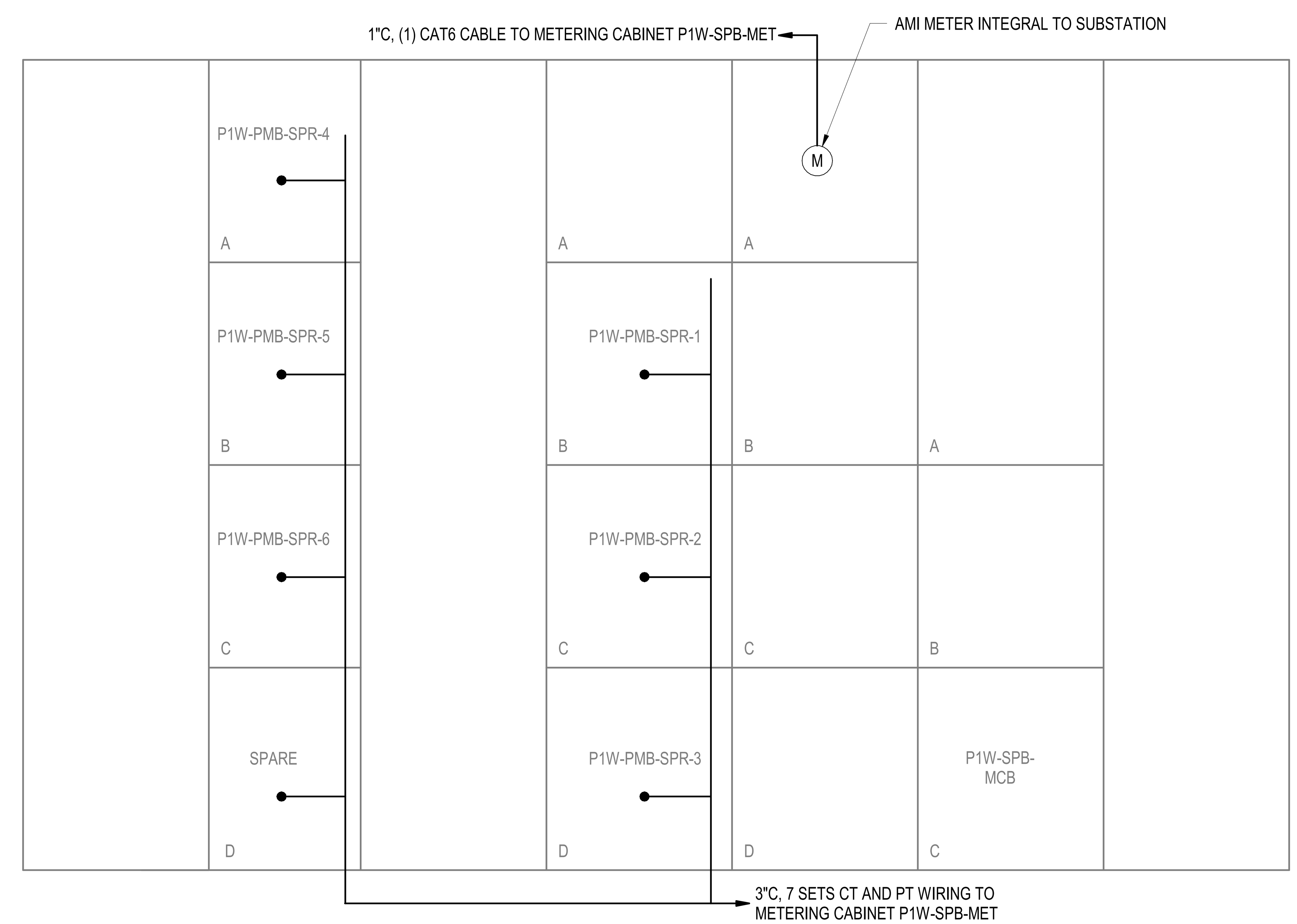
C1 P1W METERING INTERCONNECTION DIAGRAM

NOT TO SCALE



A1 ELEVATION - SWITCHGEAR P1W-SPA - METERING

NOT TO SCALE



A3 ELEVATION - SWITCHGEAR P1W-SPB - METERING

NOT TO SCALE

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS DAVID P FERGUSON ELECTRICAL No. 52642 PROFESSIONAL ENGINEER

Stantec

APPROVED FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

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CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
METERING SYSTEM DETAILS - SHEET 3 OF 3

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936787
SHEET 218 OF 247

E-507

DRAWFORM REVISION: 25 AUGUST 2020

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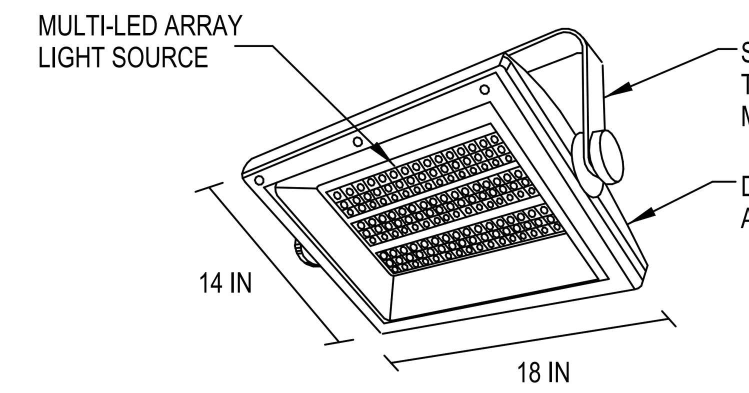
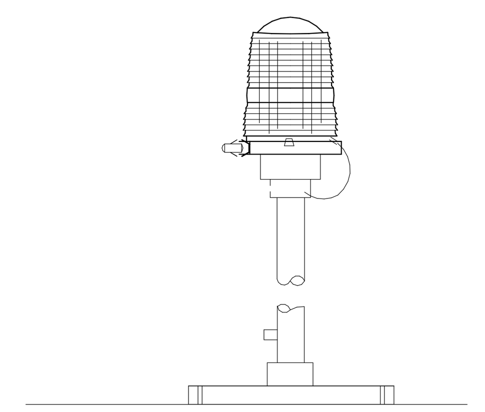


LUMINAIRE REQUIREMENTS:

1. HOUSING - DIE CAST ALUMINUM WITH INTEGRAL PASSIVE COOLING MECHANISM. HEAT SINK MUST BE INCORPORATED DIRECTLY INTO HOUSING TO ENSURE MAXIMUM HEAT TRANSFER AND DISSIPATION.
2. FINISH - MULTI-STAGE PRE-TREATMENT, FINISHED WITH BAKED-ON POLYESTER POWDER COAT. FINISH MUST PASS 2500 HOUR SALT SPRAY TEST PER ASTM B117. DARK BRONZE FINISH COLOR IS STANDARD.
3. POWER SUPPLY/LED DRIVER - CLASS 1 ELECTRONIC DRIVER MUST OPERATE AT 120/277 VOLTS, 50/60 HZ, WITH OTHER VOLTAGES OPTIONAL. POWER FACTOR MUST BE GREATER THAN 0.9 AND THD LESS THAN 20% AT FULL LOAD. MINIMUM EFFICACY MUST BE 60 LM/W AT MAXIMUM 600mA OPERATING CURRENT.
4. LED OPTICAL ASSEMBLY - MULTI-LED ARRAY OPTIMIZED FOR SPECIFIC DISTRIBUTION PATTERN AS INDICATED. MINIMUM COLOR RENDERING INDEX (CRI) OF 70 FOR CORRELATED COLOR TEMPERATURE OF 4000-4500 DEGREES K.
5. LENS - TEMPERED GLASS IN DIE-CAST ALUMINUM FRAME WITH SILICONE GASKET.
6. SURGE PROTECTION - 6 KV MINIMUM, COMPLIANT WITH ANSI C62.41.2.
7. CERTIFICATION - UL AND/OR ETL LISTED FOR WET LOCATIONS AND 2G VIBRATION STANDARD PER ANSI C136.32. OPTICAL ASSEMBLY MUST BE MINIMUM IP65 PER ANSI/IEC 60529.
8. OPTIONS - PHOTOCELL, FINISH COLOR, OUTPUT DISTRIBUTION TYPE AND TRUNNION OR SLIPFITTER TYPE MOUNTING.
9. OTHER - THE ABOVE SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS AND IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER'S PREFERENCE. ALL DIMENSIONS ARE NOMINAL AND VARY PER MANUFACTURER.

LED FLOOD LUMINAIRE A1 A2 A3

REVISED: MARCH 2013 | LUMINAIRE PLATE: XL-21



NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING - DIE-CAST ALUMINUM WITH POLYESTER POWDER COAT FINISH AND STAINLESS STEEL HARDWARE.
2. OPTICS - GLOBE IS TEMPERED RED OR CLEAR FRESNEL GLASS WITH STAINLESS STEEL LATCHING SYSTEM TO HOLD GLOBE FIRMLY ONTO BASE. PROVIDE CABLE TETHER WIRE.
3. LIGHT SOURCE - SOLID STATE LEDS, RED AND MINIMUM EFFICACY OF 90 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-240V, THERMAL MANAGEMENT, AND < 20% THD. ON-OFF CONTROL OR AS INDICATED IN LUMINAIRE SCHEDULE.
5. CERTIFICATION - FAA L810 PER FAA ADVISORY CIRCULAR 150/5345-43G. UL LISTED FOR WET LOCATION, ROHS COMPLIANT, DLC QUALIFIED. COMPLIES WITH LM79, LM80 AND TM21 TESTING STANDARDS.
6. MOUNTING - MOUNTING - GROUND MOUNTED WITH 3/4 IN OR 1 IN BOTTOM CONDUIT ENTRY.
7. OPTIONS - SINGLE OR TWIN GLOBE.

AVIATION OBSTRUCTION LUMINAIRE B

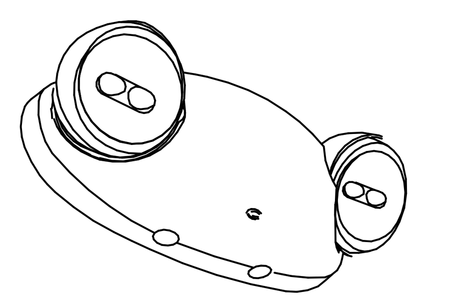
REVISED: NOVEMBER 2020 | LIGHTING PLATE: XL-13

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.

SHEET KEYNOTES:

1. OBSTRUCTION LIGHTS MUST OPERATE VIA FLOODLIGHT PHOTOCELLS.
2. TYPE "B" FIXTURES MUST BE PROVIDED BY POLE MANUFACTURER AND COMPLETELY WIRED, READY FOR FINAL CONNECTION.
3. PROVIDE FIBERGLASS HOUSING.



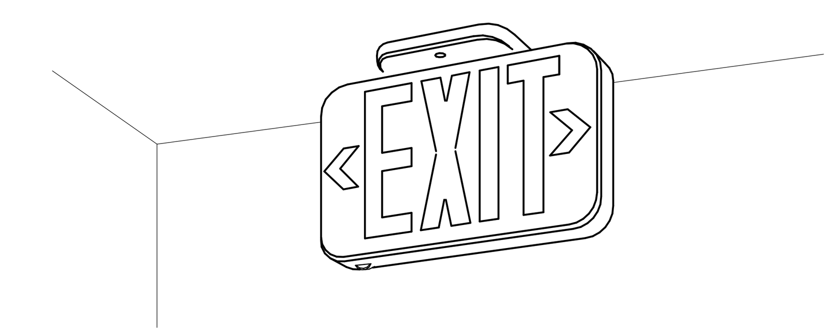
NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING - HIGH-IMPACT, UV-STABILIZED, INJECTION-MOLDED THERMOPLASTIC HOUSING.
2. LIGHT SOURCE - SOLID STATE LEDS.
3. DRIVER - INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON/OFF CONTROL AND BATTERY BACKUP INTEGRAL TO UNIT.
4. CERTIFICATION - NFPA 101. UL LISTED FOR DAMP OR WET LOCATION, ROHS COMPLIANT. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
5. MOUNTING - WALL SURFACE MOUNTED.
6. OPTIONS - WHITE OR BLACK FINISH.

LED EMERGENCY LIGHTING UNIT (ELU) E

REVISED: NOVEMBER 2020 | LIGHTING PLATE: NL-26



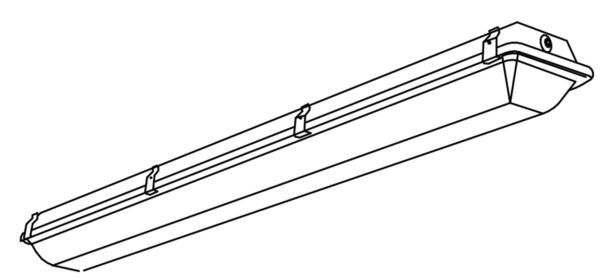
NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING - DIE-CAST ALUMINUM OR HIGH-IMPACT, UV-STABILIZED, INJECTION-MOLDED THERMOPLASTIC.
2. LIGHT SOURCE - SOLID STATE LEDS.
3. DRIVER - INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120/277V, THERMAL MANAGEMENT, AND < 20% THD.
4. CERTIFICATION - NFPA 101, UL LISTED FOR DAMP OR WET LOCATION, AND ROHS COMPLIANT.
5. MOUNTING - SURFACE MOUNTED ON CEILING AND/OR WALL.
6. OPTIONS - RED OR GREEN LETTERING, ONE- OR TWO-SIDED. ELU REMOTE HEAD CAPABILITIES. BATTERY BACKUP.

EXIT SIGN X

REVISED: NOVEMBER 2020 | LIGHTING PLATE: NL-28

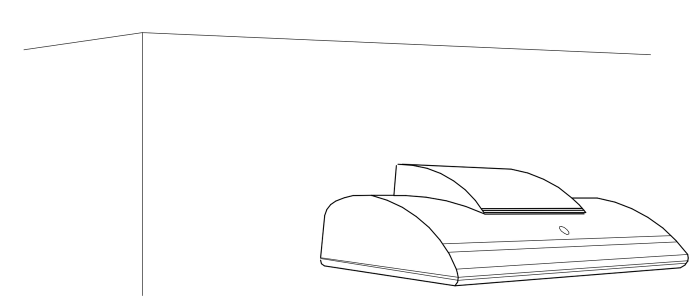


LUMINAIRE REQUIREMENTS:

1. HOUSING - FIBERGLASS OR FIBERGLASS-REINFORCED POLYESTER OUTER HOUSING, WITH ALUMINUM COMPONENT TRAY AND HEAT SINK. OPTIONAL LENGTHS OF 4FT OR 8FT.
2. LENS - IMPACT-RESISTANT ACRYLIC OR OPTIONAL POLYCARBONATE, WITH CONTINUOUS CLOSED-CELL POLYURETHANE GASKET, SECURED WITH STAINLESS STEEL OR POLYCARBONATE LATCHES.
3. LIGHT SOURCE - SOLID STATE LEDS WITH MINIMUM 50K HOURS RATED LIFE AT L70, 3500K CCT UON, MINIMUM 80 CRI, MAXIMUM 4-STEP MCADAM ELLIPSE BINNING TOLERANCE FOR COLOR CONSISTENCY, AND MINIMUM EFFICACY OF 100 LUMENS/WATT. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, < 20% TOTAL HARMONIC DISTORTION. ON-OFF CONTROL, STEP-DIMMABLE OR FULLY DIMMABLE AS INDICATED.
5. CERTIFICATION - UL 1598, WET LOCATION (IP65, IP66, IP67), DLC QUALIFIED, AND ROHS COMPLIANT. COMPLIES WITH LM79, LM80 AND TM21 TESTING STANDARDS. UL 924 WHEN EQUIPPED WITH EMERGENCY BATTERY BACK-UP.
6. MOUNTING - SURFACE-MOUNTED OR SUSPENDED FROM CEILING.
7. OPTIONS - POWER CORD, INTEGRAL MOTION SENSOR, EMERGENCY BACK-UP.
8. THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LED ENCLOSED AND GASKETED C

REVISED: APRIL 2016 | LIGHTING PLATE: NL-11



NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING - DIE-CAST OR EXTRUDED ALUMINUM WITH INTEGRAL PASSIVE COOLING MECHANISM. HEAT SINK INCORPORATED DIRECTLY INTO HOUSING OR DRIVER COMPARTMENT.
2. OPTICS - PRECISION MOLDED ACRYLIC LENS WITH TYPE II, III, OR IV DISTRIBUTIONS. BUG UPLIGHT RATING OF U0, WITH GLARE RATING AS DETERMINED BY LIGHTING ZONE INSTALLED.
3. LIGHT SOURCE - SOLID STATE LEDS, 4000K CCT UON, MINIMUM 80 CRI UON, AND MINIMUM EFFICACY OF 120 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DIMMABLE DRIVER WITH MINIMUM0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON-OFF CONTROL AND FULLY DIMMABLE DOWN TO 10% MINIMUM OR AS INDICATED INLUMINAIRE SCHEDULE.
5. CERTIFICATION - UL LISTED FOR WET LOCATIONS, ROHS COMPLIANT. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
6. MOUNTING - SURFACE MOUNTED WITH STAINLESS STEEL MOUNTING HARDWARE.
7. OPTIONS - VARIOUS LIGHT DISTRIBUTIONS. INTEGRAL MOTION SENSOR, PHOTOCELL, BATTERY BACK-UP.

LED WALL PACK D

REVISED: NOVEMBER 2020 | LUMINAIRE PLATE: XL-10

LIGHTING FIXTURE SCHEDULE

FIXTURE TYPE	NAVFAC SKETCHES, STYLE, AND TYPE DETAIL	DESCRIPTION	MOUNTING	LAMPS			REMARKS
				NO.	VOLT	TYPE	
A1	XL-21	FLOOD LIGHT LED, WITH FULL VISOR.	POLE TOP	1	480	644W 85,058 LM	WIDE FLOOD DISTRIBUTION SEE DRAWING E-509
A2	XL-21	FLOOD LIGHT LED, WITH FULL VISOR.	POLE TOP	1	480	612W 80,760 LM	FLOOD DISTRIBUTION SEE DRAWING E-509
A3	XL-21	FLOOD LIGHT LED, WITH FULL VISOR.	POLE TOP	1	480	617W 87,645 LM	SPOT DISTRIBUTION SEE DRAWING E-509
B	XL-13	OBSTRUCTION LIGHT	POLE TOP	2	120	LED	1 2
C	NL-11	INDUSTRIAL VAPOR-TIGHT LED	SURFACE	1	120	38W 4000 LM	3
D	XL-10	LED WALL PACK	WALL	1	120	15W 1950 LM	
E	NL-26	EMERGENCY LIGHTING UNIT	WALL	1	120	LED	
X	NL-28	EXIT SIGN	WALL	1	120	LED	

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
David Ferguson
SEAL

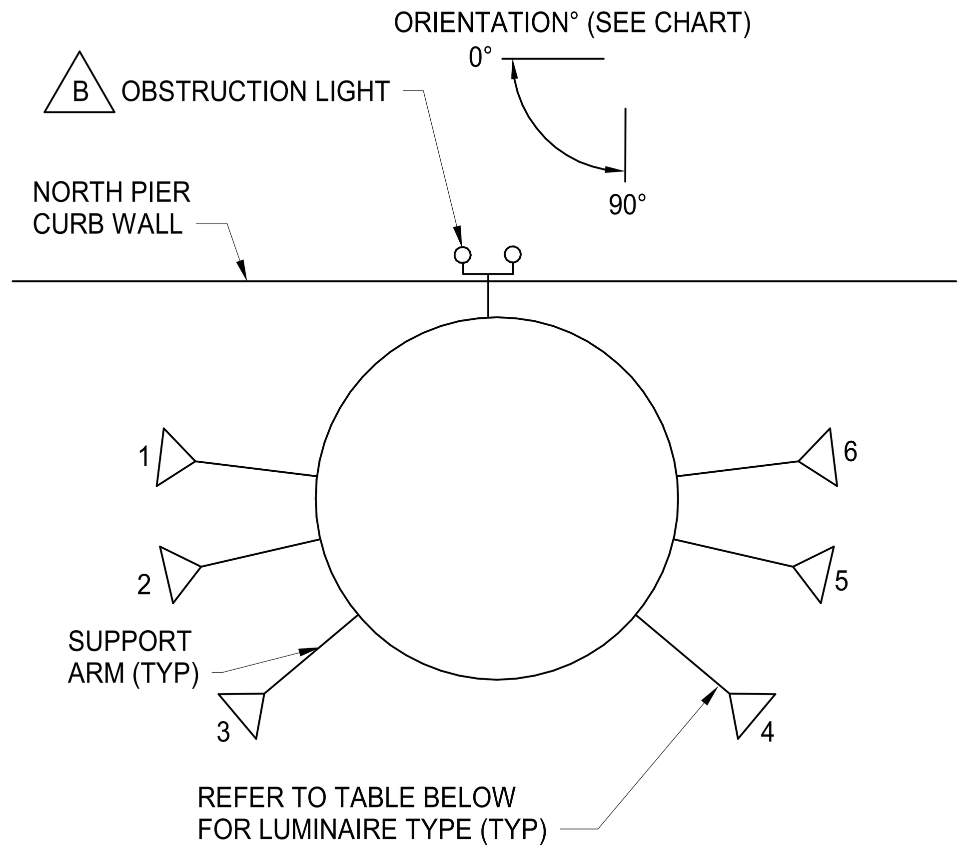
Stantec
A/E INFO

APPROVED
James E. Donahue, R.A.
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES DPF | DRW GLV | CHK RJW
PM/DM MRI/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
LUMINAIRE SCHEDULE & DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936788
SHEET 219 OF 247
E-508
DRAWFORM REVISION: 25 AUGUST 2020

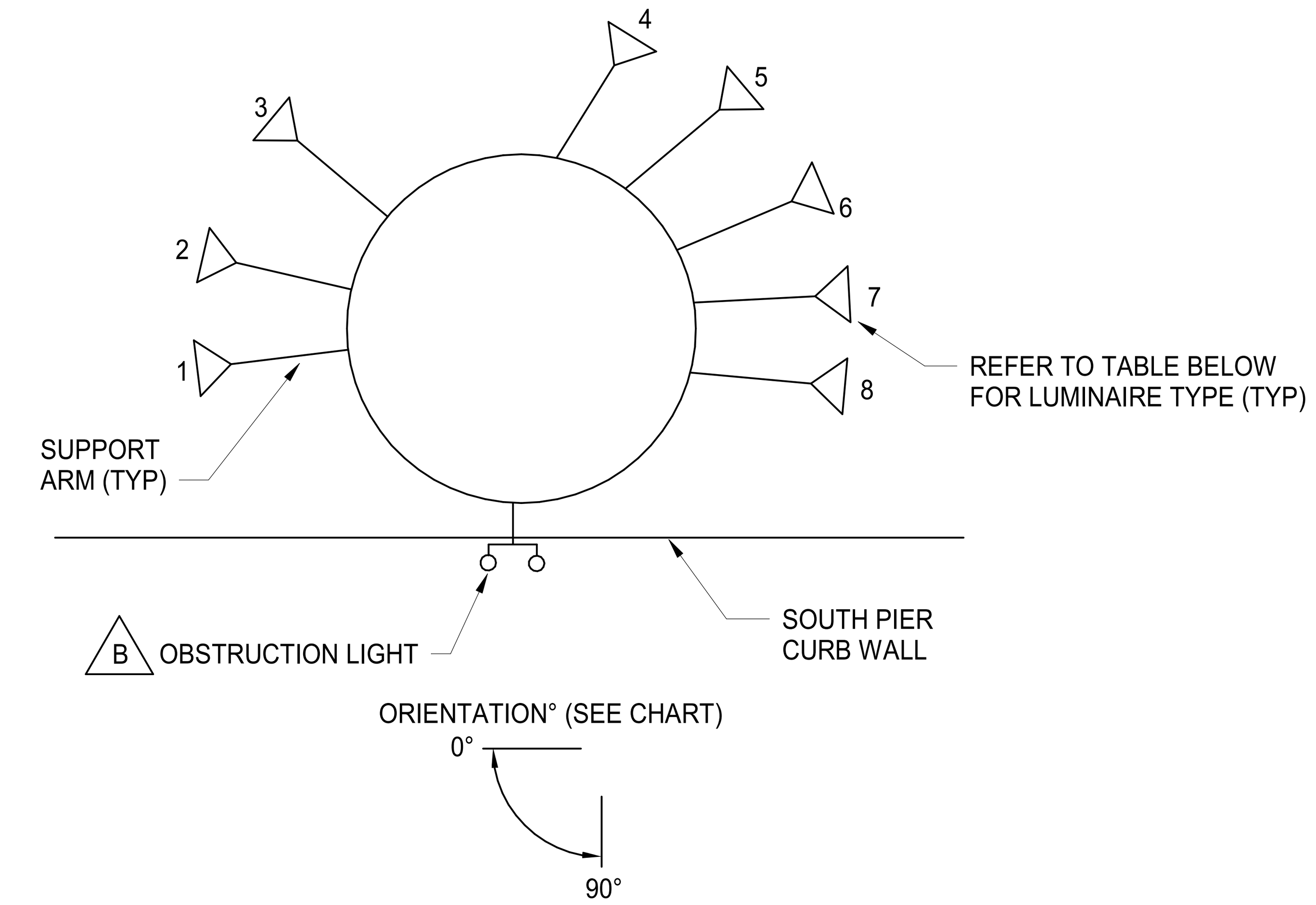
3/2/2026 5:13:57 PM E-508
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt



POLE LC1 INBOARD FIXTURE ARRANGEMENT

LC1	TYPE	ORIENTATION	TILT
1	A2	3	58
2	A2	20	60
3	A1	36	40
4	A1	142	38
5	A2	162	58
6	A3	178	66

C1 POLE LC1 DETAIL
NOT TO SCALE



POLE LC2 INBOARD FIXTURE ARRANGEMENT

LC2	TYPE	ORIENTATION	TILT
1	A3	357	64
2	A2	339	59
3	A1	315	22
4	A1	206	40
5	A2	197	60
6	A3	190	71
7	A3	186	73
8	A3	181	73

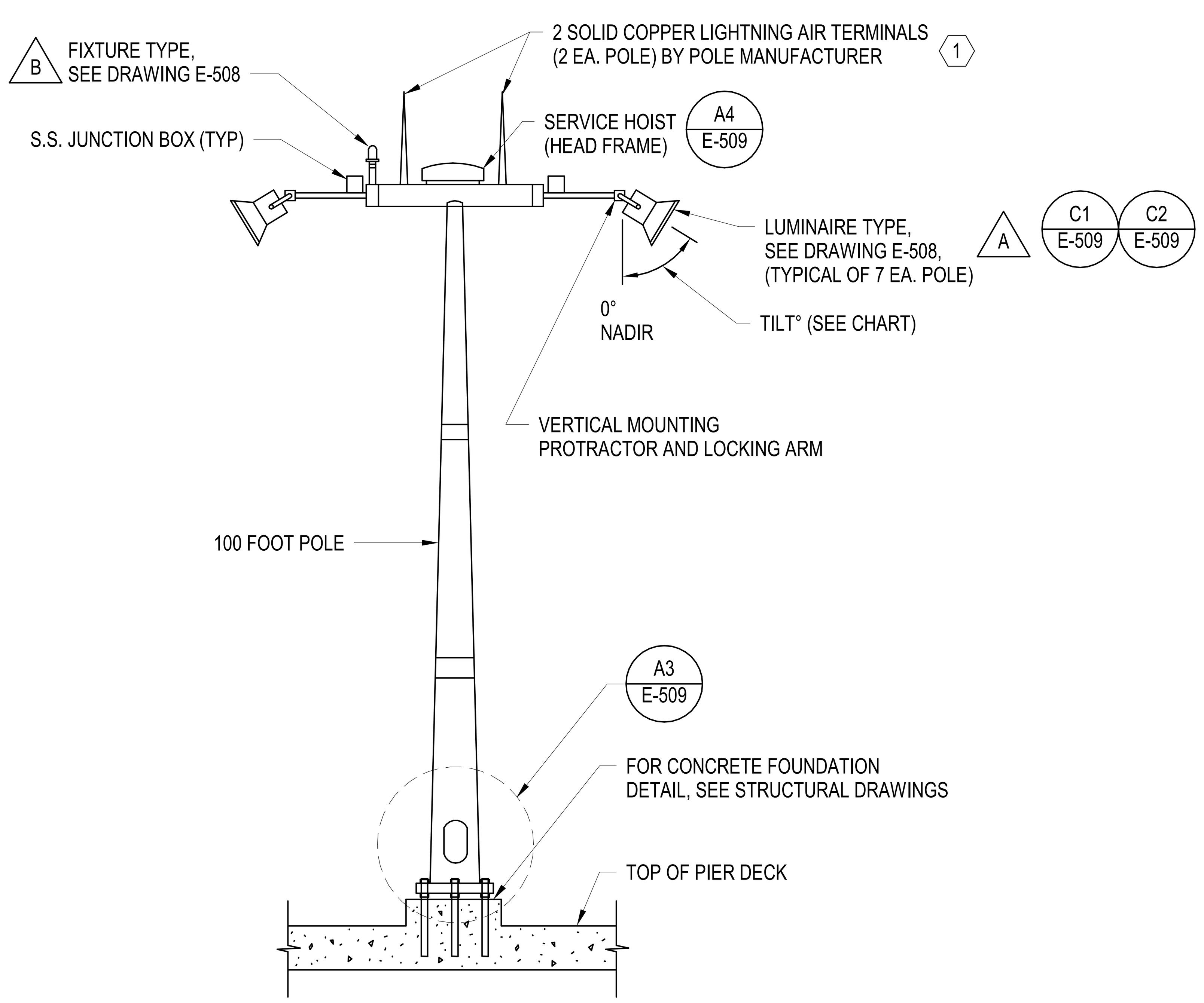
C2 POLE LC2 DETAIL
NOT TO SCALE

GENERAL SHEET NOTES:

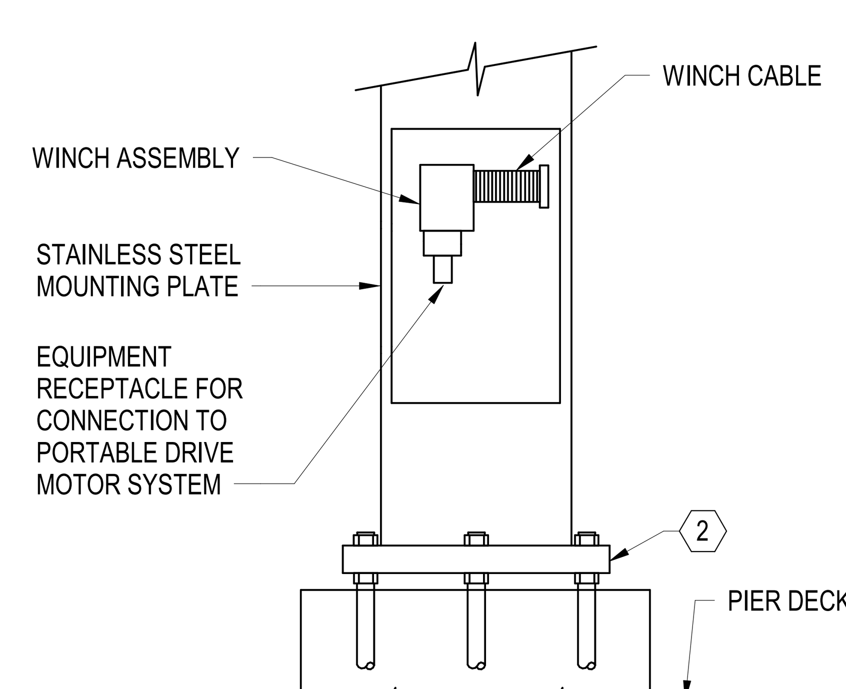
1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.
2. FOR LUMINAIRE SCHEDULE, SEE DRAWING E-508.

SHEET KEYNOTES:

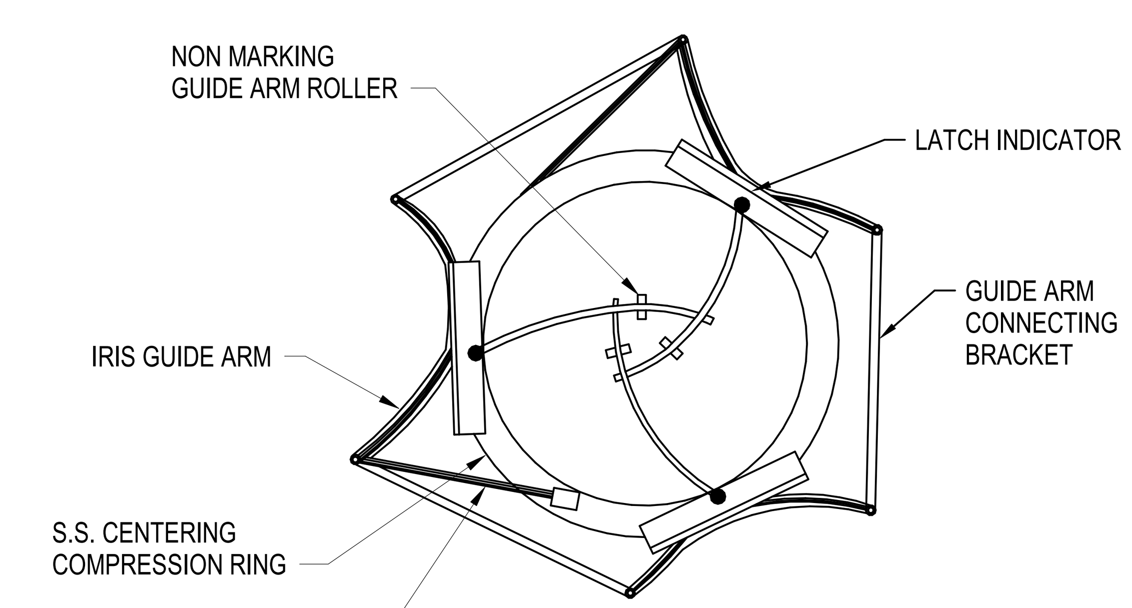
1. LIGHTNING AIR TERMINAL MUST BE 6 FT IN LENGTH WITH #4/0 BCHD CONDUCTORS FROM EACH TERMINAL TO THE BASE OF POLE AND PROVIDED BY POLE MANUFACTURER. PROVIDE 10'-0" SLACK ON EACH RUN.
2. COORDINATE EXACT POLE MOUNTING REQUIREMENTS AND BOLT PATTERNS WITH THE APPROVED POLE MANUFACTURER.



A1 HIGH MAST LIGHTING POLE - LC1 AND LC2 DETAIL
NOT TO SCALE



A3 FIXTURE LIFTING ASSEMBLY DETAIL
NOT TO SCALE



NOTE:
HEAD FRAME ASSEMBLY NOT SHOWN.

A4 FIXTURE RING ASSEMBLY DETAIL
NOT TO SCALE

SYM | DESCRIPTION | DATE | APPR

DAVID P FERGUSON ELECTRICAL No. 52642 — COMMONWEALTH OF MASSACHUSETTS — PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF | DRW GLV | CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
CI CORE - HAMPTON ROADS
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
HIGH MAST LIGHTING DETAILS

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936789

SHEET 220 OF 247

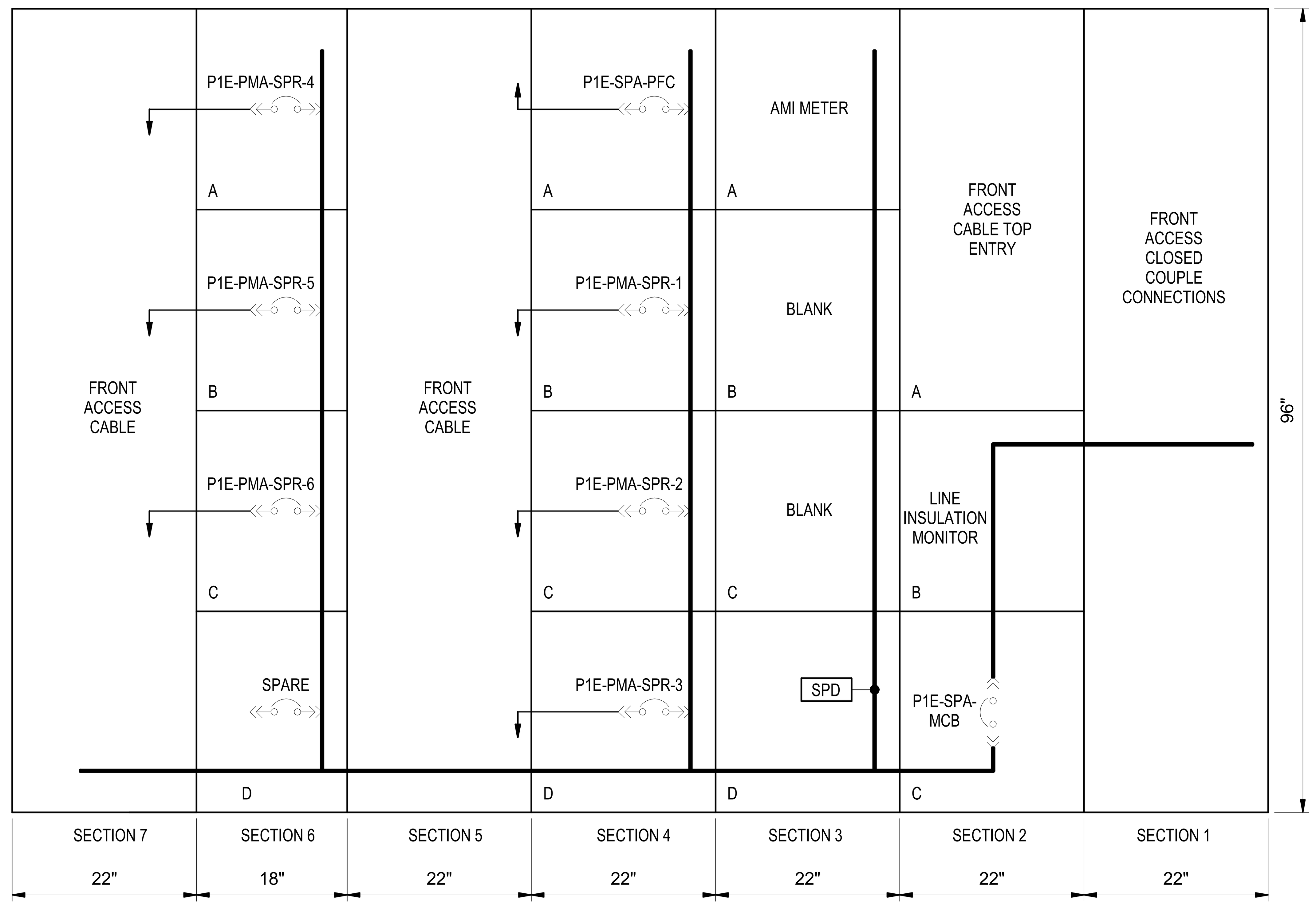
E-509

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 5:13:58 PM E-509
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt

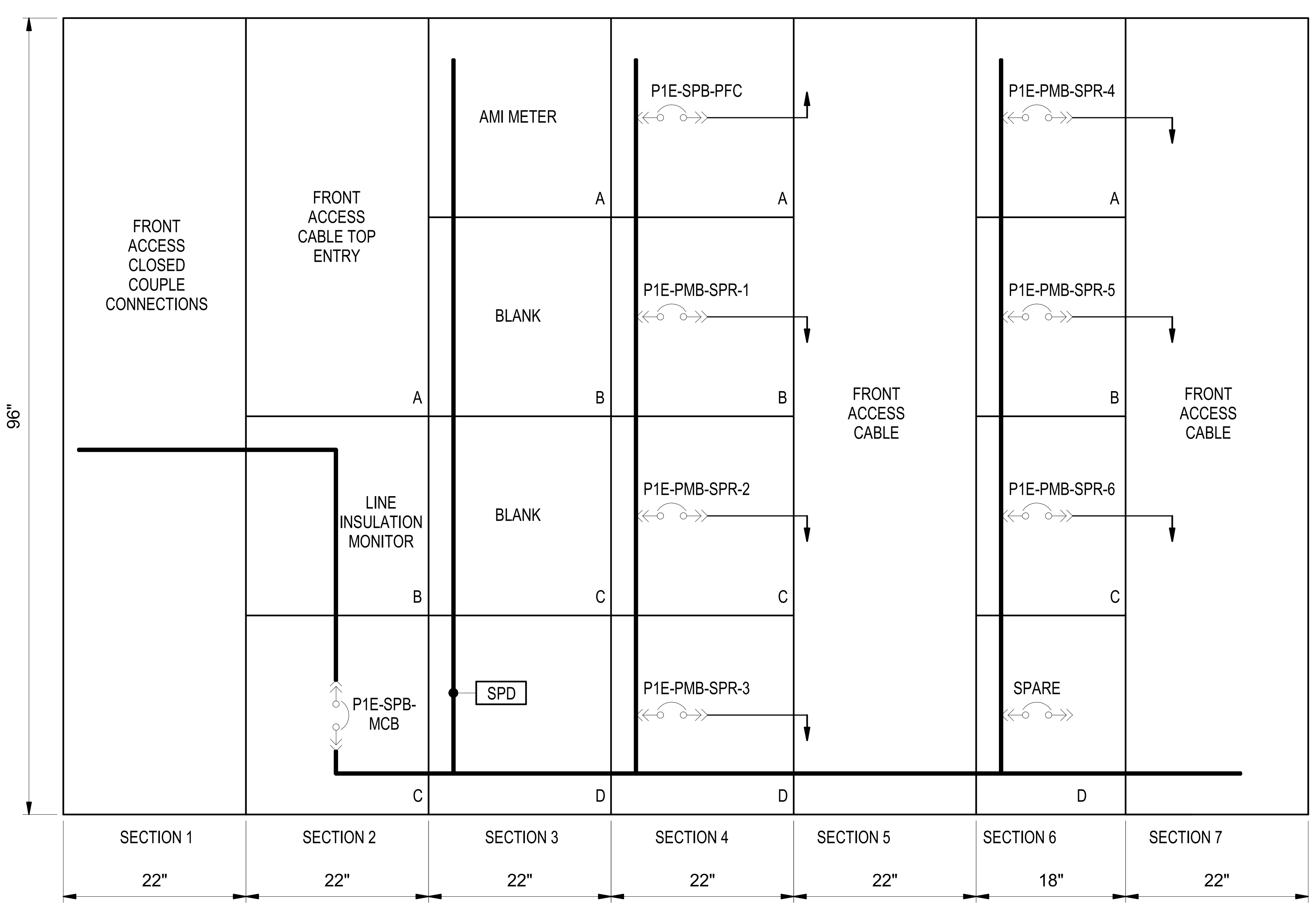
GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR POWER ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, AND E-603. FOR SWITCHGEAR SCHEDULE SEE DRAWING SEE E-610. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.



C2 ELEVATION - SWITCHGEAR P1E-SPA

NOT TO SCALE



A1 ELEVATION - SWITCHGEAR P1E-SPB

NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

DAVID P FERGUSON ELECTRICAL No. 52642 — COMMONWEALTH OF MASSACHUSETTS — PROFESSIONAL ENGINEER

SEAL

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FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
SUBSTATION P1E SWITCHGEAR ELEVATIONS

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936790

SHEET 221 OF 247

E-510

DRAWFORM REVISION: 25 AUGUST 2020

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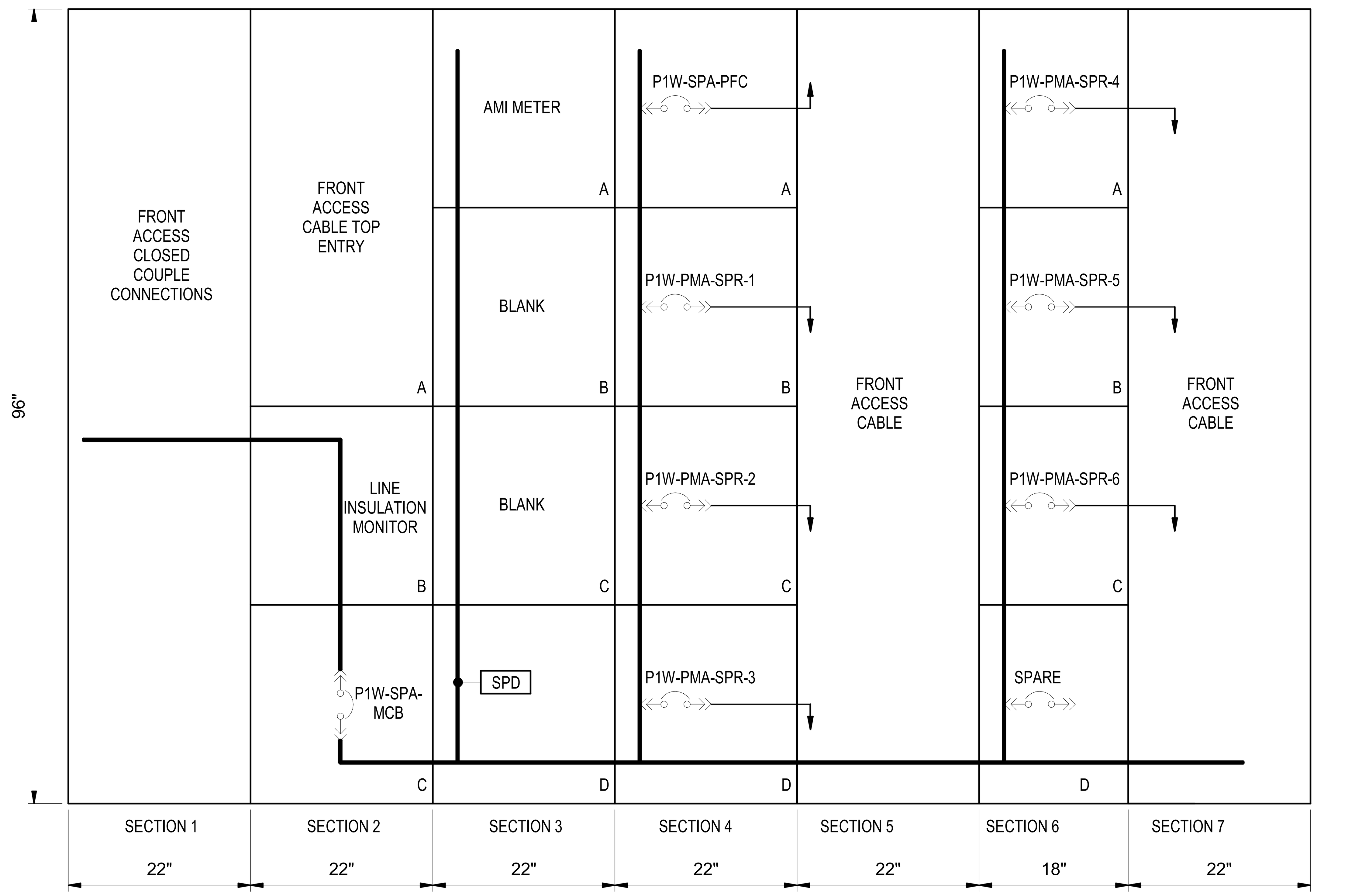
E-510

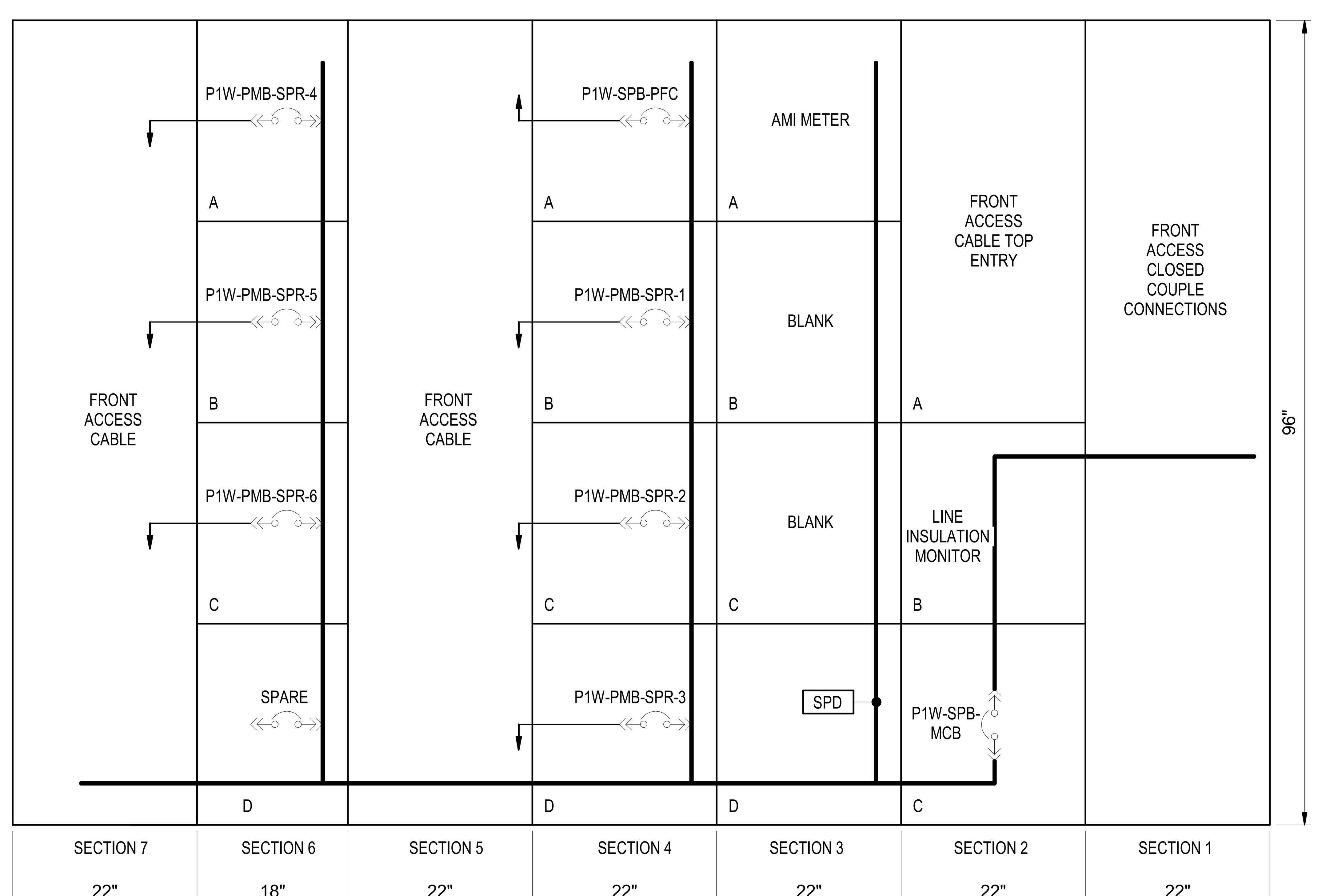
GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR POWER ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, AND E-603. FOR SWITCHGEAR SCHEDULE SEE DRAWING SEE E-611. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.



C1 **ELEVATION - SWITCHGEAR P1W-SPA**

NOT TO SCALE



A1 **ELEVATION - SWITCHGEAR P1W-SPB**

NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
David Ferguson
SEAL

Stantec

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY

Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV	CHK RJW
PM/DM		MRI/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
SUBSTATION P1W SWITCHGEAR ELEVATIONS

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936791
SHEET 222 OF 247

E-511

DRAWFORM REVISION: 25 AUGUST 2020

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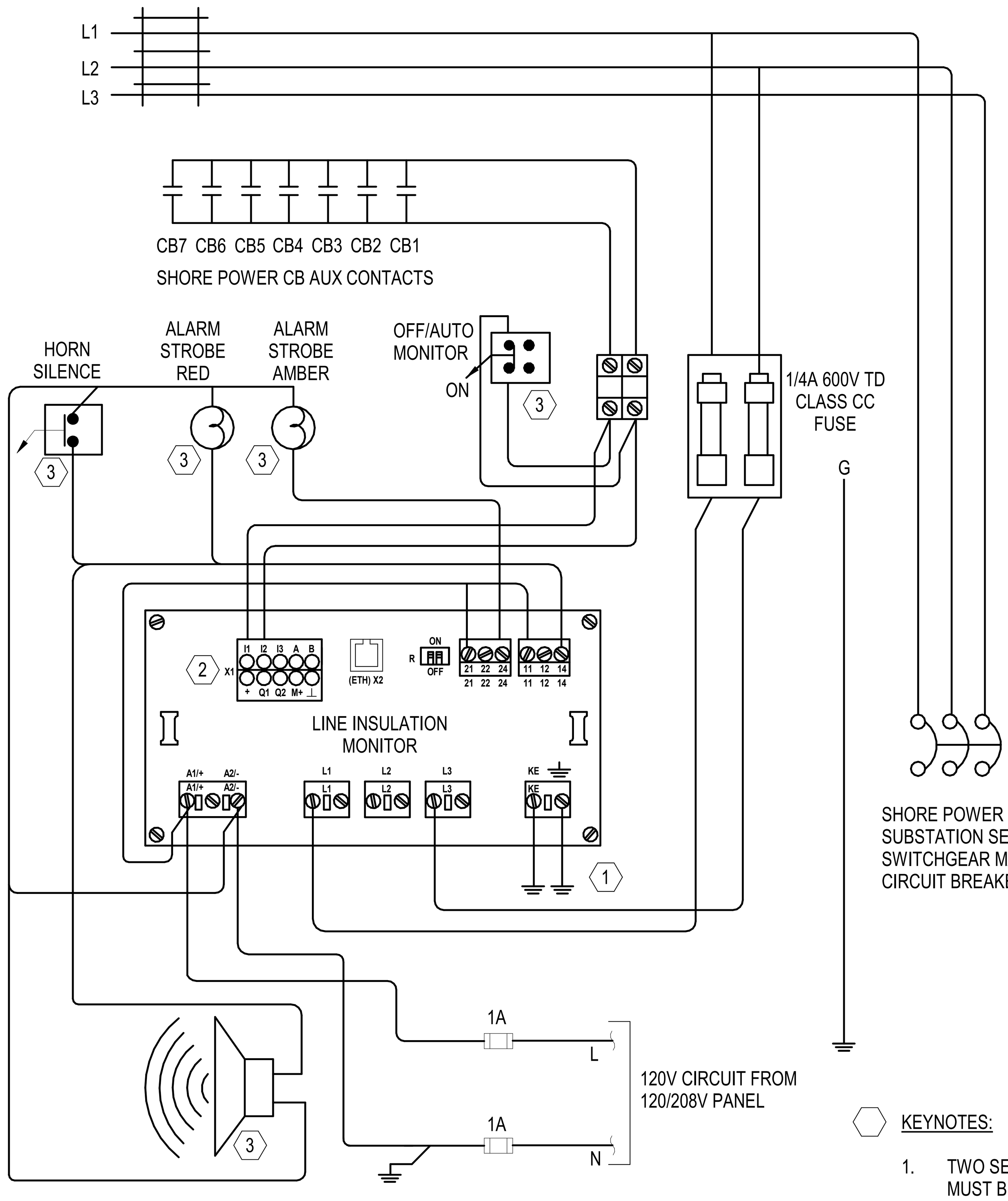


LINE INSULATION MONITOR SYSTEM NOTES:

1. LINE INSULATION MONITOR SYSTEM MUST OPERATE AS A GROUND DETECTION SYSTEM IN ACCORDANCE WITH NEC ARTICLE 250.21(B), WITH A VISUAL RED FLASHING LIGHT AND AUDIBLE ALARM INDICATION UPON DETECTION OF A GROUND FAULT.
2. MONITORING LEADS MUST BE CONNECTED DIRECTLY DOWNSTREAM OF THE SUBSTATION ISOLATION TRANSFORMER AND UPSTREAM OF THE SUBSTATION SECONDARY SWITCHGEAR MAIN CIRCUIT BREAKER.
3. UNGROUNDED SYSTEMS MUST BE LEGIBLY MARKED "CAUTION: UNGROUNDED SYSTEM OPERATING - ______ VOLTS BETWEEN CONDUCTORS" PER NEC 250.21(C).
4. PROVIDE WITH EXTERNAL PANEL MOUNTED KILOHM INDICATING INSTRUMENT, MINIMUM 96 MM SQUARE. METER MUST BE VISIBLE FROM PANEL EXTERIOR AND BE WEATHER PROTECTED.
5. PROVIDE OVERRIDE AT POWER MOUND THAT IS CONNECTED TO LIM SYSTEM AT SHORE POWER SWITCHGEAR, TO ALLOW THE SHORE SIDE SYSTEM TO BE DISABLED AND THE SHIP LIM SYSTEM TO MONITOR SHIP AND SHORE FACILITY SYSTEMS.
6. MONITOR DEVICE MUST BE BENDER ISOMETER ISO685. COORDINATE WITH USCG FOR LINE INSULATION MONITOR DEVICE SELECTION.

KEYNOTES:

1. TWO SEPARATE AND REDUNDANT GROUNDS MUST BE RUN PER MANUFACTURER.
2. MULTIFUNCTIONAL I/O INTERFACE.
3. DEVICE LOCATED AT POWER MOUND.

ALARM FUNCTION:

1. PRE ALARM WILL ONLY TRIGGER PRE-ALARM AMBER STROBE LIGHT. LIGHT WILL STAY ON UNTIL RESET ON THE MONITOR.
2. MAIN ALARM AND SYSTEM ALARM WILL TRIGGER MAIN ALARM, HORN AND RED STROBE LIGHT.
3. THE ABOVE IS A GENERAL DIAGRAM. REFER TO MANUFACTURERS LITERATURE FOR INSTALLATION INSTRUCTIONS.

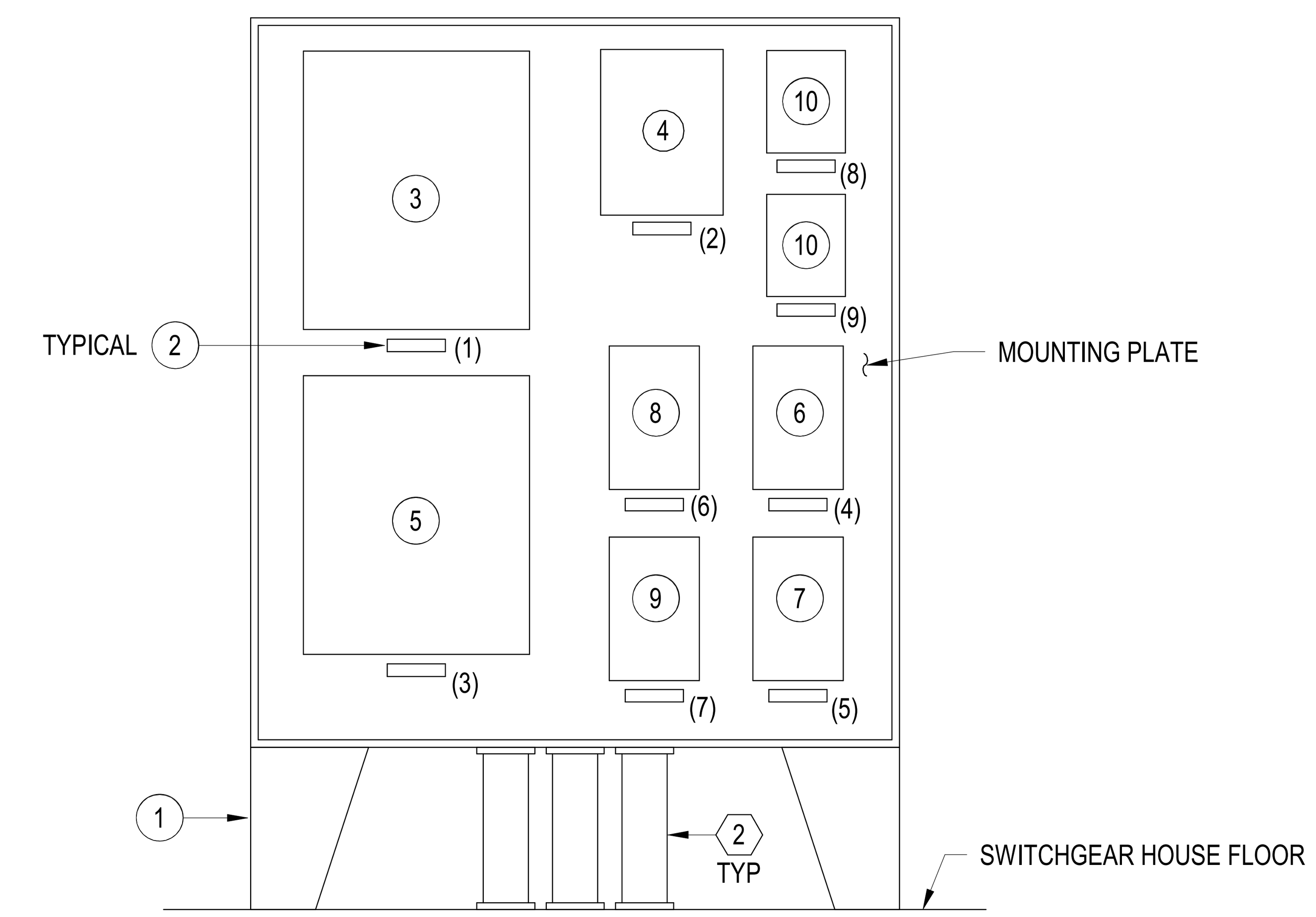
ALARM SETTINGS:

1. ALARM #1 ADJUSTABLE BETWEEN 35-200Kohms
2. ALARM #2 ADJUSTABLE BETWEEN 10-100Kohms

A. TEMPORARILY CHANGING K1 CONTACT TO TEST WILL ENERGIZE RED STROBE & HORN.
B. TEMPORARILY CHANGING K2 CONTACT TO TEST WILL ENERGIZE AMBER STROBE.

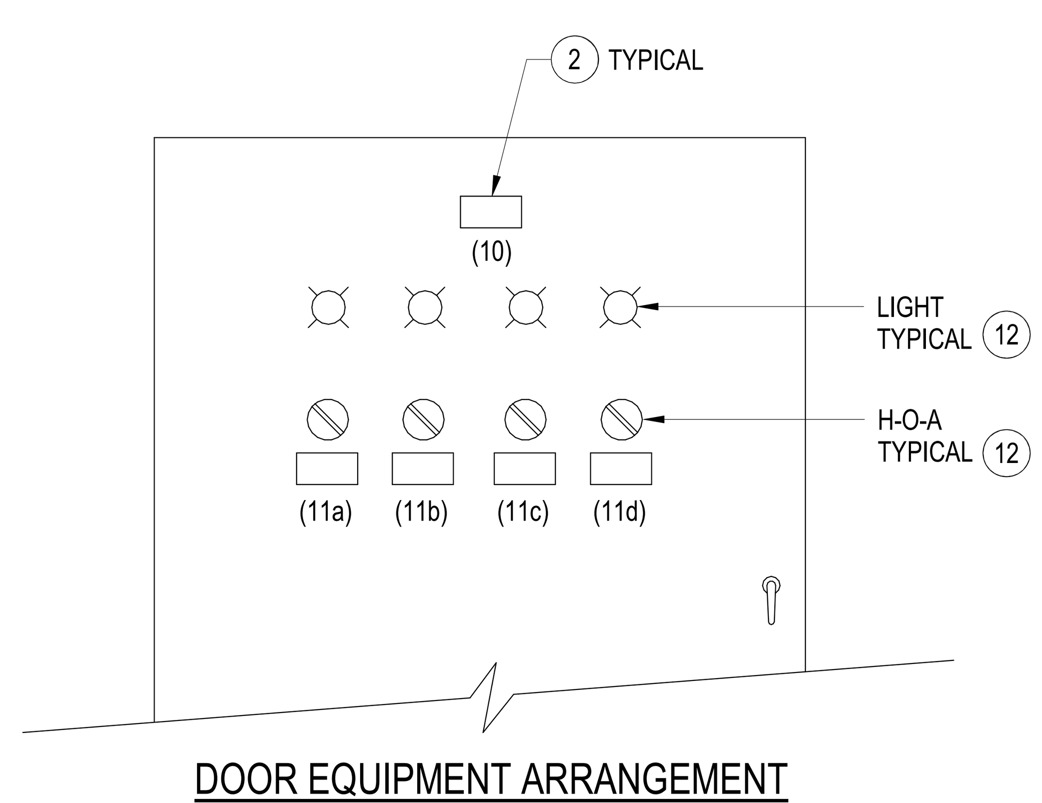
B1 LINE INSULATION MONITORING SYSTEM DETAIL

NOT TO SCALE



B3 PIER LIGHTING CONTROL CABINET INTERIOR VIEW

NOT TO SCALE



A3 PIER LIGHTING CONTROL CABINET EXTERIOR VIEW

NOT TO SCALE

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.

SHEET KEYNOTES:

1. FOR ELEMENTARY DIAGRAMS AND INTERCONNECTION DIAGRAMS SEE DRAWING E-609.
2. 3"C. SEAL CONDUIT PENETRATIONS WITH DUX SEAL. CAP SPARE CONDUIT(S) USING MANUFACTURED END PLUG. AFTER CABLES ARE INSTALLED SEAL INTERIOR OF CONDUITS WITH DUX SEAL. MOISTURE MUST NOT ENTER THE CABINET OR THE SWITCH HOUSE.

LEGEND:

1. FREE STANDING CABINET.
2. NAMEPLATES (TYPICAL) SEE SCHEDULE
3. 18 POLE SURFACE MOUNTED PANELBOARD LTG 1 (SEE PANEL SCHEDULE DWG. E-615)
4. 5 KVA DRY TYPE XFMR. 480-120/240V., 1Ø, 3W.
5. 12 POLE SURFACE MOUNTED PANELBOARD LTG 2 (SEE PANEL SCHEDULE DWG. E-615).
6. LIGHTING CONTACTOR "LC1", NEMA TYPE 1 ENCL. 8 POLE, 20A, 120V. COIL (ELECTRICALLY HELD) 1 N.O. AUX. CONTACT.
7. LIGHTING CONTACTOR "LC2" (SAME AS "LC1")
8. LIGHTING CONTACTOR "OC1" 20A, 120V. NEMA TYPE 1 ENCL.
9. LIGHTING CONTACTOR "OC2" (SAME AS "OC1")
10. 24 HOUR ASTRONOMICAL TIMER-WEST POLE.
11. 24 HOUR ASTRONOMICAL TIMER-EAST POLE.
12. H-O-A SWITCHES AND INDICATING LIGHTS LOCATED ON CABINET DOOR
    a. HIGH MAST POLE-WEST
    b. HIGH MAST POLE-EAST
    c. OBSTRUCTION LIGHT-WEST
    d. OBSTRUCTION LIGHT-EAST

NAMEPLATE SCHEDULE:

(1) LIGHTING PANEL LTG1
(2) 5 KVA XFMR (480-120/240V)
(3) LIGHTING PANEL LTG2
(4) LIGHTING CONTACTOR LC1-WEST POLE
(5) LIGHTING CONTACTOR LC2-EAST POLE
(6) LIGHTING CONTACTOR OC1-WEST POLE
(7) LIGHTING CONTACTOR OC2-EAST POLE
(8) TIMER-WEST POLE
(9) TIMER-EAST POLE
(10) (CABINET DOOR) LIGHTING CONTROL CABINET-PIER 1
(11) a. HIGH MAST POLE-WEST
b. HIGH MAST POLE-EAST
c. OBSTRUCTION LIGHT-WEST
d. OBSTRUCTION LIGHT-EAST
— CABINET DOOR

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
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BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
MISCELLANEOUS DETAILS - SHEET 1 OF 2

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936792
SHEET 223 OF 247
E-512

DRAWFORM REVISION: 20 AUGUST 2020

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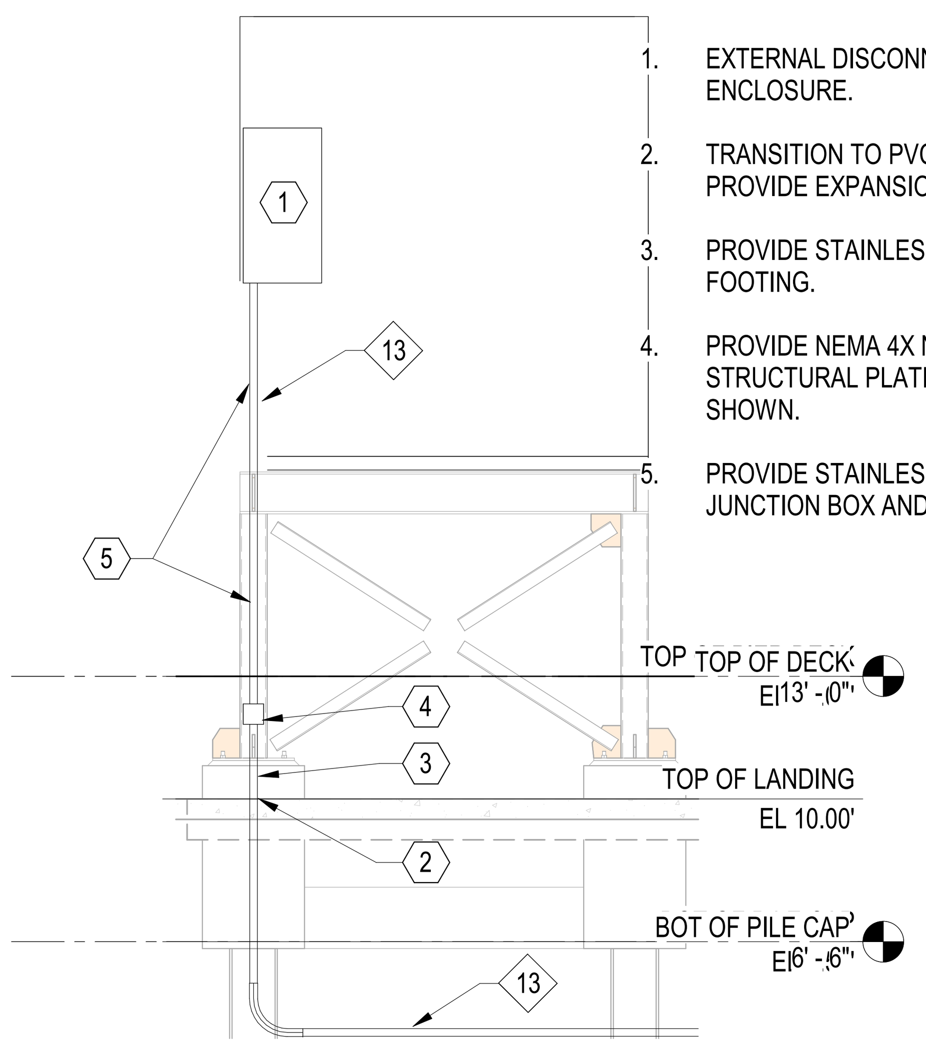
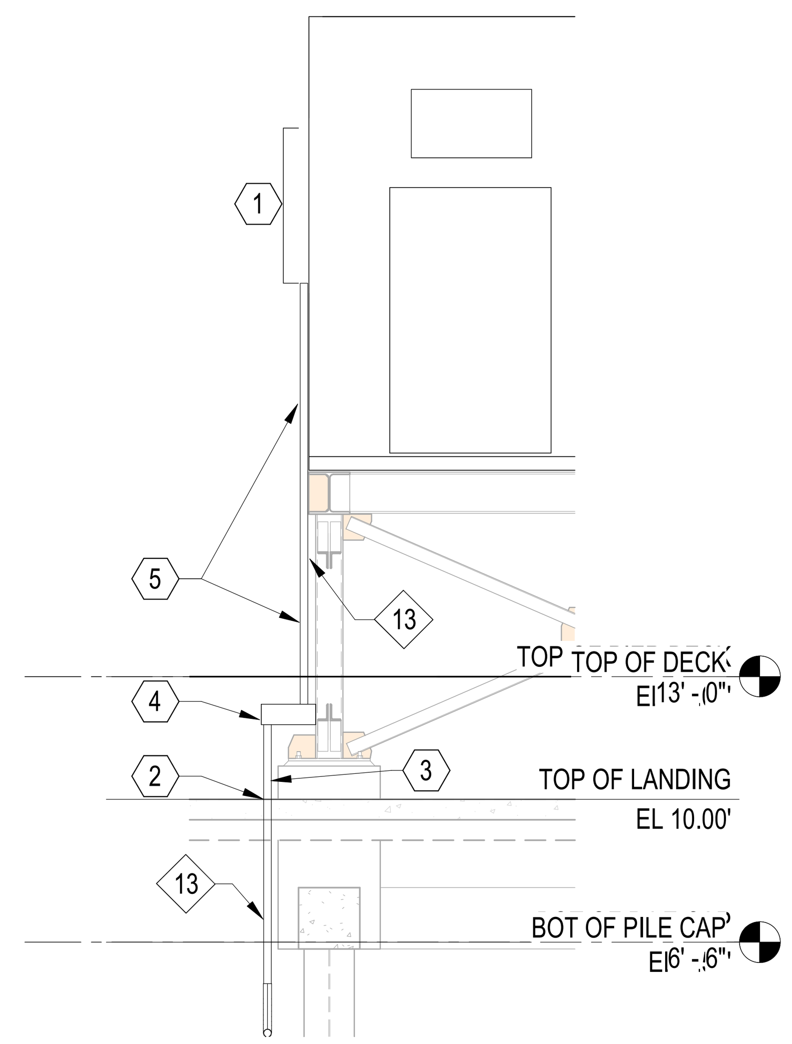
E-512

CA ENCLOSURE DETAILS KEYNOTES:

1. EXTERNAL DISCONNECT SWITCH FURNISHED WITH COMPRESSED AIR ENCLOSURE.
2. TRANSITION TO PVC COATED RGS CONDUIT FOR ABOVE GROUND CONDUIT. PROVIDE EXPANSION FITTING WHERE CONDUIT EXTENDS UP ABOVE GRADE.
3. PROVIDE STAINLESS STEEL HARDWARE TO SUPPORT CONDUIT FROM FOOTING.
4. PROVIDE NEMA 4X NONMETALLIC JUNCTION BOX. SECURE BOX TO STRUCTURAL PLATE. FOR VISUAL CLARITY STRUCTURAL PLATE NOT SHOWN.
5. PROVIDE STAINLESS STEEL HARDWARE TO SUPPORT CONDUIT WITHIN 3' OF JUNCTION BOX AND DISCONNECT SWITCH.



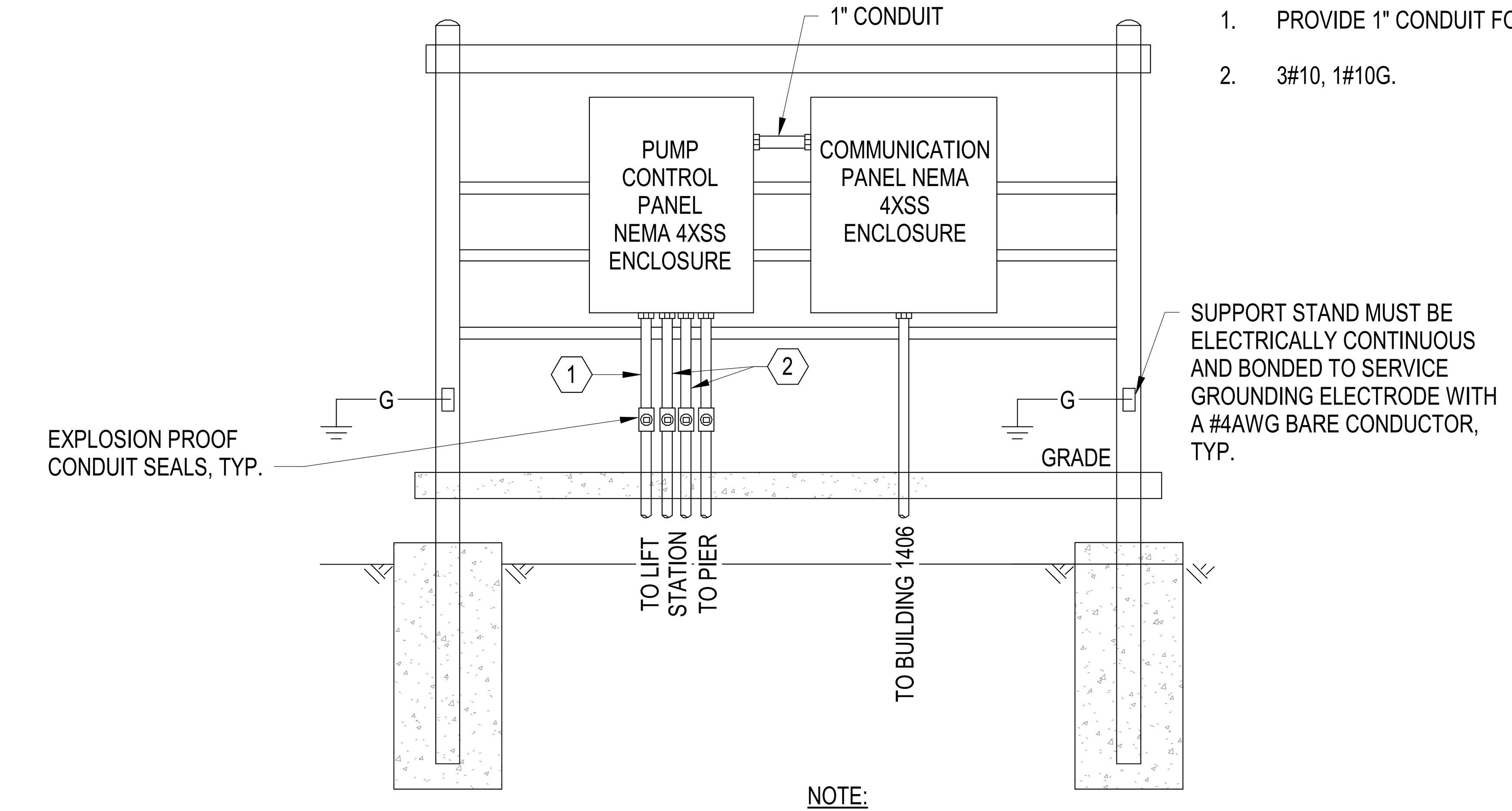
C1 CA ENCLOSURE SOUTH ELEVATION
1/4" = 1'-0" (E-107)



C2 CA ENCLOSURE EAST ELEVATION
1/4" = 1'-0" (E-107)

KEYNOTES:

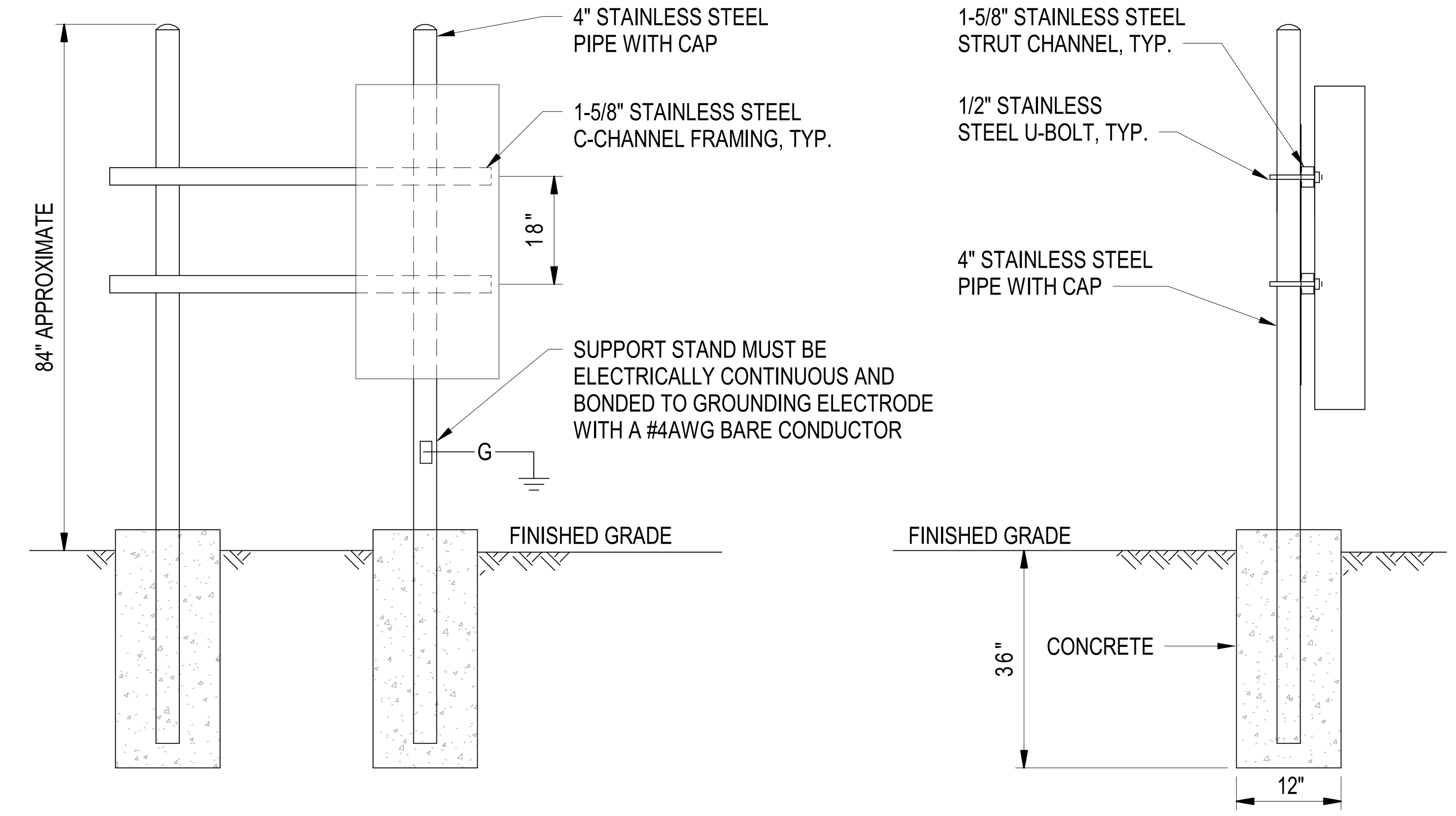
1. PROVIDE 1" CONDUIT FOR CONTROLS.
2. 3#10, 1#10G.



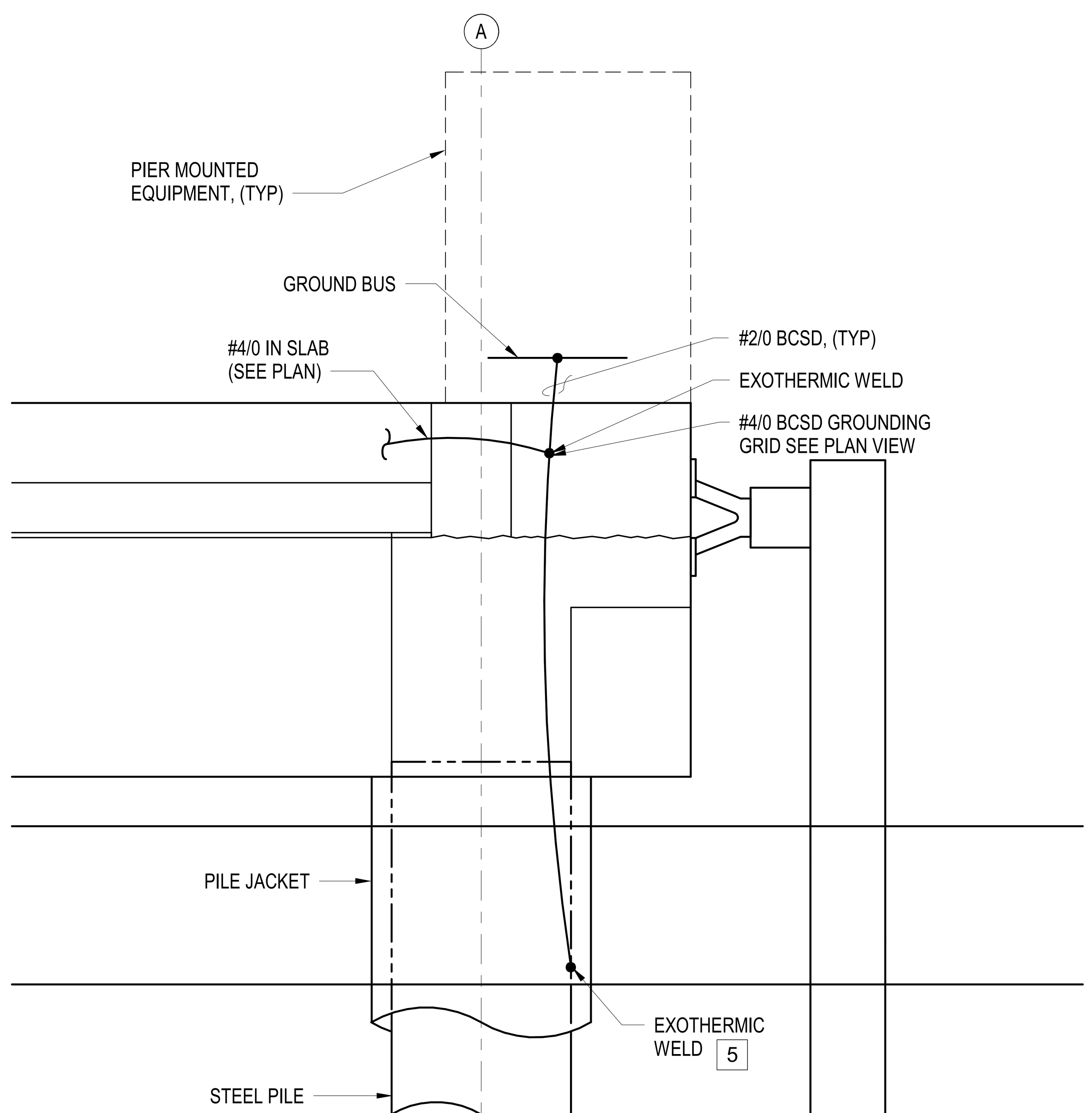
NOTE:

1. ABOVE GROUND CONDUIT MUST BE PVC COATED RGS.

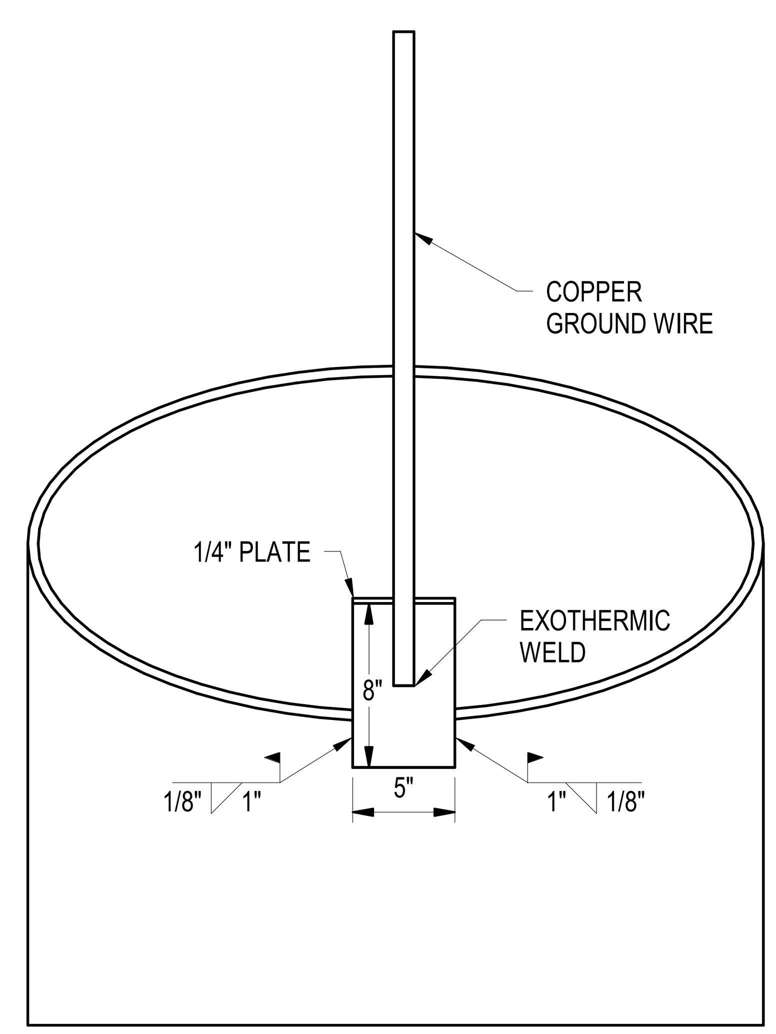
C4 LIFT STATION EQUIPMENT RACK DETAIL
NOT TO SCALE



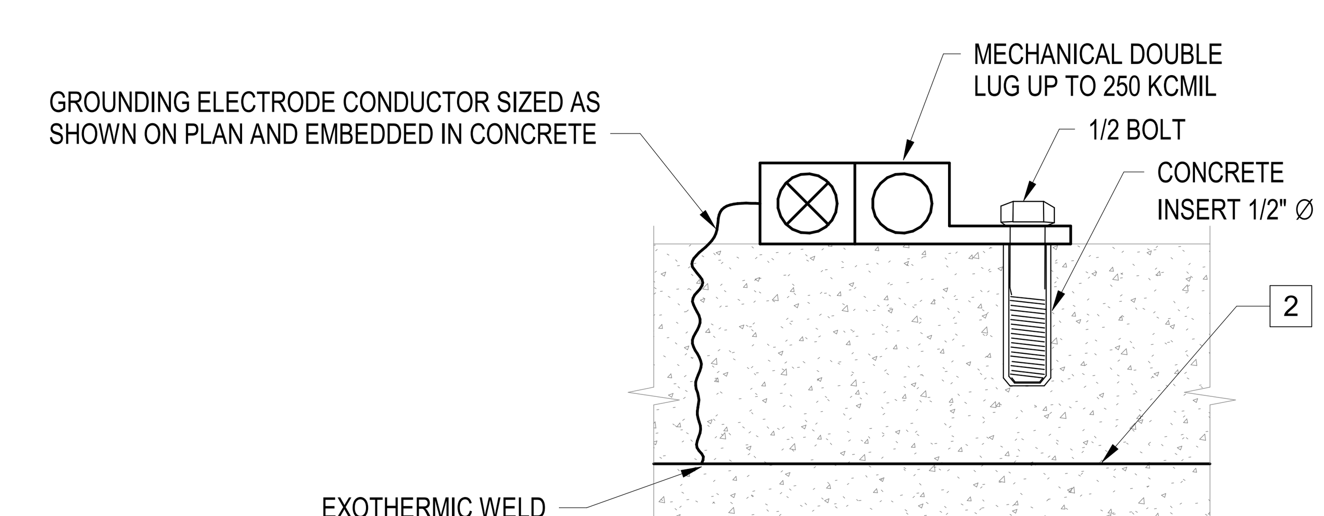
B4 TYPICAL EXTERIOR EQUIPMENT RACK
NOT TO SCALE



A1 DETAIL - TYPICAL NORTH & SOUTH
NOT TO SCALE



A3 DETAIL - PILE CAP CONNECTION
NOT TO SCALE



NOTES:

1. PLACE AT 50' INTERVALS ON OUTER PERIMETER OF PIER (12" FROM CURB).
2. PROVIDE GROUND CONDUCTOR #4/0 BARE COPPER ENCASED IN CONCRETE.

A4 DETAIL - GROUNDING INSERT
NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER

Stantec

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV	CHK RJW
PM/DM		MRI/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
MISCELLANEOUS DETAILS - SHEET 2 OF 2

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936793
SHEET 224 OF 247

E-513

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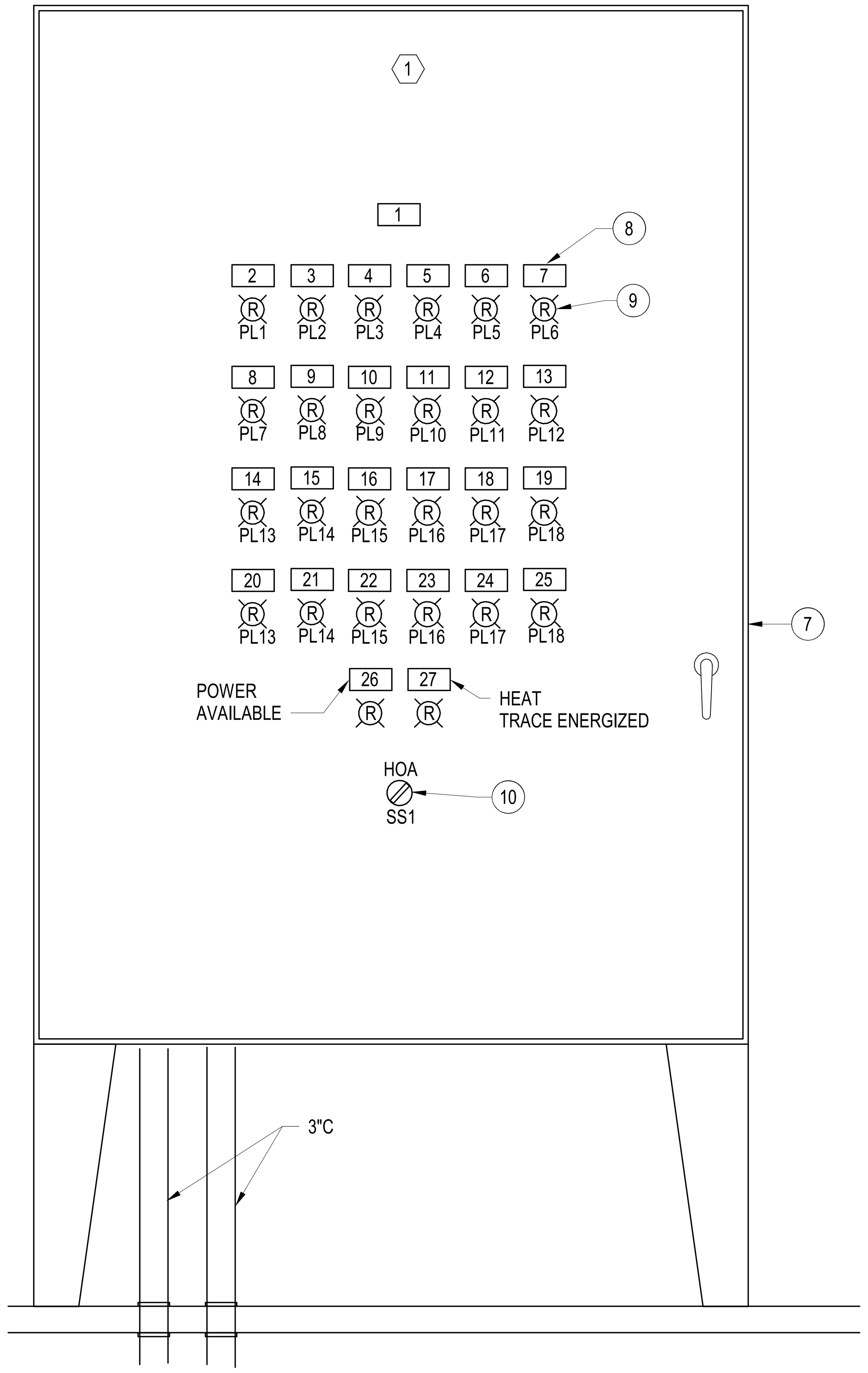
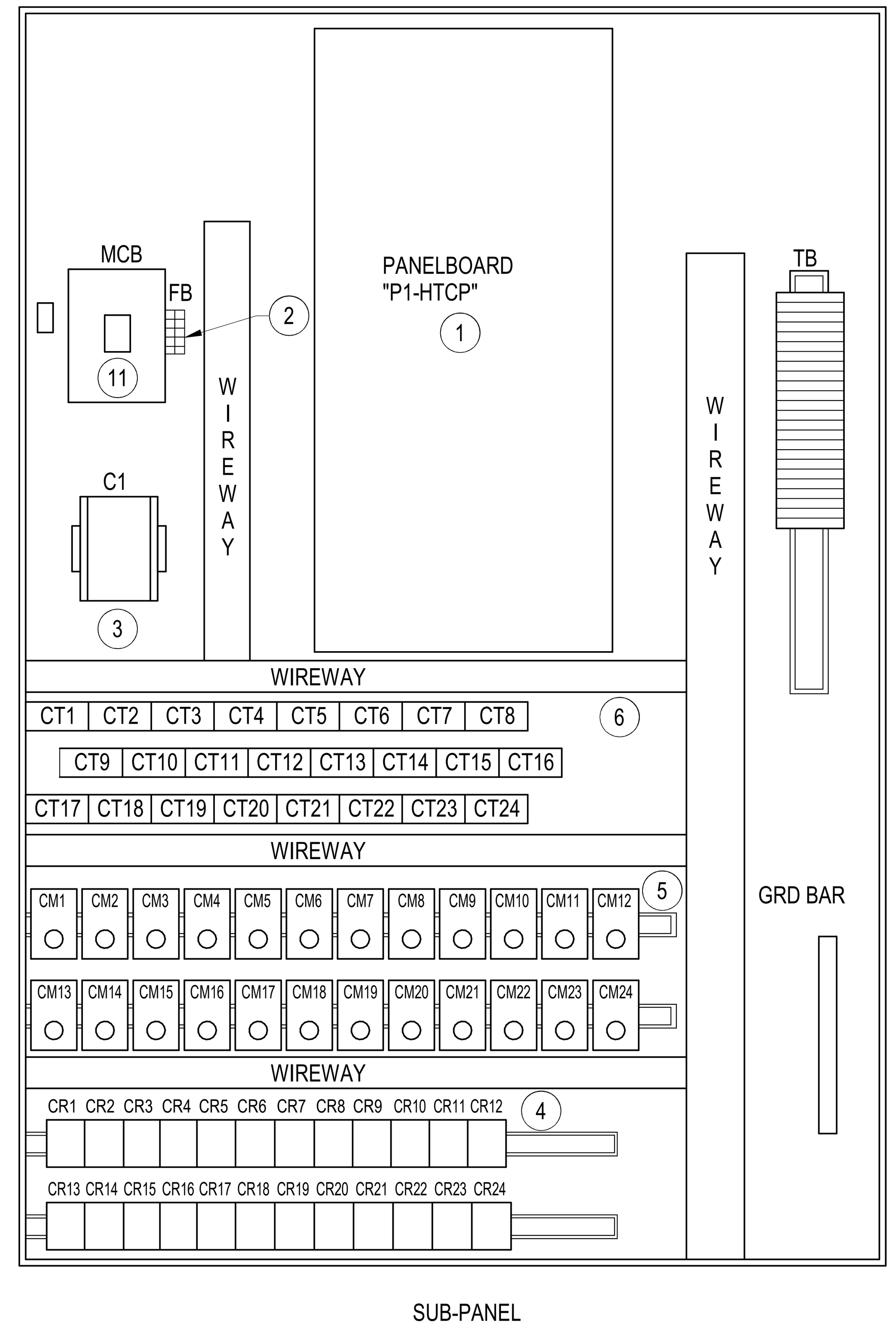
E-513



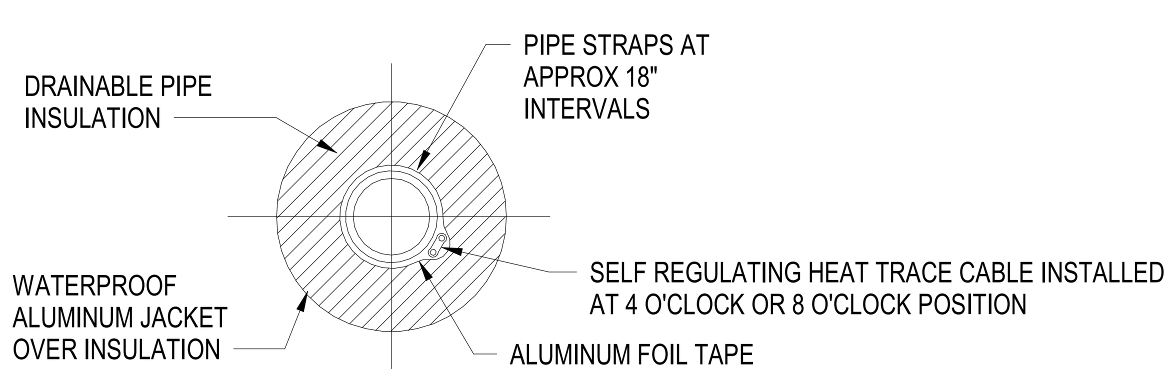
EXTERIOR VIEW



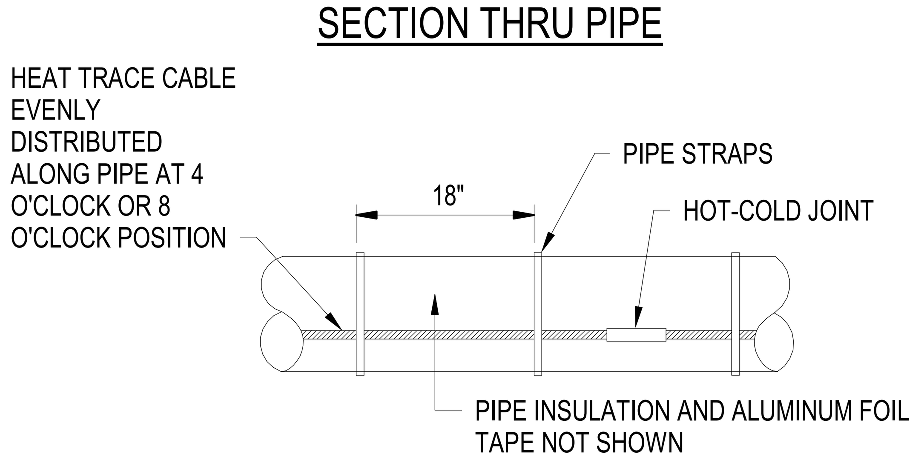
INTERIOR VIEW

B1 HEAT TRACE CONTROL CABINET

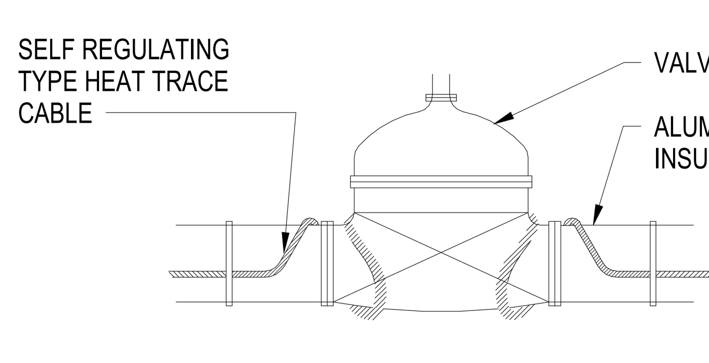
NOT TO SCALE



SECTION THRU PIPE



INSTALLATION DETAIL



INSTALLATION AT VALVES

A1 ELECTRICAL HEAT TRACE

NOT TO SCALE

NAMEPLATE SCHEDULE:

(1) HEAT TRACE CONTROL CABINET

(2) LOSS OF CONTINUITY CIRCUIT #1
SEWER - BENT 12-0 - NORTH

(3) LOSS OF CONTINUITY CIRCUIT #2
WATER - BENT 28-19 - SOUTH

(4) LOSS OF CONTINUITY CIRCUIT #3
SEWER - BENT 12-0 - SOUTH

(5) LOSS OF CONTINUITY CIRCUIT #4
WATER - BENT 43-34 - NORTH

(6) LOSS OF CONTINUITY CIRCUIT #5
SEWER - BENT 24-12 - NORTH

(7) LOSS OF CONTINUITY CIRCUIT #6
WATER - BENT 37-28 - SOUTH

(8) LOSS OF CONTINUITY CIRCUIT #7
SEWER - BENT 24-12 - SOUTH

(9) LOSS OF CONTINUITY CIRCUIT #8
WATER - BENT 43-37 - SOUTH

(10) LOSS OF CONTINUITY CIRCUIT #9
SEWER - BENT 35-24 - NORTH

(11) SPARE

(12) LOSS OF CONTINUITY CIRCUIT #11
SEWER - BENT 35-24 - SOUTH

(13) SPARE

(14) LOSS OF CONTINUITY CIRCUIT #13
SEWER - CA ENCLOSURE RISER

(15) SPARE

(16) LOSS OF CONTINUITY CIRCUIT #15
WATER - BENT 14-8 - NORTH

(17) SPARE

(18) LOSS OF CONTINUITY CIRCUIT #17
WATER - BENT 11-08 - SOUTH

(19) SPARE

(20) LOSS OF CONTINUITY CIRCUIT #19
WATER - BENT 24-14 - NORTH

(21) SPARE

(22) LOSS OF CONTINUITY CIRCUIT #21
WATER - BENT 19-11 - SOUTH

(23) SPARE

(24) LOSS OF CONTINUITY CIRCUIT #23
WATER - BENT 34-24 - NORTH

(25) SPARE

(26) POWER AVAILABLE

(27) HEAT TRACE ENERGIZED

SHEET KEYNOTES:

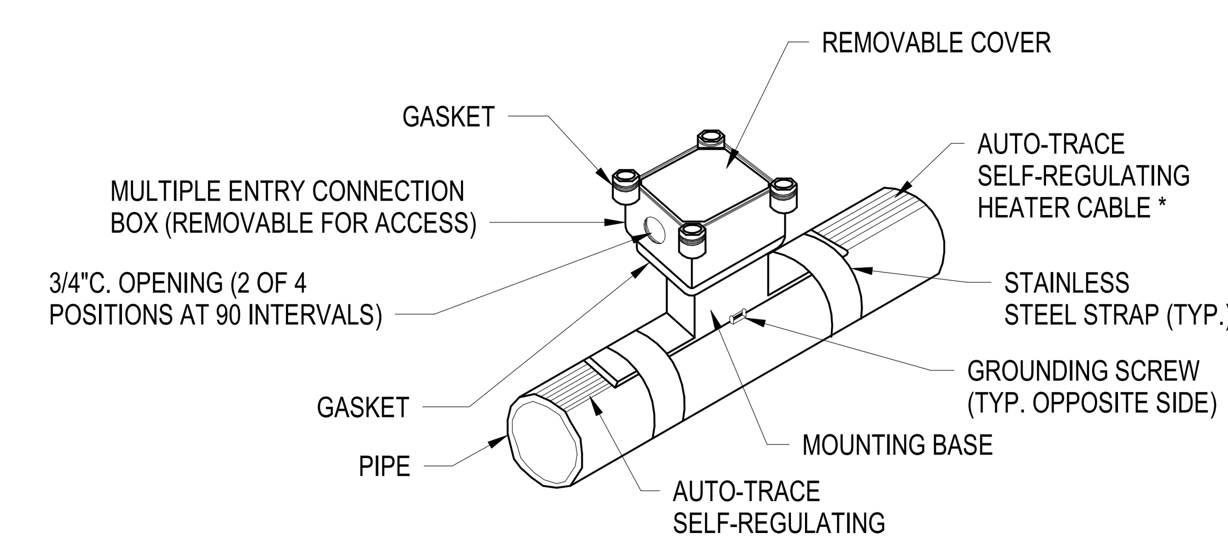
1. PROVIDE HEAT TRACE CONTROL PANEL. HEAT TRACE CONTROL CABINET MUST BE ASSEMBLED AND COMPLETELY WIRED BY ONE FABRICATOR AND MUST BE SUBMITTED FOR APPROVAL BEFORE ASSEMBLY. THE HEAT TRACE CONTROL CABINET MUST BE MICROPROCESSOR BASED DUAL-POINT HEAT TRACE SYSTEM TO PROVIDE CONTROL AND MONITORING OF TWO INDEPENDENT ELECTRICAL HEAT-TRACING CIRCUITS FOR BOTH FREEZE PROTECTION AND TEMPERATURE MAINTENANCE AND CAN BE SET TO MONITOR AND ALARM FOR HIGH AND LOW TEMPERATURE, HIGH AND LOW CURRENT, GROUND-FAULT LEVEL AND VOLTAGE ON EACH OF ITS CONTROL POINTS.

LEGEND:

1. 480Y/277V PANELBOARD
2. FUSE BLOCK (10 AMP).
3. CONTACTOR - 600 VOLT, 3 POLE, 100AMP (2-N.O. AUX CONTACTS).
4. CONTROL RELAYS, 120V DPDT.
5. CURRENT MONITOR RELAYS.
6. CURRENT TRANSFORMERS.
7. FREE STANDING NEMA TYPE 4X (S.S.-316) CABINET. (SIZE TO SUIT EQUIPMENT FURNISHED AND APPROVED BY THE GOVERNMENT.)
8. NAMEPLATES, SEE SCHEDULE.
9. PILOT LIGHT (LED) PROVIDE PUSH-TO-TEST TYPE RED LENS.
10. HAND-OFF-AUTO. THREE POSITION SELECTOR SWITCH.
11. CIRCUIT BREAKER 3P, 100A.

NOTES:

1. FOR LEGEND AND ABBREVIATION, SEE DRAWING E-001.



* = INSTALL HEATER CABLE AT 4 O'CLOCK OR 8 O'CLOCK POSITION ON PIPING (NOT AS SHOWN).

A4 HEAT TRACE POWER CONNECTION (DUAL ENTRY) DETAIL

NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
SEAL

Stantec

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES	DPF	DRW	GLV	CHK	RJW
PM/DM					MRI/CWJ
BRANCH MANAGER					WRB
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
HEAT TRACE CABINET & DETAILS

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936794
SHEET 225 OF 247
E-514

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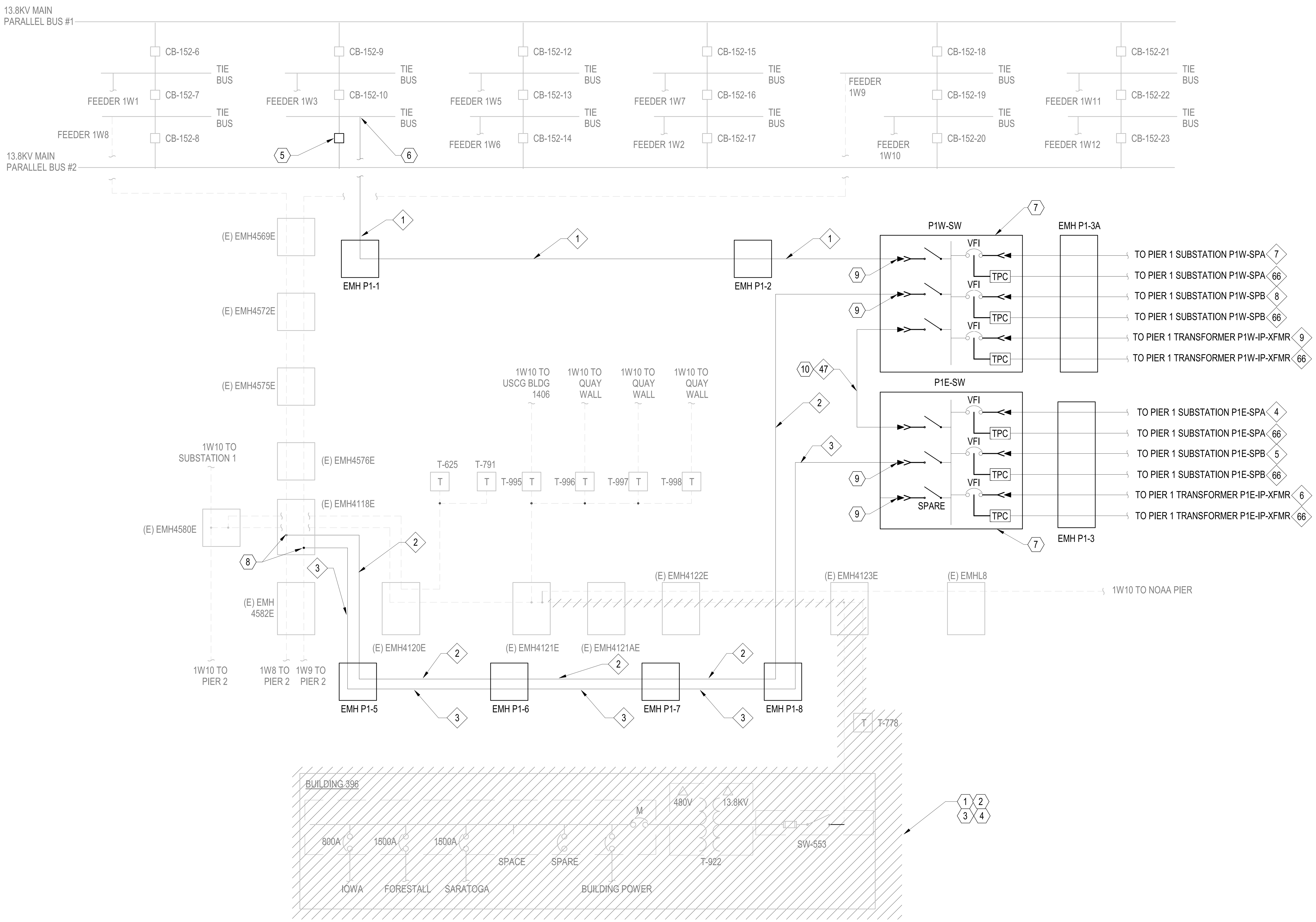
E-514

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615. FOR PIER ONE LINE DIAGRAMS SEE DRAWINGS E-602 AND E-603.
2. SEE ES SERIES DRAWINGS FOR ADDITIONAL INFORMATION ON WORK REQUIRED IN SUBSTATION 1.
3. SEE DRAWING G-102 FOR LOCATIONS OF EXISTING PIER 1 SYSTEMS TO BE DEMOLISHED.

SHEET KEYNOTES:

1. ON EXISTING PIER 1 DEMOLISH BUILDING 396 AND ITS ELECTRICAL EQUIPMENT, RACEWAYS AND DEVICES. DEMOLISH THE PIERSIDE SHIP CONNECTION STATIONS, RACEWAYS AND WIRING BACK TO BUILDING 396.
2. DEMOLISH FEEDER 1W10 CABLING FROM BUILDING 396 TO MANHOLE EMH4121E. DEMOLISH RACEWAYS FOR FEEDER 1W10 FROM BUILDING 396 TO MANHOLE EMH4123E.
3. DEMOLISH TRANSFORMER T-778 AND ASSOCIATED RACEWAYS AND WIRING BACK TO MANHOLE EMH4123E.
4. DEMOLISH EXISTING ELECTRICAL EQUIPMENT AND SYSTEMS ON PIER 1 NOT SPECIFICALLY SHOWN ON THE DRAWINGS. EQUIPMENT MUST BE FIELD VERIFIED.
5. PROVIDE 15KV CIRCUIT BREAKER IN EXISTING OPEN CUBICLE IN SUBSTATION 1. SEE DWG ES-601 AND ASSOCIATED SUBSTATION 1 ES DWGS FOR DETAILS ON INSTALLATION AND CONNECTION OF BREAKER AND FEEDER.
6. CONNECT FEEDER WIRING TO EXISTING MAIN PARALLEL BUS #2, SEE DWG ES-601 AND ASSOCIATED SUBSTATION 1 ES DWGS FOR DETAILS ON CONNECTION.
7. PROVIDE 15KV, 600A, SF6 GAS SWITCH WITH LOAD INTERRUPTER AND FAULT INTERRUPTER WAYS AS INDICATED. PROVIDE 3-15KV, 600A, NON-LOAD BREAK ELBOWS WITH CAPACITIVE TEST POINTS FOR EACH WAY. PROVIDE FAULT INDICATOR FOR EACH WAY. PROVIDE TRI-PHASE CONTROLLERS AS SHOWN.
8. CUT EXISTING 15KV (350KCMIL) CIRCUIT IN MANHOLE AND PROVIDE WATERTIGHT TAPE T-SPLICES ONTO EXISTING FEEDER.
9. USCG TO OWN DISTRIBUTION FROM 15KV SWITCHES DOWNSTREAM TO PIER. NAVFAC TO OWN DISTRIBUTION UPSTREAM FROM 15KV SWITCHES.
10. CROSS-TIE FEEDER MUST BE OPEN WHEN SWITCHES UTILIZE FEEDERS 1W8 AND 1W9 FOR PIER 1.



A1 PRIMARY & DEMOLITION ONE LINE DIAGRAM

NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS DAVID P. FERGUSON ELECTRICAL No. 52642 REGISTERED PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PRIMARY & DEMOLITION ONE LINE DIAGRAM

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936795

SHEET 226 OF 247

E-601

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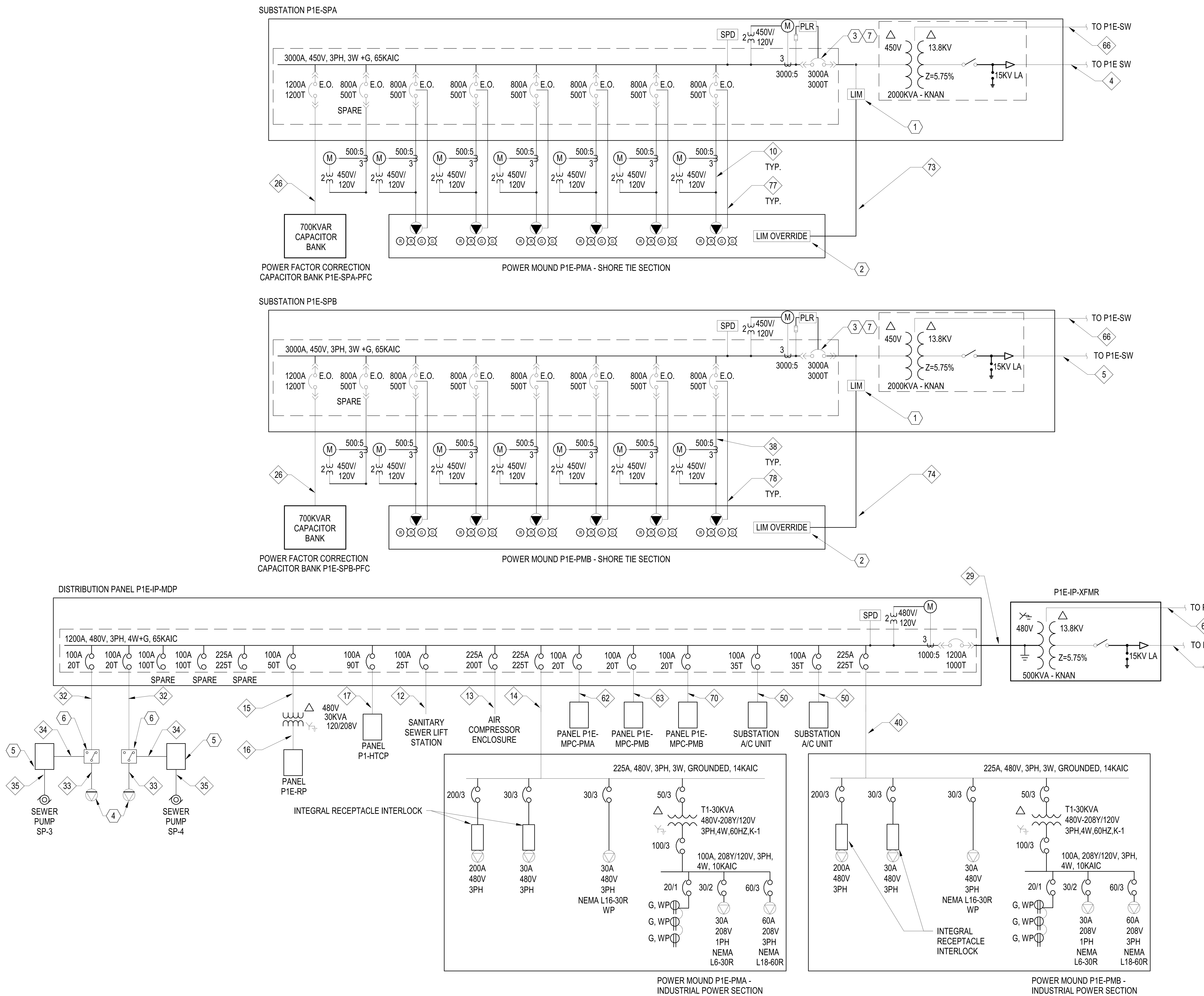

A1 **PIER ONE LINE DIAGRAM - EAST**

NOT TO SCALE

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615. FOR PRIMARY ONE LINE DIAGRAM SEE DRAWING E-601.
2. FOR METERING SYSTEM INFORMATION SEE DRAWINGS E-505 AND E-506.

SHEET KEYNOTES:

1. FOR LINE INSULATION MONITOR SYSTEM DETAIL SEE SHEET E-512.
2. FOR LINE INSULATION MONITOR OVERRIDE DETAIL SEE SHEET E-504.
3. SUBSTATION MAIN CIRCUIT BREAKER MUST TRIP UPON TRIPPING OF ANY DOWNSTREAM SHORE POWER FEEDER CIRCUIT BREAKER.
4. PROVIDE 30A, 480V, 3PH, 3P, 4W, MARINE GRADE PORTABLE EQUIPMENT RECEPTACLE.
5. SEWER PUMP CONTROLLER. SEE MECHANICAL DRAWINGS FOR ADDITIONAL INFORMATION.
6. PROVIDE 30A, 480V, 3PH, 3P, NEMA 4X MANUAL TRANSFER SWITCH.
7. SEE SPECIFICATIONS FOR MAIN CIRCUIT BREAKER PHASE LOSS PROTECTION REQUIREMENTS.

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS DAVID P FERGUSON ELECTRICAL No. 52642 PROFESSIONAL ENGINEER

SEAL

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A/E INFO

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FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
CI CORE - HAMPTON ROADS
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER ONE LINE DIAGRAM - EAST

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936796
SHEET 227 OF 247

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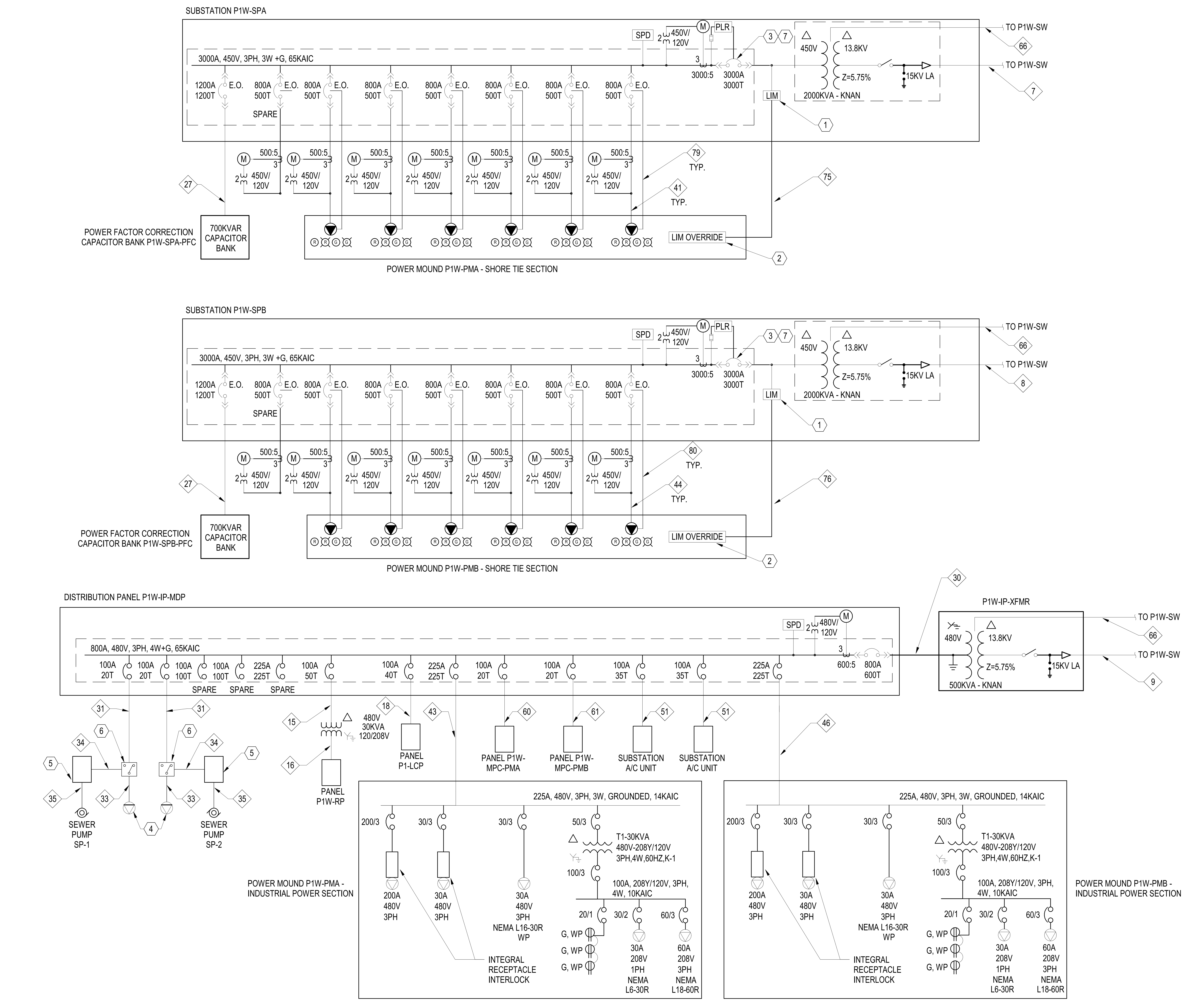
E-602



A1 **PIER ONE LINE DIAGRAM - WEST**

NOT TO SCALE

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615. FOR PRIMARY ONE LINE DIAGRAM SEE DRAWING E-601.
2. FOR METERING SYSTEM INFORMATION SEE DRAWINGS E-505 AND E-507.

SHEET KEYNOTES:

1. FOR LINE INSULATION MONITOR SYSTEM DETAIL SEE SHEET E-512.
2. FOR LINE INSULATION MONITOR OVERRIDE DETAIL SEE SHEET E-504.
3. SUBSTATION MAIN CIRCUIT BREAKER MUST TRIP UPON TRIPPING OF ANY DOWNSTREAM SHORE POWER FEEDER CIRCUIT BREAKER.
4. PROVIDE 30A, 480V, 3PH, 3P, 4W, MARINE GRADE PORTABLE EQUIPMENT RECEPTACLE.
5. SEWER PUMP CONTROLLER. SEE MECHANICAL DRAWINGS FOR ADDITIONAL INFORMATION.
6. PROVIDE 30A, 480V, 3PH, 3P, NEMA 4X MANUAL TRANSFER SWITCH.
7. SEE SPECIFICATIONS FOR MAIN CIRCUIT BREAKER PHASE LOSS PROTECTION REQUIREMENTS.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS DAVID P FERGUSON ELECTRICAL No. 52642 PROFESSIONAL ENGINEER

SEAL

Stantec

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC

CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

PIER ONE LINE DIAGRAM - WEST

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936797

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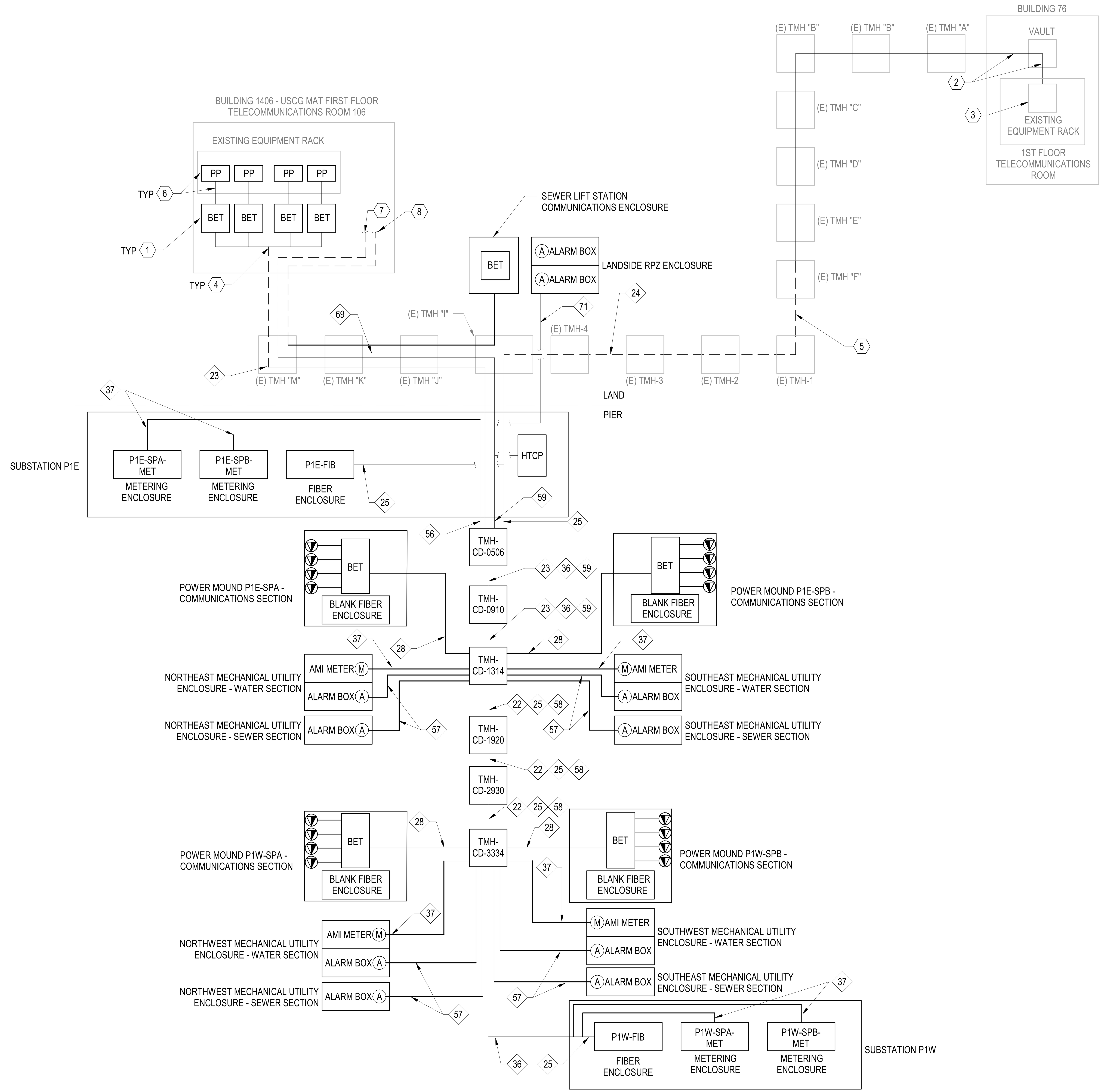

A1 COMMUNICATIONS ONE LINE DIAGRAM

NOT TO SCALE

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR FEEDER SCHEDULES SEE DRAWINGS E-614 AND E-615.

SHEET KEYNOTES:

1. PROVIDE WALL MOUNTED TELEPHONE ENTRANCE ENCLOSURE FOR PHONE CABLES FROM P1E-PMA, P1E-PMB, P1W-PMA, AND P1W-PMB. MATCH EXISTING ENCLOSURE TYPE AND INTERNAL COMPONENTS AS EXISTING ENCLOSURES SERVING FEEDERS TO IDA LEWIS, WILLOW, JUNIPER AND TIGER SHARK. MOUNT ENCLOSURES ON SAME WALL AS EXISTING ENTRANCE ENCLOSURES. PROVIDE 1#6G BONDING JUMPER FROM ENCLOSURE TO EXISTING GROUND BAR IN ROOM 106.
2. WITHIN VAULT ROUTE CABLE ON EXISTING RACK ARMS AND UTILIZE EXISTING CONDUIT PENETRATIONS TO EXTEND CABLE UP INTO FIRST FLOOR TELECOMMUNICATIONS ROOM.
3. COORDINATE WITH NAVSTA NEWPORT TO TERMINATE FIBER OPTIC CABLING ON EXISTING FIBER RACK. COORDINATE WITH NAVSTA NEWPORT FOR RACK LOCATION AND RACEWAY ROUTING THROUGH BUILDING.
4. EXISTING 10-WAY DUCTBANK FROM TMH "M" TURNS UP IN ROOM 106. EXTEND (2) 25 PAIR CAT5E CABLE FROM DUCT(S) TO EACH ENCLOSURE.
5. FROM TMH "I" TO TMH "F" ROUTE COMMUNICATIONS CABLING THROUGH EXISTING SPARE DUCTS.
6. PROVIDE PATCH PANEL ON EXISTING EQUIPMENT RACK. PROVIDE (1) 25 PAIR CAT 5E CABLE FROM ENTRANCE ENCLOSURE TO PATCH PANEL, UTILIZING EXISTING CABLE TRAY. COORDINATE PATCH PANEL REQUIREMENTS WITH USCG.
7. EXISTING 10-WAY DUCTBANK FROM TMH "M" TURNS UP IN ROOM 106. COORDINATE WITH USCG FOR LOCATION IN BUILDING 1406 TO TERMINATE ALARM WIRING FROM PIER MECHANICAL ENCLOSURE ALARM BOXES AND HEAT TRACE CONTROL PANEL. EXTEND WIRING IN 2"C WITHIN BUILDING TO TERMINATION LOCATION.
8. EXISTING 10-WAY DUCTBANK FROM TMH "M" TURNS UP IN ROOM 106. COORDINATE WITH USCG FOR LOCATION IN BUILDING 1406 TO TERMINATE ALARM WIRING FROM LIFT STATION COMMUNICATIONS ENCLOSURE. EXTEND WIRING IN 2"C WITHIN BUILDING TO TERMINATION LOCATION.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

DAVID P FERGUSON ELECTRICAL No. 52642 COMMONWEALTH OF MASSACHUSETTS PROFESSIONAL ENGINEER

Stantec

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES	DPF	DRW	GLV	CHK	RJW
PM/DM	MRI/CWJ				
BRANCH MANAGER	WRB				
CHIEF ENG/ARCH	JWJ				
FIRE PROTECTION	HPN				

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
COMMUNICATIONS ONE LINE DIAGRAM

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936798
SHEET 229 OF 247
E-604

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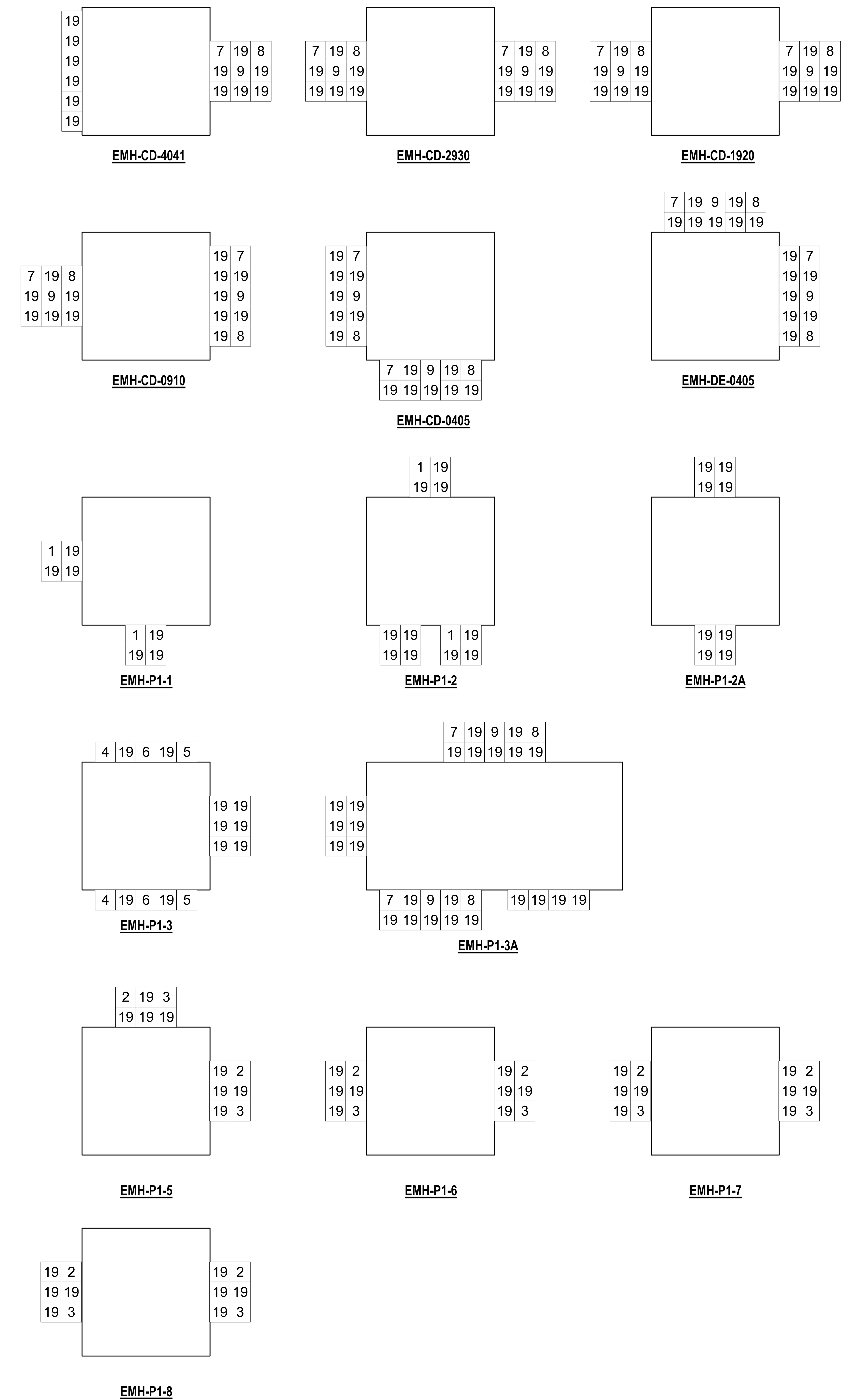
MANHOLE SCHEDULE

MANHOLE ID	INTERIOR DIMENSIONS	RISER DEPTH	TYPE PER DETAIL B3/E-501	LOCATION	REMARKS
EMH-BC-4142	18'-0"L x 11'-6"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS B&C, 41&42	POWER - SHORE AND INDUSTRIAL POWER
EMH-CD-4041	11'-6"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS C&D, 40&41	POWER - MEDIUM VOLTAGE
TMH-CD-3334	18'-0"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS C&D, 33&34	COMMUNICATIONS
EMH-BC-3334	11'-6"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS B&C, 33&34	POWER - SHORE AND INDUSTRIAL POWER
EMH-DE-3334	11'-6"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS D&E, 33&34	POWER - SHORE AND INDUSTRIAL POWER
EMH-CD-2930	18'-0"L x 4'-5"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS C&D, 29&30	POWER - MEDIUM VOLTAGE
TMH-CD-2930	18'-0"L x 4'-5"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS C&D, 29&30	COMMUNICATIONS
EMH-CD-1920	18'-0"L x 4'-5"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS C&D, 19&20	POWER - MEDIUM VOLTAGE
TMH-CD-1920	18'-0"L x 4'-5"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS C&D, 19&20	COMMUNICATIONS
TMH-CD-1314	18'-0"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS C&D, 13&14	COMMUNICATIONS
EMH-BC-1314	11'-6"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS B&C, 13&14	POWER - SHORE AND INDUSTRIAL POWER
EMH-DE-1314	11'-6"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS D&E, 13&14	POWER - SHORE AND INDUSTRIAL POWER
EMH-BC-0910	11'-6"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS B&C, 9&10	POWER - SHORE AND INDUSTRIAL POWER
EMH-DE-0910	11'-6"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS D&E, 9&10	POWER - SHORE AND INDUSTRIAL POWER
EMH-CD-0910	18'-0"L x 4'-5"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS C&D, 9&10	POWER - MEDIUM VOLTAGE
TMH-CD-0910	18'-0"L x 4'-5"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS C&D, 9&10	COMMUNICATIONS
EMH-DE-0607	18'-0"L x 11'-6"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS D&E, 6&7	POWER - SHORE AND INDUSTRIAL POWER
EMH-EF-0607	8'-9"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS E&F, 6&7	POWER - SHORE AND INDUSTRIAL POWER
TMH-CD-0506	18'-0"L x 5'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS C&D, 5&6	COMMUNICATIONS
EMH-CD-0405	11'-6"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS C&D, 4&5	POWER - MEDIUM VOLTAGE
EMH-DE-0405	11'-6"L x 6'-0"W x 7'-6"D	2'-5"	N/A	PIER - BETWEEN BENTS D&E, 4&5	POWER - MEDIUM VOLTAGE
EMH-P1-1	6'-0"L x 6'-0"W x 6'-6"D	N/A	3	SUBSTATION 1	POWER - MEDIUM VOLTAGE
EMH-P1-2	6'-0"L x 6'-0"W x 6'-6"D	N/A	3	LANDSIDE - BURMA ROAD	POWER - MEDIUM VOLTAGE
EMH-P1-2A	6'-0"L x 6'-0"W x 6'-6"D	N/A	3	LANDSIDE - PIER 1 LANDING	POWER - MEDIUM VOLTAGE
EMH-P1-3	6'-0"L x 6'-0"W x 6'-6"D	N/A	3	LANDSIDE - PIER 1 LANDING	POWER - MEDIUM VOLTAGE
EMH-P1-3A	8'-0"L x 8'-0"W x 6'-6"D	N/A	4A	LANDSIDE - PIER 1 LANDING	POWER - MEDIUM VOLTAGE
EMH-P1-4	6'-0"L x 6'-0"W x 6'-6"D	N/A	3	LANDSIDE - PIER 1 LANDING	INDUSTRIAL POWER
EMH-P1-5	6'-0"L x 6'-0"W x 6'-6"D	N/A	4A	LANDSIDE - PIER 2 LANDING	POWER - MEDIUM VOLTAGE
EMH-P1-6	6'-0"L x 6'-0"W x 6'-6"D	N/A	3	LANDSIDE - QUAY WALL	POWER - MEDIUM VOLTAGE
EMH-P1-7	6'-0"L x 6'-0"W x 6'-6"D	N/A	3	LANDSIDE - QUAY WALL	POWER - MEDIUM VOLTAGE
EMH-P1-8	6'-0"L x 6'-0"W x 6'-6"D	N/A	4A	LANDSIDE - BUOY YARD	POWER - MEDIUM VOLTAGE



A3 MANHOLE FOLDOUT DIAGRAMS

NOT TO SCALE

DAVID P FERGUSON ELECTRICAL No. 52642 COMMONWEALTH OF MASSACHUSETTS PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

James E. Donahue, R.A.

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV	CHK RJW

PM/DM	MRI/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT
NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

MANHOLE SCHEDULE & FOLDOUT DIAGRAMS

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936799

SHEET 230 OF 247

E-605

DRAWFORM REVISION: 25 AUGUST 2020

SYM	DESCRIPTION	DATE	APPR

3/2/2026 5:14:16 PM
E-605
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt

Stantec — Panel

Name:	**P1E-MPC-PMA**	Volts:	120/240V	Mains Type:	MCB	Type:	
Location:	PIER NEAR NE SHORE TIES	Phases:	1	Mains Rating:	25 A	AIC Rating:	25KAIC
Supply From:	P1E-IP-MDP	Wires:	3	MCB Rating:	25 A	Mounting:	Surface
Serves:				Lugs:	Single Lugs	Enclosure:	NEMA 4X

CKT	Circuit Description	Conduit & Wire Size	Trip	Poles	A		B		Poles	Trip	Conduit & Wire Size	Circuit Description	CKT
1	SEWER ENCL ALARM & HEATER	3/4"C, 2#12, 1#12G	20 A	1	1680	0			1	20 A	--	SPARE	2
3	WATER ENCL ALARM & HEATER	3/4"C, 2#12, 1#12G	20 A	1			1680	0	1	20 A	--	SPARE	4
5	SPARE	--	20 A	1	0	0			1	20 A	--	SPARE	6
7	SPARE	--	20 A	1			0	0	1	20 A	--	SPARE	8
9	SPARE	--	20 A	1	0	0			1	20 A	--	SPARE	10
11	SPARE	--	20 A	1			0	0	1	20 A	--	SPARE	12
13	SPACE	--	--	1	--	--			1	--	--	SPACE	14
15	SPACE	--	--	1			--	--	1	--	--	SPACE	16
17	SPACE	--	--	1	--	--			1	--	--	SPACE	18
				Total Load:	**1.68 kVA**		**1.68 kVA**						
				Total Amps:	**14 A**		**14 A**						

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Power	3360 VA	100.00%	3360 VA		
				Total Conn. Load:	**3360 VA**
				Total Est. Demand:	**3360 VA**
				Total Conn.:	**14 A**
				Total Est. Demand:	**14 A**

CB Legend (blank = circuit breaker):
G = GFCI S = Shunt Trip D = Switching Duty A = AFCI H = HID Rated C = HACR Rated † = Existing Circuit ‡ = Revised Circuit

Notes:
PANEL INTEGRAL TO 5KVA, 480V TO 120/240V, 1PH MINI POWER CENTER (MPC). MPC MUST BE BOLT-ON TYPE WITH COPPER WOUND TRANSFORMER AND COPPER BUSSING. MPC TO BE UL 1062 LISTED. MPC TRANSFORMER TO HAVE 180 DEGREES C INSULATION WITH 115 DEGREES C RISE. MAIN AND BRANCH CIRCUIT BREAKERS MUST BE BOLT-ON THERMAL MAGNETIC TYPE AND UL 489 LISTED.

Stantec — Panel

Name:	**P1W-MPC-PMA**	Volts:	120/240V	Mains Type:	MCB	Type:	
Location:	PIER NEAR NW SHORE TIES	Phases:	1	Mains Rating:	25 A	AIC Rating:	25KAIC
Supply From:	P1W-IP-MDP	Wires:	3	MCB Rating:	25 A	Mounting:	Surface
Serves:				Lugs:	Single Lugs	Enclosure:	NEMA 4X

CKT	Circuit Description	Conduit & Wire Size	Trip	Poles	A		B		Poles	Trip	Conduit & Wire Size	Circuit Description	CKT
1	SEWER ENCL ALARM & HEATER	3/4"C, 2#12, 1#12G	20 A	1	1680	0			1	20 A	--	SPARE	2
3	WATER ENCL ALARM & HEATER	3/4"C, 2#12, 1#12G	20 A	1			1680	0	1	20 A	--	SPARE	4
5	SPARE	--	20 A	1	0	0			1	20 A	--	SPARE	6
7	SPARE	--	20 A	1			0	0	1	20 A	--	SPARE	8
9	SPARE	--	20 A	1	0	0			1	20 A	--	SPARE	10
11	SPARE	--	20 A	1			0	0	1	20 A	--	SPARE	12
13	SPACE	--	--	1	--	--			1	--	--	SPACE	14
15	SPACE	--	--	1			--	--	1	--	--	SPACE	16
17	SPACE	--	--	1	--	--			1	--	--	SPACE	18
				Total Load:	**1.68 kVA**		**1.68 kVA**						
				Total Amps:	**14 A**		**14 A**						

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Power	3360 VA	100.00%	3360 VA		
				Total Conn. Load:	**3360 VA**
				Total Est. Demand:	**3360 VA**
				Total Conn.:	**14 A**
				Total Est. Demand:	**14 A**

CB Legend (blank = circuit breaker):
G = GFCI S = Shunt Trip D = Switching Duty A = AFCI H = HID Rated C = HACR Rated † = Existing Circuit ‡ = Revised Circuit

Notes:
PANEL INTEGRAL TO 5KVA, 480V TO 120/240V, 1PH MINI POWER CENTER (MPC). MPC MUST BE BOLT-ON TYPE WITH COPPER WOUND TRANSFORMER AND COPPER BUSSING. MPC TO BE UL 1062 LISTED. MPC TRANSFORMER TO HAVE 180 DEGREES C INSULATION WITH 115 DEGREES C RISE. MAIN AND BRANCH CIRCUIT BREAKERS MUST BE BOLT-ON THERMAL MAGNETIC TYPE AND UL 489 LISTED.

Stantec — Panel

Name:	**P1E-MPC-PMB**	Volts:	120/240V	Mains Type:	MCB	Type:	
Location:	PIER NEAR SE SHORE TIES	Phases:	1	Mains Rating:	25 A	AIC Rating:	25KAIC
Supply From:	P1E-IP-MDP	Wires:	3	MCB Rating:	25 A	Mounting:	Surface
Serves:				Lugs:	Single Lugs	Enclosure:	NEMA 4X

CKT	Circuit Description	Conduit & Wire Size	Trip	Poles	A		B		Poles	Trip	Conduit & Wire Size	Circuit Description	CKT
1	SEWER ENCL ALARM & HEATER	3/4"C, 2#12, 1#12G	20 A	1	1680	0			1	20 A	--	SPARE	2
3	WATER ENCL ALARM & HEATER	3/4"C, 2#12, 1#12G	20 A	1			1680	0	1	20 A	--	SPARE	4
5	SPARE	--	20 A	1	0	0			1	20 A	--	SPARE	6
7	SPARE	--	20 A	1			0	0	1	20 A	--	SPARE	8
9	SPARE	--	20 A	1	0	0			1	20 A	--	SPARE	10
11	SPARE	--	20 A	1			0	0	1	20 A	--	SPARE	12
13	SPACE	--	--	1	--	--			1	--	--	SPACE	14
15	SPACE	--	--	1			--	--	1	--	--	SPACE	16
17	SPACE	--	--	1	--	--			1	--	--	SPACE	18
				Total Load:	**1.68 kVA**		**1.68 kVA**						
				Total Amps:	**14 A**		**14 A**						

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Power	3360 VA	100.00%	3360 VA		
				Total Conn. Load:	**3360 VA**
				Total Est. Demand:	**3360 VA**
				Total Conn.:	**14 A**
				Total Est. Demand:	**14 A**

CB Legend (blank = circuit breaker):
G = GFCI S = Shunt Trip D = Switching Duty A = AFCI H = HID Rated C = HACR Rated † = Existing Circuit ‡ = Revised Circuit

Notes:
PANEL INTEGRAL TO 5KVA, 480V TO 120/240V, 1PH MINI POWER CENTER (MPC). MPC MUST BE BOLT-ON TYPE WITH COPPER WOUND TRANSFORMER AND COPPER BUSSING. MPC TO BE UL 1062 LISTED. MPC TRANSFORMER TO HAVE 180 DEGREES C INSULATION WITH 115 DEGREES C RISE. MAIN AND BRANCH CIRCUIT BREAKERS MUST BE BOLT-ON THERMAL MAGNETIC TYPE AND UL 489 LISTED.

Stantec — Panel

Name:	**P1W-MPC-PMB**	Volts:	120/240V	Mains Type:	MCB	Type:	
Location:	PIER NEAR SW SHORE TIES	Phases:	1	Mains Rating:	25 A	AIC Rating:	25KAIC
Supply From:	P1W-IP-MDP	Wires:	3	MCB Rating:	25 A	Mounting:	Surface
Serves:				Lugs:	Single Lugs	Enclosure:	NEMA 4X

CKT	Circuit Description	Conduit & Wire Size	Trip	Poles	A		B		Poles	Trip	Conduit & Wire Size	Circuit Description	CKT
1	SEWER ENCL ALARM & HEATER	3/4"C, 2#12, 1#12G	20 A	1	1680	0			1	20 A	--	SPARE	2
3	WATER ENCL ALARM & HEATER	3/4"C, 2#12, 1#12G	20 A	1			1680	0	1	20 A	--	SPARE	4
5	SPARE	--	20 A	1	0	0			1	20 A	--	SPARE	6
7	SPARE	--	20 A	1			0	0	1	20 A	--	SPARE	8
9	SPARE	--	20 A	1	0	0			1	20 A	--	SPARE	10
11	SPARE	--	20 A	1			0	0	1	20 A	--	SPARE	12
13	SPACE	--	--	1	--	--			1	--	--	SPACE	14
15	SPACE	--	--	1			--	--	1	--	--	SPACE	16
17	SPACE	--	--	1	--	--			1	--	--	SPACE	18
				Total Load:	**1.68 kVA**		**1.68 kVA**						
				Total Amps:	**14 A**		**14 A**						

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Power	3360 VA	100.00%	3360 VA		
				Total Conn. Load:	**3360 VA**
				Total Est. Demand:	**3360 VA**
				Total Conn.:	**14 A**
				Total Est. Demand:	**14 A**

CB Legend (blank = circuit breaker):
G = GFCI S = Shunt Trip D = Switching Duty A = AFCI H = HID Rated C = HACR Rated † = Existing Circuit ‡ = Revised Circuit

Notes:
PANEL INTEGRAL TO 5KVA, 480V TO 120/240V, 1PH MINI POWER CENTER (MPC). MPC MUST BE BOLT-ON TYPE WITH COPPER WOUND TRANSFORMER AND COPPER BUSSING. MPC TO BE UL 1062 LISTED. MPC TRANSFORMER TO HAVE 180 DEGREES C INSULATION WITH 115 DEGREES C RISE. MAIN AND BRANCH CIRCUIT BREAKERS MUST BE BOLT-ON THERMAL MAGNETIC TYPE AND UL 489 LISTED.

Stantec — Panel

Name:	**P1E-MPC-RPZ**	Volts:	120/240V	Mains Type:	MCB	Type:	
Location:	LANDSIDE NEAR RPZ ENCLOSURE	Phases:	1	Mains Rating:	25 A	AIC Rating:	25KAIC
Supply From:	P1E-IP-MDP	Wires:	3	MCB Rating:	25 A	Mounting:	Surface
Serves:				Lugs:	Single Lugs	Enclosure:	NEMA 4X

CKT	Circuit Description	Conduit & Wire Size	Trip	Poles	A		B		Poles	Trip	Conduit & Wire Size	Circuit Description	CKT
1	RPZ ENCL HEATER & ALARM	3/4"C, 2#12, 1#12G	20 A	1	1680	0			1	20 A	--	SPARE	2
3	RPZ ENCL HEATER & ALARM	3/4"C, 2#12, 1#12G	20 A	1			1680	0	1	20 A	--	SPARE	4
5	SEWER COMMS ENCLOSURE	3/4"C, 2#12, 1#12G	20 A	1	180	0			1	20 A	--	SPARE	6
7	SPARE	--	20 A	1			0	0	1	20 A	--	SPARE	8
9	SPARE	--	20 A	1	0	0			1	20 A	--	SPARE	10
11	SPARE	--	20 A	1			0	0	1	20 A	--	SPARE	12
13	SPACE	--	--	1	--	--			1	--	--	SPACE	14
15	SPACE	--	--	1			--	--	1	--	--	SPACE	16
17	SPACE	--	--	1	--	--			1	--	--	SPACE	18
				Total Load:	**1.86 kVA**		**1.68 kVA**						
				Total Amps:	**16 A**		**14 A**						

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Power	3540 VA	100.00%	3540 VA		
				Total Conn. Load:	**3540 VA**
				Total Est. Demand:	**3540 VA**
				Total Conn.:	**15 A**
				Total Est. Demand:	**15 A**

CB Legend (blank = circuit breaker):
G = GFCI S = Shunt Trip D = Switching Duty A = AFCI H = HID Rated C = HACR Rated † = Existing Circuit ‡ = Revised Circuit

Notes:
PANEL INTEGRAL TO 5KVA, 480V TO 120/240V, 1PH MINI POWER CENTER (MPC). MPC MUST BE BOLT-ON TYPE WITH COPPER WOUND TRANSFORMER AND COPPER BUSSING. MPC TO BE UL 1062 LISTED. MPC TRANSFORMER TO HAVE 180 DEGREES C INSULATION WITH 115 DEGREES C RISE. MAIN AND BRANCH CIRCUIT BREAKERS MUST BE BOLT-ON THERMAL MAGNETIC TYPE AND UL 489 LISTED. ABOVE GROUND CONDUITS SERVING MPC MUST BE PVC COATED RGS CONDUIT.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED
James E. Donahue, R.A.
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV	CHK RJW
PM/DM		MRI/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER

MINI POWER CENTER PANEL SCHEDULES

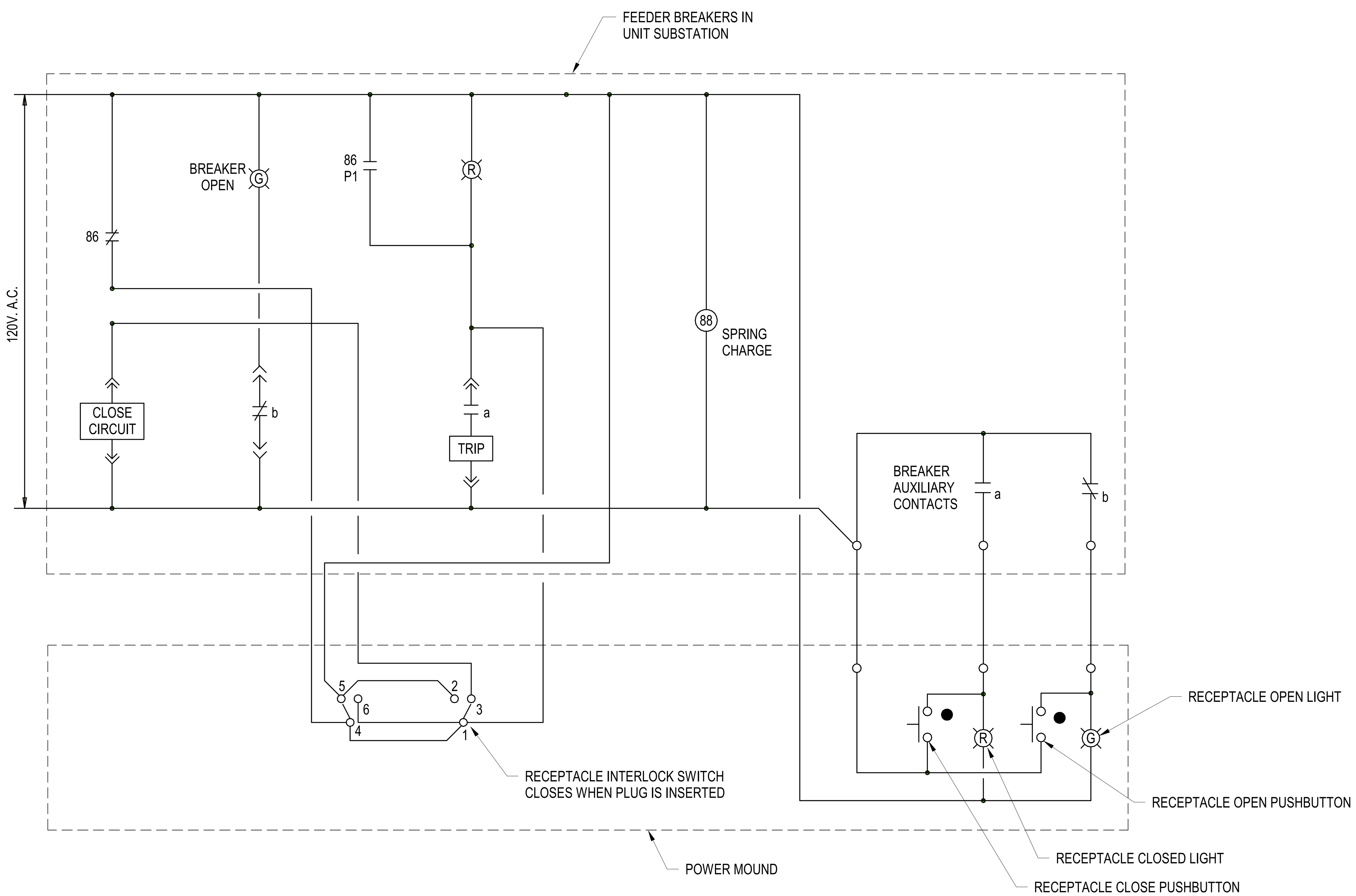
SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936800
SHEET 231 OF 247

E-606

DRAWFORM REVISION: 25 AUGUST 2020

CONTROL SYSTEM NOTES:

1. SHORE POWER RECEPTACLE INTEGRAL INTERLOCK MUST BLOCK THE INSERTION AND REMOVAL OF THE PLUG WHEN ENERGIZED AND WHEN THE SOURCE CIRCUIT BREAKER IS CLOSED.
2. SHORE POWER RECEPTACLE INTEGRAL INTERLOCK MUST DE-ENERGIZE THE RECEPTACLE WHEN THE COVER IS OPEN WITHOUT THE PLUG INSTALLED.



B1 CONTROL - POWER MOUND SHORE POWER RECEPTACLE FEEDER CIRCUIT BREAKER

NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
David Ferguson
SEAL

Stantec
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES DPF DRW GLV CHK RJW
PM/DM MRI/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
SWITCHGEAR BREAKER CONTROL DIAGRAM

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936801
SHEET 232 OF 247

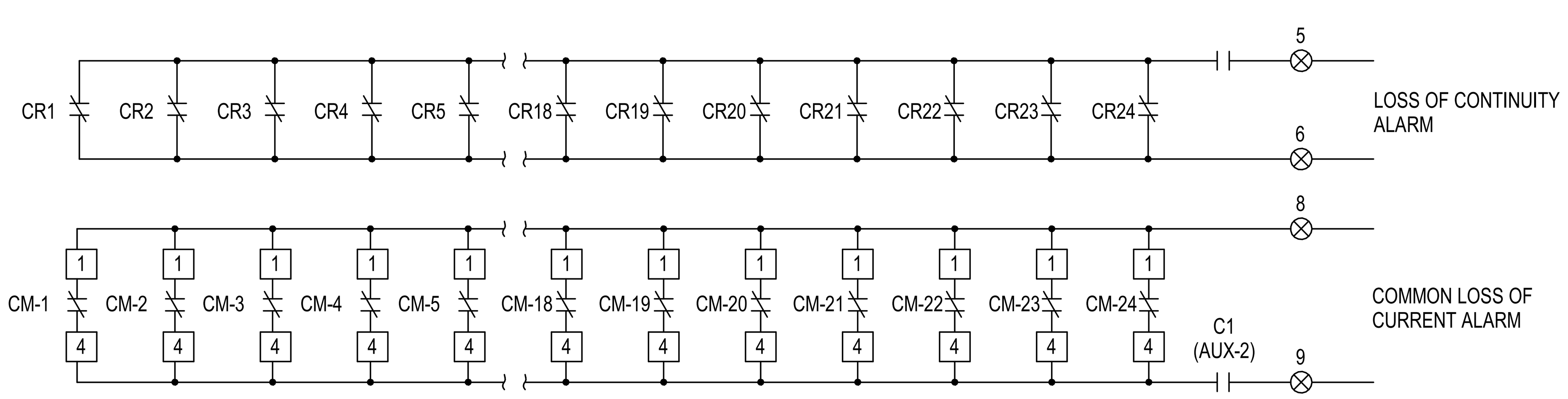
E-607

HEAT TRACE CABLE SCHEDULE

CKT. NO.	PIPE ID	SYMBOL	CKT. NO.	WATTS/FT	ESTIMATED CKT LENGTH	CKT. BKR	CKT. LOAD WATTS	VOLTAGE	WIRE + GND + CONTROLS	NOTES
1	SSFM - B12-0 - NORTH	SSFM	1	5	400'	30A	2000	277	2#8 + 1#8 + 4#12	9
2	FW - B28-19 - SOUTH	FW	2	8	330'	30A	2640	277	2#6 + 1#6 + 4#12	2
3	SSFM - B12-0 - SOUTH	SSFM	3	5	400'	30A	2000	277	2#8 + 1#8 + 4#12	10
4	FW - B43-34 - NORTH	FW	4	8	370'	30A	2960	277	2#3 + 1#3 + 4#12	5, 8
5	SSFM - B24-12 - NORTH	SSFM	5	5	465'	30A	2325	277	2#8 + 1#8 + 4#12	4, 9
6	FW - B37-28 - SOUTH	FW	6	8	335'	30A	2680	277	2#4 + 1#4 + 4#12	2, 6
7	SSFM - B24-12 - SOUTH	SSFM	7	5	465'	30A	2325	277	2#8 + 1#8 + 4#12	4, 10
8	FW - B43-37 - SOUTH	FW	8	8	265'	30A	2120	277	2#4 + 1#4 + 4#12	2. 5
9	SSFM - B35-24 - NORTH	SSFM	9	5	390'	30A	1950	277	2#6 + 1#6 + 4#12	4, 9
10	SPARE		10			30A		277		
11	SSFM - B35-24 - SOUTH	SSFM	11	5	390'	30A	1950	277	2#6 + 1#6 + 4#12	4, 10
12	SPARE		12			30A		277		
13	SSFM - CA ENCLOSURE RISER	SSFM	13	5	30'	30A	150	277	2#10 + 1#10 + 4#12	3
14	SPARE		14			30A		277		
15	FW - B14-8 - NORTH	FW	15	8	375'	30A	3000	277	2#6 + 1#6 + 4#12	1, 6, 7
16	SPARE		16			30A		277		
17	FW - B11-0 - SOUTH	FW	17	8	370'	30A	2960	277	2#8 + 1#8 + 4#12	2
18	SPARE		18			30A		277		
19	FW - B24-14 - NORTH	FW	19	8	370'	30A	2960	277	2#6 + 1#6 + 4#12	1
20	SPARE		20			30A		277		
21	FW - B19-11 - SOUTH	FW	21	8	320'	30A	2560	277	2#8 + 1#8 + 4#12	2, 6
22	SPARE		22			30A		277		
23	FW - B34-24 - NORTH	FW	23	8	385'	30A	3080	277	2#4 + 1#4 + 4#12	1, 6
24	SPARE		24			30A		277		

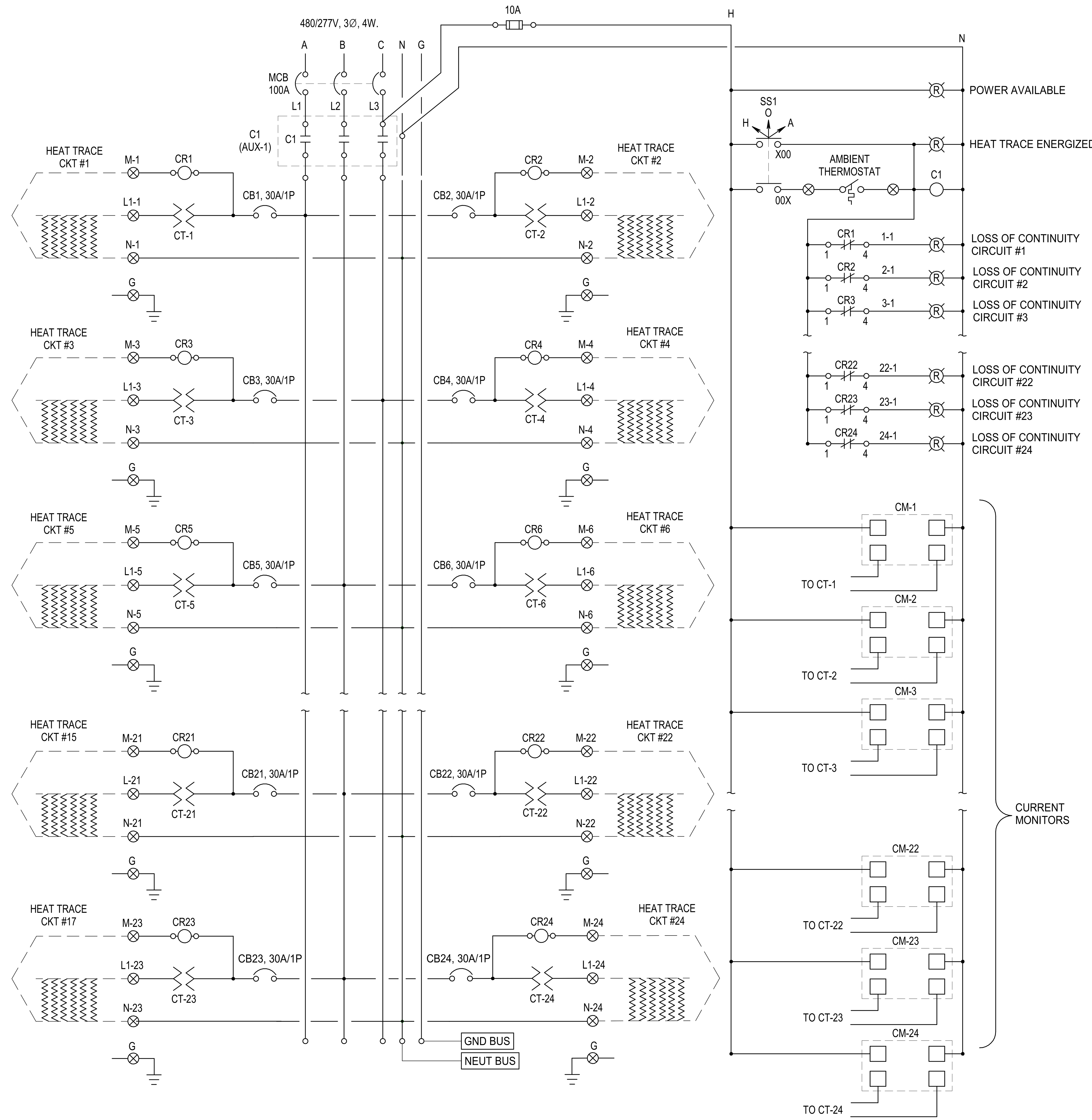
SCHEDULE NOTES:

1. ROUTE CABLING IN FEEDER 21 DUCT TO EMH-BC-1314. EXTEND CABLING IN 1-1/4"C FROM EMH-BC-1314 TO NORTHERN UTILITY TRENCH AND WITHIN UTILITY TRENCH ALONG OUTBOARD WALL TO HEAT TRACE CONNECTION KIT.
2. ROUTE CABLING IN FEEDER 48 DUCT TO EMH-DE-1314. EXTEND CABLING IN 1-1/4"C FROM EMH-DE-1314 TO SOUTHERN UTILITY TRENCH AND WITHIN UTILITY TRENCH ALONG OUTBOARD WALL TO HEAT TRACE CONNECTION KIT.
3. ROUTE CABLING IN FEEDER 48 DUCT TO EMH-DE-0607. EXTEND CABLING IN FEEDER 49 DUCT TO SEWER PIPING RISER AT COMPRESSED AIR ENCLOSURE.
4. CIRCUIT INCLUDES PIPING RUN TO SEWER UTILITY ENCLOSURE
5. CIRCUIT INCLUDES TOPSIDE PIPING AT BENT 43 FROM UTILITY TRENCH TO DECK CENTERLINE.
6. CIRCUIT INCLUDES PIPING RUN TO FRESH WATER UTILITY ENCLOSURE
7. CIRCUIT INCLUDES PIPING LATERAL TRENCH BETWEEN BENTS 8&9
8. ROUTE CABLING IN FEEDER 21 DUCT TO EMH-BC-1314. EXTEND CABLING IN 1-1/2"C FROM EMH-BC-1314 TO NORTHERN UTILITY TRENCH AND WITHIN UTILITY TRENCH ALONG OUTBOARD WALL TO HEAT TRACE CONNECTION KIT.
9. ROUTE CABLING IN FEEDER 21 DUCT TO EMH-BC-1314. EXTEND CABLING IN 1-1/4"C FROM EMH-BC-1314 TO NORTHERN OUTBOARD FACE OF PIER AND ALONG OUTBOARD FACE TO HEAT TRACE CONNECTION KIT.
10. ROUTE CABLING IN FEEDER 48 DUCT TO EMH-DE-1314. EXTEND CABLING IN 1-1/4"C FROM EMH-DE-1314 TO SOUTHERN OUTBOARD FACE OF PIER AND ALONG OUTBOARD FACE TO HEAT TRACE CONNECTION KIT.



A1 HEAT TRACE ELEMENTARY INTERCONNECTION DIAGRAM

NOT TO SCALE



GENERAL SHEET NOTES:

1. FOR LEGEND AND ABBREVIATIONS, SEE DRAWING E-001.
2. FOR MECHANICAL INFORMATION, SEE DRAWINGS M-101 - M-107.
3. FOR HEAT TRACE CABINET DETAILS, EQUIPMENT AND NP SCHEDULE, SEE DRAWING E-514.
4. HEAT TRACE CONTROLLER MUST PROVIDE OUTPUT ALARM SIGNAL FOR FAILURE OF HEAT TRACING SYSTEM.

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
REGISTERED PROFESSIONAL ENGINEER
David Ferguson

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF | DRW GLV | CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

HEAT TRACE ONE LINE DIAGRAM & SCHEDULES

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936802

SHEET 233 OF 247

E-608

DRAWFORM REVISION: 26 AUGUST 2020

3/2/2026 5:14:21 PM
E-608
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt

GENERAL SHEET NOTES:

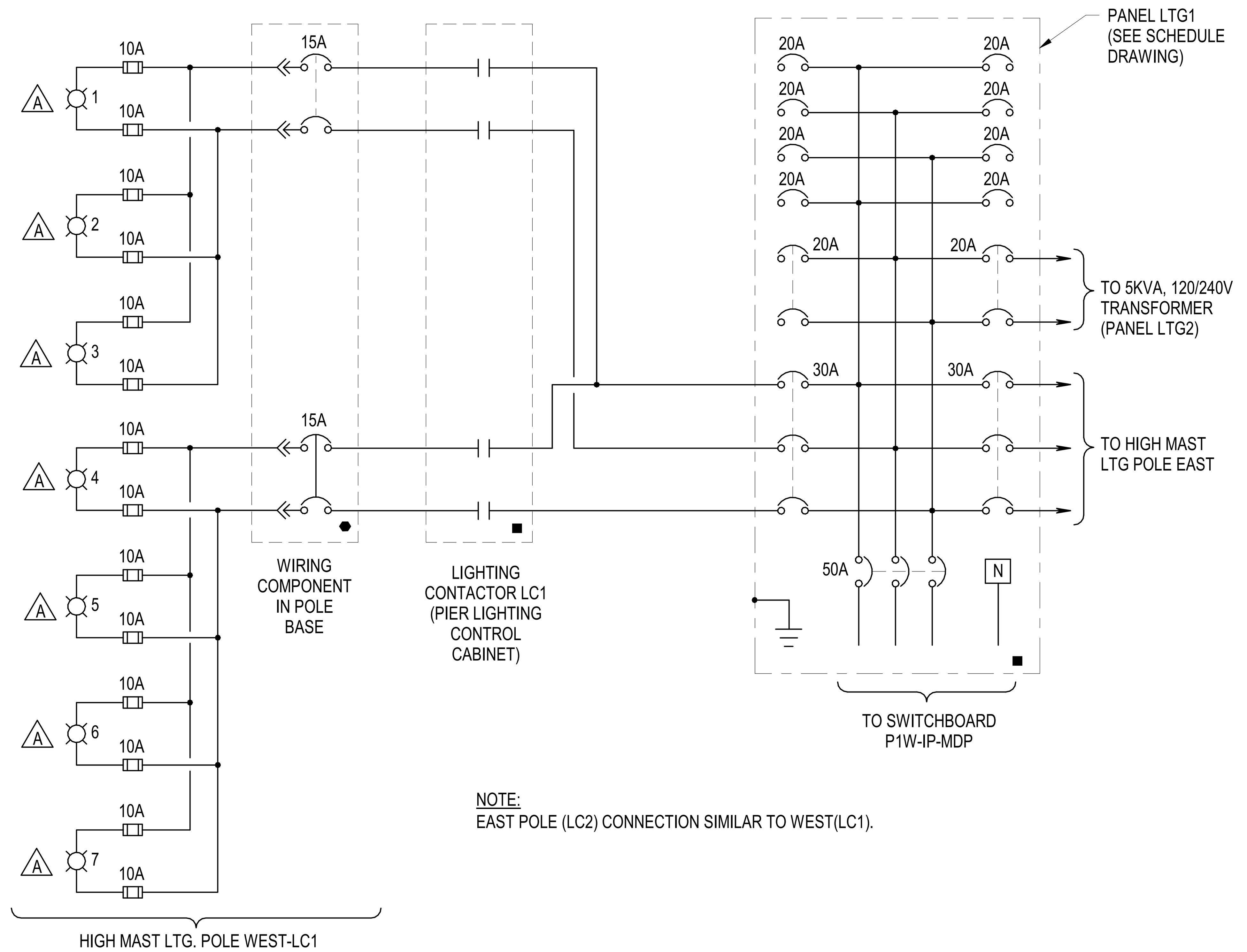
1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR LIGHTING SCHEDULE, SEE DRAWING E-509.
2. LIGHTING CONTROL CABINET IS LOCATED IN THE P1W SUBSTATION HOUSE.

SHEET KEYNOTES:

1. PHOTOCELL CONTACTS NORMALLY CLOSED AT NIGHT AND UPON PHOTOCELL FAILURE.

LOCATION LEGEND:

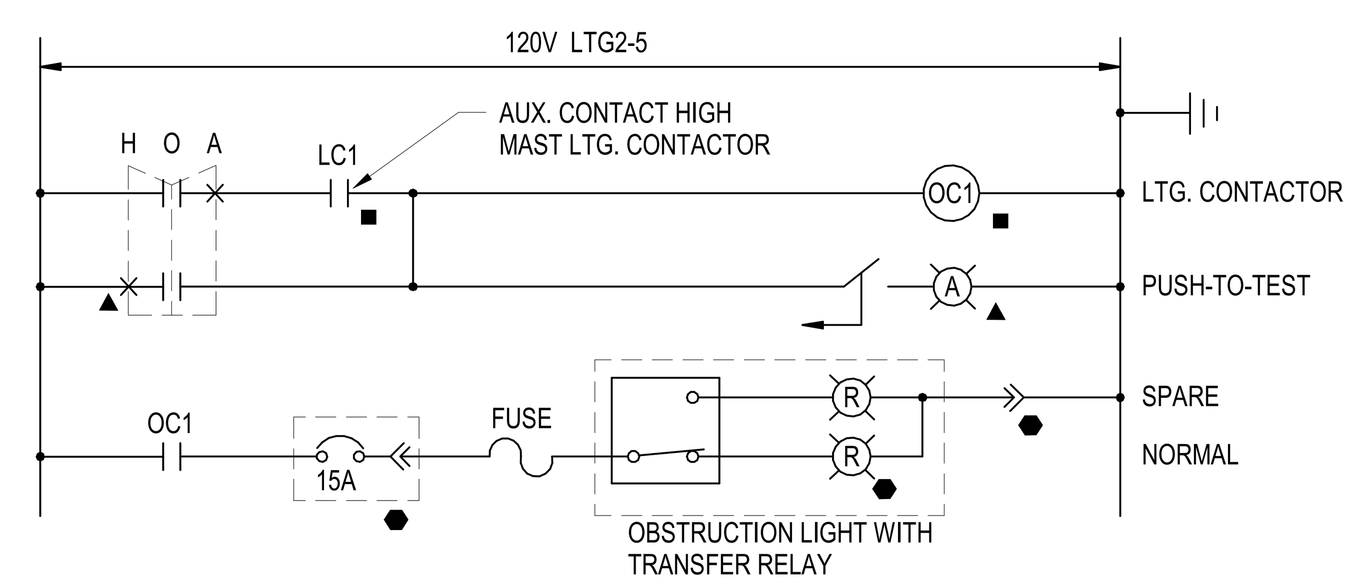
- ▲ CABINET DOOR
- ■ CABINET INTERIOR
- ● SUBSTATION P1W ENCLOSURE ROOF
- ⬢ AT POLE



NOTE:
EAST POLE (LC2) CONNECTION SIMILAR TO WEST(LC1).

C1 HIGH MAST LIGHTING INTERCONNECTION DIAGRAM

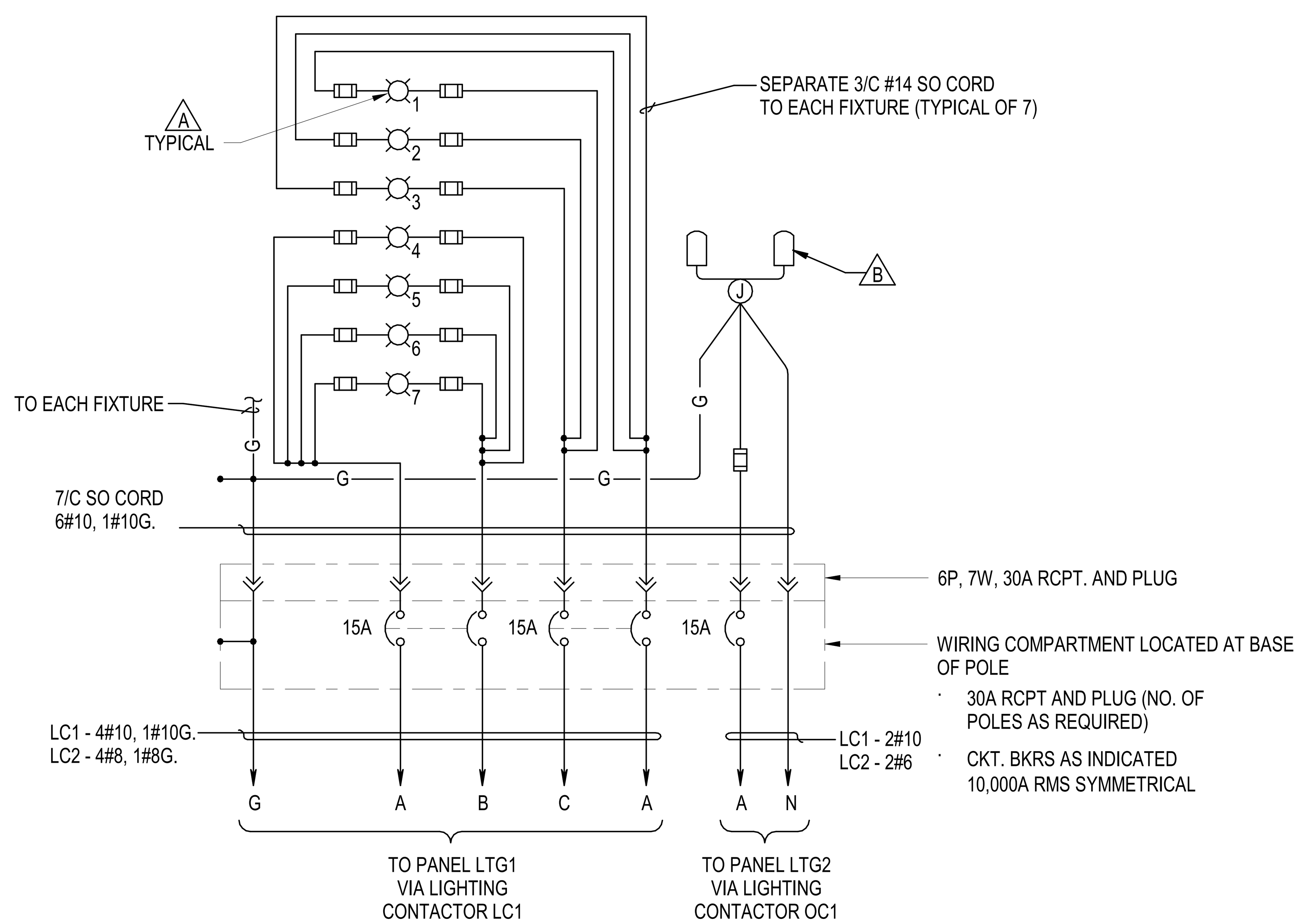
NOT TO SCALE



NOTE:
HIGH MAST POLE WEST, LC1 SHOWN, EAST SIMILAR (LC2) CKT. LTG2-7.

B3 OBSTRUCTION LIGHT ELEMENTARY DIAGRAM

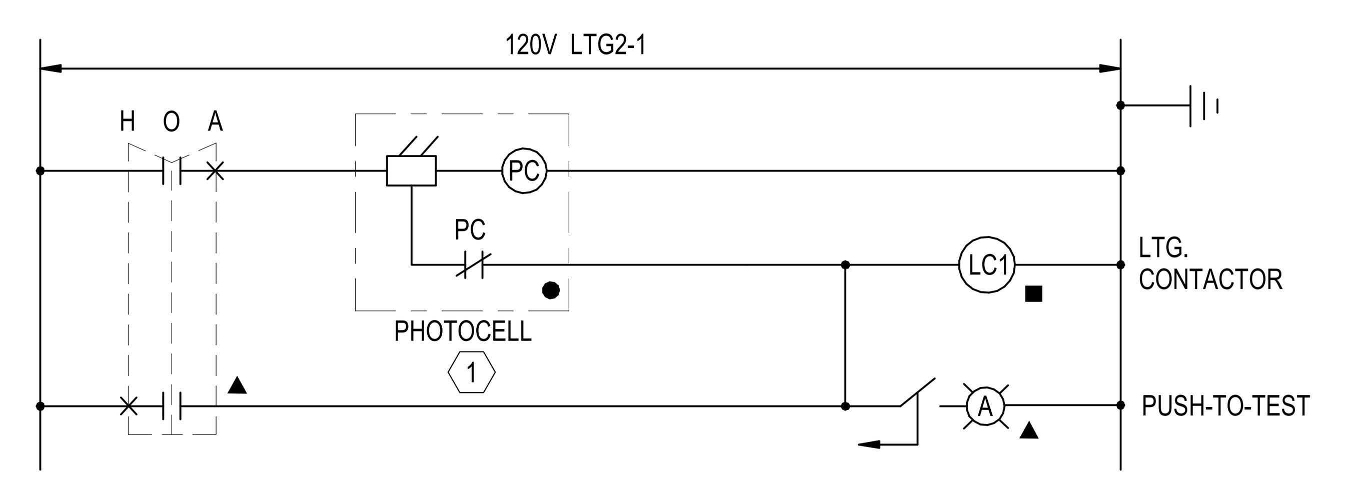
NOT TO SCALE



NOTE:
WEST POLE (LC1) SHOWN, EAST POLE SIMILAR (LC2).

A1 HIGH MAST LIGHTING POLE INTERCONNECTION DIAGRAM

NOT TO SCALE



NOTE:
POLE LC2 SIMILAR, CKT. LTG2-3.

A3 HIGH MAST LIGHTING POLE LC1 ELEMENTARY DIAGRAM

NOT TO SCALE

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
REGISTERED PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER

LIGHTING CONTROL DIAGRAMS

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936803

SHEET 234 OF 247

E-609

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR SUBSTATION PART PLAN SEE DRAWING E-402. FOR SWITCHGEAR ELEVATION SEE DRAWING E-510. FOR POWER ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602 AND E-603.
2. SWITCHGEAR CONTROLS TO OPERATE AT 120V, WHICH IS DERIVED FROM INTEGRAL CONTROL POWER TRANSFORMERS.
3. LONG-TIME DELAY SETTING FOR FEEDER CIRCUIT BREAKERS SERVING SHORE POWER RECEPTACLES MUST BE NO GREATER THAN THE SMALLER OF THE FOLLOWING:
   A. TOTAL AMPACITY OF THE SHORE-TO-SHIP CABLES CONNECTED TO IT.
   B. THE AMPACITY OF THE FEEDER BETWEEN THE ASSOCIATED SHORE POWER RECEPTACLE AND THE IN-HULL CIRCUIT BREAKER, INCLUDING THE RATING OF ANY STEP-DOWN TRANSFORMER.

UNIT SUBSTATION P1E-SPA SWITCHGEAR SCHEDULE
3000A, 450V, 3 PHASE, 3 WIRE + GROUND, 60HZ, 65,000AIC

UNIT NO.	NAMEPLATE	DEVICE DESCRIPTION	CIRCUIT BREAKER				TRIPS WITH LOW VOLTAGE BREAKERS				COMMENTS
			FRAME	SENSOR	RATING PLUG	TRIP	LONG	SHORT	INST	GF	
1	INCOMING SUPPLY	CLOSED COUPLED CONNECTIONS	-	-	-	-	-	-	-	-	
2A	FRONT ACCESS CABLE TOP ENTRY	FRONT ACCESS SECTION	-	-	-	-	-	-	-	-	
2B	LINE INSULATION MONITOR	LINE INSULATION MONITOR	-	-	-	-	-	-	-	-	
2C	MAIN BREAKER P1E-SPA-MCB	LV POWER CIRCUIT BREAKER	3000	3000	3000	3000	X	X	-	-	PROVIDE WITH ENERGY REDUCING MAINTENANCE SWITCH
3A	AMI METER	AMI METER	-	-	-	-	-	-	-	-	
3B	BLANK	BLANK	-	-	-	-	-	-	-	-	
3C	BLANK	BLANK	-	-	-	-	-	-	-	-	
3D	SURGE PROTECTIVE DEVICE	SURGE PROTECTIVE DEVICE	-	-	-	-	-	-	-	-	
4A	POWER FACTOR CORRECTION BANK P1E-SPA-PFC	LV POWER CIRCUIT BREAKER	1200	1200	1200	1200	X	X	X	-	
4B	POWER MOUND RECEPT P1E-PMA-SPR-1	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
4C	POWER MOUND RECEPT P1E-PMA-SPR-2	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
4D	POWER MOUND RECEPT P1E-PMA-SPR-3	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
5	FRONT ACCESS CABLE SECTION	FRONT ACCESS CABLE SECTION	-	-	-	-	-	-	-	-	
6A	POWER MOUND RECEPT P1E-PMA-SPR-4	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6B	POWER MOUND RECEPT P1E-PMA-SPR-5	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6C	POWER MOUND RECEPT P1E-PMA-SPR-6	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6D	SPARE	LV POWER CIRCUIT BREAKER	800	800	800	800	X	X	X	-	
7	FRONT ACCESS CABLE SECTION	FRONT ACCESS CABLE SECTION	-	-	-	-	-	-	-	-	

UNIT SUBSTATION P1E-SPB SWITCHGEAR SCHEDULE
3000A, 450V, 3 PHASE, 3 WIRE + GROUND, 60HZ, 65,000AIC

UNIT NO.	NAMEPLATE	DEVICE DESCRIPTION	CIRCUIT BREAKER				TRIPS WITH LOW VOLTAGE BREAKERS				COMMENTS
			FRAME	SENSOR	RATING PLUG	TRIP	LONG	SHORT	INST	GF	
1	INCOMING SUPPLY	CLOSED COUPLED CONNECTIONS	-	-	-	-	-	-	-	-	
2A	FRONT ACCESS CABLE TOP ENTRY	FRONT ACCESS SECTION	-	-	-	-	-	-	-	-	
2B	LINE INSULATION MONITOR	LINE INSULATION MONITOR	-	-	-	-	-	-	-	-	
2C	MAIN BREAKER P1E-SPB-MCB	LV POWER CIRCUIT BREAKER	3000	3000	3000	3000	X	X	-	-	PROVIDE WITH ENERGY REDUCING MAINTENANCE SWITCH
3A	AMI METER	AMI METER	-	-	-	-	-	-	-	-	
3B	BLANK	BLANK	-	-	-	-	-	-	-	-	
3C	BLANK	BLANK	-	-	-	-	-	-	-	-	
3D	SURGE PROTECTIVE DEVICE	SURGE PROTECTIVE DEVICE	-	-	-	-	-	-	-	-	
4A	POWER FACTOR CORRECTION BANK P1E-SPB-PFC	LV POWER CIRCUIT BREAKER	1200	1200	1200	1200	X	X	X	-	
4B	POWER MOUND RECEPT P1E-PMB-SPR-1	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
4C	POWER MOUND RECEPT P1E-PMB-SPR-2	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
4D	POWER MOUND RECEPT P1E-PMB-SPR-3	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
5	FRONT ACCESS CABLE SECTION	FRONT ACCESS CABLE SECTION	-	-	-	-	-	-	-	-	
6A	POWER MOUND RECEPT P1E-PMB-SPR-4	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6B	POWER MOUND RECEPT P1E-PMB-SPR-5	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6C	POWER MOUND RECEPT P1E-PMB-SPB-6	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6D	SPARE	LV POWER CIRCUIT BREAKER	800	800	800	800	X	X	X	-	
7	FRONT ACCESS CABLE SECTION	FRONT ACCESS CABLE SECTION	-	-	-	-	-	-	-	-	

SYM | DESCRIPTION | DATE | APPR

DAVID P FERGUSON
ELECTRICAL
No. 52642
COMMONWEALTH OF MASSACHUSETTS
PROFESSIONAL ENGINEER
SEAL

Stantec
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY

Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES DPF | DRW GLV | CHK RJW
PM/DM MRI/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
SUBSTATION P1E SWITCHGEAR SCHEDULE

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936804
SHEET 235 OF 247
E-610
DRAWFORM REVISION: 25 AUGUST 2020

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001. FOR SUBSTATION PART PLAN SEE DRAWING E-402. FOR SWITCHGEAR ELEVATION SEE DRAWING E-511. FOR POWER ONE LINE DIAGRAMS SEE DRAWINGS E-601, E-602, AND E-603.
2. SWITCHGEAR CONTROLS TO OPERATE AT 120V, WHICH IS DERIVED FROM INTEGRAL CONTROL POWER TRANSFORMERS.

UNIT SUBSTATION P1W-SPA SWITCHGEAR SCHEDULE
3000A, 450V, 3 PHASE, 3 WIRE + GROUND, 60HZ, 65,000AIC

UNIT NO.	NAMEPLATE	DEVICE DESCRIPTION	CIRCUIT BREAKER				TRIPS WITH LOW VOLTAGE BREAKERS				COMMENTS
			FRAME	SENSOR	RATING PLUG	TRIP	LONG	SHORT	INST	GF	
1	INCOMING SUPPLY	CLOSED COUPLED CONNECTIONS	-	-	-	-	-	-	-	-	
2A	FRONT ACCESS CABLE TOP ENTRY	FRONT ACCESS SECTION	-	-	-	-	-	-	-	-	
2B	LINE INSULATION MONITOR	LINE INSULATION MONITOR	-	-	-	-	-	-	-	-	
2C	MAIN BREAKER P1W-SPA-MCB	LV POWER CIRCUIT BREAKER	3000	3000	3000	3000	X	X	-	-	PROVIDE WITH ENERGY REDUCING MAINTENANCE SWITCH
3A	AMI METER	AMI METER	-	-	-	-	-	-	-	-	
3B	BLANK	BLANK	-	-	-	-	-	-	-	-	
3C	BLANK	BLANK	-	-	-	-	-	-	-	-	
3D	SURGE PROTECTIVE DEVICE	SURGE PROTECTIVE DEVICE	-	-	-	-	-	-	-	-	
4A	POWER FACTOR CORRECTION BANK P1W-SPA-PFC	LV POWER CIRCUIT BREAKER	1200	1200	1200	1200	X	X	X	-	
4B	POWER MOUND RECEPT P1W-PMA-SPR-1	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
4C	POWER MOUND RECEPT P1W-PMA-SPR-2	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
4D	POWER MOUND RECEPT P1W-PMA-SPR-3	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
5	FRONT ACCESS CABLE SECTION	FRONT ACCESS CABLE SECTION	-	-	-	-	-	-	-	-	
6A	POWER MOUND RECEPT P1W-PMA-SPR-4	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6B	POWER MOUND RECEPT P1W-PMA-SPR-5	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6C	POWER MOUND RECEPT P1W-PMA-SPR-6	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6D	SPARE	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	
7	FRONT ACCESS CABLE SECTION	FRONT ACCESS CABLE SECTION	-	-	-	-	-	-	-	-	

UNIT SUBSTATION P1W-SPB SWITCHGEAR SCHEDULE
3000A, 450V, 3 PHASE, 3 WIRE + GROUND, 60HZ, 65,000AIC

UNIT NO.	NAMEPLATE	DEVICE DESCRIPTION	CIRCUIT BREAKER				TRIPS WITH LOW VOLTAGE BREAKERS				COMMENTS
			FRAME	SENSOR	RATING PLUG	TRIP	LONG	SHORT	INST	GF	
1	INCOMING SUPPLY	CLOSED COUPLED CONNECTIONS	-	-	-	-	-	-	-	-	
2A	FRONT ACCESS CABLE TOP ENTRY	FRONT ACCESS SECTION	-	-	-	-	-	-	-	-	
2B	LINE INSULATION MONITOR	LINE INSULATION MONITOR	-	-	-	-	-	-	-	-	
2C	MAIN BREAKER P1W-SPB-MCB	LV POWER CIRCUIT BREAKER	3000	3000	3000	3000	X	X	-	-	PROVIDE WITH ENERGY REDUCING MAINTENANCE SWITCH
3A	AMI METER	AMI METER	-	-	-	-	-	-	-	-	
3B	BLANK	BLANK	-	-	-	-	-	-	-	-	
3C	BLANK	BLANK	-	-	-	-	-	-	-	-	
3D	SURGE PROTECTIVE DEVICE	SURGE PROTECTIVE DEVICE	-	-	-	-	-	-	-	-	
4A	POWER FACTOR CORRECTION BANK P1W-SPB-PFC	LV POWER CIRCUIT BREAKER	1200	1200	1200	1200	X	X	X	-	
4B	POWER MOUND RECEPT P1W-PMB-SPR-1	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
4C	POWER MOUND RECEPT P1W-PMB-SPR-2	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
4D	POWER MOUND RECEPT P1W-PMB-SPR-3	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
5	FRONT ACCESS CABLE SECTION	FRONT ACCESS CABLE SECTION	-	-	-	-	-	-	-	-	
6A	POWER MOUND RECEPT P1W-PMB-SPR-4	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6B	POWER MOUND RECEPT P1W-PMB-SPR-5	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6C	POWER MOUND RECEPT P1W-PMB-SPR-6	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	REMOTE OPERATION
6D	SPARE	LV POWER CIRCUIT BREAKER	800	800	600	500	X	X	X	-	
7	FRONT ACCESS CABLE SECTION	FRONT ACCESS CABLE SECTION	-	-	-	-	-	-	-	-	

SYM	DESCRIPTION	DATE	APPR

NAVFAC

DAVID P FERGUSON ELECTRICAL No. 52642 — COMMONWEALTH OF MASSACHUSETTS — PROFESSIONAL ENGINEER
SEAL

Stantec
A/E INFO

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FOR COMMANDER NAVFAC 3/5/2026
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FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
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USCG OCR HOMEPORT INITIATIVE OPC PIER
SUBSTATION P1W SWITCHGEAR SCHEDULE

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936805
SHEET 236 OF 247
E-611
DRAWFORM REVISION: 20 AUGUST 2020

3/2/2026 5:14:24 PM E-611
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-E.rvt

Stantec — Switchboard

Name: **P1E-IP-MDP**	Volts: 480/277V Wye	Mains Type: MCB	Type:
Location: SUBSTATION P1E	Phases: 3	Mains Rating: 1200 A	AIC Rating: 65KAIC
Supply From: P1E-IP-XFMR	Wires: 4	MCB Rating: 1000 A	Mounting: Floor
Serves:		Lugs: Single Lugs	Enclosure: TYPE 1 SS

Notes:

#	Description	Load	Trip	Poles	Type	Remarks
1	POWER MOUND P1E-PMA	149650 VA	225 A	3		
2	POWER MOUND P1E-PMB	149650 VA	225 A	3		
3	COMPRESSED AIR ENCLOSURE	129802 VA	200 A	3		
4	SANITARY SEWER LIFT STATION	13628 VA	25 A	3		
5	PANEL P1-HTCP	37660 VA	100 A	3		
6	PANEL P1E-RP	2611 VA	60 A	3		
7	SANITARY SEWER PUMP SP-3	7981 VA	20 A	3		
8	SANITARY SEWER PUMP SP-4	7981 VA	20 A	3		
9	SPARE	0 VA	225 A	3	--	
10	SPARE	0 VA	100 A	3	--	
11	SUBSTATION A/C	20762 VA	35 A	3		
12	SUBSTATION A/C	20762 VA	35 A	3		
13	P1E-MPC-PMA	3360 VA	20 A	2		
14	P1E-MPC-PMB	3360 VA	20 A	2		
15	P1E-MPC-RPZ	3360 VA	20 A	2		
16	SPARE	0 VA	100 A	3	--	
	Total Conn. Load:	**548.21 kVA**				
	Total Amps:	**659 A**				

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
HVAC	79184 VA	100.00%	79184 VA		
Lighting	1364 VA	100.00%	1364 VA	Total Conn. Load:	548209 VA
Motor	157223 VA	100.00%	157223 VA	Total Est. Demand:	548209 VA
Power	309580 VA	100.00%	309580 VA	Total Conn.:	659 A
Receptacle	1080 VA	100.00%	1080 VA	Total Est. Demand:	659 A

Legend:
CB = Circuit Breaker C = HACR Rated F = Fused Switch G = GFCI H = HID Rated M = Motor Circuit Protection S = Shunt Trip # = See Notes † = Existing Circuit ‡ = Revised Circuit

Notes:

Stantec — Panel

Name: **P1-HTCP**	Volts: 480/277V Wye	Mains Type: MCB	Type:
Location: SUBSTATION P1E	Phases: 3	Mains Rating: 100 A	AIC Rating: 42KAIC
Supply From: P1E-IP-MDP	Wires: 4	MCB Rating: 100 A	Mounting: Surface
Serves:		Lugs: Single Lugs	Enclosure: TYPE 1 SS

CKT	Circuit Description	Conduit & Wire Size	Trip	Poles	A		B		C		Poles	Trip	Conduit & Wire Size	Circuit Description	CKT
1	SSFM - B12-0 - NORTH	SEE DWG E-608	30 A	1	2000 VA	2640 VA					1	30 A	SEE DWG E-608	FW - B28-19 - SOUTH	2
3	SSFM - B12-0 - SOUTH	SEE DWG E-608	30 A	1			2000 VA	2960 VA			1	30 A	SEE DWG E-608	FW - B43-34 - NORTH	4
5	SSFM - B24-12 - NORTH	SEE DWG E-608	30 A	1					2325 VA	2680 VA	1	30 A	SEE DWG E-608	FW - B37-28 - SOUTH	6
7	SSFM - B24-12 - SOUTH	SEE DWG E-608	30 A	1	2325 VA	2120 VA					1	30 A	SEE DWG E-608	FW - B43-37 - SOUTH	8
9	SSFM - B35-24 - NORTH	SEE DWG E-608	30 A	1			1950 VA	0 VA			1	30 A	--	SPARE	10
11	SSFM - B35-24 - SOUTH	SEE DWG E-608	30 A	1					1950 VA	0 VA	1	30 A	--	SPARE	12
13	SSFM - CA ENCLOSURE RISER	SEE DWG E-608	30 A	1	150 VA	0 VA					1	30 A	--	SPARE	14
15	FW - B14-8 - NORTH	SEE DWG E-608	30 A	1			3000 VA	0 VA			1	30 A	--	SPARE	16
17	FW - B11-0 - SOUTH	SEE DWG E-608	30 A	1					2960 VA	0 VA	1	30 A	--	SPARE	18
19	FW - B24-14 - NORTH	SEE DWG E-608	30 A	1	2960 VA	0 VA					1	30 A	--	SPARE	20
21	FW - B19-11 - SOUTH	SEE DWG E-608	30 A	1			2560 VA	0 VA			1	30 A	--	SPARE	22
23	FW - B34-24 - NORTH	SEE DWG E-608	30 A	1					3080 VA	0 VA	1	30 A	--	SPARE	24
				Total Load:	**12.20 kVA**		**12.47 kVA**		**13.00 kVA**						
				Total Amps:	**44 A**		**45 A**		**47 A**						

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
HVAC	37660 VA	100.00%	37660 VA		
				Total Conn. Load:	37660 VA
				Total Est. Demand:	37660 VA
				Total Conn.:	45 A
				Total Est. Demand:	45 A

CB Legend (blank = circuit breaker):
G = GFCI S = Shunt Trip D = Switching Duty A = AFCI H = HID Rated C = HACR Rated † = Existing Circuit ‡ = Revised Circuit

Notes:
PANEL IS INTEGRAL TO HEAT TRACE CONTROL CABINET. BRANCH CIRCUIT BREAKERS ARE GFCI TYPE.

Stantec — Panel

Name: **P1E-RP**	Volts: 120/208V	Mains Type: MCB	Type:
Location: SUBSTATION P1E	Phases: 3	Mains Rating: 100 A	AIC Rating: 10KAIC
Supply From: P1E 30KVA XFMR	Wires: 4	MCB Rating: 100 A	Mounting: Surface
Serves:		Lugs: Single Lugs	Enclosure: TYPE 1 SS

CKT	Circuit Description	Conduit & Wire Size	Trip	Poles	A		B		C		Poles	Trip	Conduit & Wire Size	Circuit Description	CKT
1	SUBSTATION LIGHTING	3/4"C, 2#10, 1#10G	20 A	1	684 VA	0 VA					1	20 A	--	SPARE	2
3	SUBSTATION RECEPTACLES	3/4"C, 2#12, 1#12G	20 A	1			720 VA	0 VA			1	20 A	--	SPARE	4
5	VAULT LIGHTING	3/4"C, 2#10, 1#10G	20 A	1					646 VA	0 VA	1	20 A	--	SPARE	6
7	P1E-SPA LIM	3/4"C, 2#12, 1#12G	20 A	1	100 VA	0 VA					1	20 A	--	SPARE	8
9	P1E-SPB LIM	3/4"C, 2#12, 1#12G	20 A	1			100 VA	0 VA			1	20 A	--	SPARE	10
11	EXTERIOR & EGRESS LIGHTS	3/4"C, 2#12, 1#12G	20 A	1					34 VA	0 VA	1	20 A	--	SPARE	12
13	EXTERIOR RECEPTACLES	3/4"C, 2#12, 1#12G	20 A	1	360 VA	0 VA					1	20 A	--	SPARE	14
15	SPARE	--	20 A	1			0 VA	0 VA			1	20 A	--	SPARE	16
17	SPARE	--	20 A	1					0 VA	0 VA	1	20 A	--	SPARE	18
19	SPARE	--	20 A	1	0 VA	0 VA					1	20 A	--	SPARE	20
21	SPARE	--	20 A	1			0 VA	0 VA			1	20 A	--	SPARE	22
23	SPARE	--	20 A	1					0 VA	0 VA	1	20 A	--	SPARE	24
25	SPARE	--	20 A	1	0 VA	0 VA					1	20 A	--	SPARE	26
27	SPARE	--	20 A	1			0 VA	0 VA			1	20 A	--	SPARE	28
29	SPARE	--	20 A	1					0 VA	0 VA	1	20 A	--	SPARE	30
31	SPACE	--	--	1	--	--					1	--	--	SPACE	32
33	SPACE	--	--	1			--	--			1	--	--	SPACE	34
35	SPACE	--	--	1					--	--	1	--	--	SPACE	36
37	SPACE	--	--	1	--	--					1	--	--	SPACE	38
39	SPACE	--	--	1			--	--			1	--	--	SPACE	40
41	SPACE	--	--	1					--	--	1	--	--	SPACE	42
				Total Load:	**1.13 kVA**		**0.82 kVA**		**0.68 kVA**						
				Total Amps:	**10 A**		**7 A**		**6 A**						

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Lighting	1364 VA	100.00%	1364 VA		
Power	200 VA	100.00%	200 VA	Total Conn. Load:	2611 VA
Receptacle	1080 VA	100.00%	1080 VA	Total Est. Demand:	2611 VA
				Total Conn.:	7 A
				Total Est. Demand:	7 A

CB Legend (blank = circuit breaker):
G = GFCI S = Shunt Trip D = Switching Duty A = AFCI H = HID Rated C = HACR Rated † = Existing Circuit ‡ = Revised Circuit

Notes:

SYM | DESCRIPTION | DATE | APPR

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
SEAL

Stantec
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES DPF	DRW GLV	CHK RJW
PM/DM		MRI/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
SUBSTATION P1E PANEL SCHEDULES

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936806
SHEET 237 OF 247
E-612
DRAWFORM REVISION: 25 AUGUST 2020

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E-612

1	2	3	4	5

Stantec — Switchboard

Name:	**P1W-IP-MDP**	Volts:	480/277V Wye	Mains Type:	MCB	Type:	
Location:	SUBSTATION P1W	Phases:	3	Mains Rating:	800 A	AIC Rating:	65KAIC
Supply From:	PIW-IP-XFMR	Wires:	4	MCB Rating:	600 A	Mounting:	Floor
Serves:				Lugs:	Single Lugs	Enclosure:	TYPE 1 SS

Notes:

#	Description	Load	Trip	Poles	Type	Remarks
1	PANEL P1W-RP	2573 VA	60 A	3		
2	PANEL LTG1 IN LIGHTING CONTROL CABINET	12721 VA	40 A	3		
3	POWER MOUND P1W-PMA	149650 VA	225 A	3		
4	POWER MOUND P1W-PMB	149650 VA	225 A	3		
5	SANITARY SEWER PUMP SP-1	7981 VA	20 A	3		
6	SANITARY SEWER PUMP SP-2	7981 VA	20 A	3		
7	SPARE	0 VA	225 A	3	--	
8	SPARE	0 VA	100 A	3	--	
9	SPARE	0 VA	100 A	3	--	
10	SUBSTATION A/C	20762 VA	35 A	3		
11	SUBSTATION A/C	20762 VA	35 A	3		
12	P1W-MPC-PMA	3360 VA	20 A	2		
13	P1W-MPC-PMB	3360 VA	20 A	2		
14	SPACE	--	--	3	--	
	Total Conn. Load:	**376.48 kVA**				
	Total Amps:	**453 A**				

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
HVAC	41524 VA	100.00%	41524 VA		
Lighting	13987 VA	100.00%	13987 VA	**Total Conn. Load:**	**376476 VA**
Motor	15963 VA	100.00%	15963 VA	**Total Est. Demand:**	**376476 VA**
Power	306220 VA	100.00%	306220 VA	**Total Conn.:**	**453 A**
Receptacle	1080 VA	100.00%	1080 VA	**Total Est. Demand:**	**453 A**

Legend:
CB = Circuit Breaker C = HACR Rated F = Fused Switch G = GFCI H = HID Rated M = Motor Circuit Protection S = Shunt Trip # = See Notes † = Existing Circuit ‡ = Revised Circuit

Notes:

Stantec — Panel

Name:	**P1W-RP**	Volts:	120/208V	Mains Type:	MCB	Type:	
Location:	SUBSTATION P1W	Phases:	3	Mains Rating:	100 A	AIC Rating:	10KAIC
Supply From:	P1W 30KVA XFMR	Wires:	4	MCB Rating:	100 A	Mounting:	Surface
Serves:				Lugs:	Single Lugs	Enclosure:	TYPE 1 SS

CKT	Circuit Description	Conduit & Wire Size	Trip	Poles	A		B		C		Poles	Trip	Conduit & Wire Size	Circuit Description	CKT
1	P1W AREA LIGHTING	3/4"C, 2#10, 1#10G	20 A	1	684 VA	0 VA					1	20 A	--	SPARE	2
3	INTERIOR RECEPTACLES	3/4"C, 2#12, 1#12G	20 A	1			720 VA	0 VA			1	20 A	--	SPARE	4
5	VAULT LIGHTING	3/4"C, 2#12, 1#12G	20 A	1					608 VA	0 VA	1	20 A	--	SPARE	6
7	P1W-SPA LIM	3/4"C, 2#12, 1#12G	20 A	1	100 VA	0 VA					1	20 A	--	SPARE	8
9	P1W-SPB LIM	3/4"C, 2#12, 1#12G	20 A	1			100 VA	0 VA			1	20 A	--	SPARE	10
11	EXTERIOR & EGRESS LIGHTS	3/4"C, 2#12, 1#12G	20 A	1					34 VA	0 VA	1	20 A	--	SPARE	12
13	EXTERIOR RECEPTACLES	3/4"C, 2#12, 1#12G	20 A	1	360 VA	0 VA					1	20 A	--	SPARE	14
15	SPARE	--	20 A	1			0 VA	0 VA			1	20 A	--	SPARE	16
17	SPARE	--	20 A	1					0 VA	0 VA	1	20 A	--	SPARE	18
19	SPARE	--	20 A	1	0 VA	0 VA					1	20 A	--	SPARE	20
21	SPARE	--	20 A	1			0 VA	0 VA			1	20 A	--	SPARE	22
23	SPARE	--	20 A	1					0 VA	0 VA	1	20 A	--	SPARE	24
25	SPARE	--	20 A	1	0 VA	0 VA					1	20 A	--	SPARE	26
27	SPARE	--	20 A	1			0 VA	0 VA			1	20 A	--	SPARE	28
29	SPARE	--	20 A	1					0 VA	0 VA	1	20 A	--	SPARE	30
31	SPACE	--	--	1	--	--					1	--	--	SPACE	32
33	SPACE	--	--	1			--	--			1	--	--	SPACE	34
35	SPACE	--	--	1					--	--	1	--	--	SPACE	36
37	SPACE	--	--	1	--	--					1	--	--	SPACE	38
39	SPACE	--	--	1			--	--			1	--	--	SPACE	40
41	SPACE	--	--	1					--	--	1	--	--	SPACE	42
			Total Load:		**1.13 kVA**		**0.82 kVA**		**0.64 kVA**						
			Total Amps:		**10 A**		**7 A**		**5 A**						

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Lighting	1326 VA	100.00%	1326 VA		
Power	200 VA	100.00%	200 VA	**Total Conn. Load:**	**2573 VA**
Receptacle	1080 VA	100.00%	1080 VA	**Total Est. Demand:**	**2573 VA**
				Total Conn.:	**7 A**
				Total Est. Demand:	**7 A**

CB Legend (blank = circuit breaker):
G = GFCI S = Shunt Trip D = Switching Duty A = AFCI H = HID Rated C = HACR Rated † = Existing Circuit ‡ = Revised Circuit

Notes:

Stantec — Panel

Name:	**LTG1**	Volts:	480/277V Wye	Mains Type:	MCB	Type:	
Location:	SUBSTATION P1W	Phases:	3	Mains Rating:	100 A	AIC Rating:	14KAIC
Supply From:	P1W-IP-MDP	Wires:	4	MCB Rating:	40 A	Mounting:	Surface
Serves:				Lugs:	Single Lugs	Enclosure:	TYPE 1 SS

CKT	Circuit Description	Conduit & Wire Size	Trip	Poles	A		B		C		Poles	Trip	Conduit & Wire Size	Circuit Description	CKT
1	LIGHTING POLE LC1	SEE DWG E-609	30 A	3	1247 VA	0 VA					1	20 A	--	SPARE	2
3	--	--	--	--			1247 VA	0 VA			1	20 A	--	SPARE	4
5	--	--	--	--					1247 VA	0 VA	1	20 A	--	SPARE	6
7	LIGHTING POLE LC2	SEE DWG E-609	30 A	3	1660 VA	0 VA					1	20 A	--	SPARE	8
9	--	--	--	--			1660 VA	0 VA			1	20 A	--	SPARE	10
11	--	--	--	--					1660 VA	0 VA	1	20 A	--	SPARE	12
13	LTG2 VIA TRANSFORMER	3/4"C, 3#12, 1#12G	20 A	3	2000 VA	0 VA					1	20 A	--	SPARE	14
15	--	--	--	--			1800 VA	0 VA			1	20 A	--	SPARE	16
17	--	--	--	--					200 VA	0 VA	1	20 A	--	SPARE	18
			Total Load:		**4.91 kVA**		**4.71 kVA**		**3.11 kVA**						
			Total Amps:		**19 A**		**18 A**		**11 A**						

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Lighting	12721 VA	100.00%	12721 VA		
				Total Conn. Load:	**12721 VA**
				Total Est. Demand:	**12721 VA**
				Total Conn.:	**15 A**
				Total Est. Demand:	**15 A**

CB Legend (blank = circuit breaker):
G = GFCI S = Shunt Trip D = Switching Duty A = AFCI H = HID Rated C = HACR Rated † = Existing Circuit ‡ = Revised Circuit

Notes:

Stantec — Panel

Name:	**LTG2**	Volts:	120/208V	Mains Type:	MLO	Type:	
Location:	SUBSTATION P1W	Phases:	3	Mains Rating:	100 A	AIC Rating:	10KAIC
Supply From:	PANEL LTG1 VIA XFMR	Wires:	4	Max Rating:	100 A	Mounting:	Surface
Serves:				Lugs:	Single Lugs	Enclosure:	TYPE 1 SS

CKT	Circuit Description	Conduit & Wire Size	Trip	Poles	A		B		C		Poles	Trip	Conduit & Wire Size	Circuit Description	CKT
1	HIGH MAST LC1 CONTACTOR	3/4"C, 2#12, 1#12G	20 A	1	1800 VA	--					1	--	--	SPACE	2
3	HIGH MAST LC2 CONTACTOR	3/4"C, 2#12, 1#12G	20 A	1			1800 VA	--			1	--	--	SPACE	4
5	LC1 OBSTRUCTION LIGHT	SEE DWG E-609	20 A	1					200 VA	--	1	--	--	SPACE	6
7	LC2 OBSTRUCTION LIGHT	SEE DWG E-609	20 A	1	200 VA	--					1	--	--	SPACE	8
9	SPARE	--	20 A	1			0 VA	--			1	--	--	SPACE	10
11	SPARE	--	20 A	1					0 VA	--	1	--	--	SPACE	12
13	SPARE	--	20 A	1	0 VA	--					1	--	--	SPACE	14
15	SPARE	--	20 A	1			0 VA	--			1	--	--	SPACE	16
17	SPARE	--	20 A	1					0 VA	--	1	--	--	SPACE	18
			Total Load:		**2.00 kVA**		**1.80 kVA**		**0.20 kVA**						
			Total Amps:		**19 A**		**17 A**		**2 A**						

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals	
Lighting	4000 VA	100.00%	4000 VA		
				Total Conn. Load:	**4000 VA**
				Total Est. Demand:	**4000 VA**
				Total Conn.:	**11 A**
				Total Est. Demand:	**11 A**

CB Legend (blank = circuit breaker):
G = GFCI S = Shunt Trip D = Switching Duty A = AFCI H = HID Rated C = HACR Rated † = Existing Circuit ‡ = Revised Circuit

Notes:

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P
FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
David Ferguson

SEAL

Stantec

A/E INFO

APPROVED
James E. Donahue, R.A.
FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES	DPF	DRW	GLV	CHK	RJW

PM/DM MRI/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
SUBSTATION P1W PANEL SCHEDULES

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936807
SHEET 238 OF 247

E-613

DRAWFORM REVISION: 25 AUGUST 2020

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E-613

FEEDER SCHEDULE

FEEDER	CONDUCTORS	CONDUIT	FROM	TO	REMARKS
1	3#350KCMIL (15KV), 1#4/0G (600V)	5"	SUBSTATION 1	SWITCH P1E-SW	15KV FEEDER TO SWITCH P1E-SW
2	3#350KCMIL (15KV), 1#4/0G (600V)	5"	EMH 4118E	SWITCH P1W-SW	15KV FEEDER 1W8 TO SWITCH P1W-SW
3	3#350KCMIL (15KV), 1#4/0G (600V)	5"	EMH 4118E	SWITCH P1E-SW	15KV FEEDER 1W9 TO SWITCH P1E-SW
4	3#4/0 (15KV), 1#3G (600V)	5"	15KV SWITCH P1E-SW	SUBSTATION P1E-SPA	15KV FEEDER TO SUBSTATION P1E-SPA
5	3#4/0 (15KV), 1#3G (600V)	5"	15KV SWITCH P1E-SW	SUBSTATION P1E-SPB	15KV FEEDER TO SUBSTATION P1E-SPB
6	3#4/0 (15KV), 1#3G (600V)	5"	15KV SWITCH P1E-SW	TRANSFORMER P1E-IP-XFMR	15KV FEEDER TO TRANSFORMER P1E-IP-XFMR
7	3#4/0 (15KV), 1#3G (600V)	5"	15KV SWITCH P1W-SW	SUBSTATION P1W-SPA	15KV FEEDER TO SUBSTATION P1W-SPA
8	3#4/0 (15KV), 1#3G (600V)	5"	15KV SWITCH P1W-SW	SUBSTATION P1W-SPB	15KV FEEDER TO SUBSTATION P1W-SPB
9	3#4/0 (15KV), 1#3G (600V)	5"	15KV SWITCH P1W-SW	TRANSFORMER P1W-IP-XFMR	15KV FEEDER TO TRANSFORMER P1W-IP-XFMR
10	3#777KCMIL (DLO), 1#2G	5"	SUBSTATION P1E-SPA	P1E-PMA	SHORE POWER RECEPTACLE
11	8#12 (SHIELDED), 6-12/C#12 (SHIELDED)	4"	SUBSTATION P1E-SPA	P1E-PMA	LIM OVERRIDE (8#12), SHORE POWER RECEPTACLE REMOTE CONTROL (6-12#12)
12	3#8, 1#8G	1"/4"	PANEL P1E-IP-MDP	SANITARY SEWER LIFT STATION	4" FROM SUBSTATION TO LANDSIDE MANHOLE, 1" FROM MANHOLE TO LIFT STATION
13	3#3/0, 1#6G	2"/4"	PANEL P1E-IP-MDP	AIR COMPRESSOR ENCLOSURE	4" FROM SUBSTATION TO LANDSIDE MANHOLE, 2" FROM MANHOLE TO ENCLOSURE
14	3#4/0, 1#4G	4"	PANEL P1E-IP-MDP	P1E-PMA	INDUSTRIAL POWER RECEPTACLES
15	3#6, 1#10G	1"	INDUSTRIAL POWER MAIN DISTRIBUTION PANEL	30KVA, 480V TO 120/208V TRANSFORMER	
16	4#2, 1#6G	1-1/2"	30KVA, 480V TO 120/208V TRANSFORMER	120/208V PANEL	
17	4#3, 1#8G	1-1/4"	PANEL P1E-IP-MDP	PANEL P1-HTCP	HEAT TRACE CONTROL PANEL
18	4#8, 1#10G	1"	PANEL P1W-IP-MDP	PANEL P1-LCP	LIGHTING CONTROL PANEL
19		5"	SEE PLANS	SEE PLANS	SPARE
20		4"	SEE PLANS	SEE PLANS	SPARE
21	SEE DRAWING E-608	4"	PANEL P1-HTCP	EMH-BC-1314	NORTH UTILITY HEAT TRACE CIRCUITS
22	(4) 25PR CAT5E COPPER	4"	TMH-CD-1314	TMH-CD-3334	PHONE CABLE TO POWER MOUND
23	(8) 25PR CAT5E COPPER	4"	MAT BUILDING 1406	TMH-CD-1314	PHONE CABLE TO POWER MOUND
24	(2) 24STR SM FIBER	4"	BUILDING 76	SUBSTATION P1E	FIBER TO PIER. UTILIZE EXISTING SPARE DUCTS FROM TMH-"I" TO TMH-"F".
25	(1) 24STR SM FIBER	4"	SUBSTATION P1E	SUBSTATION P1W	FIBER TO SUBSTATION
26	(4) SETS OF 3#350KCMIL, 1#3/0G	(4) - 3"C	P1E-SPA, P1E-SPB UNIT SUBSTATIONS	POWER FACTOR CORRECTION CAPACITOR BANK	
27	(4) SETS OF 3#350KCMIL, 1#3/0G	(4) - 3"C	P1W-SPA, P1W-SPB UNIT SUBSTATIONS	POWER FACTOR CORRECTION CAPACITOR BANK	
28	(2) 25PR CAT5E COPPER	4"	ADJACENT COMMUNICATIONS MANHOLE	POWER MOUND	PHONE CABLE TO POWER MOUND
29	(3) SETS OF 4#500KCMIL, 1#2/0G	(2) - 4"C	P1E-IP-XFMR	P1E-IP-MDP	
30	(2) SETS OF 4#600KCMIL, 1#1/0G	(2) - 4"C	P1W-IP-XFMR	P1W-IP-MDP	
31	3#10, 1#10G	1"/4"	PANEL P1W-IP-MDP	MANUAL TRANSFER SWITCH	4" FROM SUBSTATION TO PIER MANHOLE, 1" FROM MANHOLE TO TRANSFER SWITCH.
32	3#10, 1#10G	1"/4"	PANEL P1E-IP-MDP	MANUAL TRANSFER SWITCH	4" FROM SUBSTATION TO PIER MANHOLE, 1" FROM MANHOLE TO TRANSFER SWITCH.
33	3#10, 1#10G	1"	PORTABLE EQUIPMENT RECEPTACLE	MANUAL TRANSFER SWITCH	
34	3#10, 1#10G	1"	MANUAL TRANSFER SWITCH	PIER SEWER PUMP CONTROLLER	
35	PUMP MANUFACTURER SUPPLIED CABLE	1"	PIER SEWER PUMP CONTROLLER	PIER SEWER PUMP	
36	(1) 24STR SM FIBER, (2) CAT6 COPPER	4"	SUBSTATION P1E	TMH-CD-1314	FIBER TO SUBSTATION, CAT6 FOR WATER ENCLOSURE AMI METER
37	(1) CAT6 COPPER	1"	ADJACENT COMMUNICATIONS MANHOLE	PIER WATER UTILITY ENCLOSURE	CAT6 FOR WATER ENCLOSURE AMI METER
38	3#777KCMIL (DLO), 1#2G	5"	SUBSTATION P1E-SPB	P1E-PMB	SHORE POWER RECEPTACLE
39	8#12 (SHIELDED), 6-12/C#12 (SHIELDED)	4"	SUBSTATION P1E-SPB	P1E-PMB	LIM OVERRIDE (8#12), SHORE POWER RECEPTACLE REMOTE CONTROL (6-12#12)
40	3#4/0, 1#4G	4"	PANEL P1E-IP-MDP	P1E-PMB	INDUSTRIAL POWER RECEPTACLES
41	3#777KCMIL (DLO), 1#2G	5"	SUBSTATION P1W-SPA	P1W-PMA	SHORE POWER RECEPTACLE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

DAVID P FERGUSON ELECTRICAL No. 52642 COMMONWEALTH OF MASSACHUSETTS PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES DPF DRW GLV CHK RJW

PM/DM MRI/CWJ

BRANCH MANAGER WRB

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

FEEDER SCHEDULE - SHEET 1 OF 2

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936808

SHEET 239 OF 247

E-614

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 5:14:26 PM E-614
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FEEDER SCHEDULE

FEEDER	CONDUCTORS	CONDUIT	FROM	TO	REMARKS
42	8#12 (SHIELDED), 6-12/C#12 (SHIELDED)	4"	SUBSTATION P1W-SPA	P1W-PMA	LIM OVERRIDE (8#12), SHORE POWER RECEPTACLE REMOTE CONTROL (6-12#12)
43	3#4/0, 1#4G	4"	PANEL P1W-IP-MDP	P1W-PMA	INDUSTRIAL POWER RECEPTACLES
44	3#777KCMIL (DLO), 1#2G	5"	SUBSTATION P1W-SPB	P1W-PMB	SHORE POWER RECEPTACLE
45	8#12 (SHIELDED), 6-12/C#12 (SHIELDED)	4"	SUBSTATION P1W-SPB	P1W-PMB	LIM OVERRIDE (8#12), SHORE POWER RECEPTACLE REMOTE CONTROL (6-12#12)
46	3#4/0, 1#4G	4"	PANEL P1W-IP-MDP	P1W-PMB	INDUSTRIAL POWER RECEPTACLES
47	3#350KCMIL (15KV), 1#4/0G (600V)	5"	SWITCH P1W-SW	SWITCH P1E-SW	TIE FEEDER BETWEEN SWITCHES
48	SEE DRAWING E-608	4"	PANEL P1-HTCP	EMH-DE-1314	SOUTH UTILITY HEAT TRACE CIRCUITS
49	SEE DRAWING E-608	4"/1"	PANEL P1-HTCP	SEWER PIPING RISER AT COMPRESSED AIR ENCLOSURE	RUN CABLE IN 4"C FROM P1-HTCP TO EMH-P1-4. RUN CABLE IN 1"C FROM EMH-P1-4 TO RISER
50	3#10, 1#10G	3/4"	PANEL P1E-IP-MDP	SUBSTATION A/C	SUBSTATION ENCLOSURE AIR CONDITIONING
51	3#10, 1#10G	3/4"	PANEL P1W-IP-MDP	SUBSTATION A/C	SUBSTATION ENCLOSURE AIR CONDITIONING
52	SEE FEEDERS 32 AND 62	4"	SUBSTATION P1E	EMH-BC-1314	POWER CIRCUITS FOR HEATERS, ALARM BOXES, AND SEWER PUMPS AT NORTHEAST MECHANICAL UTILITY ENCLOSURES. SEE E-105 FOR CONTINUATION OF CIRCUITS TO EQUIPMENT FROM EMH-BC-1314.
53	SEE FEEDERS 32 AND 63	4"	SUBSTATION P1E	EMH-DE-1314	POWER CIRCUITS FOR HEATERS, ALARM BOXES, AND SEWER PUMPS AT SOUTHEAST MECHANICAL UTILITY ENCLOSURES. SEE E-105 FOR CONTINUATION OF CIRCUITS TO EQUIPMENT FROM EMH-DE-1314.
54	SEE FEEDERS 31 AND 60	4"	PANEL P1W-IP-MDP	EMH-BC-3334	POWER CIRCUITS FOR MINI POWER CENTER AND SEWER PUMPS AT NORTHWEST MECHANICAL UTILITY ENCLOSURES. SEE E-102 FOR CONTINUATION OF CIRCUITS TO EQUIPMENT FROM EMH-BC-3334.
55	SEE FEEDERS 31 AND 61	4"	SUBSTATION P1W	EMH-DE-3334	POWER CIRCUITS FOR HEATERS, ALARM BOXES, AND SEWER PUMPS AT SOUTHWEST MECHANICAL UTILITY ENCLOSURES. SEE E-102 FOR CONTINUATION OF CIRCUITS TO EQUIPMENT FROM EMH-DE-3334.
56	(2) CAT6 COPPER	4"	TMH-CD-0506	SUBSTATION P1E	CAT6 FOR WATER ENCLOSURE AMI METER
57	2#10	3/4"	MECHANICAL UTILITY ENCLOSURE ALARM BOX	NEAREST COMMUNICATIONS MANHOLE	SIGNAL WIRING FOR MECHANICAL UTILITY ENCLOSURE ALARM BOX FAILURE SIGNAL
58	8#10	4"	TMH-CD-3334	TMH-CD-1314	SIGNAL WIRING FOR MECHANICAL UTILITY ENCLOSURE ALARM BOX FAILURE SIGNALS
59	16#10	4"	TMH-CD-1314	SUBSTATION P1E	SIGNAL WIRING FOR MECHANICAL UTILITY ENCLOSURE ALARM BOX FAILURE SIGNALS
60	2#10, 1#10G	4"/1"	PANEL P1W-IP-MDP	PANEL P1W-MPC-PMA	4" FROM SUBSTATION TO PIER MANHOLE, 1" FROM MANHOLE TO TRANSFER SWITCH.
61	2#10, 1#10G	4"/1"	PANEL P1W-IP-MDP	PANEL P1W-MPC-PMB	4" FROM SUBSTATION TO PIER MANHOLE, 1" FROM MANHOLE TO TRANSFER SWITCH.
62	2#10, 1#10G	4"/1"	PANEL P1E-IP-MDP	PANEL P1E-MPC-PMA	4" FROM SUBSTATION TO PIER MANHOLE, 1" FROM MANHOLE TO TRANSFER SWITCH.
63	2#10, 1#10G	4"/1"	PANEL P1E-IP-MDP	PANEL P1E-MPC-PMB	4" FROM SUBSTATION TO PIER MANHOLE, 1" FROM MANHOLE TO TRANSFER SWITCH.
64	6#10, 1#10G	4"/1-1/4"	LIGHTING CONTROL CABINET	HIGH MAST LIGHTING POLE LC1	4" FROM SUBSTATION TO PIER MANHOLE, 1-1/4" FROM MANHOLE TO POLE.
65	2#6, 4#8, 1#8G	4"/1-1/2"	LIGHTING CONTROL CABINET	HIGH MAST LIGHTING POLE LC2	4" FROM SUBSTATION TO PIER MANHOLE, 1-1/2" FROM MANHOLE TO POLE.
66	2-2/C#TSP	4"/1"	SUBSTATION TRANSFORMER	15KV SWITCH CONTROLLER	4" FROM SUBSTATION TO SWITCHGEAR STRUCTURE. 1" FOR STUB-UPS INTO EQUIPMENT. TRIPS SWITCH BREAKER ON TRANSFORMER OVER-PRESSURE.
67	6-2/C#TSP	4"	SUBSTATION P1W	SUBSTATION P1E	SIGNAL WIRING FOR 15KV SWITCH TRIP ON TRANSFORMER OVER-PRESSURE.
68	12-2/C#TSP	4"	SUBSTATION P1E	15KV SWITCHGEAR STRUCTURE	SIGNAL WIRING FOR 15KV SWITCH TRIP ON TRANSFORMER OVER-PRESSURE.
69	22#10	4"	SUBSTATION P1E	MAT BUILDING	SIGNAL WIRING FOR MECHANICAL UTILITY ENCLOSURE ALARM BOX FAILURE SIGNALS AND HEAT TRACE CONTROL PANEL FAILURE SIGNAL
70	2#8, 1#8G	4"/1"	PANEL P1E-IP-MDP	P1E-MPC-RPZ	4" FROM SUBSTATION TO LANDSIDE MANHOLE, 1" FROM MANHOLE TO MINI POWER CENTER
71	4#10	4"/1"	RPZ ENCLOSURE	SUBSTATION P1E	4" FROM SUBSTATION TO LANDSIDE MANHOLE, 1" FROM MANHOLE TO RPZ ENCLOSURE. CONNECT 2#10 TO EACH ALARM BOX.
72	SEE FEEDERS 70 AND 71	4"	SUBSTATION P1E	EMH P1-4	ALARM AND POWER FEEDER WIRING FOR LANDSIDE RPZ ENCLOSURE
73	8#12 (SHIELDED)	4"	SUBSTATION P1E-SPA	P1E-PMA	LIM OVERRIDE
74	8#12 (SHIELDED)	4"	SUBSTATION P1E-SPB	P1E-PMB	LIM OVERRIDE
75	8#12 (SHIELDED)	4"	SUBSTATION P1W-SPA	P1W-PMA	LIM OVERRIDE
76	8#12 (SHIELDED)	4"	SUBSTATION P1W-SPB	P1W-PMB	LIM OVERRIDE
77	6-12/C#12 (SHIELDED)	4"	SUBSTATION P1E-SPA	P1E-PMA	SHORE POWER RECEPTACLE REMOTE CONTROL
78	6-12/C#12 (SHIELDED)	4"	SUBSTATION P1E-SPB	P1E-PMB	SHORE POWER RECEPTACLE REMOTE CONTROL
79	6-12/C#12 (SHIELDED)	4"	SUBSTATION P1W-SPA	P1W-PMA	SHORE POWER RECEPTACLE REMOTE CONTROL
80	6-12/C#12 (SHIELDED)	4"	SUBSTATION P1W-SPB	P1W-PMB	SHORE POWER RECEPTACLE REMOTE CONTROL
81	12 PAIR COPPER PHONE	4"/1"	LIFT STATION COMMS ENCLOSURE	MAT BUILDING	PHONE LINE FOR LIFT STATION ALARMS. UTILIZE 1" FROM COMMS ENCLOSURE TO TMH "I". UTILIZE 4" FROM TMH "I" TO MAT BUILDING

SYM	DESCRIPTION	DATE	APPR

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER
David Ferguson
SEAL

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES DPF DRW GLV CHK RJW
PM/DM MRI/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
FEEDER SCHEDULE - SHEET 2 OF 2

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936809
SHEET 240 OF 247
E-615

DRAWFORM REVISION: 20 AUGUST 2020

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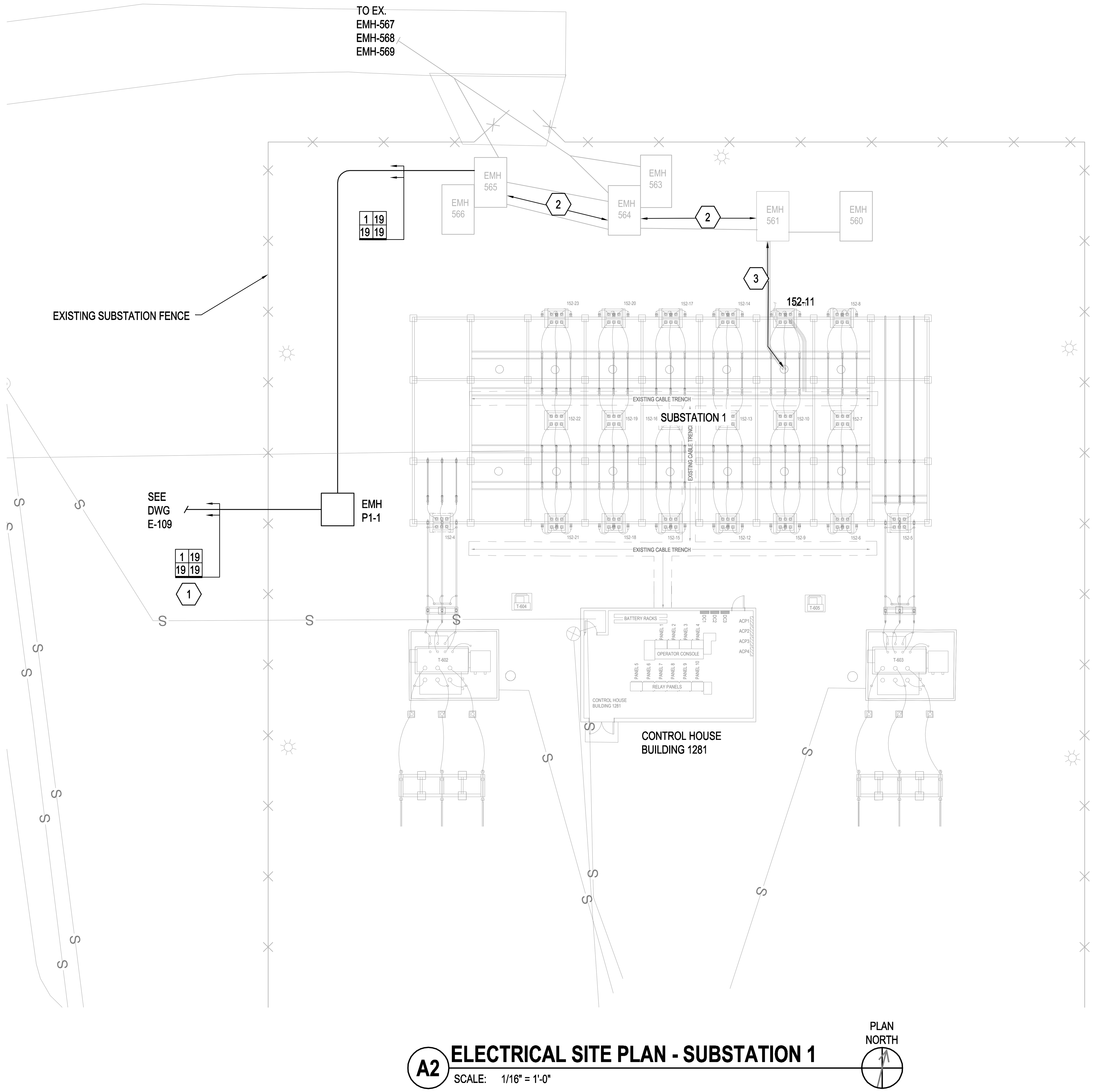
E-615

GENERAL SHEET NOTES:

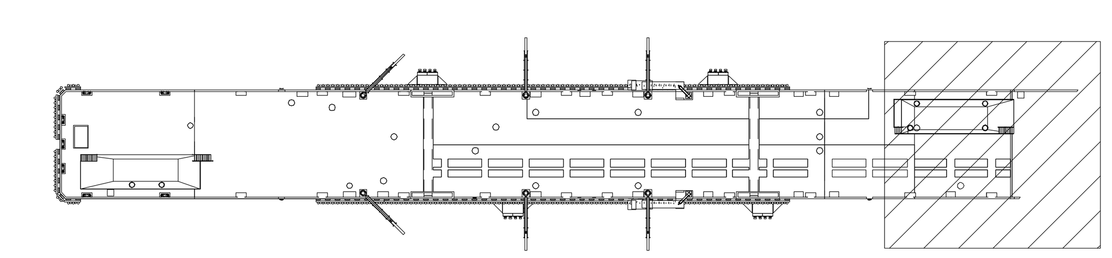
1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.

SHEET KEYNOTES:

1. PROVIDE A CONCRETE ENCASED DUCT BANK, SEE DRAWING E-109 FOR CONTINUATION.
2. ROUTE DUCT BANK INTO EXISTING MANHOLE EMH-565. ROUTE NEW FEEDER TO BREAKER 152-11 THROUGH SPARE CONDUITS BETWEEN MANHOLES EMH-564, EMH-564 AND EMH-561.
3. SEE DRAWING ES501 FOR FEEDER INSTALLATION TO BREAKER 152-11.



A2 **ELECTRICAL SITE PLAN - SUBSTATION 1**
SCALE: 1/16" = 1'-0"



KEY PLAN

1/16" = 1'-0" — 4' 0 8' 16' 32' 64'

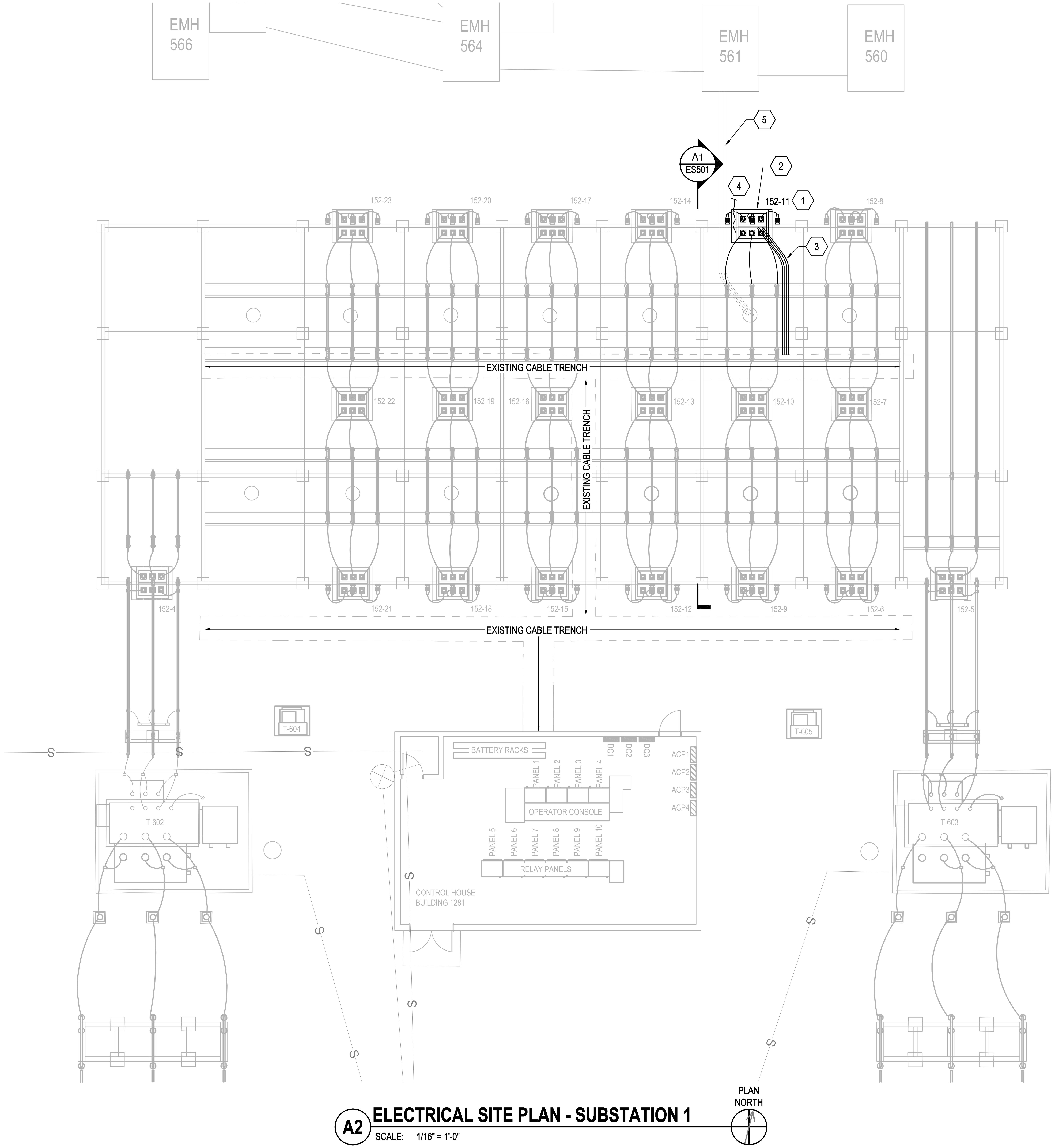
APPROVED	James E. Donahue, R.A.
FOR COMMANDER NAVFAC	3/5/2026
ACTIVITY	Approved by Frank Cole USCG MASI
SATISFACTORY TO	DATE 06 FEB 2026
DES CSA	DRW HBF, CHK RJW
PM/DM	MRI/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
ELECTRICAL SITE PLAN - SUBSTATION 1

SCALE:	AS NOTED
EPROJECT NO.:	1826479
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12936810
SHEET	241 OF 247

ES101

DRAWFORM REVISION: 25 AUGUST 2020



A2 ELECTRICAL SITE PLAN - SUBSTATION 1

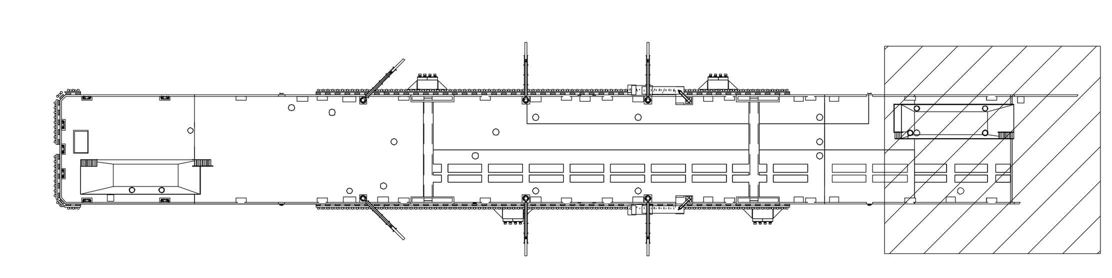
SCALE: 1/16" = 1'-0"

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001

SHEET KEYNOTES:

1. PROVIDE A 15 kV, 1200A, 40 kAIC EXTERIOR, PAD-MOUNTED DISTRIBUTION CIRCUIT BREAKER IN SPARE LOCATION 152-11. BASIS-OF-DESIGN IS SIEMENS SDV7™.
2. PROVIDE AN 8" CONCRETE PAD BENEATH CIRCUIT BREAKER SUPPORT LEGS.
3. PROVIDE (1) 30-CONDUCTOR #12 AWG CU CONTROL CABLE AND (4) 4-CONDUCTOR #10 AWG CU POWER CABLES TO BUILDING 1281. ROUTE CABLES IN SEPARATE 2" RGS UNDERGROUND CONDUITS FROM BREAKER CONTROL SECTION TO EXISTING CABLE TRENCH. ROUTE THROUGH TRENCH INTO BUILDING 1281 RAISED FLOOR.
4. PROVIDE #4/0 BARE CU GROUND WIRE FROM CIRCUIT BREAKER TO EXISTING SUBSTATION GROUND GRID, CONNECT TO GROUND GRID WITH AN EXOTHERMIC WELD.
5. EXISTING 5" CONDUITS TO MANHOLE, ROUTE 13.8 kV FEEDER TO MANHOLE IN EXISTING CONDUIT. SEE DRAWING ES101 FOR CONTINUATION.



KEY PLAN

1/8" = 1'-0"

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON ELECTRICAL No. 52642
REGISTERED PROFESSIONAL ENGINEER

Stantec

A/E INFO

APPROVED James E. Donahue, R.A.
FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES CSA	DRW HBF	CHK RJW
PM/DM		MRI/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

SUBSTATION 1 PLAN VIEW

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936811
SHEET 242 OF 247

ES102

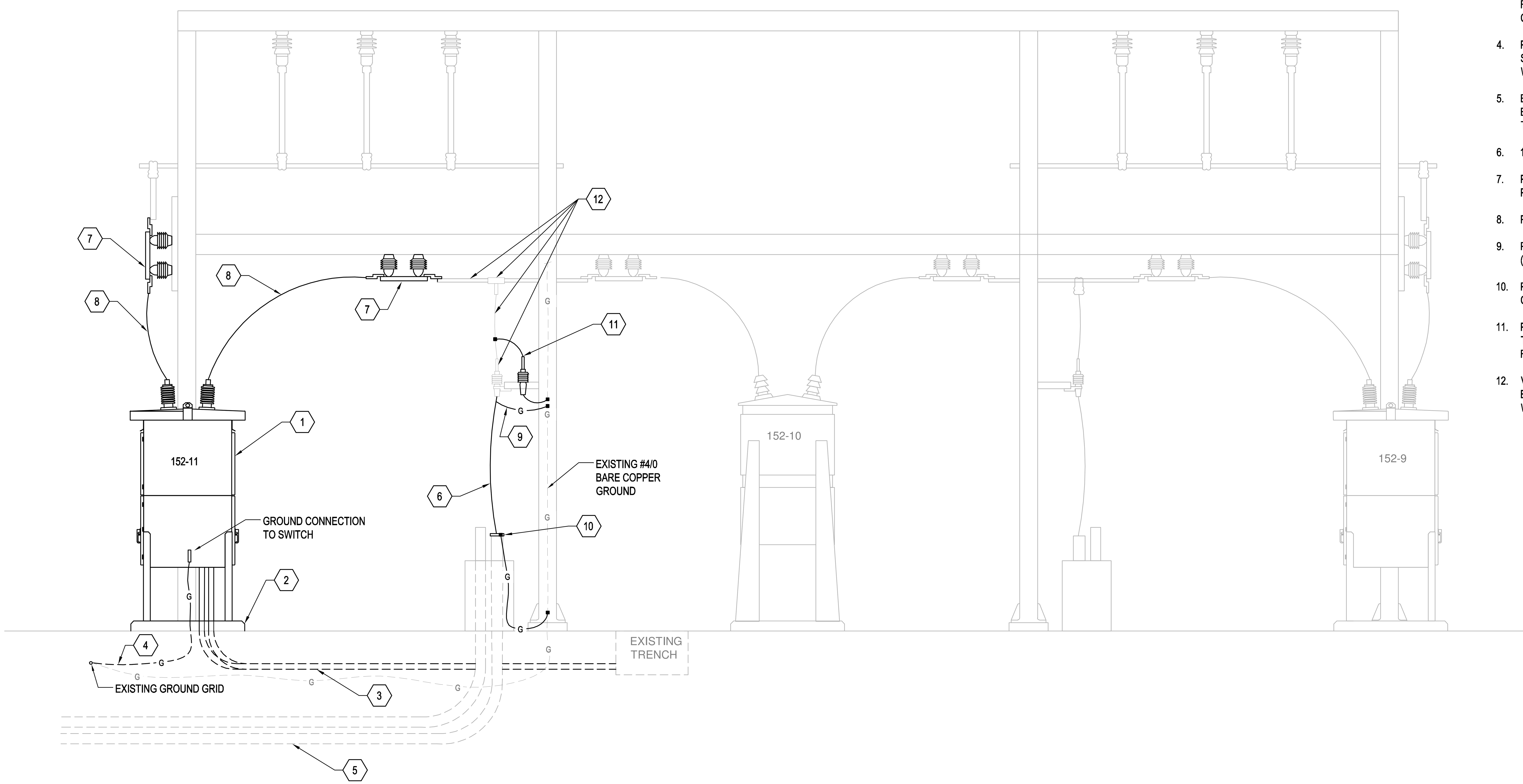
DRAWFORM REVISION: 25 AUGUST 2020

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.
2. PROVIDE SUITABLE CONNECTORS AND HARDWARE FOR ALL CONDUCTOR TERMINATIONS INDICATED.

SHEET KEYNOTES:

1. PROVIDE A 15 kV, 3000A, 40 kAIC EXTERIOR, PAD-MOUNTED DISTRIBUTION CIRCUIT BREAKER IN SPARE LOCATION 152-11. BASIS-OF-DESIGN IS SIEMENS SDV7™.
2. PROVIDE AN 8" CONCRETE PAD BENEATH CIRCUIT BREAKER SUPPORT LEGS.
3. PROVIDE (1) 30-CONDUCTOR #12 AWG CU CONTROL CABLE AND (4) 4-CONDUCTOR #10 AWG CU POWER CABLES TO BUILDING 1281. ROUTE CABLES IN SEPARATE 2" RGS UNDERGROUND CONDUITS FROM BREAKER CONTROL SECTION TO EXISTING CABLE TRENCH. ROUTE THROUGH TRENCH INTO BUILDING 1281 RAISED FLOOR.
4. PROVIDE #4/0 BARE CU GROUND WIRE FROM CIRCUIT BREAKER TO EXISTING SUBSTATION GROUND GRID, CONNECT TO GROUND GRID WITH AN EXOTHERMIC WELD.
5. EXISTING 5" CONDUITS TO MANHOLE, ROUTE 13.8 kV FEEDER TO MANHOLE IN EXISTING CONDUIT. SEE ELECTRICAL SITE PLAN FOR CONTINUATION OF FEEDER TO PIER 1.
6. 15 kV FEEDER TO PIER 1, SEE SINGLE LINE DIAGRAM.
7. PROVIDE 15 kV, 1200A, SPST, 40 kAIC, HOOKSTICK OPERATED CUTOUT WITH PORCELAIN INSULATORS AND 4-HOLE TERMINAL PADS, (TYP 3).
8. PROVIDE 1000 KCMIL, ASTMB3 SOFT OR ANNEALED, BARE COPPER WIRE, (TYP 3).
9. PROVIDE CONNECTION FROM CABLE SHIELD TO BARE COPPER GROUND WIRE, (TYP 3).
10. PROVIDE A 5" GROUNDING BUSHING ON EXISTING CONDUIT WITH A #4/0 BARE COPPER GROUND WIRE CONNECTION TO EXISTING GROUND WIRE.
11. PROVIDE (3) 12 KV LIGHTNING ARRESTORS AND AND CONNECTIONS AS SHOWN TO PHASE CONDUCTOR AND GROUND. PROVIDE ALL REQUIRED ACCESSORIES FOR MOUNTING AND CONNECTIONS.
12. VERIFY THE INTEGRITY OF EXISTING BUSBAR, INSULATORS, AND SUPPORTS BEFORE ENERGIZING. REPORT ISSUES TO THE ENGINEER BEFORE PROCEEDING WITH CABLE CONNECTIONS.



A1 BREAKER 152-11 ELEVATION VIEW
SCALE: 3/8" = 1'-0"

3/8" = 1'-0"   4' 0 4' 8'

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
CI CORE - HAMPTON ROADS
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
BREAKER 152-11 ELEVATION VIEW

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
PROFESSIONAL ENGINEER

Stantec

FOR COMMANDER NAVFAC 3/5/2026
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES CSA	DRW HBF / CHK RJW
PM/DM	MRI/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN
SCALE:	AS NOTED
EPROJECT NO.:	1826479
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12936812
SHEET	243 OF 247

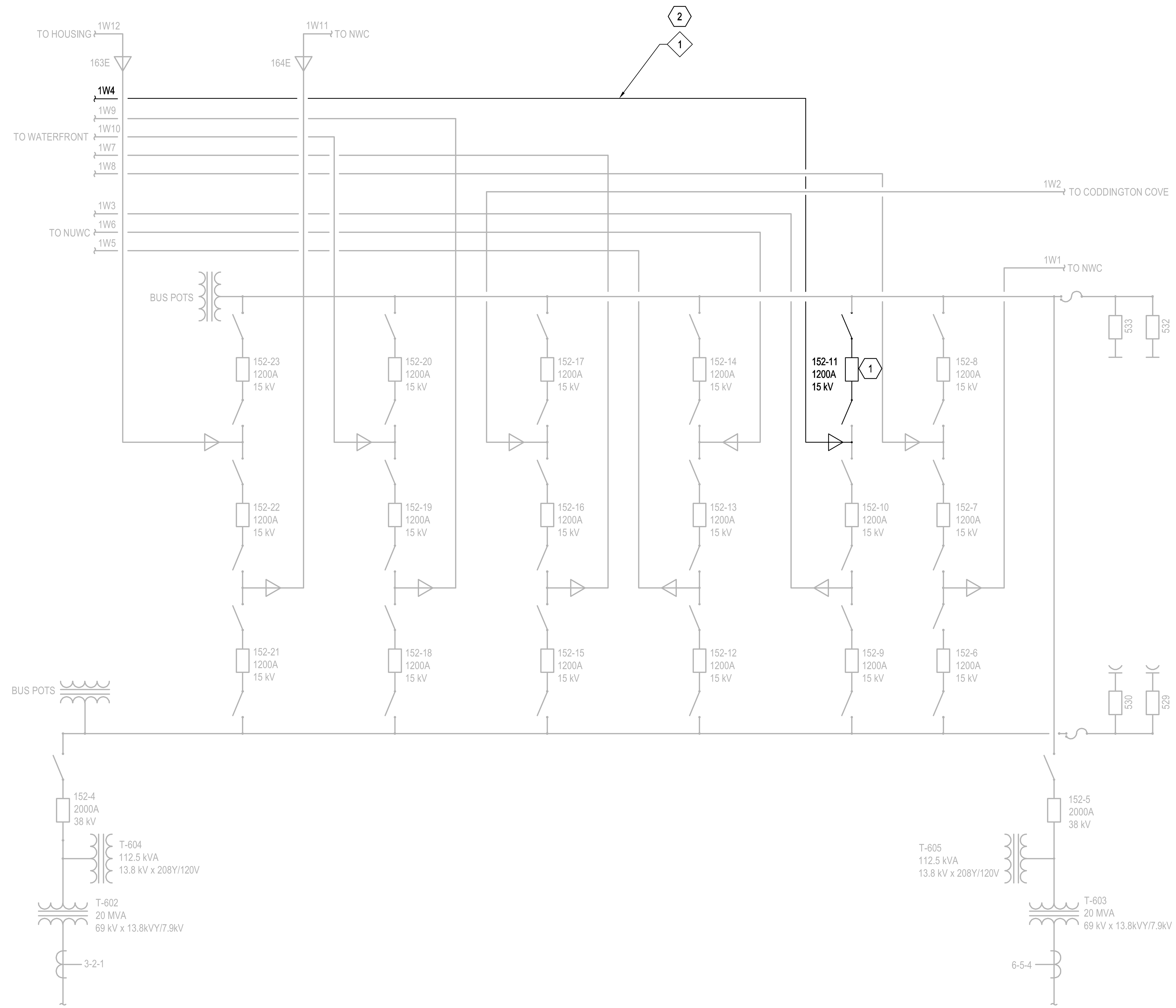
ES501

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.
2. OBTAIN THE SERVICES OF SEL (SCHWEITZER ENGINEERING LABORATORIES) TO PROVIDE TECHNICAL SUPPORT, SUBMITTALS AND PROGRAMMING OF ALL CONTROLS AND WIRING ASSOCIATED WITH THE BREAKER 152-11 INSTALLATION.

SHEET KEYNOTES:

1. PROVIDE A 15 kV, 3000A, 40 kAIC EXTERIOR, PAD-MOUNTED DISTRIBUTION CIRCUIT BREAKER IN SPARE LOCATION 152-11. BASIS-OF-DESIGN IS SIEMENS SDV7™.
2. PROVIDE FEEDER TO BREAKER PER DRAWING E-601, SEE DRAWING E-614 FOR FEEDER SIZE. ROUTE THROUGH MANHOLES AS SHOWN ON DRAWING ES101.



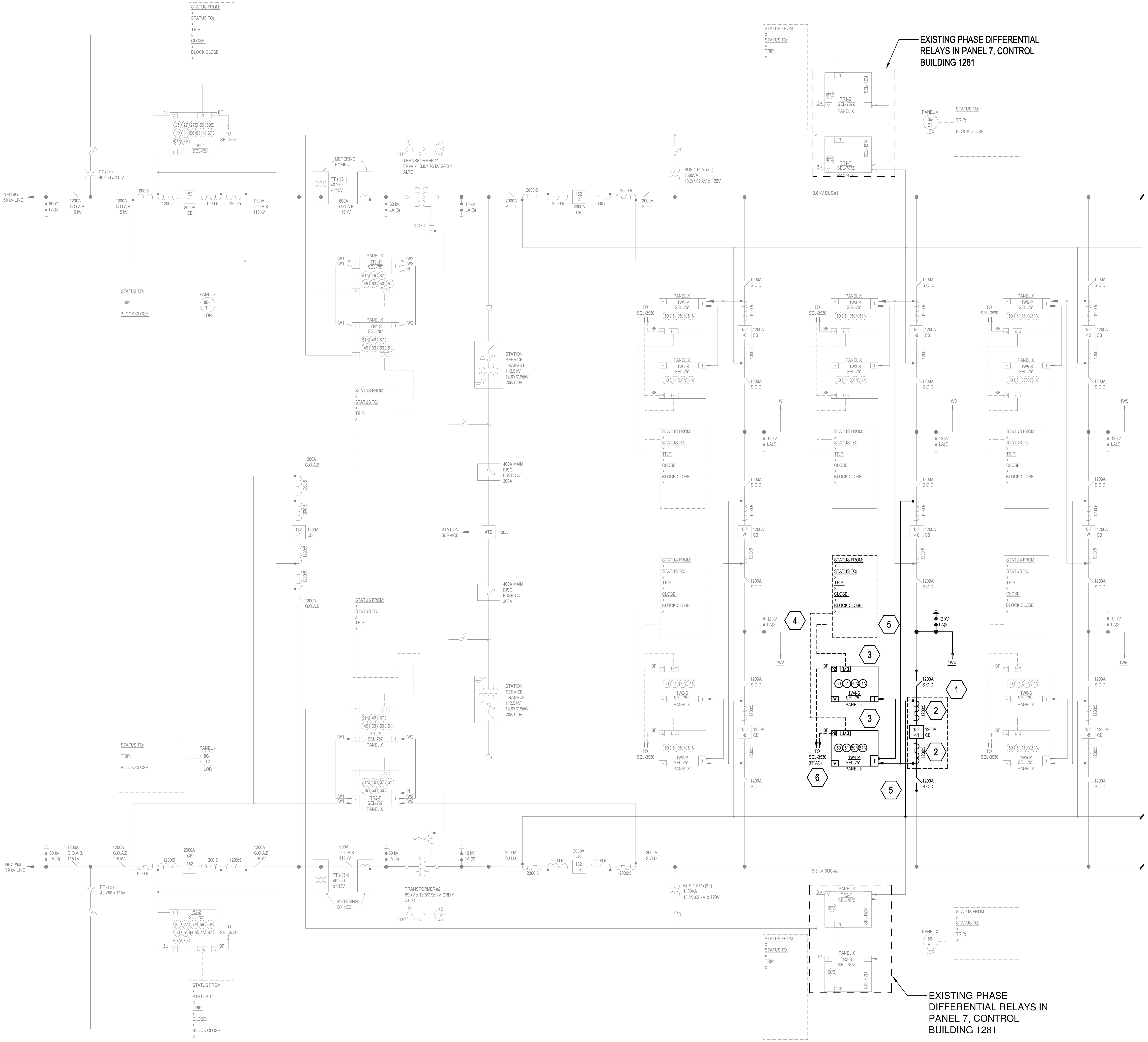
A1 SUBSTATION 1 SINGLE LINE DIAGRAM
NOT TO SCALE

APPROVED	FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY	Approved by Frank Cole USCG MASI
SATISFACTORY TO	DATE 06 FEB 2026
DES CSA	DRW HBF CHK RJW
PM/DM	MRI/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
SUBSTATION 1 SINGLE LINE DIAGRAM

SCALE:	AS NOTED
EPROJECT NO.:	1826479
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12936813
SHEET	244 OF 247

ES601



A1 **BREAKER 152-11 SCHEMATIC DIAGRAM**

NOT TO SCALE

GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.
2. OBTAIN THE SERVICES OF SEL (SCHWEITZER ENGINEERING LABORATORIES) TO PROVIDE TECHNICAL SUPPORT, SUBMITTALS AND PROGRAMMING OF ALL CONTROLS AND WIRING ASSOCIATED WITH THE BREAKER 152-11 INSTALLATION.

SHEET KEYNOTES:

1. PROVIDE A 15 kV, 3000A, 40 kAIC EXTERIOR, PAD-MOUNTED DISTRIBUTION CIRCUIT BREAKER IN SPARE LOCATION 152-11. BASIS-OF-DESIGN IS SIEMENS SDV7™.
2. PROVIDE BUSHING CURRENT TRANSFORMERS INTEGRAL TO 15 KV CIRCUIT BREAKER LINE AND LOAD BUSHINGS (TYP 6).
3. PROVIDE (2) SCHWEITZER SEL-751 RELAYS IN PANEL 5 IN BUILDING 1281, SEE DRAWING ES102 FOR LOCATION OF PANEL 5.
4. SEE DRAWING ES603 FOR DETAILED WIRING DIAGRAM OF SEL-751 RELAY IO.
5. DISCONNECT AND REMOVE EXISTING CT WIRING AS NEEDED TO INTEGRATE CT'S FOR CIRCUIT BREAKER 152-11 INTO EXISTING BUS DIFFERENTIAL RELAY PROTECTION IN RELAYS TR1-P, TR1-S, TR2-P & TR2-S, MATCH EXISTING WIRE TYPE AND QUANTITY BETWEEN CT'S.
6. SEE DRAWING ES604 FOR COMMUNICATIONS WIRING DIAGRAM OF SEL-751 RELAY.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
REGISTERED PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES CSA	DRW HBF	CHK RJW
PM/DM		MRI/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI

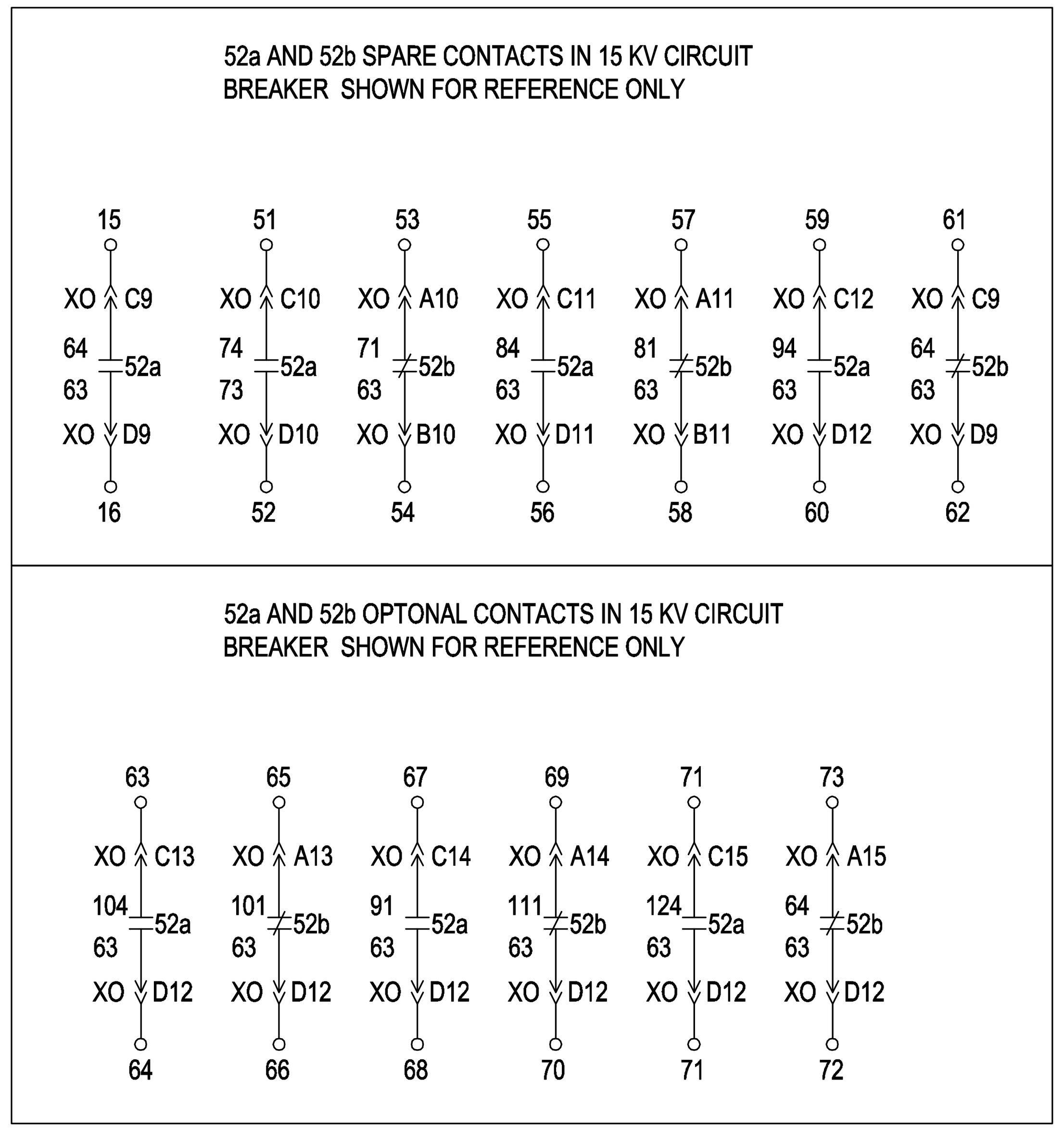
USCG OCR HOMEPORT INITIATIVE OPC PIER

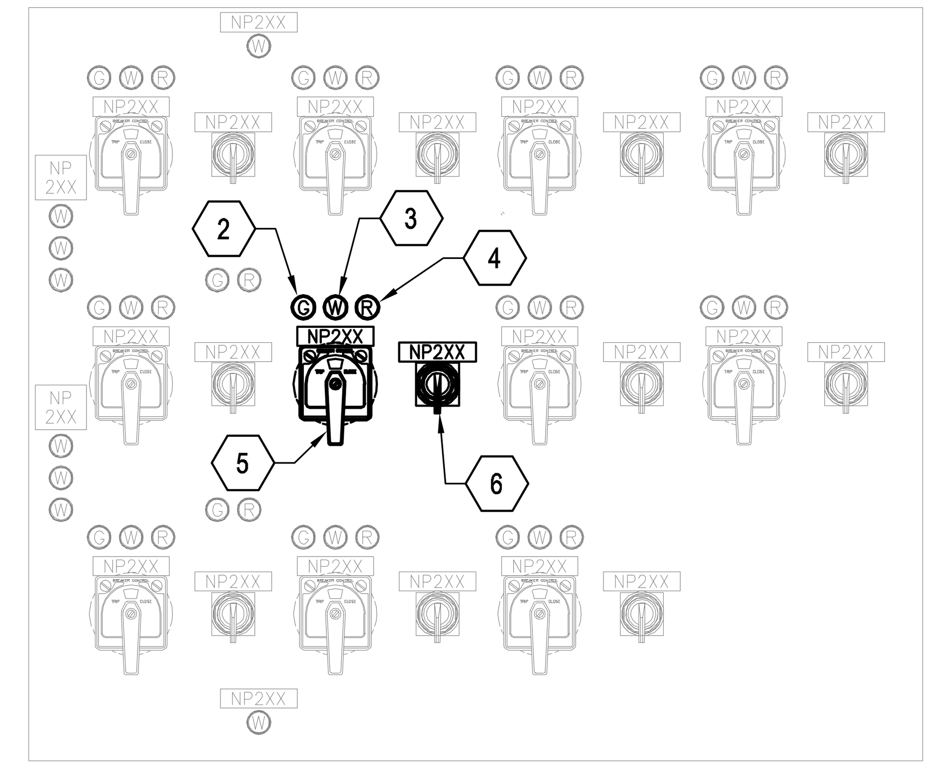
BREAKER 152-11 SCHEMATIC DIAGRAM

SCALE:	AS NOTED
EPROJECT NO.:	1826479
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12936814
SHEET 245 OF	247

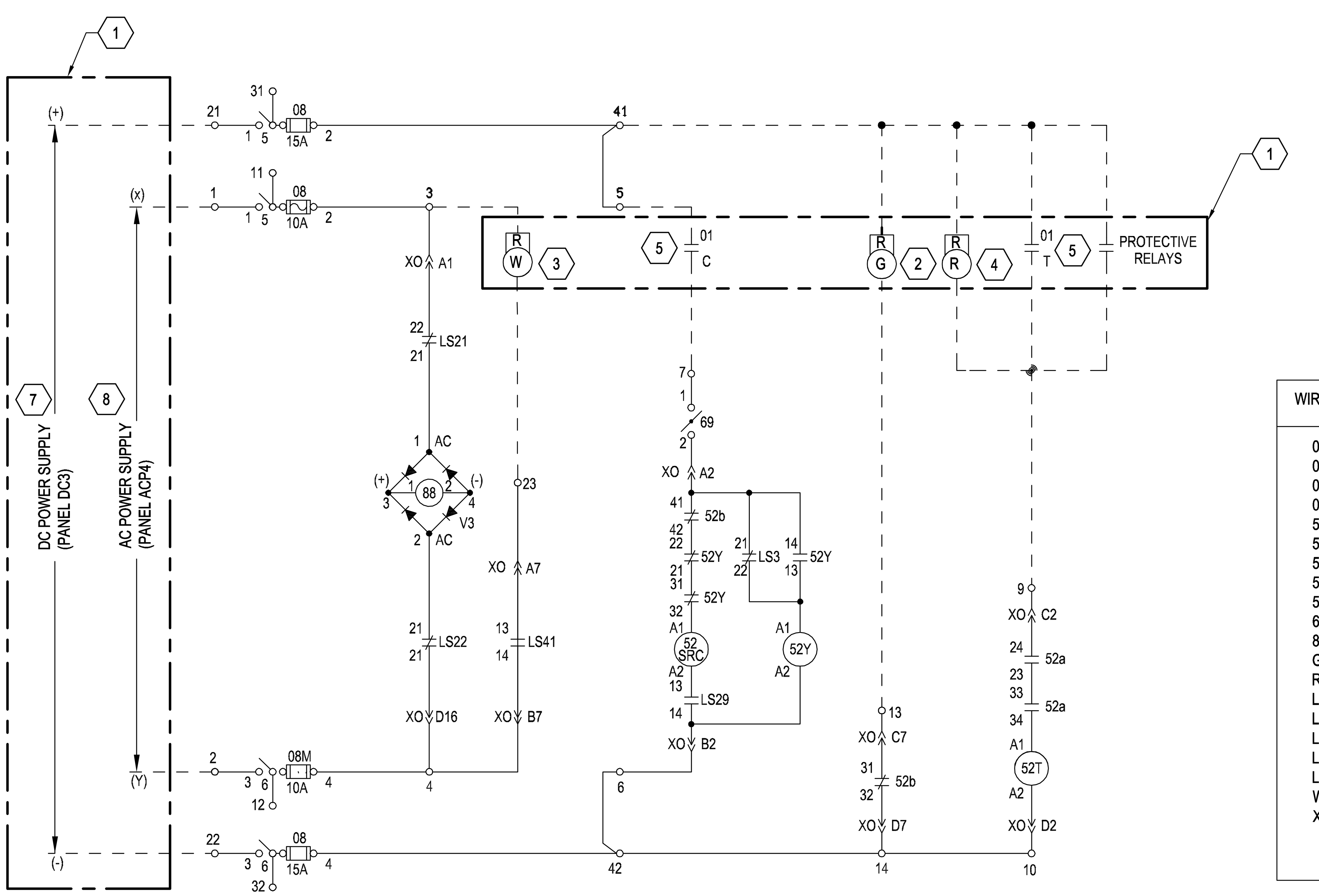
ES602

DRAWFORM REVISION: 25 AUGUST 2020





EXISTING PANEL 2- PLATE 2



GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.
2. OBTAIN THE SERVICES OF SEL (SCHWEITZER ENGINEERING LABORATORIES) TO PROVIDE TECHNICAL SUPPORT, SUBMITTALS AND PROGRAMMING OF ALL CONTROLS AND WIRING ASSOCIATED WITH THE BREAKER 152-11 INSTALLATION.

SHEET KEYNOTES:

1. ITEMS WITHIN THIS AREA ARE LOCATED IN BUILDING 1281. DASHED LINES INDICATE FIELD WIRING BETWEEN 15 KV CIRCUIT BREAKER 152-11 AND BUILDING 1281 CONTROL PANELS AND POWER PANELS, SEE KEYNOTE 3, DRAWING ES501.
2. PROVIDE AN LED INDICATING LAMP WITH A GREEN LENS CAP, "GE" MODEL # ET16. INSTALL IN EXISTING PANEL 2 PLATE #2 IN CONTROL BUILDING 1281.
3. PROVIDE AN LED INDICATING LAMP WITH A WHITE LENS CAP, "GE" MODEL # ET16. INSTALL IN EXISTING PANEL 2 PLATE #2 IN CONTROL BUILDING 1281.
4. PROVIDE AN LED INDICATING LAMP WITH A RED LENS CAP, "GE" MODEL # ET16. INSTALL IN EXISTING PANEL 2 PLATE #2 IN CONTROL BUILDING 1281.
5. PROVIDE AN "ELECTROSWITCH" SERIES 24 BREAKER CONTROL SWITCH IN EXISTING PANEL 2 PLATE #2 IN CONTROL BUILDING 1281.
6. PROVIDE A "SCHNEIDER ELECTRIC" 9001 SERIES SELECTOR SWITCH IN EXISTING PANEL 2 PLATE #2 IN CONTROL BUILDING 1281. COORDINATE WIRING WITH THE SEL TECHNICIAN, PROVIDE AS RECOMMENDED.
7. PROVIDE CIRCUIT FROM EXISTING PANEL DC3, COORDINATE VOLTAGE/AMPERAGE WITH BREAKER 152-11 CONTROL CIRCUIT REQUIREMENTS.
8. PROVIDE CIRCUIT FROM EXISTING PANEL ACP4, COORDINATE VOLTAGE/AMPERAGE WITH BREAKER 152-11 CONTROL CIRCUIT REQUIREMENTS.

WIRING DIAGRAM LEGEND

01/C	CONTROL SWITCH CLOSE (REMOTE)
01/T	CONTROL SWITCH TRIP (REMOTE)
08	CLOSE AND TRIP POWER DISCONNECT
08M	MOTOR POWER DISCONNECT
52a	AUXILIARY SWITCH, OPEN WHEN BREAKER IS OPEN
52b	AUXILIARY SWITCH, CLOSED WHEN BREAKER IS OPEN
52SR	CLOSING SPRING RELEASE COIL
52T	OPENING SPRING RELEASE COIL
52Y	ANTI-PUMP RELAY
69	CLOSING CUTOUT SWITCH
88	SPRING CHARGE MOTOR
G	GREEN INDICATING LIGHT (REMOTE)
R	RED INDICATING LIGHT (REMOTE)
LS3	CLOSING SPRING POSITION SWITCH, CLOSED WHEN CLOSING SPRING IS DISCHARGED
LS9	CLOSING SPRING POSITION SWITCH, OPEN WHEN CLOSING SPRING IS DISCHARGED
LS21	MOTOR CUTOFF SWITCH, CLOSED WHEN CLOSING SPRING IS DISCHARGED
LS22	MOTOR CUTOFF SWITCH, CLOSED WHEN CLOSING SPRING IS DISCHARGED
LS41	CLSOING SPRING POSITION SWITCH, OPEN WHEN CLOSING SPRING IS DISCHARGED
W	WHITE INDICATING LIGHT (REMOTE)
XO	PLUG CONNECTOR (OPERATOR CONNECTIONS)

NOTE:
SCHEMATIC SHOWN WITH CLOSING SPRINGS DISCHARGED AND CIRCUIT BREAKER OPEN

A1 BREAKER 152-11 DIGITAL WIRING DIAGRAM

NOT TO SCALE

NAVFAC

COMMONWEALTH OF MASSACHUSETTS — DAVID P FERGUSON ELECTRICAL No. 52642 — PROFESSIONAL ENGINEER

Stantec

APPROVED — FOR COMMANDER NAVFAC 3/5/2026

Approved by Frank Cole USCG MASI

SATISFACTORY TO — DATE 06 FEB 2026

DES CSA — DRW HBF — CHK RJW
PM/DM MRI/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

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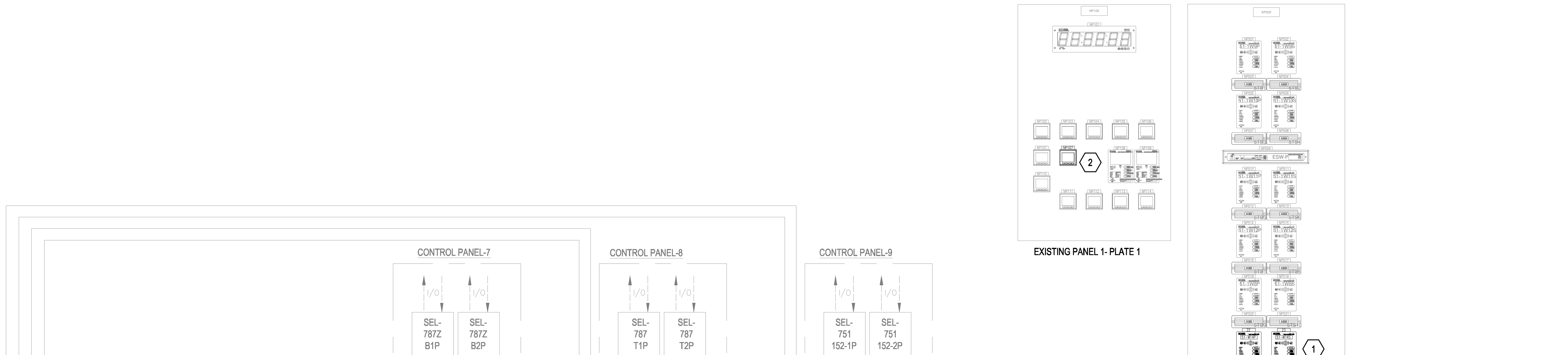
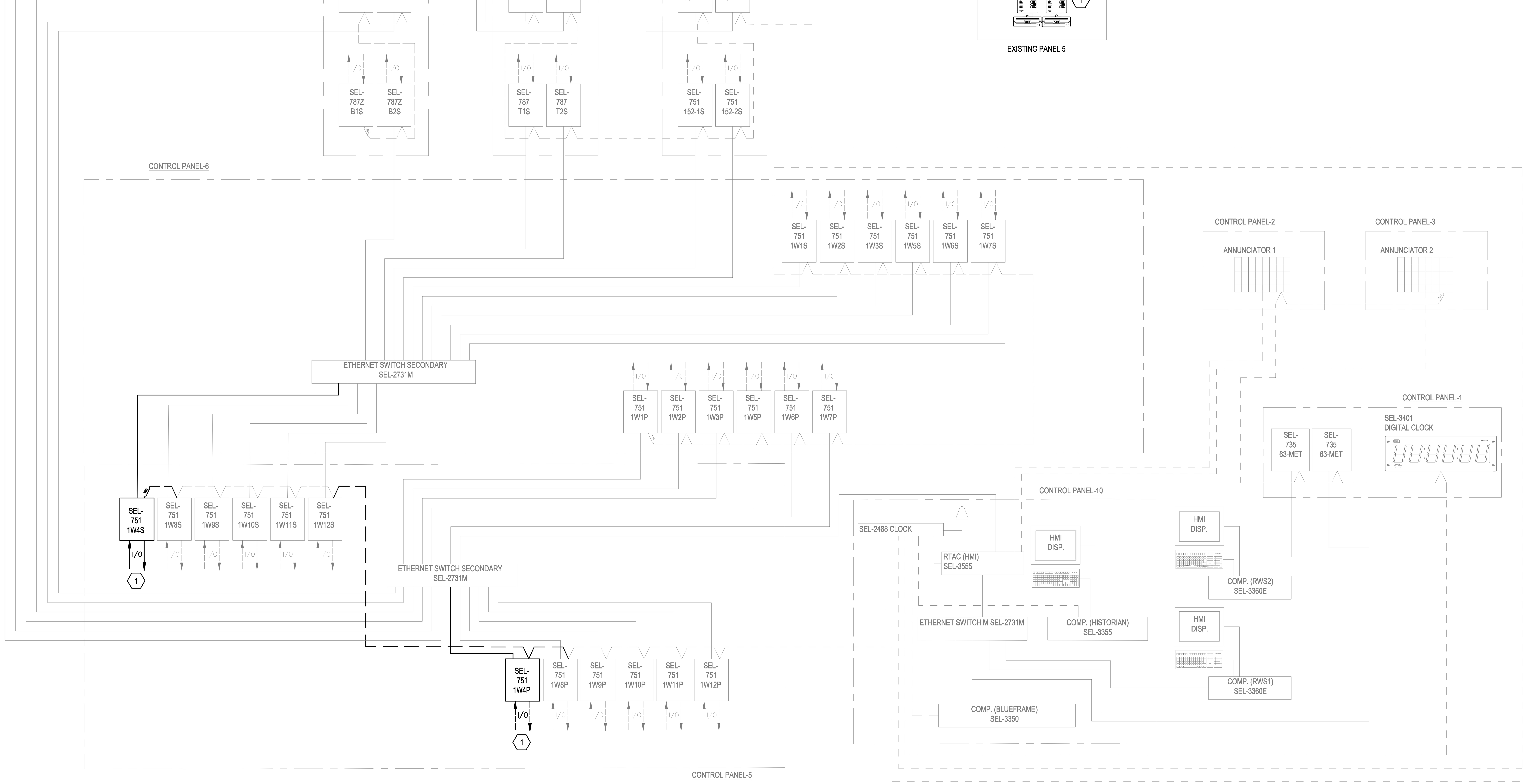
USCG OCR HOMEPORT INITIATIVE OPC PIER
BREAKER 152-11 DIGITAL CONTROL DIAGRAM

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936815
SHEET 246 OF 247

ES603





GENERAL SHEET NOTES:

1. FOR ELECTRICAL ABBREVIATIONS, LEGEND & GENERAL NOTES, SEE DRAWING E-001.
2. OBTAIN THE SERVICES OF SEL (SCHWEITZER ENGINEERING LABORATORIES) TO PROVIDE TECHNICAL SUPPORT, SUBMITTALS AND PROGRAMMING OF ALL CONTROLS AND WIRING ASSOCIATED WITH THE BREAKER 152-11 INSTALLATION.

SHEET KEYNOTES:

1. PROVIDE AND INSTALL A SCHWEITZER SEL-751 RELAY IN SPARE LOCATION IN CONTROL PANEL 5. INTERCONNECT TO EXISTING COMMUNICATIONS SYSTEM AS SHOWN, MATCH EXISTING CONDUCTOR TYPES AND QUANTITIES. CONNECT TO SPARE PORT IN EXISTING ETHERNET SWITCH. SEE DRAWING ES603 FOR DIGITAL WIRING DIAGRAM.
2. PROVIDE A "PHOENIX CONTACT" EEM-MA770- MULTI-FUNCTIONAL ENERGY MEASURING DEVICE TO DISPLAY BREAKER 152-11 STATISTICS IN EXISTING PANEL 1 PLATE #1. COORDINATE WIRING WITH THE SEL TECHNICIAN, PROVIDE AS RECOMMENDED.

A1 BREAKER 152-11 COMMUNICATIONS WIRING DIAGRAM

NOT TO SCALE

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
DAVID P FERGUSON
ELECTRICAL
No. 52642
REGISTERED PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES CSA	DRW HBF / CHK RJW
PM/DM	MRI/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

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BREAKER 152-11 COMMUNICATIONS DIAGRAM

SCALE:	AS NOTED
EPROJECT NO.:	1826479
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12938716
SHEET 247 OF	247

ES604

DRAWFORM REVISION: 25 AUGUST 2020